

UWO MATH 9144B WINTER 2026

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These are “notes” written for the course MATH 9144B Homological Algebra at the University of Western Ontario during the Winter 2026 term. The main text for the course is Weibel’s *An Introduction to Homological Algebra* [Wei94], and I intend for the course to roughly follow parts of this text. This document aims to supplement the presentation of loc. cit. , which focuses much of its attention on left/right R -modules where R is an “associative ring”¹, by discussing general definitions.

Readers are recommended to read this document using a PDF reader (e.g. Adobe Acrobat/Reader) that supports Link Annotations (i.e. the boxes, usually red/green, that are rendered when using `ref` and `cite` commands).

¹The introduction of [Wei94] assumes that the reader has a background in graduate algebra "based on a text such as *Jacobson’s Basic Algebraic I*". Jacobson defines a ring to be both associative and unital (but not necessarily commutative), so ostensibly [Wei94] adopts this same convention.

My “notes” for Homological Algebra, Category Theory, Abstract Algebra, etc. may be of interest.

0.1. **Disclaimer.** The definitions and statements in this document are generally written less as basic introductions to any given concept and more so to present them in generality. As such, they are not necessarily linearly presented. Readers should judiciously decide what to read and what to skip.

Many statements are based on AI generated ones, and errors may be abound due to my own lack of care when verifying them. Readers are advised to exercise caution when citing claims, which may be erroneous, in these notes.

The contents of these notes may change constantly.

1. BASIC CATEGORY THEORY

1.1. Categories.

Definition 1.1.1 (Category). A *category* $\mathcal{C}$ consists of the following data:

- A class of *objects* denoted $\text{Ob}(\mathcal{C})$.
- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, a class

$$\text{Hom}_{\mathcal{C}}(X, Y)$$

of *morphisms* (also called *arrows* or *homs*). If the category $\mathcal{C}$ is clear, then this *hom-class* is also denoted by $\text{Hom}(X, Y)$. It may also be denoted by $\text{hom}_{\mathcal{C}}(X, Y)$ or $\text{hom}(X, Y)$, especially to distinguish from other types of hom’s (e.g. internal hom’s)

- For each triple of objects X, Y, Z , a composition law

$$\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z),$$

denoted $(g, f) \mapsto g \circ f$.

- For each object X , an *identity morphism*

$$\text{id}_X \in \text{Hom}_{\mathcal{C}}(X, X).$$

These data satisfy the following axioms:

- (Associativity) For all morphisms $f \in \text{Hom}_{\mathcal{C}}(X, Y)$, $g \in \text{Hom}_{\mathcal{C}}(Y, Z)$, and $h \in \text{Hom}_{\mathcal{C}}(Z, W)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- (Identity) For all $f \in \text{Hom}_{\mathcal{C}}(X, Y)$,

$$\text{id}_Y \circ f = f = f \circ \text{id}_X.$$

One often writes $X \in \mathcal{C}$ synonymously with $X \in \text{Ob}(\mathcal{C})$, i.e. to denote that X is an object of $\mathcal{C}$.

We may call a category as above an *ordinary category* to distinguish this notion from the notions of *categories enriched in monoidal categories* or higher/ n -categories. (♠ TODO: TODO: define n -categories)

A category as defined above may be called a *large category* or a *class category* to emphasize that the hom-classes may be proper classes rather than sets (note, however, that the possibility that hom-classes are sets is not excluded for large categories). Accordingly, a *category* may often refer to a locally small category (Definition 1.1.2), which is a category whose hom-classes are all sets.

Definition 1.1.2 (Locally small category). A (large) category (Definition 1.1.1) $\mathcal{C}$ is called a *locally small category* if for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a (small) *set* (as opposed to a proper class). In other words, each hom-class is a set and may even be called a *hom-set*.

In some contexts, a locally small category may simply be called a *category*, especially when genuinely large categories are not considered.

A category $\mathcal{C}$ is called a *small category* if it is a locally small category and the class $\text{Ob}(\mathcal{C})$ of objects is a set.

Given a universe (Definition A.0.3) U , we can define the notion of a *U -locally small category* and of a *U -small category* similarly [?, Exposé I, Définition 1.1.]. More explicitly,

1. a U -locally small category is a category such that for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a U -set.
2. a U -small category is a category such that $\text{Ob}(\mathcal{C})$ is a U -set and for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a U -set; in particular the collection of all objects and morphisms in a U -small category is a U -set.

A *U -category* refers to a U -locally small category.

Remark 1.1.3. Many “concrete” categories considered in “classical mathematics” or outside of more “abstract” category theory tend to be locally small. For example, the categories of sets, groups, R -modules, vector spaces, topological spaces, schemes, manifolds, sheaves on “small enough” sites are all locally small.

Example 1.1.4. Here is an example of a “boring” category:

1. There is only one object, say X .
2. There is only one morphism, the identity $\text{id}_X : X \rightarrow X$.

The composed morphism $\text{id}_X \circ \text{id}_X$ is then just id_X , and associativity automatically holds.

Example 1.1.5. Any poset (Definition C.0.17) $(P, \leq)$ induces a category (Definition 1.1.1) — let the objects be the elements of the set P , and let the morphisms/arrows be given as follows: there is a unique arrow $a \rightarrow b$ whenever $a \leq b$. Composition of arrows works as follows: given arrows $a \rightarrow b$ and $b \rightarrow c$, the composed arrow $a \rightarrow b \rightarrow c$ will be the unique arrow $a \rightarrow c$ corresponding to $a \leq c$. See Lemma C.0.18

Example 1.1.6. Here are common examples of categories:

1. The category of sets (Definition 1.1.7), whose objects are sets and whose morphisms are set maps/functions (Definition C.0.1).
2. The category of groups (Definition 1.1.8), whose objects are groups (Definition C.0.3) and whose morphisms are group homomorphisms (Definition C.0.4).
3. The category of abelian groups (Definition 1.1.8), whose objects are abelian groups (Definition C.0.3) and whose morphisms are group homomorphisms (Definition C.0.4).
4. The category of topological spaces (Definition 1.1.9), whose objects are topological spaces (Definition C.0.5) and whose morphisms are continuous maps between topological spaces (Definition C.0.6).
5. The category of pointed topological spaces (Definition 1.1.10), whose objects (X, x) are pointed topological spaces (Definition 1.1.10) and whose morphisms $(X, x) \rightarrow (Y, y)$ are continuous maps (Definition C.0.6) $f : X \rightarrow Y$ such that $f(x) = y$.
6. The category of vector spaces (Definition 1.1.11) over a fixed field (Definition C.0.12) k whose objects are vector spaces (Definition C.0.14) over k and whose morphisms are k -linear maps (Definition C.0.15).
7. The category of finite dimensional vector spaces (Definition 1.1.11) over a fixed field (Definition C.0.12) k whose objects are finite dimensional (Definition C.0.16) vector spaces (Definition C.0.14) over k and whose morphisms are k -linear maps (Definition C.0.15).
8. Given a ring (Definition C.0.7) R , the category of (either left or right) R -modules (Definition 2.1.3) whose objects are R -modules (Definition 2.1.1) and whose morphisms are R -module homomorphisms (Definition 2.1.2).
9. The category of rings (Definition 1.1.12) whose objects are rings (Definition C.0.7) and whose morphisms are ring homomorphisms (Definition C.0.13).
10. The category of commutative rings (Definition 1.1.12) whose objects are commutative rings (Definition C.0.9) and whose morphisms are ring homomorphisms (Definition C.0.13).
11. The category of small categories (Definition 1.4.1), whose objects are the small categories (Definition 1.1.2) and whose morphisms are functors (Definition 1.2.2).
12. Given a fixed topological space X , the category $\text{Open}(X)$ of open subsets of X (Definition 7.1.2), whose objects are the open subsets of X and whose morphisms $U \rightarrow V$ are given exactly by inclusions $U \subseteq V$. More precisely, for each inclusion $U \subseteq V$ of open subsets of X , there is a unique morphism $U \rightarrow V$, and the composition $U \rightarrow V \rightarrow W$ is the unique morphism $U \rightarrow W$ corresponding to the inclusion $U \subseteq W$.

Definition 1.1.7. The category of sets is the (locally small) (Definition 1.1.2) category (Definition 1.1.1)

- whose objects are sets, and
- whose morphisms $X \rightarrow Y$ are set functions (Definition C.0.1) $X \rightarrow Y$.

The category of sets is often denoted by notations such as **Set**, **Set**, **Sets**, **Sets**, **(Set)**, **(Set)**, **(Sets)**, **(Sets)**.

Given a Grothendieck universe (Definition A.0.3) $\mathcal{U}$, the category of (Definition 2.1.3) $\mathcal{U}$ -sets (Definition A.0.3) may be denoted by notations such as **Set** $_{\mathcal{U}}$. The category of $\mathcal{U}$ -sets is $\mathcal{U}$ -locally small (Definition 1.1.2) and is $\mathcal{V}$ -small (Definition 1.1.2) for any Grothendieck universe $\mathcal{V}$ containing $\mathcal{U}$.

Definition 1.1.8. 1. The *category of groups* is the locally small (Definition 1.1.2) category (Definition 1.1.1) whose objects are groups (Definition C.0.3) and whose morphisms are group homomorphisms (Definition C.0.4). It is often denoted by notations such as **Grp**.

2. The *category of abelian groups* is the locally small (Definition 1.1.2) category (Definition 1.1.1) whose objects are abelian groups (Definition C.0.3) and whose morphisms are group homomorphisms (Definition C.0.4). It is often denoted by notations such as **Ab**. **Ab** is an abelian category.

Definition 1.1.9. The *category of topological spaces* is the (locally small) (Definition 1.1.2) category (Definition 1.1.1)

- whose objects are topological spaces (Definition C.0.5), and
- whose morphisms are continuous maps (Definition C.0.6).

The category of topological spaces is often denoted by notations such as **Top**, **Top**, etc.

Definition 1.1.10 (Pointed topological space). Let X be a topological space (Definition C.0.5) and let $x_0 \in X$ be a chosen element of X . A *pointed/based (topological) space* is a pair (X, x_0) consisting of the space X together with the distinguished point x_0 , called the *base point of X* . If the base point of a pointed space (X, x_0) is understood, then it may be suppressed from notation; in particular, X may be written as a pointed space as opposed to the full notation of (X, x_0) .

A *morphism of pointed spaces* (or *based map*) or *continuous map* between pointed spaces (X, x_0) and (Y, y_0) is a continuous map (Definition C.0.6)

$$f : X \rightarrow Y$$

such that $f(x_0) = y_0$.

The collection of pointed spaces with their morphisms form a locally small (Definition 1.1.2) category (Definition 1.1.1), often called the *category of pointed spaces*. This category is often denoted by notations such as **Top** $_{\bullet}$, **Top** $_{\bullet}$, **Top** $_{\bullet}$, **Top** $_{\bullet}$, etc. The set of continuous maps from pointed spaces X to Y may be denoted by notations such as $C_*(X, Y)$, $C_{\bullet}(X, Y)$, $\text{Top}_*(X, Y)$, $\text{Top}_{\bullet}(X, Y)$, $\text{Hom}_{\text{Top}_{\bullet}}(X, Y)$, etc.

Definition 1.1.11. Let k be a field (Definition C.0.12). The *category of vector spaces over k* is the locally small (Definition 1.1.2) category (Definition 1.1.1)

- whose objects are vector spaces over k (Definition C.0.14), and
- whose morphisms are k -linear maps (Definition C.0.15).

The F -vector spaces of finite dimension (Definition C.0.16) form a full subcategory (Definition 1.3.7), called the *category of finite dimensional vector spaces over k* . Notations such as

$\mathbf{Vec}_k$ or $\mathbf{Vec}_k$ are often used to denote either of these categories; when both categories are considered, notations such as $\mathbf{FinVec}_k$ or $\mathbf{FinVec}_k$ may be used to distinguish the category of finite dimensional k -vector spaces from the category of all k -vector spaces.

Definition 1.1.12. 1. The *category of rings* is the locally small (Definition 1.1.2) category (Definition 1.1.1) whose objects are rings (Definition C.0.7) R and whose morphisms $R \rightarrow S$ are ring homomorphisms (Definition C.0.13). The category of rings over R is often denoted by notations such as **Ring**.
 2. The *category of commutative rings* is the full subcategory (Definition 1.3.7) of **Ring** consisting of the commutative rings (Definition C.0.9). It is denoted by notations such as **CommRing** or **CRing**.

Definition 1.1.13 (Isomorphism in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1), and let $x, y \in \text{Ob}(\mathcal{C})$. A morphism $f \in \mathcal{C}(x, y)$ is called an *isomorphism* if there exists a morphism $g \in \mathcal{C}(y, x)$ such that

$$g \circ f = 1_x \quad \text{and} \quad f \circ g = 1_y.$$

In this case, g is called the *inverse of f* , and x and y are said to be *isomorphic objects* in $\mathcal{C}$. It is standard to write $x \cong y$ if there exists an isomorphism $f : x \rightarrow y$.

In practice, isomorphisms in specific categories may be defined in different, yet equivalent, ways.

1.2. Functors between categories.

Definition 1.2.1 (Opposite category). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1). The *opposite category* of $\mathcal{C}$, denoted $\mathcal{C}^{\text{op}}$, is defined as follows:

- The objects of $\mathcal{C}^{\text{op}}$ are the same as those of $\mathcal{C}$.
- For any pair of objects $X, Y \in \mathcal{C}$, the morphisms from X to Y in $\mathcal{C}^{\text{op}}$ are given by the morphisms from Y to X in $\mathcal{C}$:

$$\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) := \text{Hom}_{\mathcal{C}}(Y, X).$$

- Composition in $\mathcal{C}^{\text{op}}$ is defined by reversing the order of composition in $\mathcal{C}$. That is, for morphisms $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y)$ and $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(Y, Z)$, their composition is

$$g \circ_{\mathcal{C}^{\text{op}}} f := f \circ_{\mathcal{C}} g.$$

Intuitively, the category $\mathcal{C}^{\text{op}}$ thus "reverses" the direction of all morphisms in $\mathcal{C}$.

Definition 1.2.2. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.1.1).

1. A *functor $F : \mathcal{C} \rightarrow \mathcal{D}$ (from $\mathcal{C}$ to $\mathcal{D}$)* consists of :
 - For each object X in $\mathcal{C}$, an object $F(X)$ in $\mathcal{D}$.
 - For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F(f) : F(X) \rightarrow F(Y)$ in $\mathcal{D}$, such that:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } X \text{ in } \mathcal{C},$$

$$F(g \circ f) = F(g) \circ F(f) \quad \text{for all } X, Y, Z \in \text{Ob}(\mathcal{C}) \text{ and all } f : X \rightarrow Y, g : Y \rightarrow Z \text{ in } \mathcal{C}.$$

Functors as defined above are also referred to as *covariant functors* to distinguish them from contravariant functors

2. A *contravariant functor from $\mathcal{C}$ to $\mathcal{D}$* refers to a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. Equivalently, such a functor consists of

- For each object X in $\mathcal{C}$, an object $F(X)$ in $\mathcal{D}$.
- For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F(f) : F(Y) \rightarrow F(X)$ in $\mathcal{D}$,

such that:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } X \text{ in } \mathcal{C},$$

$$F(g \circ f) = F(f) \circ F(g) \quad \text{for all } X, Y, Z \in \text{Ob}(\mathcal{C}) \text{ and all } f : X \rightarrow Y, g : Y \rightarrow Z \text{ in } \mathcal{C}.$$

A synonym for a “contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ ” is a “presheaf on $\mathcal{C}$ with values in $\mathcal{D}$ (Definition 7.1.1)”.

Note that declarations such as “Let $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ be a contravariant functor” can be common; such declarations usually mean “Let F be a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ ” as opposed to “Let F be a contravariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$ ”. further note that a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is equivalent to a covariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$.

Example 1.2.3. Here are some examples of functors (Definition 1.2.2):

1. For any category $\mathcal{C}$, its identity functor (Definition 1.2.4).
2. “forgetful functors”; some forgetful functors (Definition 1.2.5) include
 - (a) The forgetful functor $F : \mathbf{Grp} \rightarrow \mathbf{Sets}$ (Definition 1.1.8) (Definition 1.1.7) sending a group (Definition C.0.3) G to the underlying set of G , and sending a group homomorphism (Definition C.0.4) $G_1 \rightarrow G_2$ to the set function (Definition C.0.1) $G_1 \rightarrow G_2$. One can verify that

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } G \text{ in } \mathbf{Grp},$$

$$F(g \circ f) = F(g) \circ F(f) \quad \text{for all } G_1, G_2, G_3 \in \text{Ob}(\mathbf{Grp}) \text{ and all } f : G_1 \rightarrow G_2, g : G_2 \rightarrow G_3 \text{ in } \mathbf{Grp}.$$

- (b) Similarly, the forgetful functor $F : \mathbf{Top} \rightarrow \mathbf{Sets}$ (Definition 1.1.9) (Definition 1.1.7) sending a topological space (Definition 1.1.9) X to the underlying set of X , and sending a continuous map (Definition C.0.6) $X \rightarrow Y$ to the set function (Definition C.0.1) $X \rightarrow Y$.
 - (c) The forgetful functor $F : \mathbf{Ab} \rightarrow \mathbf{Grp}$ (Definition 1.1.8) sending an abelian group A to itself, and sending a group homomorphism $A_1 \rightarrow A_2$ to itself.
3. “Free” functors
 - (a) There is a functor $\mathbf{Sets} \rightarrow \mathbf{Grp}$ sending a set S to the free group (Definition C.0.19) $\langle S \rangle$ generated by S . The functor sends the morphism $f : S_1 \rightarrow S_2$ of sets, to the unique group homomorphism (Definition C.0.4) $\langle S_1 \rangle \rightarrow \langle S_2 \rangle$ given by sending $s \in S_1$ to $f(s) \in S_2$.
 - (b) Similarly, there is a functor $\mathbf{Sets} \rightarrow \mathbf{Ab}$ sending a set S to the free abelian group (Definition C.0.20) $\mathbb{Z}S$ generated by S . The functor sends the morphism $f : S_1 \rightarrow S_2$ of sets, to the unique group homomorphism (Definition C.0.4) $\mathbb{Z}S_1 \rightarrow \mathbb{Z}S_2$ given by sending $s \in S_1$ to $f(s) \in S_2$.

4. The fundamental group functor $\mathbf{Top}_\bullet \rightarrow \mathbf{Grp}$ (Definition 1.1.10); given a pointed topological space (Definition 1.1.10) (X, x) , its associated fundamental group (Definition C.0.21) $\pi_1(X, x)$ is the group of homotopy classes of loops $\gamma : [0, 1] \rightarrow X$. Given a morphism $(X, x) \rightarrow (Y, y)$ of pointed topological spaces, there is an (functorially) induced morphism $f : \pi_1(X, x) \rightarrow \pi_1(Y, y)$ sending the homotopy class $[\gamma]$ of the loop $\gamma : [0, 1] \rightarrow X$ to the homotopy class $[f \circ \gamma]$.

Definition 1.2.4. Let $\mathcal{C}$ be a category (Definition 1.1.1). The *identity functor on $\mathcal{C}$* is the functor $1_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ (also denoted by $\text{id}_{\mathcal{C}}$) defined by the following data:

- For every object $X \in \text{Ob}(\mathcal{C})$, $1_{\mathcal{C}}(X) = X$.
- For every morphism $f : X \rightarrow Y$ in $\mathcal{C}$, $1_{\mathcal{C}}(f) = f$.

It satisfies the functor axioms trivially: $1_{\mathcal{C}}(f \circ g) = f \circ g = 1_{\mathcal{C}}(f) \circ 1_{\mathcal{C}}(g)$ and $1_{\mathcal{C}}(\text{id}_X) = \text{id}_X$.

Definition 1.2.5. Let $\mathcal{C}$ and $\mathcal{D}$ be categories (Definition 1.1.1). A functor (Definition 1.2.2) $U : \mathcal{C} \rightarrow \mathcal{D}$ is called a *forgetful functor* if it maps an object in $\mathcal{C}$ to an object in $\mathcal{D}$ by discarding some of its structure or properties, and maps morphisms accordingly. Common examples include the functor from the category of groups to the category of sets, or from the category of topological spaces to the category of sets.

Example 1.2.6. Here are some examples of contravariant functors (Definition 1.2.2).

1. The dual of a vector space: given a vector space (Definition C.0.14) V over a field k , the dual (Definition 2.1.12) is defined as $V^\vee := \text{Hom}_k(V, k)$. The assignment $V \mapsto V^\vee$ specifies a contravariant functor $\text{Vec}_k \rightarrow \text{Vec}_k$ — given a linear map (Definition C.0.15) $f : V_1 \rightarrow V_2$, there is an induced linear map $V_2^\vee \rightarrow V_1^\vee$ given by sending the linear map $\phi : V_2 \rightarrow k$, which is an element of $V_2^\vee$, to the linear map $V_2 \circ f : V_1 \rightarrow k$, which is an element of $V_1^\vee$.
2. A representable functor (Definition 1.3.12): given any locally small category (Definition 1.1.2) $\mathcal{C}$ and any object X of $\mathcal{C}$, there is a contravariant functor $h_X : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$, $T \mapsto \text{Hom}_{\mathcal{C}}(T, X)$; given a morphism $f : T_1 \rightarrow T_2$ in $\mathcal{C}$, there is an induced map $h_X(T_2) \rightarrow h_X(T_1)$ of sets, i.e. a map $\text{Hom}_{\mathcal{C}}(T_2, X) \rightarrow \text{Hom}_{\mathcal{C}}(T_1, X)$, given by $\phi \mapsto \phi \circ f$.
3. There is the contravariant power set functor $\mathcal{P} : \mathbf{Sets}^{\text{op}} \rightarrow \mathbf{Sets}$ that sends a set S to its power set (Definition C.0.28) $\mathcal{P}(S)$ and that sends a set morphism $f : S \rightarrow T$ to the set morphism $f^* : \mathcal{P}(T) \rightarrow \mathcal{P}(S)$ given by $B \mapsto f^{-1}(B)$.
4. The functor of continuous functions: let $\mathbf{Top}$ be the category of topological spaces and $\mathbb{R}\text{-Alg}$ the category of $\mathbb{R}$ -algebras (Definition C.0.29). The assignment $X \mapsto C(X, \mathbb{R})$ of a space to its algebra of continuous real-valued functions is a contravariant functor. For any continuous map $g : X \rightarrow Y$, the induced algebra homomorphism $g^* : C(Y, \mathbb{R}) \rightarrow C(X, \mathbb{R})$ is given by the pullback $g^*(\phi) = \phi \circ g$ for $\phi \in C(Y, \mathbb{R})$.

1.3. Natural transformation. One overarching philosophy in various categories is that we only really care about objects “up to equivalence”; intuitively, we consider objects to be equivalent when they are isomorphic (Definition 1.1.13). Similarly, we only really care about categories “up to equivalence” as well; there is a notion of equivalence (Definition 1.3.4) between categories. To define it, we first need to define the notion of natural transformations

(Definition 1.3.1) between functors — a natural transformation is like a “morphism” between functors in a sense.

Definition 1.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.1.1). Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors (Definition 1.2.2).

A *natural transformation η between F and G* is a family of morphisms $\eta_X : F(X) \rightarrow G(X)$ in $\mathcal{D}$, one for each object X in $\mathcal{C}$, such that for every morphism $f : X \rightarrow Y$ in $\mathcal{C}$,

$$G(f) \circ \eta_X = \eta_Y \circ F(f)$$

in $\mathcal{D}$. In other words, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \eta_X \downarrow & & \downarrow \eta_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

We write such a natural transformation by $\eta : F \Rightarrow G$. We might also write $\eta : F \Rightarrow G : \mathcal{C} \rightarrow \mathcal{D}$ to fully specify that η is a natural transformation from F to G , which are in turn functors from $\mathcal{C}$ to $\mathcal{D}$.

If η_X is an isomorphism (Definition 1.1.13) for all objects X of $\mathcal{C}$, then η is said to be a *natural isomorphism*.

Example 1.3.2. Let k be a field (Definition C.0.12). Note that there is a “double dual” functor $((-)^{\vee})^{\vee} : \text{Vec}_k \rightarrow \text{Vec}_k$ given by $V \mapsto (V^{\vee})^{\vee}$ (Definition 2.1.12). Recall that (Example 1.2.6) $V \mapsto V^{\vee}$ is a contravariant functor (Definition 1.2.2), so the double dual functor is covariant. Recall that there is also the identity functor (Definition 1.2.4) $\text{id}_{\text{Vec}_k} : \text{Vec}_k \rightarrow \text{Vec}_k$ given by $V \mapsto V$.

For general vector spaces V , there is an injective k -linear map (Definition C.0.15)

$$\alpha_V : V \rightarrow (V^{\vee})^{\vee}, \quad v \mapsto v^*,$$

where $v^* : V^{\vee} \rightarrow k$ is the k -linear map given by $\phi \mapsto \phi(v)$. We note that if V is finite dimensional (Definition C.0.16), then α_V is an isomorphism. In fact, α is a natural transformation $\text{id}_{\text{Vec}_k} \Rightarrow ((-)^{\vee})^{\vee}$ — one should verify that, for every k -linear map $f : V_1 \rightarrow V_2$, the following diagram commutes:

$$\begin{array}{ccc} V_1 & \xrightarrow{f} & V_2 \\ \alpha_{V_1} \downarrow & & \downarrow \alpha_{V_2} \\ (V_1^{\vee})^{\vee} & \xrightarrow{(f^{\vee})^{\vee}} & (V_2^{\vee})^{\vee} \end{array}$$

Example 1.3.3. Let CRing be the category of commutative rings (Definition 1.1.12) and Group the category of groups (Definition 1.1.8). For a fixed $n \geq 1$, we have a covariant functor (Definition 1.2.2) $\text{GL}_n : \text{CRing} \rightarrow \text{Group}$ assigning a ring R to the general linear group $\text{GL}_n(R)$, and a functor $(\cdot)^{\times} : \text{CRing} \rightarrow \text{Group}$ assigning a ring to its group of units (Definition C.0.10).

The determinant $\det : \mathrm{GL}_n \Rightarrow (\cdot)^\times$ is a natural transformation (Definition 1.3.1). Its component at a ring R is the group homomorphism $\det_R : \mathrm{GL}_n(R) \rightarrow R^\times$. For any ring homomorphism $f : R \rightarrow S$, the following diagram commutes:

$$\begin{array}{ccc} \mathrm{GL}_n(R) & \xrightarrow{\mathrm{GL}_n(f)} & \mathrm{GL}_n(S) \\ \det_R \downarrow & & \downarrow \det_S \\ R^\times & \xrightarrow{f|_{R^\times}} & S^\times \end{array}$$

This commutativity expresses that the determinant is defined by the same universal formula (a polynomial in the matrix entries) regardless of the ring R , and is thus preserved by the "change of scalars" f .

Definition 1.3.4. An *equivalence of categories* between two (large) categories (Definition 1.1.1) $\mathcal{C}$ and $\mathcal{D}$ consists of a pair of functors (Definition 1.2.2)

$$F : \mathcal{C} \rightarrow \mathcal{D} \quad \text{and} \quad G : \mathcal{D} \rightarrow \mathcal{C}$$

together with natural isomorphisms (Definition 1.3.1)

$$\eta : \mathrm{Id}_{\mathcal{C}} \xrightarrow{\sim} G \circ F \quad \text{and} \quad \epsilon : F \circ G \xrightarrow{\sim} \mathrm{Id}_{\mathcal{D}}.$$

(Definition 1.2.4) Such functors F and G may be called *(natural) inverses of each other*.

When $\mathcal{C}$ and $\mathcal{D}$ are locally small categories (Definition 1.1.2), F is an equivalence of categories if and only if F is fully faithful (Definition 1.3.5) and essentially surjective (Definition 1.3.9)

The "correct" notion for one category to embed into another is the notion of a fully faithful functor (Definition 1.3.5).

Definition 1.3.5. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.1.1). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.2.2).

1. F is called *full* if for every pair of objects $x, y \in \mathrm{Ob}(\mathcal{C})$, the induced rule/assignment/class function

$$F_{x,y} : \mathrm{Hom}_{\mathcal{C}}(x, y) \rightarrow \mathrm{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is "surjective", i.e. for all morphisms $g : F(x) \rightarrow F(y)$, there exists some morphism $f : x \rightarrow y$ such that $F(f) = g$.

2. F is called *faithful* if for every pair of objects $x, y \in \mathrm{Ob}(\mathcal{C})$, the induced class function (assignment)

$$F_{x,y} : \mathrm{Hom}_{\mathcal{C}}(x, y) \rightarrow \mathrm{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is "injective", i.e., for any morphisms $f_1, f_2 \in \mathrm{Hom}_{\mathcal{C}}(x, y)$, if $F(f_1) = F(f_2)$ in $\mathrm{Hom}_{\mathcal{D}}(F(x), F(y))$, then $f_1 = f_2$.

3. F is called *fully faithful* if it is both full and faithful.

Definition 1.3.6 (Subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1). A *subcategory* $\mathcal{D}$ of $\mathcal{C}$ consists of:

- a subclass of objects $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(\mathcal{C})$,

- for each pair of objects $X, Y \in \text{Ob}(\mathcal{D})$, a subclass of morphisms

$$\text{Hom}_{\mathcal{D}}(X, Y) \subseteq \text{Hom}_{\mathcal{C}}(X, Y),$$

such that

- for every object $X \in \text{Ob}(\mathcal{D})$, the identity morphism id_X of X in $\mathcal{C}$ lies in $\text{Hom}_{\mathcal{D}}(X, X)$,
- the composition of morphisms in $\mathcal{D}$ is inherited from $\mathcal{C}$ and is closed in $\mathcal{D}$: for morphisms $f \in \text{Hom}_{\mathcal{D}}(X, Y)$ and $g \in \text{Hom}_{\mathcal{D}}(Y, Z)$, their composition $g \circ f$, which is a priori in $\text{Hom}_{\mathcal{C}}(X, Z)$, is in $\text{Hom}_{\mathcal{D}}(X, Z)$.

Definition 1.3.7 (Full subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1). A *full subcategory* $\mathcal{D}$ of $\mathcal{C}$ is a subcategory (Definition 1.3.6) such that for every pair of objects $X, Y \in \text{Ob}(\mathcal{D})$, the morphism classes coincide:

$$\text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y).$$

In other words, a full subcategory includes all morphisms between its objects that exist in the ambient category $\mathcal{C}$.

The “correct” notion for one category to “surject” onto another is the notion of a essentially surjective functor (Definition 1.3.9).

Definition 1.3.8 (Essential image of a functor). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition 1.1.1). The *essential image of F* is the full subcategory (Definition 1.3.7) of $\mathcal{D}$ whose objects are those $d \in \text{Ob}(\mathcal{D})$ for which there exists an object $c \in \text{Ob}(\mathcal{C})$ such that

$$F(c) \cong d.$$

(Definition 1.1.13) Equivalently, the essential image is given by

$$\text{EssIm}(F) = \{ d \in \text{Ob}(\mathcal{D}) \mid \exists c \in \text{Ob}(\mathcal{C}), F(c) \cong d \},$$

endowed with all morphisms $\mathcal{D}(d, d')$ between such objects.

Definition 1.3.9. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition 1.1.1). It is said to be essentially surjective if its essential image (Definition 1.3.8) coincides with $\mathcal{D}$.

Example 1.3.10. Below are examples illustrating various properties of functors:

1. **Faithful but not full:** The forgetful functor (Definition 1.2.5) $U : \text{Group} \rightarrow \text{Set}$. It is faithful (Definition 1.3.5) because group homomorphisms are distinct if they are distinct as functions. It is not full because not every function between groups is a group homomorphism (e.g., the constant function $x \mapsto g$ for $g \neq e$).
2. **Full but not faithful:** The canonical functor $H : \text{Top} \rightarrow h\text{Top}$ from the category of topological spaces (Definition 1.1.9) to the homotopy category (Definition C.0.32) of topological spaces. It is the identity on objects and sends a continuous map f to its homotopy class $[f]$. This is full (Definition 1.3.5) by the definition of morphisms in $h\text{Top}$, but not faithful (Definition 1.3.5) because it identifies distinct but homotopic maps (Definition C.0.30) (e.g., any two paths in $\mathbb{R}^n$ with the same endpoints).

3. **Fully faithful:** The inclusion functor $\iota : \mathbf{Ab} \rightarrow \mathbf{Group}$. It is faithful (Definition 1.3.5) (it is an embedding) and it is full (Definition 1.3.5) because any group homomorphism between two abelian groups is, by definition, a morphism in $\mathbf{Ab}$.
4. **Essentially surjective:** Let $\mathbf{S}$ be the category whose objects are the standard sets $\underline{n} = \{0, \dots, n-1\}$ for each $n \in \mathbb{N}$ (and whose morphisms are the set functions between these sets), and $\mathbf{FinSet}$ be the category of all finite sets. The inclusion functor $I : \mathbf{S} \rightarrow \mathbf{FinSet}$ is essentially surjective because every finite set X is isomorphic to $\underline{n}$ where n is the cardinality of X .

Lemma 1.3.11. Let $\mathcal{C}$ and $\mathcal{D}$ be locally small categories (Definition 1.1.2), and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.2.2).

F is an equivalence of categories (Definition 1.3.4) if and only if F is fully faithful (Definition 1.3.5) and essentially surjective (Definition 1.3.9)

1.3.1. *Yoneda lemma.* The Yoneda lemma basically expresses the idea that an object of a (locally small) category is essentially determined by its morphisms to other objects.

Definition 1.3.12. Let $\mathcal{C}$ be a locally small category (Definition 1.1.2). Given an object X of $\mathcal{C}$, the *functor of points* h_X is the functor (Definition 1.2.2)/presheaf (Definition 7.1.1) $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$ given by

1. sending an object T of $\mathcal{C}$ to the set $\text{Hom}_{\mathcal{C}}(T, X)$, and
2. sending a morphism $f : T_1 \rightarrow T_2$ in $\mathcal{C}$ to the set map (Definition C.0.1)

$$\text{Hom}_{\mathcal{C}}(T_2, X) \rightarrow \text{Hom}_{\mathcal{C}}(T_1, X), \quad \phi \mapsto \phi \circ f.$$

A functor $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$ (or equivalently, a presheaf on $\mathcal{C}$ valued in $\mathbf{Sets}$) is said to be *representable* if it is naturally isomorphic (Definition 1.3.1) to some functor h_X of points for an object X of $\mathcal{C}$.

Dually, a functor $\mathcal{C} \rightarrow \mathbf{Sets}$ is called *co-representable* if it is naturally isomorphic to a functor $h^X : \mathcal{C} \rightarrow \mathbf{Sets}$ given by $T \mapsto \text{Hom}_{\mathcal{C}}(X, T)$.

Note that the above notions of representability/co-representability are special case of those of Definition A.0.1, where the monoidal category $\mathcal{V}$ is the symmetric monoidal category (Definition A.0.2) $\mathbf{Sets}$ (Definition 1.1.7).

Theorem 1.3.13 (Yoneda Lemma). Let $\mathcal{C}$ be a locally small category (Definition 1.1.2). Let A be an object of $\mathcal{C}$, and let $F : \mathcal{C} \rightarrow \mathbf{Set}$ be a covariant functor (Definition 1.2.2) to the category of sets (Definition 1.1.7). Let $h^A : \mathcal{C} \rightarrow \mathbf{Set}$ denote the covariant representable functor (Definition 1.3.12) defined by $h^A(X) = \text{Hom}_{\mathcal{C}}(A, X)$.

There exists a bijection

$$y_{A,F} : \text{Nat}(h^A, F) \xrightarrow{\cong} F(A)$$

between the set of natural transformations (Definition 1.3.1) from h^A to F and the set $F(A)$. This bijection is given by the mapping

$$\alpha \mapsto \alpha_A(\text{id}_A),$$

where $\alpha : h^A \rightarrow F$ is a natural transformation, $\alpha_A : h^A(A) \rightarrow F(A)$ is its component at A , and $\text{id}_A \in h^A(A) = \text{Hom}_{\mathcal{C}}(A, A)$ is the identity morphism.

Furthermore, this isomorphism is natural in both A and F . Explicitly:

1. For any morphism $f : A \rightarrow B$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} \text{Nat}(h^B, F) & \xrightarrow{y_{B,F}} & F(B) \\ \downarrow -\circ h^f & & \downarrow F(f) \\ \text{Nat}(h^A, F) & \xrightarrow{y_{A,F}} & F(A) \end{array}$$

where $h^f : h^B \rightarrow h^A$ is the natural transformation induced by pre-composition with f .

2. For any natural transformation $\eta : F \rightarrow G$, the following diagram commutes:

$$\begin{array}{ccc} \text{Nat}(h^A, F) & \xrightarrow{y_{A,F}} & F(A) \\ \downarrow \eta \circ - & & \downarrow \eta_A \\ \text{Nat}(h^A, G) & \xrightarrow{y_{A,G}} & G(A) \end{array}$$

Corollary 1.3.14 (Yoneda Embedding). Let $\mathcal{C}$ be a locally small category (Definition 1.1.2). The functor

$$h^\bullet : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}^{\mathcal{C}}$$

(Definition 1.2.1) (Definition 2.2.6) defined on objects by $A \mapsto h^A = \text{Hom}_{\mathcal{C}}(A, -)$ (Definition 1.3.12) and on morphisms by $f \mapsto h^f = (- \circ f)$ is fully faithful (Definition 1.3.5). That is, for any objects A, B in $\mathcal{C}$, the map

$$\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Nat}(h^B, h^A)$$

given by sending a morphism $f : A \rightarrow B$ to the natural transformation (Definition 1.3.1) $h^f : h^B \rightarrow h^A$ (pre-composition by f) is a bijection.

Consequently, $\mathcal{C}^{\text{op}}$ embeds as a full subcategory (Definition 1.3.7) of the functor category $\mathbf{Set}^{\mathcal{C}}$.

Theorem 1.3.15 (Contravariant Yoneda Lemma). Let $\mathcal{C}$ be a locally small category (Definition 1.1.2). Let A be an object of $\mathcal{C}$, and let $G : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ be a contravariant functor (Definition 1.2.2) (i.e. a presheaf (Definition 7.1.1)). Let $h_A : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ denote the contravariant representable functor defined by $h_A(X) = \text{Hom}_{\mathcal{C}}(X, A)$.

There exists a bijection natural in A and G :

$$\text{Nat}(h_A, G) \cong G(A)$$

given by $\alpha \mapsto \alpha_A(\text{id}_A)$.

Corollary 1.3.16 (Contravariant Yoneda Embedding). Let $\mathcal{C}$ be a locally small category (Definition 1.1.2). The functor

$$h_\bullet : \mathcal{C} \rightarrow \mathbf{Set}^{\mathcal{C}^{\text{op}}}$$

defined on objects by $A \mapsto h_A = \text{Hom}_{\mathcal{C}}(-, A)$ and on morphisms by $f \mapsto h_f = (f \circ -)$ is fully faithful (Definition 1.3.5). That is, for any objects A, B in $\mathcal{C}$, the map

$$\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Nat}(h_A, h_B)$$

given by sending a morphism $f : A \rightarrow B$ to the natural transformation (Definition 1.3.1) $h_f : h_A \rightarrow h_B$ (post-composition by f) is a bijection.

Consequently, $\mathcal{C}$ embeds as a full subcategory (Definition 1.3.7) of the category of presheaves $\mathbf{Set}^{\mathcal{C}^{\text{op}}}$.

Example 1.3.17. The algebra of smooth functions (Definition C.0.35) $C^\infty(M)$ on a smooth manifold (Definition C.0.37) M is represented by the real line $\mathbb{R}$ in the category of smooth manifolds $\mathbf{Man}$.

1. **The Functor:** Consider the covariant representable functor (Definition 1.3.12) $h^{\mathbb{R}} = \text{Hom}_{\mathbf{Man}}(-, \mathbb{R})$. For any manifold M , we have

$$\text{Hom}_{\mathbf{Man}}(M, \mathbb{R}) = C^\infty(M),$$

i.e. the representable functor $h^{\mathbb{R}}$ assigns M to the set of all smooth maps $f : M \rightarrow \mathbb{R}$, which is exactly $C^\infty(M)$.

2. **The Yoneda Correspondence:** By the Yoneda Lemma (Theorem 1.3.13), there is a natural bijection between the points of M and the natural transformations from h^M to $h^{\mathbb{R}}$:

$$\text{Nat}(h^M, h^{\mathbb{R}}) \cong h^{\mathbb{R}}(M) = C^\infty(M)$$

3. **Geometric Interpretation:** This identifies a smooth function $\psi \in C^\infty(M)$ with a "natural" way to turn M -valued points of any other smooth manifold X into real-valued points. Specifically, if we have a "probe" $\alpha : X \rightarrow M$, the function ψ induces a map:

$$\alpha \mapsto \psi \circ \alpha$$

mapping $\text{Hom}(X, M)$ to $\text{Hom}(X, \mathbb{R})$.

This illustrates that the "global" algebraic data of a manifold (its functions) is entirely captured by its relationship to the "simplest" non-trivial manifold, $\mathbb{R}$.

1.4. **The category of categories.** Intuitively, one might think of a functor (Definition 1.2.2) as a "morphism" between two categories. Indeed, one can consider a category of categories in which morphisms are functors.

Definition 1.4.1. The *category of small categories* is the category defined by the following data:

- The objects are all small categories (Definition 1.1.2).
- The morphisms between two small categories $\mathcal{C}$ and $\mathcal{D}$ are the functors (Definition 1.2.2) $F : \mathcal{C} \rightarrow \mathcal{D}$.
- The composition of morphisms is the standard composition of functors.
- The identity morphism for each object $\mathcal{C}$ is the identity functor $1_{\mathcal{C}}$.

This category is a large category (Definition 1.1.1) and is denoted by **Cat**

If we allow ourselves to use Grothendieck universes (Definition A.0.3), then we can also talk about a category of categories in the sense of Definition A.0.7.

1.5. Miscellaneous categorical notions.

Definition 1.5.1 (Product Category of a Family of Categories). Let $\{\mathcal{C}_i\}_{i \in I}$ be a family of (large) categories (Definition 1.1.1) indexed by a class I . The *product category of the family*, denoted

$$\prod_{i \in I} \mathcal{C}_i,$$

is the very large category (♠ TODO: define very large categories) defined as follows:

- The class of objects is

$$\text{Ob}\left(\prod_{i \in I} \mathcal{C}_i\right) = \prod_{i \in I} \text{Ob}(\mathcal{C}_i),$$

i.e., an object is a family $(A_i)_{i \in I}$ with $A_i \in \text{Ob}(\mathcal{C}_i)$.

- For two objects $(A_i)_i$ and $(B_i)_i$, the morphism class is

$$\text{Hom}_{\prod_{i \in I} \mathcal{C}_i}((A_i)_i, (B_i)_i) = \prod_{i \in I} \text{Hom}_{\mathcal{C}_i}(A_i, B_i).$$

In other words, a morphism $(f_i)_i : (A_i)_i \rightarrow (B_i)_i$ consists of morphisms $f_i : A_i \rightarrow B_i$ in each $\mathcal{C}_i$.

- For morphisms $(f_i)_i : (A_i)_i \rightarrow (B_i)_i$ and $(g_i)_i : (B_i)_i \rightarrow (C_i)_i$, composition is defined componentwise:

$$(g_i)_i \circ (f_i)_i = (g_i \circ_i f_i)_i.$$

- For each object $(A_i)_i$, the identity morphism is given by the family

$$(\text{id}_{A_i})_i.$$

If I is a set, then $\prod_{i \in I} \mathcal{C}_i$ is a large category. If I is a set and if each $\mathcal{C}_i$ is locally small (Definition 1.1.2), then $\prod_{i \in I} \mathcal{C}_i$ is locally small.

In case that I is finite, the notation of $\boxtimes$ may be used for product categories, e.g. $\mathcal{C}_i \boxtimes \mathcal{C}_j$ denotes the product of two categories $\mathcal{C}_i \times \mathcal{C}_j$.

(♠ TODO: ordinal, U_α) If α is an ordinal such that $\mathcal{C}_i$ and I are U_α -large (i.e. they live in $U_{\alpha+1}$), then $\prod_{i \in I} \mathcal{C}_i$ is $U_{\alpha+1}$ -large.

2. ADDITIVE AND ABELIAN CATEGORIES

2.1. Rings and modules.

Definition 2.1.1. Let R be a not-necessarily commutative ring (Definition C.0.7).

1. A *left R -module* is an abelian group $(M, +)$ together with an operation $R \times M \rightarrow M$, denoted $(r, m) \mapsto rm$, such that for all $r, s \in R$ and $m, n \in M$:
 - $r(m + n) = rm + rn$,

- $(r + s)m = rm + sm$,
 - $(rs)m = r(sm)$,
 - $1_R m = m$ where 1_R is the multiplicative identity of R .
2. A **right R -module** is defined similarly as an abelian group $(M, +)$ with an operation $M \times R \rightarrow M$, denoted $(m, r) \mapsto mr$, such that for all $r, s \in R$ and $m, n \in M$:
- $(m + n)r = mr + nr$,
 - $m(r + s) = mr + ms$,
 - $m(rs) = (mr)s$,
 - $m1_R = m$.
3. Let R and S be (not necessarily commutative) rings (Definition C.0.7).
 An **R - S -bimodule** (or an **R - S -module** or an (R, S) -module, etc.) is an abelian group (Definition C.0.3) $(M, +)$ equipped with
- (a) a left action of R :

$$R \times M \rightarrow M, \quad (r, m) \mapsto r \cdot m,$$

making M a left R -module (Definition 2.1.1),

- (b) a right action of S :

$$M \times S \rightarrow M, \quad (m, s) \mapsto m \cdot s,$$

making M a right S -module,

such that the left and right actions commute; that is, for all $r \in R$, $s \in S$, and $m \in M$,

$$r \cdot (m \cdot s) = (r \cdot m) \cdot s.$$

4. A **two-sided R -module** (or **R -bimodule**) is an R - R -bimodule.

If R is a commutative ring (Definition C.0.9), then a left/right R -module can automatically be regarded as a two-sided R -module. As such, we simply talk about **R -modules** in this case.

Any abelian group is equivalent to a two-sided $\mathbb{Z}$ -module. Moreover, any left R -module is equivalent to an $R - \mathbb{Z}$ -bimodule (Definition 2.1.1) and any right R -module is equivalent to an $\mathbb{Z} - R$ -bimodule (Definition 2.1.1). Given a left/right/two-sided R -module, its **natural bimodule structure** will refer to its structure as a R - $\mathbb{Z}$ / $\mathbb{Z}$ - R / R - R bimodule. In this way, many definitions associated with the notions of left/right/two-sided R -modules can be defined as special cases for definitions for R - S -bimodules.

Definition 2.1.2. Let R, S be (not-necessarily commutative) rings (Definition C.0.7).

1. Let M and N be R - S -bimodules (Definition 2.1.1). A function $\varphi : M \rightarrow N$ is called an **R - S -bimodule homomorphism** or **R - S -linear** if it is a group homomorphism (Definition C.0.4) of the underlying abelian groups of M and N and respects the scalar actions as follows: for all $m_1, m_2 \in M$, $r \in R$, and $s \in S$,

$$\begin{aligned} \varphi(r \cdot m_1) &= r \cdot \varphi(m_1), \\ \varphi(m_1 \cdot s) &= \varphi(m_1) \cdot s. \end{aligned}$$

2. Let M and N be left/right/two-sided R -modules (Definition 2.1.1). A function $\varphi : M \rightarrow N$ is called a **left/right/two-sided R -module homomorphism** if it is an bimodule

homomorphism on the natural bimodule structures (Definition 2.1.1) of M and N . Such a function is also called **R -linear**.

Modules and homomorphisms of a fixed type (i.e. R - S -bimodules or left/right/two-sided R -modules) form a locally small (Definition 1.1.2) category (Definition 1.1.1).

Definition 2.1.3. Let R and S be (not necessarily commutative) rings (Definition C.0.7).

1. The **category of (R, S) -bimodules** (or R - S -bimodules), denoted by notations such as ${}_R\mathbf{Mod}_S$, is the category whose objects are (R, S) -bimodules (Definition 2.1.1) and whose R - S -bimodule homomorphisms (Definition 2.1.2).
2. The **category of left R -modules**, denoted by notations such as ${}_R\mathbf{Mod}$ or $R - \mathbf{Mod}$, is the category ${}_R\mathbf{Mod}_{\mathbb{Z}}$, i.e. the category whose objects are left R -modules (Definition 2.1.1) and whose morphisms are left R -linear maps (Definition 2.1.2).
3. The **category of right R -modules**, denoted by notations such as $\mathbf{Mod}_R$ or $\mathbf{Mod} - R$, is the category ${}_Z\mathbf{Mod}_R$, i.e. the category whose objects are right R -modules (Definition 2.1.1) and whose morphisms are right R -linear maps (Definition 2.1.2).

The category of bimodules can be canonically identified with module categories over tensor product rings (Definition 2.1.10):

- ${}_R\mathbf{Mod}_S$ is isomorphic to the category of left modules over the ring $R \otimes_{\mathbb{Z}} S^{\text{op}}$.
- ${}_R\mathbf{Mod}_S$ is isomorphic to the category of right modules over the ring $R^{\text{op}} \otimes_{\mathbb{Z}} S$.

Consequently, standard module-theoretic concepts (such as projective objects, injective objects, and flat objects) in ${}_R\mathbf{Mod}_S$ correspond exactly to the respective concepts in ${}_{R \otimes S^{\text{op}}}\mathbf{Mod}$.

Note that there are canonical isomorphisms of categories:

$${}_R\mathbf{Mod} \cong {}_R\mathbf{Mod}_{\mathbb{Z}} \quad \text{and} \quad \mathbf{Mod}_S \cong {}_{\mathbb{Z}}\mathbf{Mod}_S.$$

That is, left R -modules are exactly $(R, \mathbb{Z})$ -bimodules, and right S -modules are exactly $(\mathbb{Z}, S)$ -bimodules.

Definition 2.1.4. Let R, S be not-necessarily commutative rings (Definition C.0.7).

1. Let M be an R - S -bimodule whose abelian group (Definition C.0.3) structure is given by the operator $+$. An **R - S -submodule of M** is a subgroup $N \subseteq (M, +)$ if for all $r \in R$, $s \in S$, and $n \in N$, we have $rn \in N$ and $ns \in N$; in this case, N inherits an R - S bimodule structure from M .
2. If M is a left/right/two-sided R -module, then a **left/right/two-sided R -submodule of M** is a submodule of the natural bimodule structure (Definition 2.1.1) of M .

Definition 2.1.5. (♠ **TODO: define coset**) Let R, S be (not necessarily commutative) rings (Definition C.0.7).

1. Let M be an R - S -bimodule (Definition 2.1.1). Let $N \subseteq M$ be a submodule of M (Definition 2.1.4).

The quotient group M/N , which is well defined as M is an abelian group (Definition C.0.3) and hence N is a normal subgroup, has the structure of an R - S -bimodule

— the (abelian) group structure is simply the group structure of M/N , whereas the R - S -bimodule structure is given as follows: for $m \in M$, $r \in R$, $s \in S$, we have

$$r \cdot (m + N) \cdot s = r \cdot m \cdot s + N.$$

This R - S -bimodule structure on M/N is called the *quotient R - S -bimodule of M by N* and is also denoted as M/N .

The canonical projection map

$$\pi : M \rightarrow M/N, \quad m \mapsto m + N,$$

is a surjective R -module homomorphism (Definition 2.1.2) with kernel (Definition 2.1.7) N .

2. Let M be a left/right/two-sided R -module. Let $N \subseteq M$ be a submodule of M . The *quotient R -module M/N* is the quotient of M by N for their respective natural bimodule structures (Definition 2.1.1).

Definition 2.1.6 (Submodule generated by elements in an (R, S) -bimodule). Let R and S be (not necessarily commutative) rings (Definition C.0.7).

1. Let M be an (R, S) -bimodule (Definition 2.1.1).

Given a subset $X \subseteq M$, the *sub-bimodule of M generated by X* is the smallest (R, S) -sub-bimodule of M containing X . It is often denoted by notations such as $\langle X \rangle = \langle X \rangle_{R,S}$ and is more explicitly the intersection

$$\langle X \rangle_{R,S} = \bigcap_{X \subseteq T \subseteq M, T \text{ is a } (R,S)\text{-submodule of } M} T$$

of all (R, S) -submodules of M containing X .

Equivalently, $\langle X \rangle_{R,S}$ consists of all linear combinations of X .

2. If M is a left/right/two-sided R -module and given a subset $X \subseteq M$, the *submodule of M generated by X* is the submodule of the natural bimodule (Definition 2.1.1) of M generated by X . It is denoted by notations such as $\langle X \rangle = \langle X \rangle_R$.

Definition 2.1.7. Let R, S be (not-necessarily commutative) rings with unity (Definition C.0.7), and let M, N be R - S -bimodules (Definition 2.1.1). Let

$$\varphi : M \rightarrow N$$

be a homomorphism of R - S -bimodules (Definition 2.1.2). We define:

1. The *kernel of φ* is the submodule of M (Definition 2.1.4) given by

$$\ker(\varphi) := \{m \in M \mid \varphi(m) = 0\} \subseteq M.$$

2. The *image of φ* is the submodule of N given by

$$\operatorname{im}(\varphi) := \{\varphi(m) \mid m \in M\} \subseteq N.$$

3. The *cokernel of φ* is the quotient module of N (Definition 2.1.5) defined by

$$\operatorname{coker}(\varphi) := N / \operatorname{im}(\varphi).$$

4. The *coimage of φ* is the quotient module of M (Definition 2.1.5) defined by

$$\text{coim}(\varphi) := M / \ker(\varphi).$$

It is not difficult to see that each of these are indeed R - S bimodules. In case M and N are left/right/two-sided R -modules, the *kernel, image, cokernel, and coimage* of a module homomorphism $\varphi : M \rightarrow N$ are respectively defined to be the kernel, image, cokernel, and coimage for the natural bimodule structures (Definition 2.1.1) of M and N .

The kernel, cokernel, image, and coimage of f are respectively the categorical kernel, cokernel (Definition 2.3.6), image, and coimage (Definition 2.3.10) (Lemma 2.3.11).

Two fundamental functors on categories of modules are given by Hom's (Definition 2.1.8) and tensor products (Definition 2.1.9)

Definition 2.1.8 (Hom of left/right/bi-modules). Let R, S, T be (not necessarily commutative) rings (Definition C.0.7).

1. Let M and N be left R -modules (Definition 2.1.1). The *homomorphism group of left R -modules from M to N* is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_R(M, N) = \text{Hom}_{R-\mathbb{Z}}(M, N) := \{f : M \rightarrow N \mid f \text{ is a left } R\text{-module homomorphism}\}.$$

(Definition 2.1.2)

2. Let M and N be right R -modules (Definition 2.1.1). The *homomorphism group of right R -modules from M to N* is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_R(M, N) = \text{Hom}_{\mathbb{Z}-R}(M, N) := \{f : M \rightarrow N \mid f \text{ is a right } R\text{-module homomorphism}\}.$$

3. Let S be a (not necessarily commutative ring) and let M and N be R - S -bimodules (Definition 2.1.1). The *homomorphism group of R - S -bimodules from M to N* is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_{R-S}(M, N) := \{f : M \rightarrow N \mid f \text{ is a } R-S\text{-bimodule homomorphism}\}$$

In each case, $\text{Hom}(M, N)$ has a natural structure of an *abelian group* given by *pointwise addition*: for $f, g \in \text{Hom}(M, N)$,

$$(f + g)(m) := f(m) + g(m),$$

and the zero morphism $\mathbf{0}$ given by $0(m) := 0_N$ acts as the identity element. The additive inverse $-f$ is defined by $(-f)(m) := -f(m)$. Moreover, depending on bi-module structures that M and N may be carrying, $\text{Hom}(M, N)$ may itself carry additional module structures:

- In case that M is a R - S -bimodule and N is a R - T -bimodule, $\text{Hom}_R(M, N)$, the group of left R -module homomorphisms, is an S - T -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(m \cdot s) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

- Dually, in case that M is a S - R -bimodule and N is a T - R -bimodule, $\text{Hom}_R(M, N)$, the group of right R -module homomorphisms, is an S - T -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(s \cdot m) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

Some cases of interest may be when R, S , or T is in fact $\mathbb{Z}$ — these allow us to see module structures on $\text{Hom}(M, N)$ even when M and N are one-sided modules.

(♠ TODO: state this as a theorem) We furthermore note that $\text{Hom}_R(-, -)$ yields biadditive functors (Definition 2.4.2)

$$\text{Hom}_R(-, -) : {}_R\mathbf{Mod}_S^{\text{op}} \times {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T$$

$$\text{Hom}_R(-, -) : {}_S\mathbf{Mod}_R^{\text{op}} \times {}_T\mathbf{Mod}_R \rightarrow {}_S\mathbf{Mod}_T.$$

(Definition 1.2.1) (Definition 2.1.3) (Theorem 2.5.11)

Definition 2.1.9 (Tensor product of bimodules). Let R, S, T be (not necessarily commutative) rings (Definition C.0.7), let M be an R - S bimodule (Definition 2.1.1), and let N be an S - T bimodule. In the free abelian group (Definition C.0.20) $\mathbb{Z}[M \times N]$ generated by the Cartesian product $M \times N$ (Definition 2.2.2), let U be the subgroup generated by elements of the form (♠ TODO: subgroup generated)

$$\begin{aligned} (m + m', n) - (m, n) - (m', n), \\ (m, n + n') - (m, n) - (m, n'), \\ (m \cdot s, n) - (m, s \cdot n), \end{aligned}$$

for all $m, m' \in M$, $n, n' \in N$, and $s \in S$. The *tensor product of M and N over S* is the quotient abelian group

$$M \otimes_S N := \mathbb{Z}[M \times N]/U.$$

The image of an element of the form $(m, n) \in M \times N$ in $M \otimes_S N$ is denoted $m \otimes n$ and called a *pure tensor*. In general, the elements of $M \otimes_S N$ are finite sums

$$\sum_{i=1}^n m_i \otimes n_i \quad m_i \in M, n_i \in N$$

of pure tensors. Thus, the pure tensors satisfy the following relations:

$$\begin{aligned} (m + m') \otimes n &= m \otimes n + m' \otimes n \\ m \otimes (n + n') &= m \otimes n + m \otimes n' \\ (m \cdot s) \otimes n &= m \otimes (s \cdot n) \end{aligned}$$

This tensor product becomes naturally an R - T bimodule with left action and right action defined by

$$\begin{aligned} r \cdot (m \otimes n) &= (r \cdot m) \otimes n, \\ (m \otimes n) \cdot t &= m \otimes (n \cdot t), \end{aligned}$$

for all $r \in R$, $t \in T$, $m \in M$, and $n \in N$.

Inductively, given rings $R_0, \dots, R_k$ and R_{i-1} – R_i -bimodules M_i for $i = 1, \dots, k$, we may speak of the tensor product

$$M_0 \otimes_{R_1} M_1 \otimes_{R_2} \cdots \otimes_{R_{k-1}} M_k;$$

tensor products are associative(♠ **TODO:**), so parentheses are not strictly needed to notate them. Its *pure tensors* are elements of the form $m_0 \otimes m_1 \otimes \cdots \otimes m_k$ for $m_i \in M_i$, and its general elements are finite sums

$$\sum_{j=1}^n m_{0j} \otimes m_{1j} \otimes \cdots \otimes m_{kj} \quad m_{ij} \in M_i.$$

of pure tensors. It also has a natural $R_0 - R_k$ -bimodule structure.

In general, $(M_0, \dots, M_k) \mapsto M_0 \otimes_{R_1} M_1 \otimes_{R_2} \cdots \otimes_{R_{k-1}} M_k$ defines a $(k+1)$ -ary additive functor (Definition 2.4.2)

$${}_{R_0}\mathbf{Mod}_{R_1} \times \cdots \times {}_{R_{k-1}}\mathbf{Mod}_{R_k} \rightarrow {}_{R_0}\mathbf{Mod}_{R_k}$$

(Theorem 2.5.11).

Given a ring R and a two-sided R -module M , we may also speak of the *n -fold tensor product* $M^{\otimes n} = M^{\otimes_{R^n}}$

Definition 2.1.10. Let k be a not necessarily commutative ring (Definition C.0.7). Let R and S be k -rings (Definition C.0.13) (not necessarily commutative). Assume that at least one of R or S is a k -algebra (Definition C.0.29). The *tensor product ring* $R \otimes_k S$ is the k -module $R \otimes_k S$ (Definition 2.1.9) equipped with a multiplication defined on simple tensors by

$$(r_1 \otimes s_1) \cdot (r_2 \otimes s_2) = (r_1 r_2) \otimes (s_1 s_2)$$

and extended by linearity. This multiplication is well-defined and makes $R \otimes_k S$ into a k -ring under the ring homomorphism

$$k \rightarrow R \otimes_k S, \quad a \mapsto a \otimes 1 = 1 \otimes a.$$

The unit element is $1_R \otimes 1_S$.

In this ring, the subrings $R \otimes 1$ and $1 \otimes S$ commute with each other; that is, for all $r \in R$ and $s \in S$,

$$(r \otimes 1) \cdot (1 \otimes s) = r \otimes s = (1 \otimes s) \cdot (r \otimes 1).$$

If R and S are both k -algebras, then $R \otimes_k S$ is also a k -algebra.

Proposition 2.1.11 (Universal Property of the Tensor Product of Bimodules). Let R, S, T be (not necessarily commutative) rings (Definition C.0.7). Let M be an R - S bimodule (Definition 2.1.1) and let N be an S - T bimodule. Let P be an R - T bimodule. Then for every R - T bilinear map

$$\beta : M \times N \rightarrow P,$$

that is, a map satisfying

$$\begin{aligned} \beta(m + m', n) &= \beta(m, n) + \beta(m', n), \\ \beta(m, n + n') &= \beta(m, n) + \beta(m, n'), \\ \beta(r \cdot m, n) &= r \cdot \beta(m, n), \\ \beta(m, n \cdot t) &= \beta(m, n) \cdot t, \\ \beta(m \cdot s, n) &= \beta(m, s \cdot n), \end{aligned}$$

for all $m, m' \in M$, $n, n' \in N$, $r \in R$, $s \in S$, $t \in T$, there exists a unique R - T bimodule homomorphism

$$\tilde{\beta} : M \otimes_S N \rightarrow P$$

such that $\tilde{\beta}(m \otimes n) = \beta(m, n)$ for all $m \in M$, $n \in N$.

Definition 2.1.12. Let R be a (not necessarily commutative) ring (Definition C.0.7). Depending on the module (Definition 2.1.1) structure of M , we define its dual module as follows:

1. If M is a left R -module (Definition 2.1.1), then the *(right) dual module of M* is

$$M^* = M^\vee := \text{Hom}_R(M, R) = \text{Hom}_{R-\mathbb{Z}}(M, R).$$

(Definition 2.1.8) Note that it is a right R -module, as M is a $R - \mathbb{Z}$ -bimodule and R is an $R - R$ -bimodule.

2. If M is a right R -module (Definition 2.1.1), then the *(left) dual module of M* is

$${}^*M = {}^\vee M := \text{Hom}_R(M, R) = \text{Hom}_{\mathbb{Z}-R}(M, R).$$

(Definition 2.1.8) Note that it is a left R -module, as M is a $\mathbb{Z} - R$ -bimodule and R is an $R - R$ -bimodule.

3. If M is a two-sided R -module, then the *dual of M* usually refers to either the right or the left dual as above.

If a convention for the rightness or the leftness of modules is fixed, then the appropriate dual module is often denoted by $M^\vee$.

In any case, the functor $M \mapsto M^\vee$ is a contravariant functor (Definition 1.2.2) from the appropriate category of modules (Definition 2.1.3) to itself. More explicitly, the dual module constructions yield functors

$$\begin{aligned} (-)^\vee : {}_R \text{Mod}^{\text{op}} &\rightarrow \text{Mod}_R \\ (-)^\vee : \text{Mod}_R^{\text{op}} &\rightarrow {}_R \text{Mod}. \end{aligned}$$

(Definition 1.2.1) (Definition 2.1.3)

If R is a field (Definition C.0.12) F and V is an F -vector space (Definition C.0.14), then the dual module

$$V^* = V^\vee := \text{Hom}_F(V, F)$$

is called the *dual vector space of V* .

2.1.1. Extension/Restriction/co-extension of scalars.

Definition 2.1.13. Let R and S be rings (Definition C.0.7) (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism (Definition C.0.13). This homomorphism gives S the structure of an (R, R) -bimodule (Definition 2.1.1) via restriction of scalars (Definition 2.1.14). Let T be a ring (Definition C.0.7).

1. The *extension of scalars* (or *base change* or *induction*) along f is defined separately for modules:

- For an $(R - T)$ -bimodule M , the extension of scalars of M along f is the $(S - T)$ -bimodule $S \otimes_R M$ (Definition 2.1.9) where S is viewed as an (S, R) -bimodule. In particular, the action of $s' \in S$ on a simple tensor (Definition 2.1.9) $s \otimes m$ is given by $s' \cdot (s \otimes m) = (s's) \otimes m$.
 - For a $(T - R)$ -bimodule M , the extension of scalars of M along f is the $(T - S)$ -bimodule $M \otimes_R S$ where S is viewed as an (R, S) -bimodule. In particular, the action of $s' \in S$ on a simple tensor $m \otimes s$ is given by $(m \otimes s) \cdot s' = m \otimes (ss')$.
2. The **base change functor** (or **extension of scalars functor** or **induction functor**), denoted by f^* , $S \otimes_R -$, $- \otimes_R S$, or Ind_R^S , is given by:
- For left R -modules:

$$f^* : {}_R\text{Mod}_T \rightarrow {}_S\text{Mod}_T, \quad M \mapsto S \otimes_R M.$$

(Definition 2.1.3) This is the left adjoint (Definition 2.5.1) to the restriction of scalars (Definition 2.1.14) functor $f_* : {}_S\text{Mod}_T \rightarrow {}_R\text{Mod}_T$.

- For right R -modules:

$$f^* : {}_T\text{Mod}_R \rightarrow {}_T\text{Mod}_S, \quad M \mapsto M \otimes_R S.$$

This is the left adjoint to the restriction of scalars functor $f_* : {}_T\text{Mod}_S \rightarrow {}_T\text{Mod}_R$.

3. Let A be an R -ring (Definition C.0.13), i.e. a ring equipped with a ring homomorphism $R \rightarrow A$. Assume that S or A is an R -algebra (Definition C.0.29).

The **base change of the algebra A along f** is the S -ring defined as

$$A_S := S \otimes_R A$$

(Definition 2.1.10) equipped with the natural homomorphism $S \rightarrow S \otimes_R A$ given by $s \mapsto s \otimes 1_A$. As a ring, the multiplication in A_S is determined by $(s_1 \otimes a_1)(s_2 \otimes a_2) = (s_1 s_2) \otimes (a_1 a_2)$ for $s_1, s_2 \in S$ and $a_1, a_2 \in A$.

4. Then the base change construction induces functors in the following situations:

- If S is only an R -ring, then base change induces a functor $f^* : \mathbf{Alg}_R \rightarrow \mathbf{Ring}_S$.
- If S is an R -algebra, then base change induces a functor $f^* : \mathbf{Ring}_R \rightarrow \mathbf{Ring}_S$ which restricts to a functor $f^* : \mathbf{Alg}_R \rightarrow \mathbf{Alg}_S$.

In either case, the base change functor is defined as follows:

- On objects: For any R -algebra (A, φ) , $f^*(A)$ is the S -algebra $S \otimes_R A$ defined above.
- On morphisms: For any homomorphism of R -algebras $h : A \rightarrow B$, the image $f^*(h)$ is the map $\text{id}_S \otimes h : S \otimes_R A \rightarrow S \otimes_R B$, defined by $s \otimes a \mapsto s \otimes h(a)$.

(♠ TODO: comment on adjunction)

Definition 2.1.14. Let R and S be associative rings with identity (Definition C.0.7), and let $\varphi : R \rightarrow S$ be a unital ring homomorphism (Definition C.0.13). Let T be a ring.

1. Let $(M, +)$ be an abelian group (Definition C.0.3) equipped with either the structure of a $S - T$ -bimodule (Definition 2.1.1) $(M, +, \cdot_S)$ or the structure of a $T - S$ -bimodule $(M, +, \cdot_S)$.

The **restriction of scalars of M along φ** is the R -module structure on the same underlying abelian group $(M, +)$ defined as follows:

- If M is a $S - T$ -bimodule, the restriction of scalars of M along φ , often denoted by $\varphi_* M$ (or ${}_R M$), is the $R - T$ -bimodule whose R -action $\cdot_R : R \times M \rightarrow M$ is given by

$$r \cdot_R m := \varphi(r) \cdot_S m$$

for all $r \in R$ and $m \in M$.

- If M is a $T - S$ -bimodule, the restriction of scalars of M along φ , often denoted by $\varphi_* M$ (or M_R), is the $T - R$ -bimodule whose R -action $\cdot_R : M \times R \rightarrow M$ is given by

$$m \cdot_R r := m \cdot_S \varphi(r)$$

for all $r \in R$ and $m \in M$.

2. Let ${}_S \text{Mod}_T$ and ${}_T \text{Mod}_S$ be the categories of $S - T$ and $T - S$ -bimodules (Definition 2.1.3), respectively, and similarly for R .

The *restriction of scalars functor for modules* is the covariant functor (Definition 1.2.2) induced by φ , defined for both left and right modules:

- For left S -modules, it is the functor $\varphi_* : {}_S \text{Mod}_T \rightarrow {}_R \text{Mod}_T$ defined as follows:
 - (a) On objects: For any left S -module M , $\varphi_*(M)$ is the left R -module obtained by restriction of scalars along φ .
 - (b) On morphisms: For any homomorphism (Definition 2.1.2) of left S -modules $h : M \rightarrow N$, the image $\varphi_*(h) : \varphi_*(M) \rightarrow \varphi_*(N)$ is the map h itself, viewed as a homomorphism of left R -modules.
- For right S -modules, it is the functor $\varphi_* : {}_T \text{Mod}_S \rightarrow {}_T \text{Mod}_R$ defined as follows:
 - (a) On objects: For any right S -module M , $\varphi_*(M)$ is the right R -module obtained by restriction of scalars along φ .
 - (b) On morphisms: For any homomorphism of right S -modules $h : M \rightarrow N$, the image $\varphi_*(h) : \varphi_*(M) \rightarrow \varphi_*(N)$ is the map h itself, viewed as a homomorphism of right R -modules.

In either context, if φ is an inclusion map (making R a subring of S), this functor is often called the *forgetful functor*.

3. Let Ring_S (or S/Ring) denote the category of S -rings. Let Ring_R be defined similarly. The *restriction of scalars functor for rings*, denoted by $\varphi_* : \text{Ring}_S \rightarrow \text{Ring}_R$ is the functor defined as follows:

- On objects: Let (A, ψ) be an S -ring. Then $\varphi_*(A)$ is the R -ring $(A, \psi \circ \varphi)$, where the structure map is the composition $R \xrightarrow{\varphi} S \xrightarrow{\psi} A$.
- On morphisms: For any morphism of S -rings $h : (A, \psi_A) \rightarrow (B, \psi_B)$, the image $\varphi_*(h)$ is the map h itself, which satisfies $h \circ (\psi_A \circ \varphi) = \psi_B \circ \varphi$ and is thus a morphism of R -rings.

This functor simply pre-composes the structure map with φ , effectively "forgetting" the factorization through S .

The restriction of scalars functor restricts to a functor $\varphi_* : \mathbf{Alg}_S \rightarrow \mathbf{Alg}_R$ (Definition C.0.29).

Definition 2.1.15. Let R and S be rings (Definition C.0.7) (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism (Definition C.0.13). The homomorphism f endows S with two bimodule structures — that of an (R, S) -bimodule and of an (S, R) -bimodule via restriction of scalars (Definition 2.1.14). Let T be a ring.

1. The **co-extension of scalars** (or simply **co-extension**) along f is defined separately for modules:
 - For an $R - T$ -bimodule M , the co-extension of scalars of M along f is the $S - T$ -bimodule $\text{Hom}_R(S, M)$ (Definition 2.1.8) of left R -module homomorphisms from the $R - S$ -bimodule S to M .
 - For a $T - R$ -bimodule M , the co-extension of scalars of M along f is the $T - S$ -bimodule $\text{Hom}_R(S, M)$ of right R -module homomorphisms from the $S - R$ -bimodule S to M .
2. The **co-extension of scalars functor** (or **coinduction functor**), denoted by $f^!$ or CoInd_R^S , is the functor given by:
 - For left R -modules:

$$f^! : {}_R\text{Mod}_T \rightarrow {}_S\text{Mod}_T, \quad M \mapsto \text{Hom}_R(S, M).$$

This functor is the right adjoint (Definition 2.5.1) to the restriction of scalars functor $f_* : {}_S\text{Mod} \rightarrow {}_R\text{Mod}$.

- For right R -modules:

$$f^! : {}_T\text{Mod}_R \rightarrow {}_T\text{Mod}_S, \quad M \mapsto \text{Hom}_R(S, M).$$

This functor is the right adjoint to the restriction of scalars functor $f_* : \text{Mod}_S \rightarrow \text{Mod}_R$.

Theorem 2.1.16 (Extension-Restriction and Restriction-Coextension adjunction for modules). Let R and S be rings (Definition C.0.7) (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism (Definition C.0.13). Let T be a ring.

1. Extension-Restriction adjunction:

- (a) **Restriction on the Left:** The extension of scalars functor (Definition 2.1.13)

$$f^* = S \otimes_R - : {}_R\text{Mod}_T \rightarrow {}_S\text{Mod}_T$$

(Definition 2.1.3) is left adjoint (Definition 2.5.1) to the restriction of scalars functor (Definition 2.1.14)

$$f_* : {}_S\text{Mod}_T \rightarrow {}_R\text{Mod}_T.$$

Explicitly, for any $R - T$ -bimodule M and any $S - T$ -bimodule N , there is a natural isomorphism:

$$\text{Hom}_{S-T}(S \otimes_R M, N) \cong \text{Hom}_{R-T}(M, f_* N).$$

- (b) **Restriction on the Right:** The extension of scalars functor

$$f^* = - \otimes_R S : {}_T\text{Mod}_R \rightarrow {}_T\text{Mod}_S$$

is left adjoint to the restriction of scalars functor

$$f_* : {}_T\text{Mod}_S \rightarrow {}_T\text{Mod}_R.$$

Explicitly, for any $T - R$ -bimodule M and any $T - S$ -bimodule N , there is a natural isomorphism:

$$\text{Hom}_{T-S}(M \otimes_R S, N) \cong \text{Hom}_{T-R}(M, f_* N).$$

2. Restriction-Coextension adjunction:

(a) **Restriction on the Left:** The co-extension of scalars functor (Definition 2.1.15)

$$f^! = \text{Hom}_R(S, -) : {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T$$

is right adjoint (Definition 2.5.1) to the restriction of scalars functor

$$f_* : {}_S\mathbf{Mod}_T \rightarrow {}_R\mathbf{Mod}_T.$$

Explicitly, for any $S - T$ -bimodule N and any $R - T$ -bimodule M , there is a natural isomorphism:

$$\text{Hom}_{R-T}(f_*N, M) \cong \text{Hom}_{S-T}(N, \text{Hom}_R(S, M)).$$

(b) **Restriction on the Right:** The co-extension of scalars functor

$$f^! = \text{Hom}_R(S, -) : {}_T\mathbf{Mod}_R \rightarrow {}_T\mathbf{Mod}_S$$

is right adjoint to the restriction of scalars functor

$$f_* : {}_T\mathbf{Mod}_S \rightarrow {}_T\mathbf{Mod}_R.$$

Explicitly, for any $T - S$ -bimodule N and any $T - R$ -bimodule M , there is a natural isomorphism:

$$\text{Hom}_{T-R}(f_*N, M) \cong \text{Hom}_{T-S}(N, \text{Hom}_R(S, M))$$

where here $\text{Hom}_R(S, M)$ uses the left R -module structure on S and the right R -module structure on M .

2.2. Limits and colimits in categories.

2.2.1. *Categorical products, coproducts, and direct sums.* It is easier to first learn about products and coproducts (Definition 2.2.1) before learning about general limits and colimits (Definition 2.2.8).

One of the nice features about many algebraic structures such as groups and modules, is that simply taking their Cartesian product produces an object in the same category. In fact, the notion of product is a categorical one. Let us first define the categorical notion of product and the dual notion of coproducts:

Definition 2.2.1 (Product in a category). Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$ indexed by a class I .

1. A **product of the family $\{X_i\}$** is an object P of $\mathcal{C}$ together with a “universal” family of morphisms

$$\pi_i : P \rightarrow X_i, \quad \text{for each } i \in I.$$

More precisely, for any object Y and any family of morphisms $\{f_i : Y \rightarrow X_i\}_{i \in I}$, there exists a unique morphism

$$f : Y \rightarrow P$$

making the following diagram commute for all $i \in I$, i.e. $\pi_i \circ f = f_i$:

$$\begin{array}{ccc}
Y & & \\
\downarrow \exists! f & \searrow f_i & \\
\prod X_i & \xrightarrow{\pi_i} & X_i
\end{array}$$

Such a product is often denoted by $\prod_{i \in I} X_i$. If $\prod_{i \in I} X_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism by the universal property described above.

Equivalently, the product $\prod_{i \in I} X_i$ is the limit (Definition 2.2.8) of the diagram (Definition 2.2.6) $I \rightarrow \mathcal{C}, i \mapsto X_i$, where I is made into a category whose objects are the members of I and whose morphisms are just the identity morphisms.

2. A **coproduct** (or synonymously **direct sum**) of the family $\{X_i\}$ is an object C of $\mathcal{C}$ together with a “universal” family of morphisms

$$\iota_i : X_i \rightarrow C, \quad \text{for each } i \in I.$$

More precisely, for any object Y and any family of morphisms $\{g_i : X_i \rightarrow Y\}_{i \in I}$, there exists a unique morphism

$$g : C \rightarrow Y$$

making the following diagram commute for all $i \in I$, i.e. $g \circ \iota_i = g_i$:

$$\begin{array}{ccc}
X_i & \xrightarrow{\iota_i} & \prod X_i \\
& \searrow g_i & \downarrow \exists! g \\
& & Y
\end{array}$$

Such a coproduct is often denoted by $\coprod_{i \in I} X_i$ or $\oplus_{i \in I} X_i$. If $\coprod_{i \in I} X_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism by the universal property described above.

Equivalently, the coproduct $\coprod_{i \in I} X_i$ is the colimit (Definition 2.2.8) of the diagram (Definition 2.2.6) $I \rightarrow \mathcal{C}, i \mapsto X_i$, where I is made into a category whose objects are the members of I and whose morphisms are just the identity morphisms.

For concrete categories (Definition A.0.8), cartesian products often result in the categorical products:

Definition 2.2.2 (Product of Sets). Let I be a (possibly infinite but small) index set and let $\{A_i\}_{i \in I}$ be a family of sets indexed by I . The **Cartesian product of the family $\{A_i\}_{i \in I}$** , denoted by $\prod_{i \in I} A_i$, is defined as the set of all tuples/functions (Definition C.0.1)

$$\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i \text{ for all } i \in I\},$$

where $(a_i)_{i \in I}$ denotes a function from I to $\bigcup_{i \in I} A_i$ such that $(a_i)_{i \in I}(i) = a_i \in A_i$ for each $i \in I$.

The Cartesian product $\prod_{i \in I} A_i$ is the product (Definition 2.2.1) of the objects A_i in the category of sets (Definition 1.1.7).

The self product of a set A indexed by I is often denoted by A^I . Note that elements of A^I can be identified with functions (Definition C.0.1) $I \rightarrow A$. The finite self product of A taken

n times is often denoted by A^n . For finitely many sets $A_1, \dots, A_n$, their Cartesian product is denoted by $A_1 \times \dots \times A_n$. Elements of such a finite product may be written as $(a_1, \dots, a_n)$.

Definition 2.2.3 (Product of Groups). Let $\{G_i\}_{i \in I}$ be a family of groups indexed by a (possibly infinite but small) set I , each with group operation denoted multiplicatively and identity element e_i . The *(direct) product of the family $\{G_i\}_{i \in I}$* , denoted by $\prod_{i \in I} G_i$, is defined as the set of all tuples/functions

$$\prod_{i \in I} G_i := \{(g_i)_{i \in I} \mid g_i \in G_i \text{ for all } i \in I\},$$

equipped with the binary operation defined componentwise by

$$(g_i)_{i \in I} \cdot (h_i)_{i \in I} := (g_i h_i)_{i \in I}$$

for all $(g_i)_{i \in I}, (h_i)_{i \in I} \in \prod_{i \in I} G_i$. Then $\prod_{i \in I} G_i$ is a group with the identity element $(e_i)_{i \in I}$ and inverses given by

$$(g_i)_{i \in I}^{-1} = (g_i^{-1})_{i \in I}.$$

The product $\prod_{i=1}^n G_i$ is the product (Definition 2.2.1) of the objects G_i in the category of groups (Definition C.0.4). As a set, note that $\prod_{i \in I} G_i$ coincides with the product $\prod_{i \in I} G_i$ (Definition 2.2.2) of the G_i as sets.

A self product of a group G (indexed by a small set I), is often denoted by G^I . A finite self product of a group G taken n times is often denoted by G^n . In case that G is abelian, these may be written as $G^{\oplus I}$ and $G^{\oplus n}$ respectively.

The product of finitely many groups $G_1, \dots, G_n$ is often denoted by $G_1 \times \dots \times G_n$.

Definition 2.2.4 (Product of Modules). Let R and S be (not necessarily commutative) rings (Definition C.0.7), and let $\{M_i\}_{i \in I}$ be a (possibly infinite but small) family of (R, S) -bimodules (Definition 2.1.1).

left R -modules, of right R -modules, of two-sided R -modules (Definition 2.1.1), or of

The *(direct) product of the family $\{M_i\}_{i \in I}$* is defined, as a group (Definition C.0.3), as the product of sets (Definition 2.2.3):

$$\prod_{i \in I} M_i := \{(m_i)_{i \in I} \mid m_i \in M_i \text{ for all } i \in I\}.$$

$\prod_{i \in I} M_i$ inherits a natural R - S module structure defined componentwise by the following rules for all $(m_i)_{i \in I}, (n_i)_{i \in I} \in \prod_{i \in I} M_i$ and all scalars $r \in R, s \in S$:

$$(m_i)_{i \in I} + (n_i)_{i \in I} := (m_i + n_i)_{i \in I}, \quad r \cdot (m_i)_{i \in I} \cdot s := (r \cdot m_i \cdot s)_{i \in I}.$$

The zero element of $\prod_{i \in I} M_i$ is the tuple $(0)_{i \in I}$, and additive inverses are given componentwise:

$$-(m_i)_{i \in I} := (-m_i)_{i \in I}.$$

Note that we can define the product of a family $\{M_i\}_{i \in I}$ of left/right/two-sided R -modules by taking the natural bimodule structure (Definition 2.1.1) of each module.

As usual (Definition 2.2.3), $\prod_{i \in I} M_i$ is the categorical product (Definition 2.2.1) of the objects M_i in the appropriate category of modules (Definition 2.1.3). Moreover, the product of finitely many modules $M_1, \dots, M_n$ is often written as $M_1 \times \dots \times M_n$, which agrees with notation for the product of finitely many groups (Definition 2.2.3). We often write the self-product of a module M indexed by a (small) set I as M^I or by $M^{\oplus I}$. A finite self-product of a module M taken n times is often denoted by M^n or $M^{\oplus n}$; note that these all agree with the notations for abelian groups.

Definition 2.2.5 (Coproduct of Modules). Let R and S be (not necessarily commutative) rings (Definition C.0.7), and let $\{M_i\}_{i \in I}$ be a (possibly infinite but small) family of (R, S) -bimodules.

The *coproduct (direct sum) of the family $\{M_i\}_{i \in I}$* , denoted by $\bigoplus_{i \in I} M_i$, is constructed as

$$\bigoplus_{i \in I} M_i := \left\{ (m_i)_{i \in I} \in \prod_{i \in I} M_i \mid m_i = 0 \text{ for all but finitely many } i \in I \right\}$$

(Definition 2.2.4) consisting of all tuples with only finitely many nonzero entries.

Addition and scalar multiplication in $\bigoplus_{i \in I} M_i$ are defined componentwise as in the direct product (Definition 2.2.4):

$$(m_i)_{i \in I} + (n_i)_{i \in I} := (m_i + n_i)_{i \in I}, \quad r \cdot (m_i)_{i \in I} \cdot s := (r \cdot m_i \cdot s)_{i \in I}, \quad r \in R, s \in S.$$

In all cases, the zero element is $(0)_{i \in I}$, and additive inverses are given by $-(m_i)_{i \in I} := (-m_i)_{i \in I}$.

Note that we can define the coproduct of a family $\{M_i\}_{i \in I}$ of left/right/two-sided R -modules by taking the natural bimodule structure (Definition 2.1.1) of each module.

(♠ TODO: submodule) Note that $\bigoplus_{i \in I} M_i$ is a submodule of $\prod_{i \in I} M_i$. Moreover, $\bigoplus_{i \in I} M_i$ is the coproduct (Definition 2.2.1) in the appropriate category of modules (Definition 2.1.3).

For finitely many modules $M_1, \dots, M_n$, the direct sum $\bigoplus_{j=1}^n M_j$, which may also be written as $M_1 \oplus \dots \oplus M_n$, is simply the usual Cartesian product $\prod_{j=1}^n M_j$ (Definition 2.2.4) of the modules, as every tuple automatically has only finitely many nonzero entries.

2.2.2. General limits and colimits. It is probably best to keep in mind that the definitions listed in this subsection are intended as precise definitions, rather than intuitively helpful ones.

Definition 2.2.6 (Diagram in a category and category of diagrams). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1), and let I be a (large) category (Definition 1.1.1).

1. A *diagram of shape I in $\mathcal{C}$* is a functor (Definition 1.2.2) $D : I \rightarrow \mathcal{C}$. We often denote such a diagram by the family $\{D(i)\}_{i \in \text{Ob}(I)}$ with transition maps given by the functorial image of morphisms in I .

A diagram is also synonymously called a *system*. Moreover, the category I is called the *index category* or the *indexing category of the diagram D* .

- Given two diagrams $D, E : I \rightarrow \mathcal{C}$, a **morphism of diagrams** is simply a natural transformation (Definition 1.3.1) $D \Rightarrow E$ of the functors D and E .
- The **category of I -shaped diagrams in $\mathcal{C}$** or simply **diagram category (of I -shaped diagrams in $\mathcal{C}$)**, often denoted $\mathcal{C}^I$, $[I, \mathcal{C}]$, or $\mathbf{Fun}(I, \mathcal{C})$, is the (large) category whose objects are functors $I \rightarrow \mathcal{C}$ (that is, diagrams of shape I in $\mathcal{C}$) and whose morphisms are natural transformations (Definition 1.3.1) between such functors. The category $\mathcal{C}^I$ is also called the **functor category of functors $I \rightarrow \mathcal{C}$** . Equivalently, the functor category $\mathcal{C}^I$ is the category $\mathbf{PreShv}(I^{\text{op}}, \mathcal{C})$ of presheaves (Definition 7.1.1) on I^{op} with values in $\mathcal{C}$ and hence notations for presheaf categories are applicable as notations for functor categories.

If $\mathcal{C}$ is locally small (Definition 1.1.2) and I is small, then $\mathcal{C}^I$ is locally small by Lemma 2.2.7.

Lemma 2.2.7. Let $\mathcal{C}$ be a small category (Definition 1.1.2) (resp. U -small category where U is some universe (Definition A.0.3)) and let $\mathcal{A}$ be a locally small (Definition 1.1.2) category (resp. U -locally small category). The presheaf category $\mathbf{PreShv}(\mathcal{C}, \mathcal{A})$ (Definition 7.1.1) is locally small (resp. U -locally small).

Proof. A morphism $\mathcal{F} \rightarrow \mathcal{G}$ in $\mathbf{PreShv}(\mathcal{C}, \mathcal{A})$ is a natural transformation (Definition 1.3.1) of the functors $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$. Such a natural transformation is encoded by a family $(\eta_C)_C$ of morphisms (satisfying certain conditions) $\eta_C : \mathcal{F}(C) \rightarrow \mathcal{G}(C)$ in $\mathcal{A}$ over objects C of $\mathcal{C}^{\text{op}}$. The product $\prod_{C \in \text{Ob } \mathcal{C}^{\text{op}}} \text{Hom}_{\mathcal{A}}(\mathcal{F}(C), \mathcal{G}(C))$ is a product of (U -small) sets indexed by a (U -small) set, and the collection of natural transformations is a subset of this set. Therefore, $\text{Hom}_{\mathbf{PreShv}(\mathcal{C}, \mathcal{A})}(\mathcal{F}, \mathcal{G})$ is a (U -small) set. $\square$

Definition 2.2.8 (Cones, limits and colimits in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1), let I be a (large) category, and let $D : I \rightarrow \mathcal{C}$ be a diagram (Definition 2.2.6) (Definition 2.2.6).

- A **cone to the diagram D** is an object $L \in \mathcal{C}$ together with a family of morphisms

$$\{\pi_i : L \rightarrow D(i)\}_{i \in I}$$

such that for every morphism $f : i \rightarrow j$ in I , the diagram

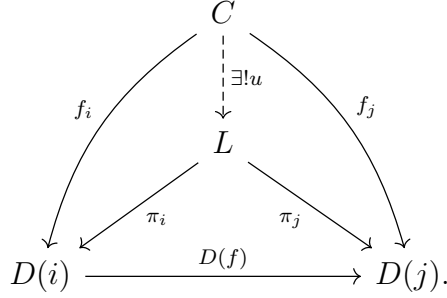
$$\begin{array}{ccc} & L & \\ \pi_i \swarrow & & \searrow \pi_j \\ D(i) & \xrightarrow{D(f)} & D(j) \end{array}$$

commutes, i.e. $D(f) \circ \pi_i = \pi_j$.

- A cone $(L, \{\pi_i\})$ is called a **limit of D** if it satisfies the following “universal property”: for any cone $(C, \{f_i\})$ over D , there exists a **unique** morphism $u : C \rightarrow L$ such that

$$\pi_i \circ u = f_i \quad \text{for all } i \in I.$$

Visually, the following diagrams commute every morphism $f : i \rightarrow j$ in I :

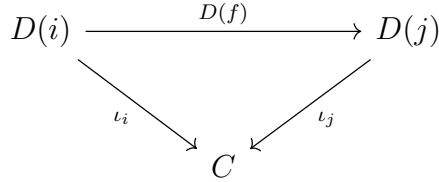


If such a cone exists, then the object L is necessarily unique up to unique isomorphism by the universal property. In this case, L is denoted by $\lim_{i \in I} D$ or $\lim D$.

3. A **cocone from the diagram D** is an object $C \in \mathcal{C}$ together with a family of morphisms

$$\{\iota_i : D(i) \rightarrow C\}_{i \in I}$$

such that for every morphism $f : i \rightarrow j$ in I , the diagram

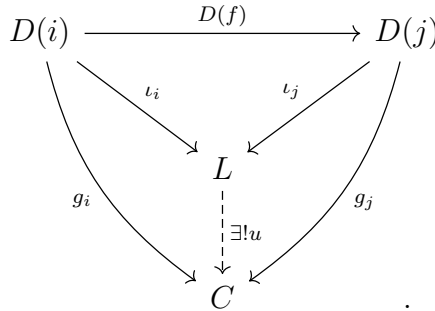


commutes, i.e. $\iota_j \circ D(f) = \iota_i$.

4. A cocone $(L, \{\iota_i\})$ is called a **colimit of D** if it satisfies the following “universal property”: for any cocone $(C, \{g_i\})$ under D , there exists a *unique* morphism $u : L \rightarrow C$ such that

$$u \circ \iota_i = g_i \quad \text{for all } i \in I.$$

Visually, the following diagrams commute every morphism $f : i \rightarrow j$ in I :



If such a cocone exists, then the object L is necessarily unique up to unique isomorphism by the universal property. In this case, L is denoted by $\text{colim}_{i \in I} D$ or $\text{colim } D$.

A limit/colimit is called **finite** (resp. **small**) if the index category I is finite (resp. small).

Some authors use the terms **projective limit** or **inverse limit** to refer to what is defined here as a limit. Similarly, the terms **inductive limit** or **direct limit** are sometimes used to mean a colimit. However, these phrases can have more specific meanings to other authors: a *projective* or *inverse limit* may refer to a limit over a codirected poset (Definition C.0.17). Likewise, an *inductive* or *direct limit* may refer to a colimit over a directed poset (Definition C.0.17) (see Definition 2.2.13).

Thus, while the terms are sometimes used interchangeably with “limit” and “colimit,” they may also emphasize particular indexing shapes and directions, distinguishing them from general limits and colimits taken over arbitrary small categories.

Example 2.2.9 (Geometric and Algebraic Limits/Colimits). Many fundamental constructions in topology and algebra are characterized by universal properties of limits and colimits.

1. **The p -adic Integers (Inverse Limit):** The ring of p -adic integers $\mathbb{Z}_p$ is the limit of the inverse system of finite cyclic rings $\mathbb{Z}/p^n\mathbb{Z}$ with quotient maps:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \{(a_n)_{n \in \mathbb{N}} \in \prod \mathbb{Z}/p^n\mathbb{Z} \mid a_{n+1} \equiv a_n \pmod{p^n}\}$$

2. **Absolute Galois Groups (Inverse Limit):** The absolute Galois group $G_K = \text{Gal}(\overline{K}/K)$ of a field K is the inverse limit of the Galois groups of all finite Galois extensions L/K contained in $\overline{K}$:

$$\text{Gal}(\overline{K}/K) \cong \varprojlim_{L/K \text{ finite}} \text{Gal}(L/K)$$

This gives G_K the structure of a profinite group.

3. **The Cone of a Space (Pushout):** Think of $X \times [0, 1]$ as a cylinder. The map $X \rightarrow X \times [0, 1]$ picks out the “top lid.” The map $X \rightarrow \{*\}$ sends that entire lid to a single point. The **pushout** CX is the result of taking the cylinder and pinching the entire top lid into a single vertex.

$$\begin{array}{ccc} X & \xrightarrow{(x,1)} & X \times [0, 1] \\ \downarrow & & \downarrow \\ \{*\} & \longrightarrow & CX = (X \times [0, 1]) / (X \times \{1\}) \end{array}$$

(♠ TODO: examples of limits and colimits, including products, coproducts, equalizer and coequalizer)

Definition 2.2.10. Let $\mathcal{C}$ be a (large) category (Definition 1.1.1), let I be a small category (Definition 1.1.2), and let $D : I \rightarrow \mathcal{C}$ be a diagram (Definition 2.2.6).

A limit or colimit (Definition 2.2.8) is called **finite** (resp. **small**) if the indexing category I has finitely many objects and morphisms (resp. if I is a small category (Definition 1.1.2)).

Definition 2.2.11 (Complete and Cocomplete Category). Let $\mathcal{C}$ be a category (Definition 1.1.1).

- The category $\mathcal{C}$ is called **complete** (resp. **finitely complete**) if all small limits (Definition 2.2.10) (resp. finite limits) exist in $\mathcal{C}$; that is, for every small diagram $D : J \rightarrow \mathcal{C}$ (with J a small category), the limit $\lim D$ exists and is an object of $\mathcal{C}$.
- The category $\mathcal{C}$ is called **cocomplete** (resp. **finitely cocomplete**) if all small colimits (Definition 2.2.10) (resp. finite colimits) exist in $\mathcal{C}$; that is, for every small diagram $D : J \rightarrow \mathcal{C}$, the colimit $\text{colim } D$ exists and is an object of $\mathcal{C}$.

Definition 2.2.12 (Filtered category). 1. A **filtered category** is a (nonempty, large) category $\mathcal{I}$ satisfying the following conditions:

- For every finite collection of objects $i_1, i_2, \dots, i_n$ in $\mathcal{I}$, there exists an object j and morphisms

$$\phi_k : i_k \rightarrow j, \quad \text{for each } k = 1, \dots, n.$$

- For every pair of morphisms $f, g : i \rightarrow j$ in $\mathcal{I}$, there exists an object k and a morphism

$$h : j \rightarrow k$$

that satisfies

$$h \circ f = h \circ g.$$

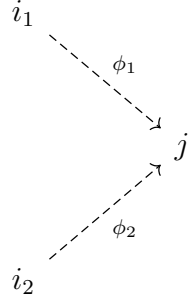


FIGURE 1. *

Condition 1: Upper Bound

$$i \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} j \xrightarrow{\quad h \quad} k$$

FIGURE 2. *

Condition 2: Coequalizing map

In other words, $\mathcal{I}$ is nonempty, any finite diagram of objects admits a cocone (Definition 2.2.8), and any pair of parallel morphisms become equal after post-composition with an appropriate morphism.

2. Dually, a *Cofiltered category* is a category whose opposite category (Definition 1.2.1) is filtered. More explicitly, A cofiltered category is a (nonempty, large) category $\mathcal{I}$ satisfying the following conditions:

- For every finite collection of objects $i_1, i_2, \dots, i_n$ in $\mathcal{I}$, there exists an object j and morphisms

$$\phi_k : j \rightarrow i_k, \quad \text{for each } k = 1, \dots, n.$$

- For every pair of morphisms $f, g : j \rightarrow i$ in $\mathcal{I}$, there exists an object k and a morphism

$$h : k \rightarrow j$$

that satisfies

$$f \circ h = g \circ h.$$

In other words, $\mathcal{I}$ is nonempty, any finite diagram of objects admits a cone, and any pair of parallel morphisms become equal after pre-composition with an appropriate morphism.

Definition 2.2.13 (Special cases of limits). Let $\mathcal{C}$ be a (large) category. Let I be a (large) category. Let $I \rightarrow \mathcal{C}$ be a diagram/system.

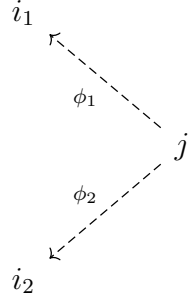


FIGURE 3. *

Condition 1: Lower Bound

$$k \overset{h}{\dashrightarrow} j \overset{f}{\underset{g}{\rightrightarrows}} i$$

FIGURE 4. *

Condition 2: Equalizing map

- Suppose that the system is a cofiltered system, i.e. I is a cofiltered category. A limit (Definition 2.2.8) of this diagram is often denoted by

$$\varprojlim_{i \in I} D(i)$$

and may be called a *cofiltered (inverse/projective) limit*. In case that the system is more specifically an inverse/projective system, i.e. I is a cofiltered poset (Definition C.0.17), the preferred term for such a limit is *inverse/projective limit*.

- Suppose that the system is a filtered system, i.e. I is a filtered category. A colimit of this diagram is often denoted by

$$\varinjlim_{i \in I} D(i)$$

and may be called a *filtered colimit* or a *direct/inductive/injective limit*. In case that the system is more specifically a direct/inductive system, i.e. I is a filtered poset (Definition C.0.17), the preferred term for such a limit is *direct/inductive limit*.

One noteworthy fact is that if a category has all (co)products (Definition 2.2.1) and (co)equalizers (Definition 2.2.14), then it has all (co)limits (Definition 2.2.8) (Theorem 2.2.15).

Definition 2.2.14 (Equalizer in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.1.1) and let $f, g : X \rightarrow Y$ be morphisms in $\mathcal{C}$.

1. An *equalizer of f and g* is an object E together with a morphism

$$e : E \rightarrow X$$

such that

$$f \circ e = g \circ e$$

and for any object Z with morphism $z : Z \rightarrow X$ satisfying

$$f \circ z = g \circ z,$$

there exists a unique morphism $u : Z \rightarrow E$ making the diagram commute:

$$e \circ u = z.$$

$$\begin{array}{ccccc}
Z & & & & \\
\downarrow \exists! u & \searrow z & & & \\
E & \xrightarrow{e} & X & \xrightleftharpoons[f]{g} & Y
\end{array}$$

If such an equalizer of f and g exists, then we say that the following **equalizer diagram** is exact:

$$E \xrightarrow{e} X \xrightleftharpoons[f]{g} Y$$

2. A **coequalizer of f and g** is an object Q together with a morphism

$$q : Y \rightarrow Q$$

such that

$$q \circ f = q \circ g$$

and for any object Z with morphism $w : Y \rightarrow Z$ satisfying

$$w \circ f = w \circ g,$$

there exists a unique morphism $v : Q \rightarrow Z$ making the diagram commute:

$$\begin{array}{ccccc}
X & \xrightleftharpoons[f]{g} & Y & \xrightarrow{q} & Q \\
& & \searrow w & & \downarrow \exists! v \\
& & & & Z
\end{array}$$

If such a coequalizer of f and g exists, then we say that the following **coequalizer diagram** is exact:

$$X \xrightleftharpoons[f]{g} Y \xrightarrow{q} Q$$

Theorem 2.2.15. Let $\mathcal{C}$ be a category (Definition 1.1.1). Let $F : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram (Definition 2.2.6) where $\mathcal{J}$ is a small category (Definition 1.1.2).

1. The limit (Definition 2.2.8) of F is constructed as the equalizer (Definition 2.2.14) of the pair of morphisms (s, t) : assuming that the products (Definition 2.2.1) and equalizers below exist in $\mathcal{C}$, the limit $\lim F$ exists and

$$\lim F \cong \text{eq} \left(\prod_{j \in \text{Ob}(\mathcal{J})} F(j) \xrightleftharpoons[t]{s} \prod_{\alpha \in \text{Mor}(\mathcal{J})} F(\text{cod}(\alpha)) \right)$$

where $\text{cod}(\alpha)$ stands for the codomain of the morphism α , and the morphisms s and t are induced by the universal property of the product, such that for any morphism $\alpha : i \rightarrow k$ in $\mathcal{J}$, the projection to the factor indexed by α is:

- $\pi_\alpha \circ s = F(\alpha) \circ \pi_i$
- $\pi_\alpha \circ t = \pi_k$

2. The colimit (Definition 2.2.8) of F is constructed as the coequalizer (Definition 2.2.14) of the pair of morphisms (s, t) : assuming that the coproducts (Definition 2.2.1) and coequalizers below exist in $\mathcal{C}$, the colimit $\text{colim } F$ exists and

$$\text{colim } F \cong \text{coeq} \left(\coprod_{\alpha \in \text{Mor}(\mathcal{J})} F(\text{dom}(\alpha)) \xrightarrow[t]{s} \coprod_{j \in \text{Ob}(\mathcal{J})} F(j) \right)$$

where $\text{dom}(\alpha)$ stands for the domain of the morphism α , and the morphisms s and t are induced by the universal property of the coproduct, such that for any morphism $\alpha : i \rightarrow k$ in $\mathcal{J}$, the injection from the summand indexed by α is:

- $s \circ \iota_\alpha = \iota_k \circ F(\alpha)$
- $t \circ \iota_\alpha = \iota_i$

In particular,

1. If $\mathcal{C}$ has all nonempty finite (resp. small) products and equalizers, then $\mathcal{C}$ has all nonempty finite (resp. small) limits.
2. If $\mathcal{C}$ has all nonempty finite (resp. small) coproducts and coequalizers, then $\mathcal{C}$ has all nonempty finite (resp. small) colimits.
3. If $\mathcal{C}$ has all finite (resp. small) products and equalizers, then $\mathcal{C}$ has all finite (resp. small) limits.
4. If $\mathcal{C}$ has all finite (resp. small) coproducts and coequalizers, then $\mathcal{C}$ has all finite (resp. small) colimits.

We can consider limits of limits or colimits of colimits as long as the diagrams are set up for such notions to make sense; under nice enough circumstances, limits commute with limits and colimits commute with colimits.

Theorem 2.2.16 (Fubini Theorem for Limits and Colimits (Definition 2.2.8)). Let I and J be categories (Definition 1.1.1) and $F : I \times J \rightarrow \mathcal{C}$ be a bifunctor.

1. **Iterated Limits:** Suppose the product (Definition 2.2.1) limit $\lim_{(i,j) \in I \times J} F(i, j)$ exists. If for each $i \in I$ the partial limit $\lim_{j \in J} F(i, j)$ exists, then the iterated limit $\lim_{i \in I} (\lim_{j \in J} F(i, j))$ necessarily exists and there is a unique natural isomorphism (Definition 1.3.1):

$$(A) \quad \lim_{i \in I} \lim_{j \in J} F(i, j) \cong \lim_{(i,j) \in I \times J} F(i, j)$$

Consequently, if the product limit exists and all partial limits $\lim_{j \in J} F(i, j)$ and $\lim_{i \in I} F(i, j)$ exist, then both iterated limits exist and commute:

$$(B) \quad \lim_{i \in I} \lim_{j \in J} F(i, j) \cong \lim_{j \in J} \lim_{i \in I} F(i, j)$$

2. **Iterated Colimits:** Suppose the product colimit $\text{colim}_{(i,j) \in I \times J} F(i, j)$ exists. If for each $i \in I$ the partial colimit $\text{colim}_{j \in J} F(i, j)$ exists, then the iterated colimit $\text{colim}_{i \in I} (\text{colim}_{j \in J} F(i, j))$ necessarily exists and there is a unique natural isomorphism:

$$(C) \quad \text{colim}_{i \in I} \text{colim}_{j \in J} F(i, j) \cong \text{colim}_{(i,j) \in I \times J} F(i, j)$$

Consequently, if the product colimit exists and all partial colimits $\operatorname{colim}_{j \in J} F(i, j)$ and $\operatorname{colim}_{i \in I} F(i, j)$ exist, then both iterated colimits exist and commute:

$$(D) \quad \operatorname{colim}_{i \in I} \operatorname{colim}_{j \in J} F(i, j) \cong \operatorname{colim}_{j \in J} \operatorname{colim}_{i \in I} F(i, j)$$

2.3. Additive categories.

2.3.1. Additive and abelian categories. Given modules M and N , notice that $\operatorname{Hom}(M, N)$ (Definition 2.1.8) is not only a set, but also an abelian group. This, along with few other nice properties, makes the category of modules into an additive category (Definition 2.3.4).

Definition 2.3.1. Let $\mathcal{C}$ be a (large) category (Definition 1.1.1).

1. An object $I \in \mathcal{C}$ is called an **initial object** if for every object $X \in \mathcal{C}$ there exists a unique morphism

$$I \rightarrow X.$$

Equivalently, an initial object is a limit (Definition 2.2.8) of the empty diagram (Definition 2.2.6), if such a limit exists.

2. An object $F \in \mathcal{C}$ is called a **final object** (or **terminal object**) if for every object $X \in \mathcal{C}$ there exists a unique morphism

$$X \rightarrow F.$$

Equivalently, a final object is a colimit (Definition 2.2.8) of the empty diagram (Definition 2.2.6), if such a colimit exists.

3. An object $Z \in \mathcal{C}$ is called a **zero object** if Z is both initial and final in $\mathcal{C}$. In particular, for every object $X \in \mathcal{C}$ there exist unique morphisms

$$Z \rightarrow X \quad \text{and} \quad X \rightarrow Z.$$

In particular, if initial/final/zero objects exist in a category, then they are unique up to unique isomorphism.

Definition 2.3.2. A **pointed category** is a category (Definition 1.1.1) that has a zero object (Definition 2.3.1).

Definition 2.3.3 (Zero morphism). Let $\mathcal{C}$ be a pointed category (Definition 2.3.2). Let X and Y be objects in $\mathcal{C}$, and let 0 be a zero object (Definition 2.3.1) of $\mathcal{C}$. The **zero morphism** from X to Y is the unique composite morphism

$$0_{XY} : X \rightarrow 0 \rightarrow Y,$$

where $X \rightarrow 0$ is the unique morphism into the terminal object 0 and $0 \rightarrow Y$ is the unique morphism out of the initial object 0 . This morphism is independent of the choice of the zero object 0 .

Definition 2.3.4 (Additive category). Let $\mathcal{A}$ be a locally small category (Definition 1.1.2).

1. $\mathcal{A}$ is said to be a **preadditive category** if the following hold:
 - For any two objects A, B in $\mathcal{A}$, the set $\operatorname{Hom}_{\mathcal{A}}(A, B)$ is an abelian group (Definition C.0.3), and composition of morphisms is bilinear.

- There is a zero object (Definition 2.3.1) 0 in $\mathcal{A}$.
2. If $\mathcal{A}$ is preadditive, then it is called **additive** if it additionally satisfies the following:
- For any two objects A, B in $\mathcal{A}$, there exists a product object $A \times B$ (Definition 2.2.1), often written $A \oplus B$, called the **direct sum of A and B** . In fact, $A \oplus B$ is not only a product but also a coproduct (Definition 2.2.5) of A and B (Lemma 2.3.5).

Given a finite collection $\{A_i\}_i$ of objects A_i in an additive category $\mathcal{A}$, we may more generally speak of the **direct sum** $\bigoplus_i A_i$; it has canonical injections from and projections to each A_i .

Lemma 2.3.5. Let $\mathcal{A}$ be a preadditive category (Definition 2.3.4). Finite products (Definition 2.2.1) in $\mathcal{A}$ coincide with finite coproducts (Definition 2.2.1). More precisely, if $\{A_i\}_{i=1}^n$ is a finite collection of objects of $\mathcal{A}$, then

1. if $\prod_{i=1}^n A_i$ exists, then so does $\coprod_{i=1}^n A_i$ and these are naturally isomorphic.
2. if $\coprod_{i=1}^n A_i$ exists, then so does $\prod_{i=1}^n A_i$ and these are naturally isomorphic.

Proof. (♠ TODO:) □

In fact, morphisms between modules have kernels, cokernels, and nice images and coimages; the same can be said of morphisms between general abelian categories. Much of our intuition about exact sequences in categories of modules carries over to exact sequences in abelian categories

Definition 2.3.6. Let $\mathcal{C}$ be a (large) (Definition 1.1.1) pointed category (Definition 2.3.2), i.e. a category with a zero object (Definition 2.3.1) 0 . Let $X, Y \in \text{Ob}(\mathcal{C})$ be an object and let $f : X \rightarrow Y$ be a morphism.

1. A morphism $i : K \rightarrow X$ is called the **kernel of f** if:
 - (a) $f \circ i = 0$, where 0 is the zero morphism (Definition 2.3.3) $K \rightarrow Y$,
 - (b) for any morphism $g : Z \rightarrow X$ such that $f \circ g = 0$, there exists a unique morphism $u : Z \rightarrow K$ such that $g = i \circ u$.

The kernel, if it exists, is unique up to unique isomorphism (Definition 1.1.13). $\ker(f)$ denotes the object K determined (up to isomorphism) by a kernel of f .

Equivalently, $\ker(f)$ is the equalizer (Definition 2.2.14) of f and the 0 morphism $X \rightarrow Y$, see Lemma 2.3.7.

2. a morphism $p : Y \rightarrow Q$ is called the **cokernel of f** if:
 - (a) $p \circ f = 0$, where 0 is the zero morphism (Definition 2.3.1) $X \rightarrow Q$,
 - (b) for any morphism $g : Y \rightarrow Z$ such that $g \circ f = 0$, there exists a unique morphism $v : Q \rightarrow Z$ such that $g = v \circ p$.

The cokernel, if it exists, is unique up to unique isomorphism. $\text{coker}(f)$ denotes the object Q determined (up to isomorphism) by a cokernel of f .

Equivalently, $\text{coker}(f)$ is the coequalizer (Definition 2.2.14) of f and the 0 morphism $X \rightarrow Y$, see Lemma 2.3.7.

Lemma 2.3.7. Let $\mathcal{C}$ be a pointed category (Definition 2.3.2). All Kernels/cokernels (Definition 2.3.6) are to equalizers/coequalizers (Definition 2.2.14) in the following sense: given

a morphism $f : X \rightarrow Y$ in $\mathcal{C}$, $\ker f$ and $\operatorname{coker} f$ are respectively isomorphic to the equalizer and the coequalizer of f and the zero map (Definition 2.3.3) $0_{XY} : X \rightarrow Y$.

Definition 2.3.8 (Monomorphism and Epimorphism in Categories). Let $\mathcal{C}$ be a category (Definition 1.1.1). For objects $A, B \in \mathcal{C}$, let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$.

- The morphism f is called a **monomorphism** (or a **monic morphism**) if for every object X and every pair of morphisms $g_1, g_2 : X \rightarrow A$, the equality $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$.
- The morphism f is called an **epimorphism** (or an **epic morphism**) if for every object Y and every pair of morphisms $h_1, h_2 : B \rightarrow Y$, the equality $h_1 \circ f = h_2 \circ f$ implies $h_1 = h_2$.

Lemma 2.3.9. Let $\mathcal{C}$ be a (large) (Definition 1.1.1) pointed category (Definition 2.3.2), i.e. a category with a zero object (Definition 2.3.1) 0 . Let $f : X \rightarrow Y$ be a morphism.

1. The kernel (Definition 2.3.6) of f is a monomorphism (Definition 2.3.8) $\ker(f) \rightarrow X$.
2. The cokernel (Definition 2.3.6) of f is an epimorphism (Definition 2.3.8) $Y \rightarrow \operatorname{coker}(f)$.

Proof. (♠ TODO:) □

Definition 2.3.10. Let $\mathcal{C}$ be a category (Definition 1.1.1), and let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$.

1. An **image of f** consists of an object $I \in \operatorname{Ob}(\mathcal{C})$ together with a factorization of f into two morphisms

$$A \xrightarrow{e} I \xrightarrow{m} B,$$

where e is an epimorphism (Definition 2.3.8) and m is a monomorphism (Definition 2.3.8), such that for any other factorization

$$A \xrightarrow{e'} I' \xrightarrow{m'} B$$

with e' epi and m' mono, there exists a unique isomorphism $\varphi : I \simeq I'$ satisfying $m = m'\varphi$ and $\varphi e = e'$.

$$\begin{array}{ccc} & I & \\ e \nearrow & \downarrow \exists! \varphi & \nwarrow m \\ A & & B \\ e' \searrow & \downarrow & \nearrow m' \\ & I' & \end{array}$$

The monomorphism $m : I \rightarrow B$ (or equivalently its subobject class) is called the **image of f in $\mathcal{C}$** .

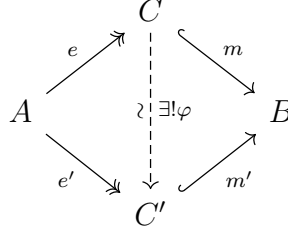
2. Let $\mathcal{C}$ be a category (Definition 1.1.1), and let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$. A **coimage of f** consists of an object $C \in \operatorname{Ob}(\mathcal{C})$ together with a factorization of f into two morphisms

$$A \xrightarrow{e} C \xrightarrow{m} B,$$

where e is an epimorphism and m is a monomorphism, such that for any other factorization

$$A \xrightarrow{e'} C' \xrightarrow{m'} B$$

with e' epi and m' mono, there exists a unique isomorphism $\varphi : C \simeq C'$ satisfying $m = m'\varphi$ and $\varphi e = e'$.



The epimorphism $e : A \rightarrow C$ (or equivalently its quotient class) is called the **coimage of f in $\mathcal{C}$** .

Lemma 2.3.11. Let R, S be (not-necessarily commutative) rings with unity (Definition C.0.7), let M, N be R - S -bimodules (Definition 2.1.1), and let $f : M \rightarrow N$ be a module homomorphism (Definition 2.1.2). The kernel, cokernel, image, and coimage (Definition 2.1.7) of f are respectively the categorical kernel, cokernel (Definition 2.3.6), image, and coimage (Definition 2.3.10).

Definition 2.3.12 (Abelian category). Let $\mathcal{A}$ be a category. The category $\mathcal{A}$ is an **abelian category** if:

- $\mathcal{A}$ is an additive category (Definition 2.3.4).
- Every morphism $f : A \rightarrow B$ has a kernel $\ker(f)$ and a cokernel $\operatorname{coker}(f)$ (Definition 2.3.6).
- For every morphism $f : A \rightarrow B$, the canonical morphism $\operatorname{coim}(f) \rightarrow \operatorname{im}(f)$ is an isomorphism, where

$$\operatorname{coim}(f) = \operatorname{coker}(\ker(f) \rightarrow A), \quad \operatorname{im}(f) = \ker(B \rightarrow \operatorname{coker}(f)).$$

(♠ TODO: I think I need to re-check this definition) (♠ TODO: coimage)

It is also worth considering Grothendieck's additional axioms for abelian categories (Definition 2.5.7).

Here are some nice categorical facts about abelian categories to keep in mind:

Lemma 2.3.13. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). All kernels (Lemma 2.3.7) are tantamount to equalizers (Definition 2.2.14) and all cokernels are tantamount to coequalizers in $\mathcal{A}$. More precisely,

1. • Given a morphism $f : X \rightarrow Y$ in $\mathcal{A}$, $\ker f$ is isomorphic to the equalizer of f and the zero morphism $0 : X \rightarrow Y$.
• Given morphisms $f, g : X \rightarrow Y$ in $\mathcal{A}$, the equalizer of f and g is isomorphic to $\ker(f - g)$.

2. • Given a morphism $f : X \rightarrow Y$ in $\mathcal{A}$, $\text{coker } f$ is isomorphic to the coequalizer of f and the zero morphism $0 : X \rightarrow Y$.
- Given morphisms $f, g : X \rightarrow Y$ in $\mathcal{A}$, the coequalizer of f and g is isomorphic to $\text{coker}(f - g)$.

Proof. All kernels and cokernels are equalizers and coequalizers in the claimed sense by Lemma 2.3.7. One can conversely see that the equalizer/coequalizer of f and g has the same universal property as $\ker(f - g)/\text{coker}(f - g)$. $\square$

Lemma 2.3.14. Any abelian category (Definition 2.3.12) admits all finite (Definition 2.2.10) limits (Definition 2.2.8) and all finite colimits.

Proof. By definition, any morphism in an abelian category has a kernel and a cokernel (Definition 2.3.6), which are tantamount to (Lemma 2.3.13) equalizers and coequalizers (Definition 2.2.14). Moreover, any additive category (Definition 2.3.4) admits finite products by definition, which coincide with finite coproducts by Lemma 2.3.5. Therefore, any abelian category admits finite limits and finite colimits by Theorem 2.2.15. $\square$

Proposition 2.3.15. Let $\mathcal{A}$ be a preadditive (Definition 2.3.4) (resp. additive (Definition 2.3.4), abelian (Definition 2.3.12)) category and let J be a small (Definition 1.1.2) category. The diagram category (Definition 2.2.6) $\mathcal{A}^J$ is preadditive (resp. additive, abelian).

Proposition 2.3.16 (Commutativity of Limits and Colimits in Abelian Categories). Let $\mathcal{A}$ be an Abelian category (Definition 2.3.12), and let $F : I \times J \rightarrow \mathcal{A}$ be a bifunctor.

1. **Finite Limits and Colimits:** If I and J are finite categories (Definition 1.1.1), then the following natural isomorphisms (Definition 1.3.1) always hold in $\mathcal{A}$:

$$(E) \quad \lim_{i \in I} \lim_{j \in J} F(i, j) \cong \lim_{(i, j) \in I \times J} F(i, j) \cong \lim_{j \in J} \lim_{i \in I} F(i, j)$$

$$(F) \quad \text{colim}_{i \in I} \text{colim}_{j \in J} F(i, j) \cong \text{colim}_{(i, j) \in I \times J} F(i, j) \cong \text{colim}_{j \in J} \text{colim}_{i \in I} F(i, j)$$

(Definition 2.2.8) In particular, for any finite family of morphisms $f_k : A_k \rightarrow B_k$ indexed by $k \in K$, we have $\ker(\bigoplus f_k) \cong \bigoplus \ker(f_k)$ and $\text{coker}(\bigoplus f_k) \cong \bigoplus \text{coker}(f_k)$ (Definition 2.3.6).

2. **Small Colimits (Axiom AB3):** If $\mathcal{A}$ satisfies axiom AB3 (Definition 2.5.7) and I, J are small categories (Definition 1.1.2), the iterated colimits (Definition 2.2.8) exist and commute:

$$(G) \quad \text{colim}_{i \in I} \text{colim}_{j \in J} F(i, j) \cong \text{colim}_{(i, j) \in I \times J} F(i, j) \cong \text{colim}_{j \in J} \text{colim}_{i \in I} F(i, j)$$

Specifically, this implies that the cokernel (Definition 2.3.6) of a small (Definition 1.1.2) coproduct of morphisms is the coproduct of the cokernels (Lemma 2.3.7): $\text{coker}(\coprod f_k) \cong \coprod \text{coker}(f_k)$.

3. **Small Limits (Axiom AB3*):** If $\mathcal{A}$ satisfies axiom AB3 (Definition 2.5.7)* and I, J are small categories (Definition 1.1.2), the iterated limits (Definition 2.2.8) exist and commute:

$$(H) \quad \lim_{i \in I} \lim_{j \in J} F(i, j) \cong \lim_{(i, j) \in I \times J} F(i, j) \cong \lim_{j \in J} \lim_{i \in I} F(i, j)$$

Specifically, this implies that the kernel (Definition 2.3.6) of a small product (Definition 2.2.1) of morphisms is the product of the kernels (Lemma 2.3.7): $\ker(\prod f_k) \cong \prod \ker(f_k)$.

Proof. These are consequences of Theorem 2.2.16. $\square$

Lemma 2.3.17. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. A morphism $f: A \rightarrow B$ is a monomorphism (Definition 2.3.8) if and only if $\ker f = 0$.
2. A morphism $f: A \rightarrow B$ is an epimorphism (Definition 2.3.8) if and only if $\operatorname{coker} f = 0$.

Proof. We prove that f is a monomorphism if and only if $\ker f = 0$. The other statement is dually true.

($\Rightarrow$) Suppose f is a monomorphism. Let $k: \ker f \rightarrow A$ be the kernel inclusion. Then $f \circ k = 0 = f \circ 0$. Since f is a monomorphism, $k = 0$, so $\ker f = 0$.

($\Leftarrow$) Suppose $\ker f = 0$. Let $g, h: B \rightarrow C$ satisfy $gf = hf$. Let $d = g - h: B \rightarrow C$. Then $df = 0$. The kernel inclusion $k: \ker f \rightarrow A$ satisfies $dk = 0$, so $k = 0$ and $\ker d = 0$. Thus $d = 0$ and $g = h$, so f is a monomorphism. $\square$

2.4. Additive functors between additive categories.

Definition 2.4.1 (Additive functor). Let $\mathcal{A}$ and $\mathcal{B}$ be pre-additive categories (Definition 2.3.4). A functor

$$F: \mathcal{A} \rightarrow \mathcal{B}$$

is an **additive functor** if for every pair of objects $A, A' \in \mathcal{A}$, the induced map

$$F_{A,A'}: \operatorname{Hom}_{\mathcal{A}}(A, A') \rightarrow \operatorname{Hom}_{\mathcal{B}}(F(A), F(A'))$$

is a group homomorphism of abelian groups, or equivalently if it is enriched over the category $\mathbf{Ab}$ of abelian groups.

In the case that $\mathcal{A}$ and $\mathcal{B}$ are additive categories (Definition 2.3.4), this condition implies that F is compatible with the additive structures on $\mathcal{A}$ and $\mathcal{B}$. More precisely,

- F sends the zero object $0_{\mathcal{A}}$ of $\mathcal{A}$ to the zero object $0_{\mathcal{B}}$ of $\mathcal{B}$, i.e.,

$$F(0_{\mathcal{A}}) = 0_{\mathcal{B}}.$$

- F preserves finite direct sums: For any finite family of objects $\{A_i\}_{i=1}^n$ in $\mathcal{A}$,

$$F\left(\bigoplus_{i=1}^n A_i\right) \cong \bigoplus_{i=1}^n F(A_i)$$

via the canonical isomorphism induced by F applied to the canonical injections and projections.

(♠ TODO: examples of additive functors)

We note that Hom's and tensor products induce bi-additive functors (Definition 2.4.2) on categories of modules.

Definition 2.4.2 (n-ary Additive Functor). Let I be a finite set with $|I| = n$. Let $\{\mathcal{A}_i\}_{i \in I}$ be additive categories (Definition 2.3.4) and let $\mathcal{B}$ be an additive category. An *n-ary additive functor* (or *multilinear functor*)

$$F : \prod_{i \in I} \mathcal{A}_i \rightarrow \mathcal{B}$$

(Definition 1.5.1) is a functor such that for each fixed collection of all but one variable, the resulting functor in the remaining variable is additive (Definition 2.4.1). Equivalently, for every $j \in I$ and objects $(A_i)_{i \in I}$ and morphisms $f_1, f_2 : A_j \rightarrow A'_j$ in $\mathcal{A}_j$, we have

$$\begin{aligned} & F(A_1, \dots, A_{j-1}, f_1 + f_2, A_{j+1}, \dots, A_n) \\ &= F(A_1, \dots, A_{j-1}, f_1, A_{j+1}, \dots, A_n) \\ &+ F(A_1, \dots, A_{j-1}, f_2, A_{j+1}, \dots, A_n), \end{aligned}$$

and F preserves zero morphisms componentwise:

$$F(A_1, \dots, 0_{A_j, A'_j}, \dots, A_n) = 0_{F(A_1, \dots), F(A'_1, \dots)}.$$

A bifunctor that satisfies this property for $n = 2$ is simply called a *biadditive functor*.

2.4.1. *Exact functors between abelian categories.* One particularly nice kind of additive functor is an exact functor.

Definition 2.4.3. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. A sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C$ in $\mathcal{A}$ is called *left exact* if f is a kernel (Definition 2.3.6) of g .
2. A sequence $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ in $\mathcal{A}$ is called *right exact* if g is a cokernel (Definition 2.3.6) of f .

Definition 2.4.4. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). A sequence

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

of morphisms in $\mathcal{A}$ is called a *short exact sequence* if the morphisms satisfy:

- $f : A \rightarrow B$ is a kernel of g (Definition 2.3.6),
- $g : B \rightarrow C$ is a cokernel of f (Definition 2.3.6),

In other words, the sequences $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C$ and $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ are left and right exact (Definition 2.4.3) respectively.

Lemma 2.4.5. Let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12). It is exact (Definition 2.4.12) if and only if it preserves kernels and cokernels (Definition 2.3.6).

Although we have defined left/right/short exact sequences in additive categories, it is probably most useful to think about left/right/short exact sequences in abelian categories, where kernels/cokernels/images/coimages are guaranteed to exist.

Definition 2.4.6. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $X' \xrightarrow{f} X \xrightarrow{g} X''$ be (a sequence of) morphisms in $\mathcal{A}$. We say that the sequence is **exact at X** if the image of f is equal to the kernel of g . More precisely, we mean that

1. $g \circ f = 0$ (which implies that there is a canonical morphism $\text{im } f \rightarrow \ker g$ (Definition 2.3.10) (Definition 2.3.6), which is in fact a monomorphism (Definition 2.3.8)), and that
2. the canonical morphism $\text{im } f \rightarrow \ker g$ is in fact an isomorphism.

Lemma 2.4.7. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $X' \xrightarrow{f} X \xrightarrow{g} X''$ be morphisms in $\mathcal{A}$.

1. The following are equivalent:
 - (a) $0 \rightarrow X' \xrightarrow{f} X \xrightarrow{g} X''$ is left exact (Definition 2.4.3) in the sense that f is a kernel of g .
 - (b) $0 \rightarrow X' \xrightarrow{f} X \xrightarrow{g} X''$ is exact (Definition 2.4.6) at X' and at X .
 - (c) f is a monomorphism (Definition 2.3.8) and $\ker g \cong \text{im } f$ (Definition 2.3.6) (Definition 2.3.10).
2. The following are equivalent:
 - (a) $X' \xrightarrow{f} X \xrightarrow{g} X'' \rightarrow 0$ is right exact (Definition 2.4.3) in the sense that g is a cokernel of f .
 - (b) $X' \xrightarrow{f} X \xrightarrow{g} X'' \rightarrow 0$ is exact (Definition 2.4.6) at X and at X'' .
 - (c) g is an epimorphism (Definition 2.3.8) and $\ker g \cong \text{im } f$ (Definition 2.3.6) (Definition 2.3.10).
3. The following are equivalent:
 - (a) $0 \rightarrow X' \xrightarrow{f} X \xrightarrow{g} X'' \rightarrow 0$ is (short) exact (Definition 2.4.4) in the sense of being left and right exact, i.e. f is a kernel of g and g is a cokernel of f ,
 - (b) $0 \rightarrow X' \xrightarrow{f} X \xrightarrow{g} X'' \rightarrow 0$ is exact (Definition 2.4.6) at X' , X , and X'' ,
 - (c) f is a monomorphism (Definition 2.3.8), g is an epimorphism, and $\ker g \cong \text{im } f$ (Definition 2.3.6) (Definition 2.3.10).

Definition 2.4.8. A short exact sequence (Definition 2.4.4) $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ in an additive category (Definition 2.3.4) $\mathcal{A}$ is **split exact** if there exists a morphism $s : C \rightarrow B$ (called a **section**) such that $g \circ s = \text{id}_C$, or equivalently (Theorem 2.4.9), a morphism $r : B \rightarrow A$ (called a **retraction**) such that $r \circ f = \text{id}_A$.

Theorem 2.4.9. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). For a short exact sequence (Definition 2.4.4) $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$, the following conditions are equivalent:

1. The sequence is split exact (Definition 2.4.8).
2. There exists a morphism (i.e. a section (Definition 2.4.8)) $s : C \rightarrow B$ such that $g \circ s = \text{id}_C$.

3. There exists a morphism (i.e. a retraction (Definition 2.4.8)) $r : B \rightarrow A$ such that $r \circ f = \text{id}_A$.
4. There exists an isomorphism $\varphi : B \xrightarrow{\cong} A \oplus C$ such that the following diagram commutes:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C & \longrightarrow & 0 \\
& & \parallel & & \varphi \downarrow & & \parallel & & \\
0 & \longrightarrow & A & \xrightarrow{\iota_1} & A \oplus C & \xrightarrow{\pi_2} & C & \longrightarrow & 0
\end{array}$$

where ι_1 is the canonical injection and π_2 is the canonical projection.

Proposition 2.4.10. In an additive category (Definition 2.3.4) $\mathcal{A}$, any split exact sequence (Definition 2.4.8) $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is preserved by any additive functor (Definition 2.4.1) $F : \mathcal{A} \rightarrow \mathcal{B}$ to another additive category $\mathcal{B}$. That is, $0 \rightarrow F(A) \xrightarrow{F(f)} F(B) \xrightarrow{F(g)} F(C) \rightarrow 0$ is also a split exact sequence.

There are notions of left/right exactness of *functors* — a left/right exact functor between additive categories is one that preserves left/right exact sequences. One can more generally discuss left/right exact functors between more general categories by the property of preserving finite limits/colimits.

Definition 2.4.11. Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

1. Assume $\mathcal{C}$ admits all finite limits (Definition 2.2.8). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called *left exact* if it preserves all finite limits (Definition 2.2.10). Explicitly, for every finite diagram $D : \mathcal{J} \rightarrow \mathcal{C}$, the limit

$$\varprojlim (F \circ D)$$

exists and the canonical map

$$F(\varprojlim D) \xrightarrow{\cong} \varprojlim (F \circ D)$$

is an isomorphism.

2. Assume $\mathcal{C}$ admits all finite colimits (Definition 2.2.8). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called *right exact* if it preserves all finite colimits (Definition 2.2.10). Explicitly, for every finite diagram $D : \mathcal{J} \rightarrow \mathcal{C}$, the colimit

$$\varinjlim (F \circ D)$$

exists and the canonical map

$$\varinjlim (F \circ D) \xrightarrow{\cong} F(\varinjlim D)$$

is an isomorphism.

3. If $\mathcal{C}$ admits both finite limits and finite colimits, a functor F is called *exact* if it is both left exact and right exact.

Definition 2.4.12. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

1. F is called *left exact* if it preserves all finite limits (Definition 2.2.8); since F is an additive functor between abelian categories, this is equivalent to saying that F preserves kernels (Definition 2.3.6)²

Equivalently, for every left exact sequence (Definition 2.4.3) in $\mathcal{A}$

$$0 \rightarrow A' \xrightarrow{f} A \xrightarrow{g} A''$$

the sequence

$$0 \rightarrow F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'')$$

is exact at $F(A')$ and $F(A)$ (i.e., F preserves monomorphisms (Definition 2.3.8) and exactness at the first two terms).

2. Dually, F is called *right exact* if it preserves all finite colimits (Definition 2.2.8); since F is an additive functor between abelian categories, this is equivalent to saying that F preserves cokernels (Definition 2.3.6).

Equivalently, for every right exact sequence (Definition 2.4.3) in $\mathcal{A}$

$$A' \xrightarrow{f} A \xrightarrow{g} A'' \rightarrow 0,$$

the sequence

$$F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'') \rightarrow 0$$

is exact at $F(A)$ and $F(A'')$ (i.e., F preserves epimorphisms (Definition 2.3.8) and exactness at the last two terms).

3. F is called *exact* if it is both left and right exact.

The additive functor F is left/right exact if and only if it is left/right exact (Definition 2.4.11) in the more general sense, i.e. if it preserves all finite (Definition 2.2.10) limits/colimits (Definition 2.2.8)

Proposition 2.4.13. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and $A \in \text{Ob}(\mathcal{A})$. Then the covariant Hom functor

$$\text{Hom}_{\mathcal{A}}(A, -) : \mathcal{A} \rightarrow \mathbf{Ab}$$

is left exact (Definition 2.4.12), and the contravariant Hom functor

$$\text{Hom}_{\mathcal{A}}(-, A) : \mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$$

(Definition 1.2.1) is also left exact. Explicitly,

1. given a left exact sequence (Definition 2.4.3)

$$0 \rightarrow X' \xrightarrow{f} X \xrightarrow{g} X'',$$

²To see this equivalence, note that

- finite products (Definition 2.2.1) coincide with finite direct sums (Definition 2.3.4) in abelian categories,
- additive functors preserve finite direct sums by definition,
- kernels (Definition 2.3.6) in a pointed category (Definition 2.3.2) are equalizers (Definition 2.2.14), and all equalizers in abelian categories are kernels,
- if all finite products and equalizers exist in a general category, then all finite limits exist in the category (Theorem 2.2.15)

we have a left exact sequence (of abelian groups)

$$0 \rightarrow \operatorname{Hom}_{\mathcal{A}}(A, X') \xrightarrow{f \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X) \xrightarrow{g \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X''),$$

and

2. dually, given a right exact sequence

$$X' \xrightarrow{f} X \xrightarrow{g} X'' \rightarrow 0$$

we have a left exact sequence

$$0 \rightarrow \operatorname{Hom}_{\mathcal{A}}(X'', A) \xrightarrow{- \circ g} \operatorname{Hom}_{\mathcal{A}}(X, A) \xrightarrow{- \circ f} \operatorname{Hom}_{\mathcal{A}}(X', A).$$

Proof. We prove that the covariant Hom functor is left exact; that the contravariant Hom functor is left exact follows dually. Suppose that

$$0 \rightarrow X' \xrightarrow{f} X \xrightarrow{g} X'',$$

is left exact, i.e. f is a kernel of g . We want to show that the sequence

$$0 \rightarrow \operatorname{Hom}_{\mathcal{A}}(A, X') \xrightarrow{f \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X) \xrightarrow{g \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X''),$$

of abelian groups is left exact. To see that

$$\operatorname{Hom}_{\mathcal{A}}(A, X') \xrightarrow{f \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X)$$

is injective, say that $h \in \operatorname{Hom}_{\mathcal{A}}(A, X')$ satisfies $f \circ h = 0$. Since f is a kernel, which are monomorphisms (Definition 2.3.8) by Lemma 2.3.9, $h = 0$. Thus, $\ker(f \circ -) = \{0\}$, so the abelian group homomorphism $f \circ -$ is injective.

Clearly, the composition

$$\operatorname{Hom}_{\mathcal{A}}(A, X') \xrightarrow{f \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X) \xrightarrow{g \circ -} \operatorname{Hom}_{\mathcal{A}}(A, X'')$$

is the zero map of abelian groups — given $h \in \operatorname{Hom}_{\mathcal{A}}(A, X')$, the composition $g \circ f \circ h$ equals 0 because f is the kernel of g . Therefore, $\operatorname{im}(f \circ -) \subseteq \ker(g \circ -)$.

It remains to show that $\operatorname{im}(f \circ -) \supseteq \ker(g \circ -)$. Let $\phi \in \ker(g \circ -)$, which means $\phi : A \rightarrow X$ is a morphism in $\mathcal{A}$ such that $g \circ \phi = 0$. By the universal property of the kernel $f : X' \rightarrow X$ of g , there exists a unique morphism $u : A \rightarrow X'$ such that $f \circ u = \phi$. In terms of our functorial maps, this says $(f \circ -)(u) = \phi$, which proves that $\phi \in \operatorname{im}(f \circ -)$. Thus $\ker(g \circ -) = \operatorname{im}(f \circ -)$, and the sequence of abelian groups is left exact.

□

Lemma 2.4.14. Let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

1. The following are equivalent:

- (a) F is left exact (Definition 2.4.11) in the sense of preserving all finite limits (Definition 2.2.8).

- (b) F is left exact (Definition 2.4.12) in the sense of preserving left exact sequences (Definition 2.4.3), i.e. for every exact sequence $0 \rightarrow X' \rightarrow X \rightarrow X''$, the sequence $0 \rightarrow F(X') \rightarrow F(X) \rightarrow F(X'')$ is also exact.
- (c) F preserves all kernels (Lemma 2.3.7).
- 2. Dually, the following are equivalent:
 - (a) F is right exact (Definition 2.4.11) in the sense of preserving all finite colimits (Definition 2.2.8).
 - (b) F is right exact (Definition 2.4.12) in the sense of preserving right exact sequences (Definition 2.4.3), i.e. for every exact sequence $X' \rightarrow X \rightarrow X'' \rightarrow 0$, the sequence $F(X') \rightarrow F(X) \rightarrow F(X'') \rightarrow 0$ is also exact.
 - (c) F preserves all cokernels (Lemma 2.3.7).

Theorem 2.4.15 (Freyd-Mitchell Embedding Theorem). Let $\mathcal{A}$ be a small (Definition 1.1.2) abelian category (Definition 2.3.12). There exists a ring (Definition C.0.7) R (which may not be commutative) and a functor $F : \mathcal{A} \rightarrow \text{Mod}_R$ (Theorem 7.6.4) such that: (♠ TODO: Show that exact functors preserve finite limits and colimits)

1. F is exact (Definition 2.4.12), meaning it preserves all finite limits and colimits (in particular, kernels, cokernels, and exact sequences).
2. F is fully faithful (Definition 1.3.5).

Consequently, any diagram-chasing argument valid for modules over a ring is also valid in any small abelian category, and by extension (using the fact that any exact diagram involves only a set of objects), in any abelian category.

The Snake lemma is an application to the Freyd-Mitchell embedding theorem (Theorem 2.4.15). More specifically, while one can prove the snake lemma for general abelian categories directly, one can alternatively first prove the snake lemma for the category of R -modules, then make use of the theorem to prove it in general.

Definition 2.4.16 (Reflecting a type of morphism). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition 1.2.2), and let $\mathcal{P}$ be a property of morphisms (or more generally a property of sequences or families of morphisms) that is stable under isomorphism (Definition 1.1.13) (e.g. monomorphism, epimorphism (Definition 2.3.8), isomorphism, etc.). We say that F *reflects $\mathcal{P}$ -morphisms* if for every morphism $f : x \rightarrow y$ in $\mathcal{C}$, whenever $F(f)$ has property $\mathcal{P}$ in $\mathcal{D}$, it follows that f has property $\mathcal{P}$ in $\mathcal{C}$.

One can speak of more general notions of reflection (e.g. reflection of limits (Definition 2.2.8), of colimits (Definition 2.2.8), of “exact sequences (Definition 3.0.2)”, say for abelian categories, etc.)

Proposition 2.4.17. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a full and faithful (Definition 1.3.5) additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12). Then F reflects exactness (Definition 2.4.16), i.e., given a sequence

$$0 \rightarrow A' \xrightarrow{f} A \xrightarrow{g} A'' \rightarrow 0$$

in $\mathcal{A}$, if the sequence

$$0 \rightarrow F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'') \rightarrow 0$$

is exact in $\mathcal{B}$, then the original sequence is exact in $\mathcal{A}$. More generally, F reflects any exact sequence (Definition 3.0.2) and all limits and colimits that exist in $\mathcal{B}$ for diagrams coming from $\mathcal{A}$.

Lemma 2.4.18. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between additive categories (Definition 2.3.4). The functor F is faithful (Definition 1.3.5) if and only if for any morphism f in $\mathcal{A}$, we have $F(f) = 0$ exactly when $f = 0$.

Proof. To say that F is faithful means that for every pair of objects X and Y in $\mathcal{A}$, the induced abelian group (Definition C.0.3) homomorphism (Definition C.0.4)

$$F_{X,Y} : \text{Hom}_{\mathcal{A}}(X, Y) \rightarrow \text{Hom}_{\mathcal{B}}(F(X), F(Y))$$

on the Hom-sets is injective. Equivalently, this means that for every morphism $f : X \rightarrow Y$ in $\mathcal{A}$, we have $F(f) = 0$ if and only if $f = 0$. $\square$

Proposition 2.4.19. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor (Definition 2.4.12) between abelian categories (Definition 2.3.12). The functor F is faithful (Definition 1.3.5) if and only if it reflects (Definition 2.4.16) short exact sequences (Definition 2.4.4).

Proof. Suppose F is faithful. Let

$$A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3$$

be a sequence in $\mathcal{A}$ such that

$$F(A_1) \xrightarrow{F(f)} F(A_2) \xrightarrow{F(g)} F(A_3)$$

is exact in $\mathcal{B}$. Because F is exact, it preserves kernels and images, so

$$F(\ker(g)) = \ker(F(g)), \quad F(\text{im}(f)) = \text{im}(F(f)).$$

Exactness in $\mathcal{B}$ gives

$$\ker(F(g)) = \text{im}(F(f)),$$

so

$$F(\ker(g)) = F(\text{im}(f)).$$

Since F is faithful, it reflects (Definition 2.4.16) monomorphisms and epimorphisms (Definition 2.3.8) (Proposition 2.4.17), and morphisms that become equal via F must be equal in $\mathcal{A}$. Hence,

$$\ker(g) = \text{im}(f).$$

Therefore, the sequence

$$A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3$$

is exact in $\mathcal{A}$, i.e., F reflects short exact sequences.

Conversely, suppose F reflects short exact sequences. Let $f : A \rightarrow B$ be a morphism in $\mathcal{A}$ such that

$$F(f) = 0.$$

Consider the sequence

$$0 \rightarrow A \xrightarrow{f} B.$$

Since $F(f) = 0$, we have the sequence

$$0 \rightarrow F(A) \xrightarrow{F(f)} F(B)$$

is exact at $F(A)$. Because F reflects short exact sequences, the original sequence must be exact at A , so f is a monomorphism with zero image, hence $f = 0$. Thus F is faithful by Lemma 2.4.18. $\square$

Lemma 2.4.20 (Snake lemma). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let Given the commutative diagram with exact rows:

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \rightarrow & 0 \\ \downarrow a & & \downarrow b & & \downarrow c & & \\ 0 & \rightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$$

there exists an exact sequence connecting the kernels and cokernels:

$$\ker(a) \xrightarrow{\bar{f}} \ker(b) \xrightarrow{\bar{g}} \ker(c) \xrightarrow{d} \operatorname{coker}(a) \xrightarrow{\bar{f}'} \operatorname{coker}(b) \xrightarrow{\bar{g}'} \operatorname{coker}(c)$$

where d is the connecting homomorphism. Furthermore, if f is monic (Definition 2.3.8), then $\bar{f}$ is monic; if g' is epi (Definition 2.3.8), then $\bar{g}'$ is epi.

Proof. We first show this in the case that $\mathcal{A}$ is the category of R -modules (Definition 2.1.3) for any (not necessarily commutative) ring R . We perform a diagram chase on the elements of the modules.

1. **Construction of d :** Let $z \in \ker(c) \subseteq C$. Since g is surjective, there exists $y \in B$ such that $g(y) = z$. By commutativity, $g'(b(y)) = c(g(y)) = c(z) = 0$. Thus $b(y) \in \ker(g')$. By exactness of the bottom row, there exists a unique $x' \in A'$ such that $f'(x') = b(y)$. Define $d(z) = [x'] \in A'/\operatorname{im}(a) = \operatorname{coker}(a)$.
We show that $[x']$ is independent of the choice of y . Suppose we choose another lift $y_1 \in B$ such that $g(y_1) = z$. Then $g(y - y_1) = g(y) - g(y_1) = z - z = 0$. By the exactness of the top row at B , there exists some $x \in A$ such that $f(x) = y - y_1$. Applying b , we have $b(y - y_1) = b(f(x))$. By the commutativity of the first square, $b(f(x)) = f'(a(x))$. Let x' and x'_1 be the unique elements in A' such that $f'(x') = b(y)$ and $f'(x'_1) = b(y_1)$. Then $f'(x' - x'_1) = b(y) - b(y_1) = b(y - y_1) = f'(a(x))$. Since f' is injective (due to the exactness of the bottom row at A'), we must have $x' - x'_1 = a(x)$. This implies $x' \equiv x'_1 \pmod{\operatorname{im}(a)}$. Therefore, $[x'] = [x'_1]$ in $\operatorname{coker}(a)$, proving that the definition of $d(z)$ is independent of the choice of the preimage y .
2. **Exactness at $\ker(b)$:** Clearly $\bar{g} \circ \bar{f} = 0$. If $y \in \ker(\bar{g})$, then $g(y) = 0$, so $y = f(x)$ for some $x \in A$. Then $f'(a(x)) = b(f(x)) = b(y) = 0$. Since f' is injective, $a(x) = 0$, so $x \in \ker(a)$ and $y \in \operatorname{im}(\bar{f})$.
3. **Exactness at $\ker(c)$:** If $z = \bar{g}(y)$ for $y \in \ker(b)$, then in the construction of $d(z)$, we can choose this y . Since $b(y) = 0$, we have $x' = 0$, so $d(z) = 0$. Conversely, if $d(z) = 0$, then the lift x' is in $\operatorname{im}(a)$, say $x' = a(x)$. Then $b(f(x)) = f'(a(x)) = f'(x') = b(y)$. Thus $y - f(x) \in \ker(b)$ and $g(y - f(x)) = g(y) = z$.
4. **Exactness at $\operatorname{coker}(a)$:** Similar diagram chasing confirms $\bar{f}' \circ d = 0$ and the corresponding inclusion.

We verify the exactness of the sequence at $\text{coker}(a)$, specifically the sequence $\text{ker}(c) \xrightarrow{d} \text{coker}(a) \xrightarrow{\bar{f}'} \text{coker}(b)$. Let $z \in \text{ker}(c)$. By the construction of the connecting homomorphism, $d(z) = [x']$ where $f'(x') = b(y)$ for some $y \in B$ with $g(y) = z$. Applying $\bar{f}'$, we get $\bar{f}'([x']) = [f'(x')] = [b(y)]$. Since $b(y) \in \text{im}(b)$, its class in $\text{coker}(b)$ is zero. Thus $\bar{f}' \circ d = 0$. **Inclusion** $\text{ker}(\bar{f}') \subseteq \text{im}(d)$: Let $[x'] \in \text{coker}(a)$ such that $\bar{f}'([x']) = [0]$ in $\text{coker}(b)$. This implies $f'(x') \in \text{im}(b)$, so there exists $y \in B$ such that $b(y) = f'(x')$. We apply g' to both sides: $g'(b(y)) = g'(f'(x'))$. Since the bottom row is exact, $g' \circ f' = 0$, so $g'(b(y)) = 0$. By commutativity of the second square, $c(g(y)) = g'(b(y)) = 0$. This means $g(y) \in \text{ker}(c)$. Let $z = g(y)$. By the definition of the connecting homomorphism, $d(z)$ is found by lifting z to B (we choose y), applying b (we get $b(y) = f'(x')$), and taking the preimage under f' (which is x'). Thus, $d(z) = [x']$, which shows that $[x'] \in \text{im}(d)$.

5. We verify the exactness of the sequence at $\text{coker}(b)$, specifically looking at the sequence $\text{coker}(a) \xrightarrow{\bar{f}'} \text{coker}(b) \xrightarrow{\bar{g}'} \text{coker}(c)$. Let $[x'] \in \text{coker}(a)$ where $x' \in A'$. Then $\bar{f}'([x']) = [f'(x')]$. Applying $\bar{g}'$, we have $\bar{g}'([f'(x')]) = [g'(f'(x'))]$. Since the bottom row is exact, $g' \circ f' = 0$, thus $[g'(f'(x'))] = [0]$, so $\bar{g}' \circ \bar{f}' = 0$. Let $[y'] \in \text{coker}(b)$ such that $\bar{g}'([y']) = [0]$ in $\text{coker}(c)$. This implies $g'(y') \in \text{im}(c)$, so there exists $z \in C$ such that $c(z) = g'(y')$. Since g is surjective, there exists $y \in B$ such that $g(y) = z$. By commutativity, $g'(b(y)) = c(g(y)) = c(z) = g'(y')$. Thus $g'(y' - b(y)) = 0$, which means $y' - b(y) \in \text{ker}(g')$. By exactness of the bottom row, there exists $x' \in A'$ such that $f'(x') = y' - b(y)$. Transitioning to the cokernel, $[y'] = [f'(x') + b(y)] = [f'(x')] + [b(y)]$. Since $b(y) \in \text{im}(b)$, its class $[b(y)] = 0$ in $\text{coker}(b)$. Therefore, $[y'] = [f'(x')] = \bar{f}'([x'])$, showing $[y'] \in \text{im}(\bar{f}')$.

Let $\mathcal{B}$ be the smallest full (Definition 1.3.7) abelian subcategory of $\mathcal{A}$ containing the objects A, B, C, A', B', C' ; it is a small category. Let choose a ring R and a functor $F : \mathcal{B} \rightarrow \text{Mod}_R$ that is exact (Definition 2.4.12) and fully faithful (Definition 1.3.5) by the Freyd-Mitchell Embedding Theorem (Theorem 2.4.15). Apply the functor F to the diagram. Since F is exact, it preserves kernels, cokernels, images, and the exactness of the rows. The result holds in Mod_R by the element-based proof above. Since F is a full and faithful embedding, the morphism $d : \text{ker}(c) \rightarrow \text{coker}(a)$ is present in $\mathcal{B}$. Moreover, by Proposition 2.4.17, F reflects (Definition 2.4.16) exactness, so the exact sequence in Mod_R remains exact in $\mathcal{B}$. In particular, the exact sequence exists and is exact in $\mathcal{A}$. $\square$

The five-lemma may also be proven using the Freyd-Mitchell embedding theorem

Lemma 2.4.21 (Five Lemma). Consider the following commutative diagram in an abelian category (Definition 2.3.12) $\mathcal{A}$ with exact rows (Definition 2.4.4):

$$\begin{array}{ccccccccc} A & \xrightarrow{f_1} & B & \xrightarrow{f_2} & C & \xrightarrow{f_3} & D & \xrightarrow{f_4} & E \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \epsilon \\ A' & \xrightarrow{g_1} & B' & \xrightarrow{g_2} & C' & \xrightarrow{g_3} & D' & \xrightarrow{g_4} & E' \end{array}$$

where $A, B, C, D, E, A', B', C', D', E'$ are objects and $\alpha, \beta, \gamma, \delta, \epsilon$ are morphisms.

In the commutative diagram defined above, the following statements hold:

1. If β and δ are monomorphisms and α is an epimorphism, then γ is a monomorphism.
2. If β and δ are epimorphisms and ϵ is a monomorphism, then γ is an epimorphism.
3. If $\alpha, \beta, \delta, \epsilon$ are isomorphisms, then γ is an isomorphism.

2.5. Adjoint functors. Adjoint functors (Definition 2.5.1) enjoy nice properties, especially in homological algebra.

Definition 2.5.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories (Definition 1.1.1). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be functors.

An *adjunction between F and G* consists of two natural transformations (Definition 1.3.1): $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ (the *unit*), and $\varepsilon : FG \Rightarrow \text{Id}_{\mathcal{D}}$ (the *counit*)

These must satisfy the triangle identities: For every object $X \in \mathcal{C}$ and $Y \in \mathcal{D}$,

$$\varepsilon_{FX} \circ F(\eta_X) = \text{id}_{FX}$$

$$G(\varepsilon_Y) \circ \eta_{GY} = \text{id}_{GY}.$$

In diagrammatic form, the triangle identities assert that the following are commutative diagrams:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(\eta_X)} & FGF(X) \\ & \searrow \text{id}_{F(X)} & \downarrow \varepsilon_{F(X)} \\ & & F(X) \end{array} \quad \begin{array}{ccc} G(Y) & \xrightarrow{\eta_{G(Y)}} & GFG(Y) \\ & \searrow \text{id}_{G(Y)} & \downarrow G(\varepsilon_Y) \\ & & G(Y) \end{array}$$

We say that F is a *left adjoint to G* and G is a *right adjoint to F* (written $F \dashv G$).

In the case that $\mathcal{C}$ and $\mathcal{D}$ are locally small (Definition 1.1.2) categories (or U -locally small categories if a universe (Definition A.0.3) U is available), we have an adjunction $F \dashv G$ if and only if for every object X in $\mathcal{C}$ and Y in $\mathcal{D}$ there is a natural isomorphism (Definition 1.3.1)

$$\text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$$

that is natural in both X and Y . More precisely, there are two functors

$$\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \mathbf{Sets}$$

given by

$$(X, Y) \mapsto \text{Hom}_{\mathcal{D}}(F(X), Y)$$

$$(X, Y) \mapsto \text{Hom}_{\mathcal{C}}(X, G(Y))$$

respectively, and there is a natural isomorphism between them.

In this case, the *unit of the adjunction* is the natural transformation $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ such that,

1. for every $X \in \mathcal{C}$, the morphism $\eta_X : X \rightarrow GF(X)$ (each called a *unit morphism*) in $\mathcal{C}$ is obtained as the image of $\text{id}_{F(X)}$ via the adjoint isomorphism

$$\text{Hom}_{\mathcal{D}}(F(X), F(X)) \cong \text{Hom}_{\mathcal{C}}(X, GF(X)).$$

2. for every $Y \in \mathcal{D}$, the morphism $\epsilon_Y : FG(Y) \rightarrow Y$ (each called a *counit morphism*) in $\mathcal{D}$ is obtained as the image of $\text{id}_{G(Y)}$ via the adjoint isomorphism

$$\text{Hom}_{\mathcal{C}}(G(Y), G(Y)) \cong \text{Hom}_{\mathcal{D}}(FG(Y), Y).$$

One of the most essential adjunction of functors in algebra would be the tensor-hom adjunction:

Theorem 2.5.2 (Tensor-Hom Adjunction for Bimodules).

1. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7). Let M be an R - S bimodule (Definition 2.1.1), let N be an S - T bimodule, and let P be an R - T bimodule. Then there is a natural isomorphism of abelian groups

$$\text{Hom}_{R-T}(M \otimes_S N, P) \cong \text{Hom}_{S-T}(N, \text{Hom}_R(M, P))$$

(Definition 2.1.9) (Definition 2.1.8). Note that Hom_{R-T} is the abelian group of R - T bimodule homomorphisms, Hom_{S-T} is the abelian group of S - T bimodule homomorphisms, and $\text{Hom}_R(M, P)$, which is the abelian group of left R -linear homomorphisms $M \rightarrow P$, is endowed with the structure of an S - T bimodule via

$$\begin{aligned} (s \cdot f)(m) &= f(m \cdot s), \\ (f \cdot t)(m) &= f(m) \cdot t, \end{aligned}$$

for all $s \in S, t \in T, f \in \text{Hom}_R(M, P), m \in M$. Intuitively, this expresses that $M \otimes_S -$ is left adjoint (Definition 2.5.1) to $\text{Hom}_R(M, -)$ in the category of bimodules.

2. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7). Let M be an R - S bimodule (Definition 2.1.1), let N be an S - T bimodule, and let P be an R - T bimodule. Then there is a natural isomorphism of abelian groups

$$\text{Hom}_{R-T}(M \otimes_S N, P) \cong \text{Hom}_{R-S}(M, \text{Hom}_T(N, P))$$

(Definition 2.1.9) (Definition 2.1.8).

Note that Hom_{R-T} is the abelian group of R - T bimodule homomorphisms, Hom_{R-S} is the abelian group of R - S bimodule homomorphisms, and $\text{Hom}_T(N, P)$, which is the abelian group of right T -linear homomorphisms $N \rightarrow P$, is endowed with the structure of an R - S bimodule via

$$\begin{aligned} (r \cdot f)(n) &= r \cdot f(n), \\ (f \cdot s)(n) &= f(n \cdot s), \end{aligned}$$

for all $r \in R, s \in S, f \in \text{Hom}_T(N, P)$, and $n \in N$.

Intuitively, this expresses that $- \otimes N$ is left adjoint (Definition 2.5.1) to $\text{Hom}_T(N, -)$ in the category of bimodules.

Another useful adjunction is a adjunction of limit and colimit functors:

Theorem 2.5.3. Let $\mathcal{C}$ be a locally small (Definition 1.1.2) category and $\mathcal{J}$ be a small index category (Definition 1.1.2).

1. If $\mathcal{C}$ admits all colimits (Definition 2.2.8) of shape (Definition 2.2.6) $\mathcal{J}$, then the colimit functor

$$\text{colim} : \mathcal{C}^{\mathcal{J}} \rightarrow \mathcal{C}$$

is left adjoint (Definition 2.5.1) to the diagonal functor Δ . That is, for any functor $F : \mathcal{J} \rightarrow \mathcal{C}$ and any object $X \in \mathcal{C}$, there is a natural bijection:

$$\text{Hom}_{\mathcal{C}}(\text{colim } F, X) \cong \text{Hom}_{\mathcal{C}^{\mathcal{J}}}(F, \Delta(X)).$$

2. If $\mathcal{C}$ admits all limits of shape $\mathcal{J}$, then the limit functor

$$\text{lim} : \mathcal{C}^{\mathcal{J}} \rightarrow \mathcal{C}$$

is right adjoint (Definition 2.5.1) to the diagonal functor Δ . That is, for any object $X \in \mathcal{C}$ and any functor $F : \mathcal{J} \rightarrow \mathcal{C}$, there is a natural bijection:

$$\text{Hom}_{\mathcal{C}}(X, \text{lim } F) \cong \text{Hom}_{\mathcal{C}^{\mathcal{J}}}(\Delta(X), F).$$

These adjunctions characterize limits and colimits via their universal properties.

Proposition 2.5.4. Let $\mathcal{A}, \mathcal{B}$ be additive categories (Definition 2.3.4) and let $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ be adjoint (Definition 2.5.1) functors (Definition 1.2.2), say with $F \dashv G$ (i.e. F is left adjoint to G).

1. Assuming that the adjunction isomorphisms

$$\alpha = \alpha_{X,Y} : \text{Hom}_{\mathcal{B}}(F(X), Y) \xrightarrow{\sim} \text{Hom}_{\mathcal{A}}(X, G(Y))$$

are isomorphisms of abelian groups for each pair of objects $X, Y \in \mathcal{A}$, the functors F and G are additive (Definition 2.4.1).

2. If $\mathcal{A}$ and $\mathcal{B}$ are abelian (Definition 2.3.12), then the left adjoint functor F is right exact (Definition 2.4.12) and the right adjoint functor G is left exact (Definition 2.4.12).

Proof. Given a morphism $f : X' \rightarrow X$ in $\mathcal{A}$, the fact that the $\alpha_{X,Y}$ are natural isomorphisms with respect to the first input means that we have the commuting diagrams

$$(I) \quad \begin{array}{ccc} \text{Hom}_{\mathcal{B}}(F(X), Y) & \xrightarrow{\alpha_{X,Y}} & \text{Hom}_{\mathcal{A}}(X, G(Y)) \\ \downarrow - \circ F(f) & & \downarrow - \circ f \\ \text{Hom}_{\mathcal{B}}(F(X'), Y) & \xrightarrow{\alpha_{X',Y}} & \text{Hom}_{\mathcal{A}}(X', G(Y)). \end{array}$$

1. We show that F is additive; that G is additive can be obtained dually. To do so, we need to show that for all objects $A_1, A_2 \in \mathcal{A}$, the induced map

$$F_{A_1, A_2} : \text{Hom}_{\mathcal{A}}(A_1, A_2) \rightarrow \text{Hom}_{\mathcal{B}}(F(A_1), F(A_2))$$

is a group homomorphism of abelian groups. In other words, given morphisms $f, g : A_1 \rightarrow A_2$, we need to show that $F(f + g) = F(f) + F(g)$.

Write $\eta : \text{Id}_{\mathcal{A}} \rightarrow GF$ for the unit natural transformation; in particular, given each object $A \in \mathcal{A}$, we write $\eta_A : A \rightarrow GF(A)$ for the unit morphism (Definition 2.5.1).

For each $f : A_1 \rightarrow A_2$, we claim that

$$(J) \quad \alpha(F(f)) = \eta_{A_2} \circ f.$$

To see this, we use (I) in the case of $X = A_2$, $Y = F(A_2)$, and $X' = A_1$. Start with $\text{id}_{\mathcal{D}} \in \text{Hom}_{\mathcal{D}}(F(A_2), F(A_2))$ in the upper left corner. Traversing to the right then down sends $\text{id}_{\mathcal{D}}$ to the unit map η_{A_2} via α and then to $\eta_{A_2} \circ f$. Traversing down then to the right sends $\text{id}_{\mathcal{D}}$ to $F(f)$ then to $\alpha(F(f))$. Therefore, (J) holds.

In particular,

$$\alpha(F(f + g)) = \eta_B \circ (f + g).$$

Since $f + g : A_1 \rightarrow A_2$ and $\eta_{A_2} : A_2 \rightarrow GF(A_2)$ are morphisms in the abelian category $\mathcal{B}$, where composition of morphisms is distributive over addition, we have

$$\eta_{A_2} \circ (f + g) = (\eta_{A_2} \circ f) + (\eta_{A_2} \circ g).$$

Applying (J), we have

$$\alpha(F(f + g)) = \alpha(F(f)) + \alpha(F(g)).$$

Since α is assumed to be an isomorphism of groups, we conclude

$$F(f + g) = F(f) + F(g)$$

2. We show that F is right exact; that G is left exact can be obtained dually. It suffices to show that F preserves cokernels (Definition 2.3.6), i.e. that if $f : A_1 \rightarrow A_2$ is a morphism in $\mathcal{A}$, then $F(\text{coker } f) \cong \text{coker } F(f)$. Let $q : A_2 \rightarrow Q$ be the cokernel of f .

$$A_1 \xrightarrow{f} A_2 \xrightarrow{q} Q$$

We show that $F(q) : F(A_2) \rightarrow F(Q)$ is the cokernel of $F(f)$, i.e. that $F(q)$ possesses the universal property of $\text{coker } F(f)$.

$$F(A_1) \xrightarrow{F(f)} F(A_2) \xrightarrow{F(q)} F(Q)$$

Equivalently, we show that $F(\text{coker } f)$ possesses the universal property of $\text{coker } F(f)$. Since F is additive, note that

$$F(q) \circ F(f) = F(q \circ f) = F(0) = 0.$$

Now suppose that $g : F(A_2) \rightarrow Z$ is a morphism such that $g \circ F(f) = 0$. We need to show that g factors uniquely through $F(q)$.

Apply the commutativity of (I) in the case that $X = A_2$, $Y = Z$, and $X' = A_1$; this yields

$$\alpha(g \circ F(f)) = \alpha(g) \circ f.$$

Since $g \circ F(f) = 0$, we have $\alpha(g) \circ f = 0$ in $\mathcal{A}$. By the universal property of $\text{coker } f$, the map $\alpha(g)$ factors uniquely through $\text{coker } f$, i.e. there is a unique map $\psi : Q \rightarrow G(Z)$ such that $\psi \circ q = \alpha(g)$.

$$\begin{array}{ccccc} A_1 & \xrightarrow{f} & A_2 & \xrightarrow{q} & Q \\ & & \searrow \alpha(g) & & \downarrow \exists! \psi \\ & & & & G(Z) \end{array}$$

Applying α^{-1} , we have

$$\alpha^{-1}(\psi \circ q) = g.$$

By the naturality of the adjunction isomorphism in the first variable again, we obtain

$$g = \alpha^{-1}(\psi \circ q) = \alpha^{-1}(\psi) \circ F(q).$$

In particular,

$$\begin{array}{ccccc}
 F(A_1) & \xrightarrow{F(f)} & F(A_2) & \xrightarrow{F(q)} & F(Q) \\
 & & \searrow g & & \downarrow \alpha^{-1}(\psi) \\
 & & & & Z
 \end{array}$$

commutes. In fact, $\alpha^{-1}(\psi)$ is the unique morphism making the above commute because α is a bijection and ψ is unique. Therefore, $F(q) : F(A_2) \rightarrow F(Q)$ is indeed the cokernel of $F(f)$ as desired.

□

Proposition 2.5.5. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7). Recall that categories of modules are abelian (Definition 2.3.12) (Theorem 2.5.11).

1. Let M be an R - S -bimodule (Definition 2.1.1). The functor $M \otimes_S - : {}_S\mathbf{Mod}_T \rightarrow {}_R\mathbf{Mod}_T$ (Definition 2.1.9) is a right exact functor (Definition 2.4.12).
2. Let N be an S - T -bimodule. The functor $- \otimes_S N : {}_R\mathbf{Mod}_S \rightarrow {}_R\mathbf{Mod}_T$ is a right exact functor (Definition 2.4.12).
3. Let M be an R - S -bimodule. The functor

$$\mathrm{Hom}(M, -) : {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T$$

(Definition 2.1.8) is a left exact functor (Definition 2.4.12).

Now let M be an S - R -bimodule. The functor

$$\mathrm{Hom}(M, -) : {}_T\mathbf{Mod}_R \rightarrow {}_S\mathbf{Mod}_T$$

is a left exact functor (Definition 2.4.12).

4. Let N be an R - T -bimodule. The functor

$$\mathrm{Hom}(-, N) : {}_R\mathbf{Mod}_S^{\mathrm{op}} \rightarrow {}_S\mathbf{Mod}_T$$

(Definition 1.2.1) is a left exact functor (Definition 2.4.12).

Now let N be an T - R -bimodule. The functor

$$\mathrm{Hom}(-, N) : {}_S\mathbf{Mod}_R^{\mathrm{op}} \rightarrow {}_S\mathbf{Mod}_T$$

is a left exact functor (Definition 2.4.12).

Proof. Note that the tensor-hom adjunction (Theorem 2.5.2) yields natural isomorphisms of abelian groups. Proposition 2.5.4 implies the additivity of the above functors and implies the right exactness of the tensor functors and the left exactness of $\mathrm{Hom}(M, -)$.

Also recall Proposition 2.4.13.

□

Corollary 2.5.6. (♠ TODO: this should be more true of more general categories) Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and $\mathcal{J}$ be a small index category (Definition 1.1.2).

1. Assume that $\mathcal{A}$ admits all colimits (Definition 2.2.8) of shape $\mathcal{J}$ (e.g. $\mathcal{A}$ is Ab3 (Definition 2.5.7)). The colimit functor

$$\mathrm{colim} : \mathcal{A}^{\mathcal{J}} \rightarrow \mathcal{A}$$

(Definition 2.2.6)³ is right exact (Definition 2.4.11).

2. Assume that $\mathcal{A}$ admits all limits (Definition 2.2.8) of shape $\mathcal{J}$ (e.g. $\mathcal{A}$ is Ab3* (Definition 2.5.7)). The limit functor

$$\lim : \mathcal{A}^{\mathcal{J}} \rightarrow \mathcal{A}$$

(Definition 2.2.6) is left exact (Definition 2.4.11).

Proof. This follows from Theorem 2.5.3 and Proposition 2.5.4 □

2.5.1. *Grothendieck's additional axioms for abelian categories.* In fact, Grothendieck posed certain additional axioms that some abelian categories might enjoy.

Definition 2.5.7 (Grothendieck's axioms for abelian categories (Ab1–Ab5)). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

Grothendieck introduced the following hierarchy of additional axioms to express stronger completeness and exactness properties in $\mathcal{A}$ — we note that Ab1, Ab2, and Ab2* are already satisfied for any abelian category:

- **Ab1**: Every morphism in $\mathcal{A}$ has a kernel and a cokernel (Definition 2.3.6).
- **Ab2**: Every monic (Definition 2.3.8) in $\mathcal{A}$ is the kernel of its cokernel.
- **Ab2***: Every epi in $\mathcal{A}$ is the cokernel of its kernel.
- **AB3**: The category $\mathcal{A}$ is cocomplete (Definition 2.2.11).
 - Since $\mathcal{A}$ is abelian (and hence admits equalizers (Definition 2.2.14) as kernels (Definition 2.1.7)), this is equivalent to requiring that $\mathcal{A}$ has all small coproducts (Definition 2.2.1) (direct sums).
- **AB4**: The category $\mathcal{A}$ satisfies AB3, and coproducts are exact (Definition 2.4.12) (when regarded as functors $\bigoplus_I : \mathcal{A}^I \rightarrow \mathcal{A}$).
 - That is, the coproduct of a family of short exact sequences is a short exact sequence. Explicitly, for any family of short exact sequences $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$ indexed by a set I , the sequence

$$0 \rightarrow \bigoplus_{i \in I} A_i \rightarrow \bigoplus_{i \in I} B_i \rightarrow \bigoplus_{i \in I} C_i \rightarrow 0$$

is exact in $\mathcal{A}$.

- **AB5**: The category $\mathcal{A}$ satisfies AB3, and filtered colimits (Definition 2.2.13) are exact (Definition 2.4.12) (when regarded as functors $\varinjlim_J : \mathcal{A}^J \rightarrow \mathcal{A}$).
 - Equivalently, for any filtered (Definition 2.2.12) index category J and any directed system of short exact sequences $0 \rightarrow A_j \rightarrow B_j \rightarrow C_j \rightarrow 0$, the colimit sequence

$$0 \rightarrow \varinjlim A_j \rightarrow \varinjlim B_j \rightarrow \varinjlim C_j \rightarrow 0$$

is exact.

- Note: AB5 implies AB4. An abelian category satisfying AB5 and having a generator (Definition 2.5.8) is called a **Grothendieck category**.

³Recall that $\mathcal{A}^{\mathcal{J}}$ is abelian (Proposition 2.3.15)

- **AB6**: The category $\mathcal{A}$ satisfies AB3, and for any object X and any family of filtered subobjects $\{F_i\}_{i \in I}$ of X (where each F_i is a filter of subobjects), the intersection commutes with the limit:

$$\bigcap_{i \in I} \left(\varinjlim_{j \in F_i} U_{i,j} \right) = \varinjlim_{(j_i) \in \prod_{i \in I} F_i} \left(\bigcap_{i \in I} U_{i,j_i} \right).$$

(This axiom is less commonly cited but appears in Grothendieck's Tohoku paper).

- **AB3***: The category $\mathcal{A}$ is complete (Definition 2.2.11) (i.e., has all small products).
- **AB4***: The category $\mathcal{A}$ satisfies AB3* and products are exact.
 - Note: This is rarely satisfied for module categories (e.g., it fails for Abelian groups), but it is satisfied for the category of sheaves on a space.
- **AB5***: The category $\mathcal{A}$ satisfies AB3* and filtered limits (inverse limits) are exact.

Notes:

- AB5 implies AB4, and AB4 implies AB3.
- AB5* implies AB4*, and AB4* implies AB3*.

Definition 2.5.8 (Generator of a category). Let $\mathcal{C}$ be a category (Definition 1.1.1).

1. An object $G \in \mathcal{C}$ is called a **generator** (or **separator**) if for every pair of distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{C}$, there exists a morphism $h : G \rightarrow X$ such that

$$f \circ h \neq g \circ h.$$

In case that $\mathcal{C}$ is locally small (Definition 1.1.2), this is equivalent to the condition that the representable functor (Definition A.0.1)

$$\mathrm{Hom}_{\mathcal{C}}(G, -) : \mathcal{C} \rightarrow \mathbf{Set}$$

is faithful (Definition 1.3.5), which in turn is equivalent to the condition that for every object $X \in \mathcal{C}$, there exists an epimorphism

$$\bigoplus_{i \in I} G \twoheadrightarrow X$$

for some indexing set I , where $\bigoplus$ denotes the coproduct (Definition 2.2.1) in $\mathcal{C}$.

2. A family $\{G_i\}_{i \in I}$ is called a **generating family** if for every pair of distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{C}$, there exists some index $i \in I$ and a morphism $h : G_i \rightarrow X$ such that

$$f \circ h \neq g \circ h.$$

In case $\mathcal{C}$ is locally small, this is equivalent to the condition that the collection of representable functors

$$\{\mathrm{Hom}_{\mathcal{C}}(G_i, -) : \mathcal{C} \rightarrow \mathbf{Set}\}_{i \in I}$$

is jointly faithful, which in turn is equivalent to the condition that for every object $X \in \mathcal{C}$, there exists a family of objects $\{G_i\}_{i \in J}$ from the generating set indexed by some set J , and an epimorphism

$$\bigoplus_{i \in J} G_i \twoheadrightarrow X.$$

Theorem 2.5.9 (Examples of Grothendieck Categories). Examples of Grothendieck categories (Definition 2.5.7) include:

- The category of abelian groups,
- The category of R - S bimodules where R, S are (not necessarily commutative) rings (Definition C.0.7) (Theorem 2.5.11)
- The category of sheaves (Definition B.0.8) of abelian groups on a site (Definition B.0.5) with a small topologically generating family (Definition B.0.5),
- The category of sheaves of $\mathcal{O}_X$ -modules (Definition B.0.18) for any ringed space (Definition 7.6.1) $(X, \mathcal{O}_X)$.
- The category of quasi-coherent sheaves on a scheme or algebraic stack. (♠ TODO: quasi-coherent sheaves) (♠ TODO: I need to figure out if for sheaves of abelian groups/sheaves of $\mathcal{O}$ -modules whether essential smallness of the site is really necessary)
- The category of sheaves (Definition B.0.8) of abelian groups on an essentially small site (Definition B.0.5) (C, J) .
- ([?, Exposé II, Proposition 6.7]) The category of sheaves of $\mathcal{O}$ -modules on an essentially small site (or an essentially $\mathcal{U}$ -small site if a universe $\mathcal{U}$ is available) (C, J) .

Lemma 2.5.10. Let $\mathcal{C}$ be a category of sets with finitary operations (e.g. the category of sets (Definition 1.1.7), the category of modules over a ring (Definition 2.1.3), the category of monoids, the category of semigroups, the category of groups, the category of Lie algebras, the category of Lie algebras, etc.) (♠ TODO: define category of sets with finitary operations). Let $F : I \rightarrow \mathcal{C}$ be a small direct system (Definition 2.2.13). The direct limit (Definition 2.2.13) $\varinjlim_{i \in I} F(i)$ can be constructed as the quotient of the coproduct (Definition 2.2.1) $\coprod_{i \in I} F(i)$ given by the following equivalence relation: for $i, j \in I$, the elements $x_i \in F(i)$ and $x_j \in F(j)$ are equivalent if there exists some k with $i \leq k$ and $j \leq k$ such that $f_{ik}(x_i) = f_{jk}(x_j)$. In particular, the operations on the objects of $\mathcal{C}$ induce natural operations on this set making it into an object of $\mathcal{C}$.

Theorem 2.5.11. Let R, S be (not necessarily commutative) rings (Definition C.0.7). The category of (Definition 2.1.3) R - S -bimodules (Definition 2.1.1) is a Grothendieck (Definition 2.5.7) category and an $AB4^*$ (Definition 2.5.7) category.

Proof. We handwave details.

Given R - S -bimodules M and N , the set $\text{Hom}_{R\text{Mod}_S}(M, N)$ is an abelian group. Moreover, there is a 0-object, namely the zero module in $R\text{Mod}_S$. Therefore, $R\text{Mod}_S$ is preadditive (Definition 2.3.4). The direct sum of finitely many R - S -bimodules is their coproduct (Definition 2.2.5). Therefore, $R\text{Mod}_S$ is additive (Definition 2.3.4).

Given a morphism (Definition 2.1.2) $f : M \rightarrow N$ be R - S -bimodules, $\ker f$ and $\text{coker } f$ (Definition 2.1.7) are the categorical kernel and cokernel (Definition 2.3.6) of f in $R\text{Mod}_S$ (Definition 2.1.3). Moreover, the monomorphisms (Definition 2.3.8) in $R\text{Mod}_S$ are the injective module homomorphisms $f : M \rightarrow N$; such an f is the kernel of its cokernel. In other words, $R\text{Mod}_S$ satisfies $AB1$ and $AB2$ and hence is an abelian category (Definition 2.3.12).

Moreover, small coproducts (Definition 2.2.1) exist in ${}_R\text{Mod}_S$ (Definition 2.2.5), and it is easy to see that they are in fact exact, so ${}_R\text{Mod}_S$ satisfies AB3 and AB4. To show that filtered colimits in ${}_R\text{Mod}_S$ are exact, we first note that small (Definition 2.2.10) colimits (Definition 2.2.13) are (Corollary 2.5.6) right exact (Definition 2.4.12); for any small index category J and any system of short exact sequences $0 \rightarrow A_j \rightarrow B_j \rightarrow C_j \rightarrow 0$, the sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ of diagrams (Definition 2.2.6) is exact by Proposition 3.0.5, so applying the colimit functor yields a right exact sequence

$$\text{colim}_{j \in J} A_j \rightarrow \text{colim}_{j \in J} B_j \rightarrow \text{colim}_{j \in J} C_j \rightarrow 0.$$

If J is additionally filtered (Definition 2.2.12) so that the system is a directed one, then we show that the sequence

$$(K) \quad \varinjlim_{j \in J} A_j \rightarrow \varinjlim_{j \in J} B_j \rightarrow \varinjlim_{j \in J} C_j \rightarrow 0.$$

is left exact as well. Take some element⁴ of the kernel of $\varinjlim_{j \in J} A_j \rightarrow \varinjlim_{j \in J} B_j$; such an element is represented by some element $a_j \in A_j$ for some $j \in J$. Since its image is zero in $\varinjlim_{j \in J} B_j$, it must be zero as an element of B_k for some $k \in J$. Since J is filtered, there exists some $k' \in J$ so that there are arrows $j \rightarrow k'$ and $k \rightarrow k'$ and so that the image of a_j in $B_{k'}$ is 0. The image of a_j in $A_{k'}$ is then 0 due to the assumption that

$$0 \rightarrow A_{k'} \rightarrow B_{k'} \rightarrow C_{k'} \rightarrow 0$$

is exact. Therefore, a_j is 0 in $\varinjlim_{j \in J} A_j$, so (K) is left exact as claimed and ${}_R\text{Mod}_S$ is AB5.

Similarly as how we argued that ${}_R\text{Mod}_S$ is AB3 and AB4, one can argue that ${}_R\text{Mod}_S$ is AB3* and AB4*. Moreover, one can show that the R - S -bimodule $R \otimes_{\mathbb{Z}} S^{\text{op}}$ (Definition 2.1.10) (Definition C.0.38) is a generator for ${}_R\text{Mod}_S$. $\square$

3. CHAIN COMPLEXES OF OBJECTS IN ADDITIVE CATEGORIES

Chain complexes (Definition 3.0.1) are made of a sequence of morphisms in an additive category such that the composition of any two consecutive morphisms is 0. The first kind of chain complexes one should keep in mind are chain complexes of modules (Definition 2.1.1) over a ring.

Definition 3.0.1 (Chain complex in a preadditive category). Let $\mathcal{A}$ be a preadditive category (Definition 2.3.4) and let I be a totally ordered set (typically $\mathbb{Z}$, but $I \subseteq \mathbb{Z}$ is also allowed).

1. A *chain complex* $(K_{\bullet}, d_{\bullet})$ in $\mathcal{A}$ indexed by I is the homological convention for sequences with decreasing degrees. It consists of:
 - Objects $\{K_i\}_{i \in I}$ in $\mathcal{A}$, called the *terms in degree i* ,
 - Morphisms $d_i : K_i \rightarrow K_{i-1}$ in $\mathcal{A}$, called the *boundary maps* or *differentials in degree i* ,

⁴Recall ??

such that for every $i \in I$, $d_{i-1} \circ d_i = 0$. That is,

$$K_\bullet : \cdots \xrightarrow{d_{i+1}} K_i \xrightarrow{d_i} K_{i-1} \xrightarrow{d_{i-1}} K_{i-2} \rightarrow \cdots$$

with $d_{i-1}d_i = 0$ for all i . We typically use the notation $K_\bullet = (K_i, d_i)_{i \in I}$.

2. Dually, a **cochain complex** $(K^\bullet, d^\bullet)$ in $\mathcal{A}$ follows the cohomological convention with increasing degrees. It consists of objects $\{K^i\}_{i \in I}$ and **coboundary maps** $d^i : K^i \rightarrow K^{i+1}$ such that $d^{i+1} \circ d^i = 0$:

$$K^\bullet : \cdots \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \xrightarrow{d^{i+1}} K^{i+2} \rightarrow \cdots$$

We typically use the notation $K^\bullet = (K^i, d^i)_{i \in I}$.

3. Let $K_\bullet = (K_i, d_i^K)$ and $L_\bullet = (L_i, d_i^L)$ be chain complexes (Definition 3.0.1) in $\mathcal{A}$ indexed by the same set I . A **morphism of chain complexes** (or **chain map**)

$$f_\bullet : K_\bullet \rightarrow L_\bullet$$

consists of morphisms $f_i : K_i \rightarrow L_i$ for all $i \in I$, such that for every $i \in I$, the following diagram commutes:

$$\begin{array}{ccc} K_i & \xrightarrow{d_i^K} & K_{i-1} \\ \downarrow f_i & & \downarrow f_{i-1} \\ L_i & \xrightarrow{d_i^L} & L_{i-1} \end{array}$$

i.e., $d_i^L \circ f_i = f_{i-1} \circ d_i^K$.

A **morphism of cochain complexes** $f^\bullet : K^\bullet \rightarrow L^\bullet$ is defined similarly, satisfying the commutativity condition $d_L^i \circ f^i = f^{i+1} \circ d_K^i$.

The collection of these objects and morphisms forms a category. Notation for these categories is as follows:

- **Ch**($\mathcal{A}$) or **Ch**($\mathcal{A}$) is often used as a general term.
- To be explicit about the indexing convention, one uses **Ch** $_\bullet$ ($\mathcal{A}$) for chain complexes and **Ch** $^\bullet$ ($\mathcal{A}$) (or sometimes **CoCh**($\mathcal{A}$)) for cochain complexes.
- The set of chain maps between two complexes is denoted by **Hom** $_{\text{Ch}(\mathcal{A})}(K_\bullet, L_\bullet)$; it is an abelian group under pointwise addition $(f + g)_i = f_i + g_i$.

Notations used to write chain complexes are *homological conventions* whereas notations used to write cochain complexes are *cohomological conventions*. In homological conventions, indices are written as subscripts and decrease when “following the differential maps”. In cohomological conventions, indices are written as superscripts and increase when “following the differential maps”. The theories of chain complexes and of cochain complexes are not genuinely different, however, as the conventions are essentially notational — starting from a chain complex $(K_\bullet, d_\bullet)$, one can obtain a cochain complex $(K^\bullet, d^\bullet)$ simply by defining $K^i = K_{-i}$ and d^i as d_{-i} and vice versa. As such, cochain complexes may be simply called **chain complexes** when careful distinctions between the conventions are unnecessary. Homological and cohomological notational conventions persist throughout homological algebra, see e.g. spectral sequences, double complexes (Definition 4.6.1), etc, but again the distinction between such conventions are merely notational.

Definition 3.0.2 (Acyclic complex). Let $\mathcal{A}$ be an additive category (Definition 2.3.4), and let $(C_\bullet, d_\bullet)$ be a complex (Definition 3.0.1) in $\mathcal{A}$:

$$\cdots \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots$$

The complex $(C_\bullet, d_\bullet)$ is called **acyclic at C_n** (or sometimes synonymously **exact at C_n** , which agrees with the terminology of Definition 2.4.6) if we have $\ker d_n \cong \operatorname{im} d_{n+1}$ (Definition 2.3.6) (Definition 2.3.10).

If $\mathcal{A}$ is an abelian category (Definition 2.3.12), then this is equivalent to the condition that the (co)homology objects (Definition 3.2.2) $H_n(C_\bullet) := \ker d_n / \operatorname{im} d_{n+1}$ are zero in $\mathcal{A}$.

We furthermore say that the complex $(C_\bullet, d_\bullet)$ is **acyclic** or **exact** if it is acyclic/exact everywhere.

Example 3.0.3. All short exact sequences (Definition 2.4.4) in an additive category can be regarded as chain complexes, once we assign indices to the objects

(♠ TODO: simple examples of chain complexes)

Example 3.0.4. Some examples of chain complexes include (♠ TODO:)

1. The singular chain complex of a topological space.

Proposition 3.0.5. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let J be a small category (Definition 1.1.2).

Given a sequence $A \rightarrow B \rightarrow C$ of objects in the diagram category (Definition 2.2.6) $\mathcal{A}^J$, the sequence is exact (Definition 3.0.2) at B if and only if all the sequences $A(j) \rightarrow B(j) \rightarrow C(j)$ are exact at $B(j)$ for every $j \in \operatorname{Ob}(J)$.

Definition 3.0.6. Let $\mathcal{A}$ be a preadditive category (Definition 2.3.4).

1. Let $C_\bullet$ be a chain complex (Definition 3.0.1) in $\mathcal{A}$. For any $k \in \mathbb{Z}$, the **translation (or shift) of $C_\bullet$ by k** , denoted $C[k]_\bullet$, is the chain complex defined by:

$$(C[k])_n = C_{n-k}$$

with differential (Definition 3.0.1) $d[k]_n : (C[k])_n \rightarrow (C[k])_{n-1}$ defined by:

$$d[k]_n = (-1)^k d_{n-k}$$

2. Let $C^\bullet$ be a cochain complex in $\mathcal{A}$. For any $k \in \mathbb{Z}$, the **translation (or shift) of $C^\bullet$ by k** , denoted $C[k]^\bullet$, is the cochain complex defined by:

$$(C[k])^n = C^{n+k}$$

with differential $d[k]^n : (C[k])^n \rightarrow (C[k])^{n+1}$ defined by:

$$d[k]^n = (-1)^k d^{n+k}$$

If $f : C \rightarrow D$ is a morphism of (co)chain complexes (Definition 3.0.1), the **translated morphism $f[k] : C[k] \rightarrow D[k]$** is defined component-wise by $f[k]_n = f_{n-k}$ for chain complexes and $f[k]^n = f^{n+k}$ for cochain complexes (Definition 3.0.1).

In fact, the translation operation $[k]$ defines an additive (Definition 2.4.1) automorphism of the category of chain complexes (Definition 3.0.1) $\mathbf{Ch}_\bullet(\mathcal{A})$ and the category of cochain complexes (Definition 3.0.1) $\mathbf{Ch}^\bullet(\mathcal{A})$. In particular, for any integers k, m , there is an identity of functors $[k] \circ [m] = [k + m]$.

3.1. Category of chain complexes as a functor category.

Definition 3.1.1. A **quiver** is a quadruple $Q = (Q_0, Q_1, s, t)$, where:

- Q_0 is a collection of **vertices**.
- Q_1 is a collection of **arrows**.
- $s, t : Q_1 \rightarrow Q_0$ are functions assigning to each arrow $\alpha \in Q_1$ its **source** $s(\alpha)$ and its **target** $t(\alpha)$.

Definition 3.1.2. Let Q be a quiver (Definition 3.1.1). The **path category generated by Q** , denoted $\mathcal{F}(Q)$, is the category (Definition 1.1.1) defined as follows:

- The objects of $\mathcal{F}(Q)$ are the vertices Q_0 .
- For any two objects $x, y \in Q_0$, the set of morphisms $\text{Hom}_{\mathcal{F}(Q)}(x, y)$ consists of all paths from x to y — A **path of length $n \geq 1$ from x to y** is a sequence of arrows $\alpha_n \dots \alpha_1$ such that $s(\alpha_1) = x$, $t(\alpha_n) = y$, and $s(\alpha_{i+1}) = t(\alpha_i)$ for all $1 \leq i < n$. Additionally, for each vertex x , there is a path e_x of length 0, which serves as the identity morphism.
- Composition of morphisms is defined by the concatenation of paths.

Definition 3.1.3. Let Q be a quiver (Definition 3.1.1) whose collection of arrows is small.

The **preadditive category generated by Q** , denoted $\mathbb{Z}Q$, is the preadditive category (Definition 2.3.4), i.e. the category enriched over the category of abelian groups (Definition 1.1.8) defined as follows:

- The objects of $\mathbb{Z}Q$ are the vertices Q_0 .
- For any objects $x, y \in Q_0$, the morphism set $\text{Hom}_{\mathbb{Z}Q}(x, y)$ is the free abelian group generated by the set of all paths from x to y in Q .
- Composition is the unique bilinear extension of the path concatenation in $\mathcal{F}(Q)$. That is, for paths u, v, w where concatenation is defined, composition satisfies $(u + v) \circ w = u \circ w + v \circ w$ and $w \circ (u + v) = w \circ u + w \circ v$.

Definition 3.1.4. Let Q_{chain} be the quiver (Definition 3.1.1) with vertex set $Q_0 = \mathbb{Z}$ and arrow set $Q_1 = \{d_n : n \rightarrow n-1 \mid n \in \mathbb{Z}\}$. (♠ TODO: quotient of a category) The **walking chain complex category**, denoted $\mathbb{Z}$, is the quotient of the preadditive category (Definition 2.3.4) $\mathbb{Z}Q_{\text{chain}}$ by the ideal generated by the relations $d_{n-1} \circ d_n = 0$ for all $n \in \mathbb{Z}$. Explicitly:

- Objects are the integers $\mathbb{Z}$.
- Morphisms are $\mathbb{Z}$ -linear combinations of paths, subject to the relation that any path containing a subsegment $d_{n-1}d_n$ is identified with the zero morphism.

Definition 3.1.5. Let $\mathcal{A}$ and $\mathcal{B}$ be preadditive categories (Definition 2.3.4) (categories enriched over the category of abelian groups (Definition 1.1.8)). The **additive functor category** $\text{Add}(\mathcal{A}, \mathcal{B})$ is the functor category where:

- Objects are additive functors $F : \mathcal{A} \rightarrow \mathcal{B}$. An additive functor (Definition 2.4.1) is a functor such that for any $x, y \in \text{Ob}(\mathcal{A})$, the map $F : \text{Hom}_{\mathcal{A}}(x, y) \rightarrow \text{Hom}_{\mathcal{B}}(F(x), F(y))$ is a group homomorphism.
- Morphisms are natural transformations (Definition 1.3.1) between additive functors (Definition 2.4.1).

Proposition 3.1.6. Let $\mathcal{B}$ be a preadditive category (Definition 2.3.4). The category $\text{Ch}(\mathcal{B})$ of chain complexes (Definition 3.0.1) in $\mathcal{B}$ is isomorphic to the category $\text{Add}(\mathcal{I}, \mathcal{B})$ (Definition 3.1.5)(Definition 3.1.4).

An additive functor $F : \mathcal{I} \rightarrow \mathcal{B}$ corresponds to the chain complex defined by $C_n = F(n)$ and differentials $\partial_n = F(d_n)$.

Lemma 3.1.7. Let $\mathcal{A}, \mathcal{B}$ be preadditive categories (Definition 2.3.4) with $\mathcal{A}$ small (Definition 1.1.2).

1. The additive functor category (Definition 3.1.5) $\text{Add}(\mathcal{A}, \mathcal{B})$ is preadditive. If $\mathcal{B}$ is additionally additive (Definition 2.3.4)/abelian (Definition 2.3.12), then so is $\text{Add}(\mathcal{A}, \mathcal{B})$.
2. If $\mathcal{B}$ is an abelian category with property ABn for $n = 3, 4, 5, 6$ or ABn^* for $n = 3, 4, 5$ (Definition 2.5.7), then $\text{Add}(\mathcal{A}, \mathcal{B})$ possesses the same property.

Proposition 3.1.8. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. The category $\text{Ch}(\mathcal{A})$ (Definition 3.0.1) of chain complexes is itself an additive category.
2. If $\mathcal{A}$ is an abelian category (Definition 2.3.12), then $\text{Ch}(\mathcal{A})$ is an abelian category.
3. If $\mathcal{A}$ is an abelian category (Definition 2.3.12) satisfying Grothendieck's axiom ABn (resp. ABn^*) (Definition 2.5.7) for $n \in \{3, 4, 5, 6\}$, then $\text{Ch}(\mathcal{A})$ also satisfies ABn (resp. ABn^*). If $\mathcal{A}$ is a Grothendieck abelian category (Definition 2.5.7), then so is $\text{Ch}(\mathcal{A})$.

Proof. Combine Proposition 3.1.6 and Lemma 3.1.7. □

3.2. Homology and cohomology of a chain complex.

Definition 3.2.1. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. Let $C_{\bullet} = (\cdots \rightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \rightarrow \cdots)$ be a chain complex (Definition 3.0.1) in $\mathcal{A}$. For each integer n , we define:
 - The object of *n -cycles*, denoted $Z_n(C)$, is the kernel (Definition 2.3.6) of the differential leaving C_n :

$$Z_n(C) := \ker(d_n : C_n \rightarrow C_{n-1}).$$

- The object of *n -boundaries*, denoted $B_n(C)$, is the image (Definition 2.3.10) of the differential entering C_n :

$$B_n(C) := \text{im}(d_{n+1} : C_{n+1} \rightarrow C_n).$$

Since $d_n \circ d_{n+1} = 0$, there is a canonical monomorphism (Definition 2.3.8) $B_n(C) \hookrightarrow Z_n(C)$.

2. Let $C^\bullet = (\dots \rightarrow C^{n-1} \xrightarrow{d^{n-1}} C^n \xrightarrow{d^n} C^{n+1} \rightarrow \dots)$ be a **cochain complex in $\mathcal{A}$** . For each integer n , we define:

- The object of **n -cocycles**, denoted $Z^n(C)$, is the kernel (Definition 2.3.6) of the differential leaving C^n :

$$Z^n(C) := \ker(d^n : C^n \rightarrow C^{n+1}).$$

- The object of **n -coboundaries**, denoted $B^n(C)$, is the image (Definition 2.3.10) of the differential entering C^n :

$$B^n(C) := \operatorname{im}(d^{n-1} : C^{n-1} \rightarrow C^n).$$

Since $d^n \circ d^{n-1} = 0$, there is a canonical monomorphism (Definition 2.3.8) $B^n(C) \hookrightarrow Z^n(C)$.

Definition 3.2.2 (Chain complexes and their (co)homology objects). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

- For a cochain complex $K^\bullet$, its **cohomology object in degree i** is defined as the quotient of the object of i -cocycles by the object of i -coboundaries:

$$H^i(K^\bullet) := Z^i(K)/B^i(K) = \ker(d^i)/\operatorname{im}(d^{i-1}).$$

In fact, each H^i is (Proposition 3.2.7) a functor $H^i : \mathbf{Ch}^\bullet(\mathcal{A}) \rightarrow \mathcal{A}$ (Definition 3.0.1); given a morphism $f^\bullet : K^\bullet \rightarrow L^\bullet$ of cochain complexes, it is thus customary to write (Definition 1.2.2) $H^i(f^\bullet) : H^i(K^\bullet) \rightarrow H^i(L^\bullet)$ for the induced morphism on the i th cohomology objects.

- For a chain complex $K_\bullet$, its **homology object in degree i** is defined as the quotient of the object of i -cycles by the object of i -boundaries:

$$H_i(K_\bullet) := Z_i(K)/B_i(K) = \ker(d_i)/\operatorname{im}(d_{i+1}).$$

In fact, each H_i is (Proposition 3.2.7) a functor $H_i : \mathbf{Ch}_\bullet(\mathcal{A}) \rightarrow \mathcal{A}$ (Definition 3.0.1); given a morphism $f_\bullet : K_\bullet \rightarrow L_\bullet$ of chain complexes, it is thus customary to write (Definition 1.2.2) $H_i(f_\bullet) : H_i(K_\bullet) \rightarrow H_i(L_\bullet)$ for the induced morphism on the i th homology objects.

Example 3.2.3 (The Singular Complex of a Point). The simplest non-zero geometric example is the chain complex of a single point $X = \{*\}$. In singular homology, there is exactly one map $\sigma_n : \Delta^n \rightarrow X$ for each $n \geq 0$. Thus, $C_n(X) \cong \mathbb{Z}$ for all $n \geq 0$. The boundary map $d_n : C_n \rightarrow C_{n-1}$ is given by the alternating sum of face maps, which results in:

$$\dots \xrightarrow{\operatorname{id}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\operatorname{id}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$$

The homology groups are $H_0(X) \cong \mathbb{Z}$ and $H_n(X) = 0$ for $n > 0$.

Example 3.2.4 (The Algebraic Complex of a Ring Element). Let R be a ring and $x \in R$ be an element. We can define a chain complex $C_\bullet$ of R -modules concentrated in degrees 1 and 0:

$$0 \rightarrow R \xrightarrow{d_1} R \rightarrow 0$$

where $d_1(r) = xr$. Since $d_0 = 0$, the condition $d_0 \circ d_1 = 0$ is trivially satisfied. The homology groups are:

- $H_1(C_\bullet) = \ker(\cdot x) = \{r \in R \mid xr = 0\}$, the x -torsion of R .
- $H_0(C_\bullet) = \operatorname{coker}(\cdot x) = R/xR$.

Example 3.2.5 (Acyclic Complexes from Exact Sequences). A chain complex $C_\bullet$ is said to be *acyclic* if $H_n(C_\bullet) = 0$ for all n . Any short exact sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ in an abelian category $\mathcal{A}$ can be viewed as an acyclic chain complex:

$$\cdots \rightarrow 0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0 \rightarrow \cdots$$

The condition $g \circ f = 0$ is guaranteed by the fact that $\operatorname{im}(f) = \ker(g)$, which simultaneously ensures that the homology at object B is zero.

Example 3.2.6 (The Simplicial Triangle). Let K be a 2-simplex (a triangle) with vertices $\{v_0, v_1, v_2\}$. There is an associated chain complex

$$\cdots \rightarrow 0 \rightarrow C_2 \rightarrow C_1 \rightarrow C_0 \rightarrow 0 \rightarrow \cdots$$

of abelian groups. The chain groups are:

- $C_0 = \mathbb{Z}\langle v_0, v_1, v_2 \rangle$ (rank 3)
- $C_1 = \mathbb{Z}\langle [v_0, v_1], [v_1, v_2], [v_0, v_2] \rangle$ (rank 3)
- $C_2 = \mathbb{Z}\langle [v_0, v_1, v_2] \rangle$ (rank 1)

The boundary map $d_2 : C_2 \rightarrow C_1$ is $d_2([v_0, v_1, v_2]) = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$. One can verify that $d_1 \circ d_2 = 0$ by calculating the boundaries of the resulting edges.

Proposition 3.2.7. 1. Let $f_\bullet : C_\bullet \rightarrow D_\bullet$ be a morphism of chain complexes (Definition 3.0.1) in an abelian category (Definition 2.3.12). This morphism sends boundaries to boundaries (Definition 3.2.1) and cycles to cycles (Definition 3.2.1). More precisely, for all $i \in \mathbb{Z}$, $f_i : C_i \rightarrow D_i$ restricts to morphisms $Z_i(C_\bullet) \rightarrow Z_i(D_\bullet)$ and $B_i(C_\bullet) \rightarrow B_i(D_\bullet)$. Therefore, there are induced morphisms $H_i(f_\bullet) : H_i(C_\bullet) \rightarrow H_i(D_\bullet)$ which are in fact functorial, i.e. respect composition.

2. Let $f^\bullet : C^\bullet \rightarrow D^\bullet$ be a morphism of cochain complexes (Definition 3.0.1) in an abelian category (Definition 2.3.12). This morphism sends coboundaries to coboundaries (Definition 3.2.1) and cocycles to cocycles (Definition 3.2.1). More precisely, for all $i \in \mathbb{Z}$, $f^i : C^i \rightarrow D^i$ restricts to morphisms $Z^i(C^\bullet) \rightarrow Z^i(D^\bullet)$ and $B^i(C^\bullet) \rightarrow B^i(D^\bullet)$. Therefore, there are induced morphisms $H^i(f^\bullet) : H^i(C^\bullet) \rightarrow H^i(D^\bullet)$ which are functorial, i.e. respect composition.

Proof. (♠ TODO: Theorem 2.4.15 (Freyd-Mitchell) should be usable) □

Definition 3.2.8 (Quasi-isomorphism). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12), and let

$$f_\bullet : (C_\bullet, d_\bullet^C) \rightarrow (D_\bullet, d_\bullet^D)$$

be a chain map between complexes (Definition 3.0.1) in $\mathcal{A}$.

The morphism $f_\bullet$ is called a *quasi-isomorphism* if it induces isomorphisms on all cohomology objects, i.e., for every integer n , the induced morphism on homology (Definition 3.2.2) (or cohomology, depending on the convention)

$$H_n(f_\bullet) : H_n(C_\bullet) \rightarrow H_n(D_\bullet)$$

is an isomorphism in $\mathcal{A}$.

Note that all of these notions are applicable to the cohomological convention as well (Definition 3.0.1).

Remark 3.2.9. Let $C_\bullet$ and $D_\bullet$ be chain complexes (Definition 3.0.1) in an abelian category (Definition 2.3.12). Even if $H_n(C_\bullet)$ and $H_n(D_\bullet)$ are isomorphic for all n , there need not exist a quasi-isomorphism (Definition 3.2.8) $C_\bullet \rightarrow D_\bullet$.

Theorem 3.2.10 (Long Exact Sequence in Homology). 1. Let $(C_\bullet, d_\bullet^C)$, $(D_\bullet, d_\bullet^D)$, and $(E_\bullet, d_\bullet^E)$ be chain complexes (Definition 3.0.1) in an abelian category (Definition 2.3.12). Recall that $\mathbf{Ch}(\mathcal{A})$ (Definition 3.0.1) is itself an abelian category (Proposition 3.1.8). Assume that

$$0 \longrightarrow C_\bullet \xrightarrow{\alpha_\bullet} D_\bullet \xrightarrow{\beta_\bullet} E_\bullet \longrightarrow 0$$

is a short exact sequence (Definition 2.4.4) of chain complexes. Equivalently, for each integer n ,

$$0 \rightarrow C_n \xrightarrow{\alpha_n} D_n \xrightarrow{\beta_n} E_n \rightarrow 0$$

is an exact sequence of R -modules.

Then there exists a natural *long exact sequence* in homology (Definition 3.2.2):

$$\cdots \longrightarrow H_{n+1}(E_\bullet) \xrightarrow{\delta_{n+1}} H_n(C_\bullet) \xrightarrow{H_n(\alpha)} H_n(D_\bullet) \xrightarrow{H_n(\beta)} H_n(E_\bullet) \xrightarrow{\delta_n} H_{n-1}(C_\bullet) \longrightarrow \cdots$$

The homomorphisms $\delta_n : H_n(E_\bullet) \rightarrow H_{n-1}(C_\bullet)$ are called the *connecting homomorphisms* induced by the short exact sequence of chain complexes.

Moreover, this long exact sequence is natural with respect to morphisms (Definition 3.0.1) of short exact sequences of chain complexes.

2. Let $(C^\bullet, d_C^\bullet)$, $(D^\bullet, d_D^\bullet)$, and $(E^\bullet, d_E^\bullet)$ be cochain complexes (Definition 3.0.1) in an abelian category (Definition 2.3.12). Recall that $\mathbf{Ch}(\mathcal{A})$ (Definition 3.0.1) is itself an abelian category (Proposition 3.1.8).

Assume that

$$0 \longrightarrow C^\bullet \xrightarrow{\alpha^\bullet} D^\bullet \xrightarrow{\beta^\bullet} E^\bullet \longrightarrow 0$$

is a short exact sequence (Definition 2.4.4) of cochain complexes. Equivalently, for each integer n ,

$$0 \rightarrow C^n \xrightarrow{\alpha^n} D^n \xrightarrow{\beta^n} E^n \rightarrow 0$$

is an exact sequence of R -modules.

Then there exists a natural *long exact sequence* in cohomology (Definition 3.2.2):

$$\cdots \longrightarrow H^{n-1}(E^\bullet) \xrightarrow{\delta^{n-1}} H^n(C^\bullet) \xrightarrow{H^n(\alpha)} H^n(D^\bullet) \xrightarrow{H^n(\beta)} H^n(E^\bullet) \xrightarrow{\delta^n} H^{n+1}(C^\bullet) \longrightarrow \cdots$$

The morphisms $\delta^n : H^n(E^\bullet) \rightarrow H^{n+1}(C^\bullet)$ are called the *connecting homomorphisms* induced by the short exact sequence of cochain complexes.

Moreover, this long exact sequence is natural with respect to morphisms of short exact sequences of cochain complexes.

Proof. (♠ TODO: why kernel and cokernel are left/right exact) Consider the following commutative diagram in $\mathcal{A}$ with exact rows, where the vertical maps are the differentials of the

respective complexes:

$$(L) \quad \begin{array}{ccccccc} 0 & \longrightarrow & C_n / \operatorname{im}(d_{n+1}^C) & \xrightarrow{\bar{\alpha}_n} & D_n / \operatorname{im}(d_{n+1}^D) & \xrightarrow{\bar{\beta}_n} & E_n / \operatorname{im}(d_{n+1}^E) \longrightarrow 0 \\ & & \downarrow \bar{d}_n^C & & \downarrow \bar{d}_n^D & & \downarrow \bar{d}_n^E \\ 0 & \longrightarrow & \ker(d_{n-1}^C) & \xrightarrow{\alpha_{n-1}} & \ker(d_{n-1}^D) & \xrightarrow{\beta_{n-1}} & \ker(d_{n-1}^E) \longrightarrow 0 \end{array}$$

The rows are exact because the original sequence of complexes is exact and the functors $\ker(d_{n-1})$ and $\operatorname{coker}(d_{n+1})$ are respectively left and right exact. Specifically, the bottom row is exact at the right because β_n is surjective and the complexes satisfy the cycle condition. (♠ TODO: why is the bottom row exact at the right and why is the top row exact at the left)

We now apply the Snake Lemma to this diagram. To identify the resulting terms, we calculate the kernel and cokernel of the vertical map $\bar{d}_n^C$ (and similarly for D and E):

- **Kernel:** The kernel of $\bar{d}_n^C : C_n / \operatorname{im}(d_{n+1}^C) \rightarrow \ker(d_{n-1}^C)$ consists of elements⁵ $x + \operatorname{im}(d_{n+1}^C)$ such that $d_n^C(x) = 0$, i.e. $x \in \ker(d_n^C)$. Therefore,

$$\ker(\bar{d}_n^C) = \ker(d_n^C) / \operatorname{im}(d_{n+1}^C) = H_n(C_\bullet).$$

- **Cokernel:** The cokernel of $\bar{d}_n^C$ is the quotient of the target by the image. The target is $\ker(d_{n-1}^C)$ and the image is $\operatorname{im}(d_n^C)$. Thus,

$$\operatorname{coker}(\bar{d}_n^C) = \ker(d_{n-1}^C) / \operatorname{im}(d_n^C) = H_{n-1}(C_\bullet).$$

Applying the Snake Lemma to (L) yields the exact sequence:

$$\ker(\bar{d}_n^C) \rightarrow \ker(\bar{d}_n^D) \rightarrow \ker(\bar{d}_n^E) \xrightarrow{\delta_n} \operatorname{coker}(\bar{d}_n^C) \rightarrow \operatorname{coker}(\bar{d}_n^D) \rightarrow \operatorname{coker}(\bar{d}_n^E)$$

Substituting the homology groups identified above, we obtain:

$$H_n(C_\bullet) \xrightarrow{H_n(\alpha)} H_n(D_\bullet) \xrightarrow{H_n(\beta)} H_n(E_\bullet) \xrightarrow{\delta_n} H_{n-1}(C_\bullet) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(D_\bullet) \xrightarrow{H_{n-1}(\beta)} H_{n-1}(E_\bullet)$$

Since this construction exists for every $n \in \mathbb{Z}$, we can splice these sequences together to form the bi-infinite long exact sequence. The naturality of the sequence follows from the naturality of the Snake Lemma and the functoriality of the kernel and cokernel constructions. $\square$

3.3. Singular homology and cohomology groups of a topological space. Singular chain complexes of topological spaces are one of the quintessential examples of chain complexes (Definition 3.0.1). Many notions, such as mapping cones (Definition 3.7.3) or chain homotopy (Definition 3.5.1), in the general theory of chain complexes, can be seen as formalized constructions of those notions from the theory of singular chain complexes.

⁵Since elements are invoked in this argument, the Freyd-Mitchell embedding theorem (Theorem 2.4.15) should technically be used for this argument; nevertheless, one can argue that $\ker(\bar{d}_n^C) = H_n(C_\bullet)$ purely categorically.

3.3.1. Singular homology definitions.

Definition 3.3.1. Let V be a real vector space of finite dimension. A *k -simplex in topology* (or a *geometric k -simplex*) is the convex hull of $k + 1$ affinely independent points $v_0, v_1, \dots, v_k \in V$, and is denoted by

$$[v_0, v_1, \dots, v_k] := \left\{ \sum_{i=0}^k t_i v_i \mid t_i \geq 0, \sum_{i=0}^k t_i = 1 \right\}.$$

It is also standard to talk of the *standard topological n -simplex* — the topological space, often denoted by notations such as Δ^n or $|\Delta^n|$ defined as the subset of Euclidean space $\mathbb{R}^{n+1}$ given by

$$|\Delta^n| = \left\{ (t_0, t_1, \dots, t_n) \in \mathbb{R}^{n+1} : \sum_{i=0}^n t_i = 1, \text{ and } t_i \geq 0 \text{ for all } i \right\}$$

equipped with the induced topology from the usual Euclidean topology on $\mathbb{R}^{n+1}$.

(♠ TODO: comment on how $|\Delta^n|$ makes sense via a geometric realization)

(♠ TODO: comment on functoriality)

Definition 3.3.2 (Singular simplices). Let X be a topological space (Definition C.0.5). For each integer $n \geq 0$, A *singular n -simplex in X* is a continuous map (Definition C.0.6)

$$\sigma : \Delta^n \rightarrow X.$$

where Δ^n is the standard topological n -simplex (Definition 3.3.1). The set of all singular n -simplices in X is denoted by $S_n(X)$.

Definition 3.3.3 (Singular chain group with coefficients). Let X be a topological space (Definition C.0.5), let $A \subseteq X$ be a subspace (Definition C.0.39), and let $S_n(X)$ be the set of singular n -simplices in X (Definition 3.3.2).

1. Let R be a ring (Definition C.0.7).

- (a) The *singular n -chain group of X with coefficients in R* is the free R -module $C_n(X; R)$ whose elements are finite formal linear combinations

$$\sum_i r_i \sigma_i, \quad \text{with } \sigma_i \in S_n(X), r_i \in R.$$

Elements of $C_n(X; R)$ are called *singular n -chains in X with coefficients in R* .

- (b) If $A \subseteq X$ is a subspace, the quotient groups

$$C_n(X, A; R) = C_n(X; R) / C_n(A; R)$$

may be referred to as the *relative singular n -chain groups of X in A with coefficients in R* and elements of this group may be referred to as *relative singular n -chains in X relative to A with coefficients in R* .

In either case, when $R = \mathbb{Z}$, the ring R may be suppressed from notation, so we may write $C_n(X)$ and $C_n(X, A)$ for $C_n(X, \mathbb{Z})$ and $C_n(X, A, \mathbb{Z})$ respectively.

2. Let G be an abelian group (Definition C.0.3).

- (a) More generally, the *singular n -chain group of X with coefficients in G* is the abelian group

$$C_n(X; G) = C_n(X; \mathbb{Z}) \otimes_{\mathbb{Z}} G.$$

(Definition 2.1.9) In particular, an element of $C_n(X; G)$ is a finite formal sum

$$\sum_i g_i \sigma_i, \quad g_i \in G, \sigma_i \in S_n(X).$$

Elements of $C_n(X; G)$ are called *singular n -chains in X with coefficients in G* .

- (b) Similarly, the *relative singular n -chain group of X in A with coefficients in G* is the quotient group

$$C_n(X, A; G) = C_n(X; G) / C_n(A; G)$$

Elements of $C_n(X, A; G)$ are called *singular n -chains in X relative to A with coefficients in G* .

When $G = R$ is a ring, note that the notions of $C_n(X; R)$ where R is taken as a ring and where R is taken as an abelian group are equivalent; similarly for $C_n(X, A; R)$.

We note that, in general $C_n(X; G) \cong C_n(X, \emptyset; G)$.

Definition 3.3.4 (Singular cycles and singular boundaries). Let X be a topological space (Definition C.0.5), $A \subseteq X$ a subspace (Definition C.0.39), and G an abelian group. Let $(C_\bullet(X; G), \partial_n)$ and $(C_\bullet(X, A; G), \partial_n)$ be the singular chain complex (Definition 3.3.5) and *relative singular chain complex*, respectively.

1. A *(absolute) singular n -cycle group in X with coefficients in G* is an element of the subgroup

$$Z_n(X; G) = \ker(\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)).$$

More generally, a *relative singular n -cycle group of X relative to A* is an element of

$$Z_n(X, A; G) = \ker(\partial_n : C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)).$$

Geometrically, these are n -chains whose boundary is zero (in the absolute case) or whose boundary lies entirely within A (in the relative case).

2. A *(absolute) singular n -boundary group in X with coefficients in G* is an element of the subgroup

$$B_n(X; G) = \text{im}(\partial_{n+1} : C_{n+1}(X; G) \rightarrow C_n(X; G)).$$

More generally, a *relative singular n -boundary group of X relative to A* is an element of

$$B_n(X, A; G) = \text{im}(\partial_{n+1} : C_{n+1}(X, A; G) \rightarrow C_n(X, A; G)).$$

By the chain complex (Definition 3.0.1) property $\partial_n \circ \partial_{n+1} = 0$, the n -boundaries are always contained within their corresponding n -cycles, i.e. $B_n(X, A; G) \subseteq Z_n(X, A; G)$.

Note that the absolute and relative singular homology groups (Definition 3.3.6) are the resulting quotient groups:

$$H_n(X; G) = Z_n(X; G) / B_n(X; G) \quad \text{and} \quad H_n(X, A; G) = Z_n(X, A; G) / B_n(X, A; G).$$

Definition 3.3.5 (Singular chain complex with coefficients). Let X be a topological space (Definition C.0.5), let $A \subseteq X$ be a subspace (Definition C.0.39), and let G be an abelian group.

For each $n \geq 1$, define the *boundary operator*

$$\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)$$

(Definition 3.3.3) by

$$\partial_n(\sigma) = \sum_{i=0}^n (-1)^i \sigma \circ \delta_i,$$

where $\delta_i : \Delta^{n-1} \rightarrow \Delta^n$ is the i -th face inclusion, and extend ∂_n to $C_n(X; G)$ linearly. For $n \leq 0$, we simply let $\partial_n = 0$

One can verify that $(C_n(X; G), \partial_n)$ forms a chain complex (Definition 3.0.1) in the abelian category (Definition 2.3.12) of abelian groups (Definition 1.1.8). This chain complex is called the *(absolute) singular chain complex of X with coefficients in G* and is often denoted by $C_\bullet(X; G)$.

Then the boundary maps above induce maps

$$C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)$$

on the relative chain groups $C_n(X, A; G)$, yielding a chain complex of abelian groups whose boundary maps, often also denoted by ∂_n , are induced by the boundary operators ∂_n above; this chain complex may be called the *relative singular chain complex of the pair (X, A) with coefficients in G* and is often denoted by $C_\bullet(X, A; G)$.

In the case that $G = R$ is a (not necessarily commutative) ring, then the boundary maps of $C_\bullet(X; R)$ and $C_\bullet(X, A; R)$ are left R -module homomorphisms (Definition 2.1.2):

$$\partial \left(\sum_i r_i \sigma_i \right) = \sum_i r_i \partial(\sigma_i)$$

Definition 3.3.6 (Singular homology with coefficients). Let X be a topological space (Definition C.0.5), let $A \subseteq X$ be a subspace (Definition C.0.39), and let G be an abelian group.

The *(absolute) n -th singular homology group of X with coefficients in G* is the homology group (Definition 3.2.2)

$$H_n(X; G) = H_n(C_*(X; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_*(X; G)$ is the singular chain complex of X with coefficients in G and $\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)$ are the boundary maps.

Given a subspace (Definition C.0.39) $A \subseteq X$, the *n -th relative singular homology group of (X, A) with coefficients in G* is defined as

$$H_n(X, A; G) = H_n(C_*(X, A; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_*(X, A; G)$ is the relative singular chain complex of (X, A) with coefficients in G and $\partial_n : C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)$ are the boundary maps.

We may denote $H_n(X; \mathbb{Z})$ and $H_n(X, A; \mathbb{Z})$ by $H_n(X)$ and $H_n(X, A)$ respectively.

In the case that $G = R$ is a ring (Definition C.0.7), $H_n(X; R)$ and $H_n(X, A; R)$ have the structure of left R -modules (Definition 2.1.1) since the corresponding chain complexes are chain complexes of left R -modules..

Note that singular chain groups (Definition 3.3.3) are generally very large, so it is difficult to carry out computations with them; in particular, it is often not feasible in practice to compute singular homology/cohomology groups using the definition. Instead, if the topological spaces are nice enough (see Proposition 3.3.28), i.e. are “homotopy equivalent” to simplicial complexes or are CW complexes (Definition 3.3.27), then the singular homology/cohomology can be computed using simplicial homology/cohomology or cellular homology (Proposition 3.3.35)/cohomology.

Proposition 3.3.7 (Functoriality of the singular chain complex). (♠ TODO:) The construction of the singular chain complex satisfies the following functorial properties:

1. **Bifunctionality:** The assignment of a singular chain complex to a pair of spaces and an abelian group defines a covariant bifunctor

$$C_\bullet(-; -) : \mathbf{Top}_{pair} \times \mathbf{Ab} \rightarrow \mathbf{Ch}(\mathbf{Ab})$$

(♠ TODO: category of pairs of topological spaces) where $\mathbf{Top}_{pair}$ is the category of pairs (Definition C.0.46) (X, A) of topological spaces and $\mathbf{Ch}(\mathbf{Ab})$ is the category of chain complexes of abelian groups.

A morphism $(f, \phi) : ((X, A), G) \rightarrow ((Y, B), H)$ consists of a continuous map of pairs $f : (X, A) \rightarrow (Y, B)$ and a group homomorphism $\phi : G \rightarrow H$. This induces a chain map:

$$(f, \phi)_\# : C_\bullet(X, A; G) \rightarrow C_\bullet(Y, B; H)$$

defined on chains by $(f, \phi)_\# (\sum_i g_i \sigma_i) = \sum_i \phi(g_i) (f \circ \sigma_i)$.

2. **Restriction to Modules:** For a fixed commutative ring (Definition C.0.9) R , the construction restricts to a covariant functor

$$C_\bullet(-; R) : \mathbf{Top}_{pair} \rightarrow \mathbf{Ch}(R\text{-Mod})$$

where $\mathbf{Ch}(R\text{-Mod})$ is the category of chain complexes of R -modules. In this case, the induced maps $f_\#$ are R -module homomorphisms.

3. **Compatibility with Ring Homomorphisms:** If $\psi : R \rightarrow S$ is a homomorphism of commutative rings, there is a natural transformation of functors from $C_\bullet(-; R)$ to $C_\bullet(-; S)$ induced by the change of coefficients.

3.3.2. Singular cohomology definitions. (♠ TODO: go through these definitions carefully)

Definition 3.3.8 (Singular cochain group with coefficients). Let X be a topological space (Definition C.0.5), $A \subseteq X$ a subspace (Definition C.0.39), and G an abelian group.

1. The *singular n -cochain group of X with coefficients in G* is the abelian group defined as the dual of the integral singular chain group:

$$C^n(X; G) = \text{Hom}(C_n(X; \mathbb{Z}), G).$$

Elements of $C^n(X; G)$ are called **singular n -cochains** and are functions $\varphi : S_n(X) \rightarrow G$ that assign an element of G to each singular n -simplex in X .

2. The **relative singular n -cochain group of (X, A) with coefficients in G** is the subgroup of $C^n(X; G)$ consisting of cochains that vanish on all singular simplices in A :

$$C^n(X, A; G) = \ker(i^* : C^n(X; G) \rightarrow C^n(A; G)) \cong \text{Hom}(C_n(X, A; \mathbb{Z}), G)$$

where $i : A \hookrightarrow X$ is the inclusion map.

When $G = R$ is a ring (Definition C.0.7), $C^n(X, A; R)$ inherits the structure of an R -module.

Definition 3.3.9 (Singular cochain complex and coboundary operator). Let X be a topological space (Definition C.0.5) and G an abelian group. For each $n \geq 0$, the **coboundary operator**

$$\delta^n : C^n(X; G) \rightarrow C^{n+1}(X; G)$$

is the dual of the singular boundary operator (Definition 3.3.5) ∂_{n+1} . Specifically, for a cochain $\varphi \in C^n(X; G)$ and a simplex $\sigma \in S_{n+1}(X)$:

$$(\delta^n \varphi)(\sigma) = \varphi(\partial_{n+1} \sigma) = \sum_{i=0}^{n+1} (-1)^i \varphi(\sigma \circ \delta_i)$$

where δ_i is the i -th face inclusion. Since $\partial_n \circ \partial_{n+1} = 0$, it follows that $\delta^{n+1} \circ \delta^n = 0$.

The sequence $(C^n(X, A; G), \delta^n)$ forms a cochain complex (Definition 3.0.1) called the **singular cochain complex** of (X, A) with coefficients in G , denoted by $C^\bullet(X, A; G)$.

Definition 3.3.10 (Singular cocycles and coboundaries). Let $(C^\bullet(X, A; G), \delta^n)$ be the singular cochain complex (Definition 3.3.9).

1. A **singular n -cocycle** is an element of the subgroup

$$Z^n(X, A; G) = \ker(\delta^n : C^n(X, A; G) \rightarrow C^{n+1}(X, A; G)).$$

2. A **singular n -coboundary** is an element of the subgroup

$$B^n(X, A; G) = \text{im}(\delta^{n-1} : C^{n-1}(X, A; G) \rightarrow C^n(X, A; G)).$$

By the property $\delta^n \circ \delta^{n-1} = 0$, we have $B^n(X, A; G) \subseteq Z^n(X, A; G)$.

Definition 3.3.11 (Singular cohomology with coefficients). The **(absolute) n -th singular cohomology group of X with coefficients in G** is the quotient group:

$$H^n(X; G) = Z^n(X; G) / B^n(X; G).$$

Given a subspace (Definition C.0.39) $A \subseteq X$, the **n -th relative singular cohomology group of (X, A) with coefficients in G** is:

$$H^n(X, A; G) = Z^n(X, A; G) / B^n(X, A; G).$$

If $G = R$ is a ring, $H^n(X, A; R)$ typically carries additional structure as an R -algebra via the (♠ TODO: define cup product).

Proposition 3.3.12 (Contravariant functoriality of cohomology). The construction of singular cohomology satisfies the following functorial properties:

1. **Contravariance:** For a fixed coefficient group G , the assignment of the n -th singular cohomology group to a pair of spaces $(X, A) \mapsto H^n(X, A; G)$ defines a contravariant functor from the category of pairs $\mathbf{Top}_{pair}$ to the category of abelian groups $\mathbf{Ab}$.
2. **Induced Maps:** A continuous map of pairs $f : (X, A) \rightarrow (Y, B)$ induces a group homomorphism, called the pullback map:

$$f^* : H^n(Y, B; G) \rightarrow H^n(X, A; G)$$

defined on the level of cochains by $f^\#(\varphi)(\sigma) = \varphi(f \circ \sigma)$.

3. **Bifunctoriality:** Viewed as a function of the coefficients, the construction is covariant. Thus, $H^n(-; -)$ is a contravariant functor in the first variable and a covariant functor in the second variable.

3.3.3. *Universal coefficient theorems of singular homology and cohomology.* The universal coefficient theorems (Theorem 3.3.14) (Theorem 3.3.13) tell us how to relate singular homology (Definition 3.3.6)/cohomology (Definition 3.3.11) with coefficients in $\mathbb{Z}$ with those with general coefficients. Such universal coefficient theorems motivate the notions of Tor (Definition 4.5.1) and Ext (Definition 4.5.2) and more generally, derived functors (Definition 4.3.1).

Theorem 3.3.13 (Universal Coefficient Theorem for Singular Homology). Let (X, A) be a pair of topological spaces and G an abelian group (Definition C.0.3). For each $n \in \mathbb{Z}$, there exists a natural short exact sequence:

$$0 \rightarrow H_n(X, A; \mathbb{Z}) \otimes_{\mathbb{Z}} G \xrightarrow{\mu} H_n(X, A; G) \xrightarrow{\beta} \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A; \mathbb{Z}), G) \rightarrow 0$$

where:

- $H_n(X, A; \mathbb{Z})$ is the integral singular homology.
- μ is the tensor product map defined by $\mu([c] \otimes g) = [c \otimes g]$.
- $\mathrm{Tor}_1^{\mathbb{Z}}$ is the first torsion product functor (♠ TODO: define Tor functor).
- β is a natural surjection that identifies the kernel of μ .

Furthermore, this sequence splits (Definition 2.4.8) (albeit non-naturally), meaning there is a non-natural isomorphism:

$$H_n(X, A; G) \cong (H_n(X, A; \mathbb{Z}) \otimes_{\mathbb{Z}} G) \oplus \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A; \mathbb{Z}), G)$$

Theorem 3.3.14 (Universal Coefficient Theorem for Cohomology). Let (X, A) be a pair of topological spaces and G an abelian group. For each $n \in \mathbb{Z}$, there exists a natural short exact sequence:

$$0 \rightarrow \mathrm{Ext}_{\mathbb{Z}}^1(H_{n-1}(X, A; \mathbb{Z}), G) \rightarrow H^n(X, A; G) \xrightarrow{h} \mathrm{Hom}_{\mathbb{Z}}(H_n(X, A; \mathbb{Z}), G) \rightarrow 0$$

where:

- h is the evaluation homomorphism defined by $h([\varphi])([c]) = \varphi(c)$.
- $\mathrm{Ext}_{\mathbb{Z}}^1$ is the first extension product functor (♠ TODO: define Ext functor).

Furthermore, this sequence splits (Definition 2.4.8), though the splitting is not natural, yielding the isomorphism:

$$H^n(X, A; G) \cong \text{Hom}_{\mathbb{Z}}(H_n(X, A; \mathbb{Z}), G) \oplus \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(X, A; \mathbb{Z}), G)$$

Theorem 3.3.15 (Relative Künneth Theorem for Singular Homology). Let (X, A) and (Y, B) be pairs of topological spaces (Definition C.0.5) such that $A \subseteq X$ and $B \subseteq Y$ are open (or (X, A) and (Y, B) are CW pairs (Definition 3.3.29)). If R is a principal ideal domain (Definition C.0.41) and $H_n(X, A; R)$ is finitely generated for all n , then for each n there exists a natural short exact sequence:

$$0 \rightarrow \bigoplus_{i+j=n} H_i(X, A; R) \otimes_R H_j(Y, B; R) \xrightarrow{\times} H_n(X \times Y, A \times Y \cup X \times B; R) \rightarrow \bigoplus_{i+j=n-1} \text{Tor}_1^R(H_i(X, A; R), H_j(Y, B; R))$$

(♠ TODO: define relative cross product) where $\times$ is the relative homological cross product. This sequence splits (Definition 2.4.8) (non-naturally), and if R is a field, the Tor terms vanish, providing a clean isomorphism:

$$H_n(X \times Y, A \times Y \cup X \times B; R) \cong \bigoplus_{i+j=n} H_i(X, A; R) \otimes_R H_j(Y, B; R)$$

Theorem 3.3.16 (Relative Künneth Theorem for Singular Cohomology). Let (X, A) and (Y, B) be pairs of topological spaces (Definition C.0.5) such that $A \subseteq X$ and $B \subseteq Y$ are open (or (X, A) and (Y, B) are CW pairs (Definition 3.3.29)). Let R be a principal ideal domain (Definition C.0.41).

If $H_n(X, A; R)$ is finitely generated (Definition C.0.42) for all n , there is a natural short exact sequence:

$$0 \rightarrow \bigoplus_{i+j=n} H^i(X, A; R) \otimes_R H^j(Y, B; R) \xrightarrow{\times} H^n(X \times Y, A \times Y \cup X \times B; R) \rightarrow \bigoplus_{i+j=n+1} \text{Tor}_1^R(H^i(X, A; R), H^j(Y, B; R))$$

(♠ TODO: define relative cohomological cross product) where $\times$ is the relative cohomological cross product. If R is a field, this simplifies to an isomorphism of graded R -algebras:

$$H^*(X \times Y, A \times Y \cup X \times B; R) \cong H^*(X, A; R) \otimes_R H^*(Y, B; R)$$

(♠ TODO: discuss graded ring structure of relative cohomology)

3.3.4. Simplicial homology. (♠ TODO: verify these definitions)

Definition 3.3.17. An *abstract simplicial complex* K is a collection of finite non-empty subsets of a set of vertices V , called *simplices*, such that for every $\sigma \in K$, every non-empty subset $\tau \subseteq \sigma$ (called a *face*) is also in K . An n -simplex is a simplex consisting of $n + 1$ vertices. The *geometric realization* $|K|$ is the union of the corresponding geometric simplices in $\mathbb{R}^V$ equipped with the *weak topology* (the finest topology such that the inclusion of each simplex is continuous).

Definition 3.3.18. A *abstract simplicial pair* (K, L) consists of an abstract simplicial complex K and a *subcomplex* $L \subseteq K$, which is a subset of K that is itself an abstract simplicial complex.

Definition 3.3.19. An *oriented simplex* is a simplex $\sigma \in K$ with a chosen ordering of its vertices, where two orderings are considered equivalent if they differ by an even permutation. We denote an oriented simplex by $[v_0, \dots, v_n]$.

Let G be an abelian group. The *group of simplicial n -chains* $C_n^\Delta(K; G)$ is the free abelian group generated by the oriented n -simplices of K with coefficients in G , subject to the relation $[v_{\sigma(0)}, \dots, v_{\sigma(n)}] = \text{sgn}(\sigma)[v_0, \dots, v_n]$.

For a abstract simplicial pair (K, L) , the *relative group of simplicial n -chains* $C_n^\Delta(K, L; G)$ is the quotient group:

$$C_n^\Delta(K, L; G) := C_n^\Delta(K; G) / C_n^\Delta(L; G)$$

Lemma 3.3.20. For an abstract simplicial complex K , the group $C_n^\Delta(K; G)$ is isomorphic to the cellular chain group of its geometric realization:

$$C_n^\Delta(K; G) \cong H_n(|K|^n, |K|^{n-1}; G)$$

where $|K|^n$ denotes the n -skeleton of the geometric realization.

Definition 3.3.21. The *simplicial boundary map* $\partial_n : C_n^\Delta(K, L; G) \rightarrow C_{n-1}^\Delta(K, L; G)$ is the map induced on the quotient by:

$$\partial_n([v_0, \dots, v_n]) = \sum_{i=0}^n (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_n]$$

This map satisfies $\partial_n \circ \partial_{n+1} = 0$, forming the *simplicial chain complex* $(C_\bullet^\Delta(K, L; G), \partial_\bullet)$.

Definition 3.3.22. The *relative simplicial homology groups* $H_n^\Delta(K, L; G)$ are defined as:

$$H_n^\Delta(K, L; G) = \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})}$$

Theorem 3.3.23. For any abstract simplicial pair (K, L) where K is an abstract simplicial complex, there is a natural isomorphism:

$$H_n^\Delta(K, L; G) \cong H_n(|K|, |L|; G)$$

Example 3.3.24. The n -simplex Δ^n is an abstract simplicial complex consisting of $n + 1$ vertices and all their subsets. Its boundary $\partial\Delta^n$ is the subcomplex of all proper subsets. The pair $(\Delta^n, \partial\Delta^n)$ provides a triangulation of the disk D^n relative to its boundary S^{n-1} . The group $C_n^\Delta(\Delta^n, \partial\Delta^n; \mathbb{Z})$ is $\mathbb{Z}$, generated by the single n -simplex. Since all $(n - 1)$ -faces are in the subcomplex $\partial\Delta^n$, they are zero in the quotient, making the boundary map zero. Thus $H_n^\Delta(\Delta^n, \partial\Delta^n; \mathbb{Z}) \cong \mathbb{Z}$.

Example 3.3.25. The 2-torus T^2 can be triangulated as an abstract simplicial complex K . While C_n^Δ contains more generators than a CW complex (requiring at least 14 triangles and 21 edges), the resulting homology groups are identical:

$$H_1^\Delta(K; \mathbb{Z}) \cong \mathbb{Z}^2, \quad H_2^\Delta(K; \mathbb{Z}) \cong \mathbb{Z}$$

3.3.5. CW-complexes.

Definition 3.3.26 (Cellular Complex). A **cellular complex** is a topological space X equipped with a partition into subspaces called **open n -cells** e_α^n , each homeomorphic to an open n -dimensional ball B^n , such that:

- The **n -skeleton** is defined inductively as $X^n = \bigcup_{k \leq n} \bigcup_\alpha e_\alpha^k$.
- For each n -cell e_α^n , there exists a continuous **characteristic map** $\Phi_\alpha : D^n \rightarrow X$ from the closed n -disk which:
 1. Restricts to a homeomorphism on the interior $\text{int}(D^n) \cong e_\alpha^n$.
 2. Maps the boundary ∂D^n into the $(n-1)$ -skeleton X^{n-1} .

Definition 3.3.27 (CW Complex). A **CW complex** is a cellular complex (Definition 3.3.26) X that satisfies the following conditions:

- **Hausdorff**: X is a Hausdorff space (Definition C.0.51).
- **C (Closure Finiteness)**: The closure of each cell $\overline{e_\alpha^n}$ is contained in a finite union of cells of dimension less than or equal to n .
- **W (Weak Topology)**: A subset $A \subseteq X$ is closed if and only if $A \cap \overline{e_\alpha^n}$ is closed for every cell e_α^n .

Proposition 3.3.28. The class of spaces admitting a CW complex structure (or the homotopy type of a CW complex) includes several fundamental families of topological spaces:

- **Manifolds**: Every compact smooth manifold is homeomorphic to a CW complex. This follows from Morse theory (**♠ TODO: define Morse function**), where a manifold is reconstructed by attaching cells at the critical points of a Morse function.
- **Simplicial Complexes**: Every simplicial complex is a CW complex, where the n -cells are the open n -simplices.
- **Algebraic Varieties**: All projective algebraic varieties over $\mathbb{C}$ admit CW structures.
- **Covering Spaces**: If X is a CW complex, then any covering space $p : \tilde{X} \rightarrow X$ inherits a natural CW structure where the cells of $\tilde{X}$ are the components of the preimages of the cells of X .
- **Function Spaces**: While the space of continuous maps $\text{Map}(X, Y)$ is generally not a finite CW complex, it has the homotopy type of a CW complex if X is a compact CW complex and Y is a CW complex.

Definition 3.3.29. A **CW pair** (X, A) consists of a CW complex (Definition 3.3.27) X and a **subcomplex** $A \subseteq X$, which by definition is a closed subspace that is a union of cells of X such that for every cell $e_\alpha^n \subseteq A$, its closure $\overline{e_\alpha^n}$ is also contained in A .

Definition 3.3.30. Let G be an abelian group. Let (X, A) be a CW pair (Definition 3.3.29).

The **relative group of cellular n -chains**, which is the free abelian group $C_n^{CW}(X, A; G)$ generated by the n -cells of $X \setminus A$ with coefficients in G .

In particular, the **absolute group of cellular n -chains of the CW complex X** is $C_n^{CW}(X; G) = C_n^{CW}(X, \emptyset; G)$.

Lemma 3.3.31. Let (X, A) be a CW pair (Definition 3.3.29) and G an abelian group. There is a canonical isomorphism:

$$C_n^{CW}(X, A; G) \cong H_n(X^n \cup A, X^{n-1} \cup A; G)$$

Definition 3.3.30 where H_n denotes relative singular homology (Definition 3.3.6). Furthermore, $H_k(X^n \cup A, X^{n-1} \cup A; G) = 0$ for $k \neq n$.

Definition 3.3.32. Let G be an abelian group. Let (X, A) be a CW pair (Definition 3.3.29).

The *relative cellular chain complex* $(C_\bullet^{CW}(X, A; G), d_\bullet^G)$ is defined as the chain complex (Definition 3.0.1) whose objects are $C_n^{CW}(X, A; G)$ (Definition 3.3.30), which are isomorphic to $H_n(X^n \cup A, X^{n-1} \cup A; G)$ by Lemma 3.3.31.

The boundary map (Definition 3.0.1) $d_n^G : C_n^{CW}(X, A; G) \rightarrow C_{n-1}^{CW}(X, A; G)$, called the *cellular boundary map*, is defined via the composition $j_{n-1} \circ \partial_n$:

$$H_n(X^n \cup A, X^{n-1} \cup A; G) \xrightarrow{\partial_n} H_{n-1}(X^{n-1} \cup A; G) \xrightarrow{j_{n-1}} H_{n-1}(X^{n-1} \cup A, X^{n-2} \cup A; G)$$

where ∂_n is the connecting homomorphism of the long exact sequence of the pair $(X^n \cup A, X^{n-1} \cup A)$ and j_{n-1} is the map induced by the inclusion of pairs.

Definition 3.3.33. Let G be an abelian group. Let (X, A) be a CW pair (Definition 3.3.29).

The *relative cellular homology groups* $H_n^{CW}(X, A; G)$ are the homology groups of the cellular chain complex:

$$H_n^{CW}(X, A; G) = \frac{\ker(d_n^G)}{\text{im}(d_{n+1}^G)}$$

When $A = \emptyset$, these are the *absolute cellular homology groups* $H_n^{CW}(X; G)$.

Theorem 3.3.34. For any CW pair (Definition 3.3.29) (X, A) and any abelian group G , there exists a natural isomorphism:

$$H_n^{CW}(X, A; G) \cong H_n(X, A; G)$$

(Definition 3.3.33) (Definition 3.3.6) for all $n \geq 0$.

Proposition 3.3.35. Let G be an abelian group. Let (X, A) be a CW pair (Definition 3.3.29).

(♠ TODO: degree) The cellular boundary map $d_n^G : C_n^{CW}(X, A; G) \rightarrow C_{n-1}^{CW}(X, A; G)$ (Definition 3.3.32) can be computed via the degree of attaching maps. For an n -cell e_α^n and its coefficient $g \in G$:

$$d_n^G(g e_\alpha^n) = \sum_{\beta} g[e_\alpha^n : e_\beta^{n-1}] e_\beta^{n-1}$$

The incidence number $[e_\alpha^n : e_\beta^{n-1}]$ is the degree of the map $S^{n-1} \rightarrow S^{n-1}$ defined by the composition:

$$S^{n-1} \cong \partial D^n \xrightarrow{\phi_\alpha} X^{n-1} \rightarrow X^{n-1} / (X^{n-1} \setminus e_\beta^{n-1}) \cong S^{n-1}$$

where ϕ_α is the restriction of the characteristic map to the boundary (♠ TODO: define topological degree).

Example 3.3.36. The n -sphere S^n for $n > 0$ can be given a CW structure with one 0-cell e^0 and one n -cell e^n . The characteristic map for e^n maps ∂D^n to e^0 . The cellular chain groups are:

$$C_k^{CW}(S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } k = 0, n \\ 0 & \text{otherwise} \end{cases}$$

If $n > 1$, the boundary maps d_k must be zero because there are no adjacent non-zero chain groups. Thus:

$$H_k(S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } k = 0, n \\ 0 & \text{otherwise} \end{cases}$$

Example 3.3.37. The Real Projective Plane $\mathbb{RP}^2$ has a CW structure with one 0-cell e^0 , one 1-cell e^1 , and one 2-cell e^2 . The 1-skeleton is S^1 . The 2-cell is attached to the 1-skeleton by a map of degree 2 (the map $z \mapsto z^2$ on the unit circle). The cellular chain complex is:

$$0 \rightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z} \xrightarrow{d_1} \mathbb{Z} \rightarrow 0$$

The map d_1 is the zero map because it maps the 1-cell to $e^0 - e^0$. The map d_2 is multiplication by the degree of the attaching map, which is 2. Therefore:

$$H_0(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}, \quad H_2(\mathbb{RP}^2; \mathbb{Z}) \cong 0$$

Example 3.3.38. The complex projective space $\mathbb{CP}^n$ has a CW structure with exactly one cell in each even dimension $0, 2, 4, \dots, 2n$. Because there are no cells in odd dimensions, all boundary maps d_k in the cellular chain complex must be zero (as either the domain or codomain is 0). Consequently, the homology groups are:

$$H_k(\mathbb{CP}^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } k \in \{0, 2, \dots, 2n\} \\ 0 & \text{otherwise} \end{cases}$$

Example 3.3.39. The 2-torus T^2 admits a CW structure with:

- One 0-cell: e^0 (the vertex where all corners of the square meet).
- Two 1-cells: a and b (the horizontal and vertical edges).
- One 2-cell: f (the interior of the square).

The cellular chain groups with integer coefficients are:

$$C_2^{CW}(T^2) \cong \mathbb{Z}\langle f \rangle, \quad C_1^{CW}(T^2) \cong \mathbb{Z}\langle a, b \rangle, \quad C_0^{CW}(T^2) \cong \mathbb{Z}\langle e^0 \rangle$$

The boundary maps are computed as follows:

- $d_1(a) = e^0 - e^0 = 0$ and $d_1(b) = e^0 - e^0 = 0$. Thus, d_1 is the zero map.
- The boundary of the 2-cell f follows the word $aba^{-1}b^{-1}$. In the abelian group C_1^{CW} , this becomes $a + b - a - b = 0$. Thus, d_2 is the zero map.

Since all boundary maps are zero, the homology groups are isomorphic to the chain groups:

$$H_0(T^2; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(T^2; \mathbb{Z}) \cong \mathbb{Z}^2, \quad H_2(T^2; \mathbb{Z}) \cong \mathbb{Z}$$

3.4. Homotopy between morphisms of topological spaces.

Definition 3.4.1 (Homotopy of maps of topological spaces). Let X and Y be topological spaces and let $K \subseteq X$ be a subset. Let $C(X, Y)$ denote the set of all continuous maps $f : X \rightarrow Y$.

1. A *homotopy between two maps $f, g \in C(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $x \in K$ and $t \in [0, 1]$,

$$H(x, t) = f(x) = g(x).$$

If such an H exists, we say f and g are *homotopic relative to K* , and we write $f \simeq g \text{ rel } K$ or $f \simeq_K g$; this is an equivalence relation.

In the case that $K = \emptyset$, a homotopy relative to K is simply called a *homotopy*. We write we write $f \simeq g$ if a homotopy between f and g exists.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.1.10) and let $K \subseteq X$ be a subset with $x_0 \in K$. Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ satisfying $f(x_0) = y_0$.

A *homotopy of based maps $f, g \in C_*(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

in particular fixing the basepoint throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

If such an H exists, we say f and g are *based homotopic relative to K* , and we write $f \simeq g \text{ rel } K$ or $f \simeq_K g$. This is an equivalence relation.

When $K = \{x_0\}$, one calls a homotopy of based maps relative to K simply as a *homotopy of based maps* or *based homotopy*. We write $f \simeq g$ if such a homotopy exists.

Definition 3.4.2 (Homotopy equivalence). Let X and Y be topological spaces and let $K \subseteq X$.

1. A continuous map (Definition C.0.6) $f : X \rightarrow Y$ is a *homotopy equivalence relative to K* if there exists a continuous map $g : Y \rightarrow X$ such that

$$g \circ f \simeq \text{id}_X \text{ rel } K \quad \text{and} \quad f \circ g \simeq \text{id}_Y \text{ rel } f(K).$$

(Definition C.0.30) That is, there exist homotopies (Definition C.0.30) $H : X \times [0, 1] \rightarrow X$ and $G : Y \times [0, 1] \rightarrow Y$ satisfying

$$\begin{aligned} H(x, 0) &= (g \circ f)(x), & H(x, 1) &= x, & H(k, t) &= k \text{ for all } k \in K, t \in [0, 1], \\ G(y, 0) &= (f \circ g)(y), & G(y, 1) &= y, & G(f(k), t) &= f(k) \text{ for all } k \in K, t \in [0, 1]. \end{aligned}$$

In this case, the map g is called a *homotopy inverse relative to K of f* , and the pairs (X, K) and $(Y, f(K))$ are said to be *homotopy equivalent relative to K* , denoted $(X, K) \simeq (Y, f(K)) \text{ rel } K$. This is an equivalence relation.

A continuous map $f : X \rightarrow Y$ is a *homotopy equivalence* if it is a homotopy equivalence relative to $\emptyset$. A *homotopy inverse of f* is then simply a homotopy inverse of f relative to $\emptyset$. In this case, and the spaces X and Y are said to be *homotopy equivalent*, denoted $X \simeq Y$. This is an equivalence relation.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.1.10) and let $K \subseteq X$ be a subset with $x_0 \in K$.

A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces is a *homotopy equivalence relative to K* if there exists a continuous map $g : (Y, y_0) \rightarrow (X, x_0)$ such that

$$g \circ f \simeq \text{id}_X \text{ rel } K \quad \text{and} \quad f \circ g \simeq \text{id}_Y \text{ rel } f(K),$$

where the homotopies (Definition C.0.30) are homotopies of based maps relative to K and $f(K)$, respectively.

That is, there exist homotopies of based maps $H : X \times [0, 1] \rightarrow X$ and $G : Y \times [0, 1] \rightarrow Y$ satisfying

$$\begin{aligned} H(x, 0) &= (g \circ f)(x), & H(x, 1) &= x, & H(k, t) &= k \text{ for all } k \in K, t \in [0, 1], \\ G(y, 0) &= (f \circ g)(y), & G(y, 1) &= y, & G(f(k), t) &= f(k) \text{ for all } k \in K, t \in [0, 1], \end{aligned}$$

with $H(x_0, t) = x_0$ and $G(y_0, t) = y_0$ for all t .

The map g is called a *homotopy inverse relative to K of f* , and the pairs (X, K) and $(Y, f(K))$ are said to be *homotopy equivalent relative to K* , denoted $(X, K) \simeq (Y, f(K)) \text{ rel } K$. This is an equivalence relation.

A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces is a *homotopy equivalence* if it is a homotopy equivalence relative to $\{x_0\}$. A *homotopy inverse of f* is then simply a homotopy inverse relative to $\{x_0\}$. In this case, the pointed spaces (X, x_0) and (Y, y_0) are said to be *homotopy equivalent*, denoted $(X, x_0) \simeq (Y, y_0)$.

In each case, topological spaces that are homotopy equivalent are said to be of the same *homotopy type*.

(♠ TODO: homotopy of maps of topological spaces results in chain homotopy for singular homology)

(♠ TODO: verify)

Theorem 3.4.3. Let (X, A) and (Y, B) be pairs of topological spaces (Definition C.0.5), and let G be an abelian group (Definition C.0.3). Let $f, g : (X, A) \rightarrow (Y, B)$ be continuous maps (Definition C.0.6) of pairs.

If f and g are homotopic (Definition C.0.30) (denoted $f \simeq g$), then the induced chain maps (Proposition 3.3.7) $f_{\#}, g_{\#} : C_{\bullet}(X, A; G) \rightarrow C_{\bullet}(Y, B; G)$ on the singular chain complexes (Definition 3.3.5) are chain homotopic (Definition 3.5.1).

Specifically, if $H : (X \times I, A \times I) \rightarrow (Y, B)$ is a homotopy from f to g , there exists a *prism operator* $P_n : C_n(X, A; G) \rightarrow C_{n+1}(Y, B; G)$ such that:

$$\partial_{n+1}P_n + P_{n-1}\partial_n = g_{\#} - f_{\#}$$

Corollary 3.4.4. Let (X, A) and (Y, B) be pairs of topological spaces, and let G be an abelian group. If $f \simeq g : (X, A) \rightarrow (Y, B)$ are homotopic maps of pairs, then the induced homomorphisms (Proposition 3.3.7) on singular homology groups (Definition 3.3.6) are equal:

$$f_* = g_* : H_n(X, A; G) \rightarrow H_n(Y, B; G)$$

Similarly, the induced homomorphisms (Proposition 3.3.12) on singular cohomology groups are equal:

$$f^* = g^* : H^n(Y, B; G) \rightarrow H^n(X, A; G)$$

Theorem 3.4.5. Let (X, A) and (Y, B) be pairs of topological spaces and G an abelian group. If (X, A) and (Y, B) are homotopy equivalent (Definition 3.4.2) as pairs, then their corresponding singular chain complexes (Definition 3.3.5) $C_{\bullet}(X, A; G)$ and $C_{\bullet}(Y, B; G)$ are chain homotopy equivalent (Definition 3.5.1).

Consequently, there are isomorphisms of homology (Definition 3.3.6) and cohomology groups:

$$H_n(X, A; G) \cong H_n(Y, B; G) \quad \text{and} \quad H^n(X, A; G) \cong H^n(Y, B; G)$$

for all $n \geq 0$.

3.5. Chain homotopy between morphisms of chain complexes.

Definition 3.5.1. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. Let $f_{\bullet}, g_{\bullet} : C_{\bullet} \rightarrow D_{\bullet}$ be chain maps between complexes (Definition 3.0.1) in $\mathcal{A}$. A *chain homotopy from $f_{\bullet}$ to $g_{\bullet}$* is a collection of morphisms $\{s_n : C_n \rightarrow D_{n+1}\}$ such that for all n ,

$$f_n - g_n = d_{n+1}^D \circ s_n + s_{n-1} \circ d_n^C.$$

If such an $s_{\bullet}$ exists, we say that $f_{\bullet}$ and $g_{\bullet}$ are *chain homotopic* and write $f_{\bullet} \simeq g_{\bullet}$.

2. Let $f_{\bullet} : C_{\bullet} \rightarrow D_{\bullet}$ be a chain map between complexes (Definition 3.0.1) in $\mathcal{A}$. A *chain contraction* is a chain homotopy from $f_{\bullet}$ to the zero complex. The chain map $f_{\bullet}$ is said to be *null homotopic* if a chain contraction of $f_{\bullet}$ exists, i.e. $f_{\bullet}$ is chain homotopic to the 0 chain complex.
3. Let $f_{\bullet} : C_{\bullet} \rightarrow D_{\bullet}$ be a chain map between complexes. We say that $f_{\bullet}$ is a *chain homotopy equivalence* if there exists a chain map and $h_{\bullet} : D_{\bullet} \rightarrow C_{\bullet}$ such that

$$fg \simeq \text{id}_{D_{\bullet}} \quad \text{and} \quad gh \simeq \text{id}_{C_{\bullet}}.$$

In this case, it is appropriate to call f and g *chain homotopy inverses of each other*.

One similarly defines the above notions for cochain complexes and their morphisms.

Lemma 3.5.2. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). Being chain homotopic (Definition 3.5.1) is an equivalence relation on chain complexes (Definition 3.0.1) of objects in $\mathcal{A}$. (♠ TODO: equivalence relation)

Lemma 3.5.3. (♠ TODO: preservation of chain homotopy under additive functor) Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between additive categories (Definition 2.3.4). Given chain maps $f_\bullet : g_\bullet : C_\bullet \rightarrow D_\bullet$ between chain complexes (Definition 3.0.1) in $\mathcal{A}$ and a chain homotopy (Definition 3.5.1) $s_\bullet : f_\bullet \simeq g_\bullet$, there is a chain homotopy $Fs_\bullet : Ff_\bullet \simeq Fg_\bullet$ whose components are simply $Fs_n : FC_n \rightarrow FD_{n+1}$.

In particular, the relation of chain complexes being chain homotopic is preserved by additive functors.

Lemma 3.5.4. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $f_\bullet : C_\bullet \rightarrow D_\bullet$ (resp. $f^\bullet : C^\bullet \rightarrow D^\bullet$) be a morphism of chain complexes (Definition 3.0.1) (resp. of cochain complexes). If $f_\bullet$ is null homotopic (Definition 3.5.1) (resp. $f^\bullet$ is null homotopic), then the induced maps

$$H_n(f_\bullet) : H_n(C_\bullet) \rightarrow H_n(D_\bullet)$$

of homology (Definition 3.2.2) (resp.

$$H^n(f^\bullet) : H^n(C^\bullet) \rightarrow H^n(D^\bullet)$$

of cohomology) are zero.

Proof. (♠ TODO:) □

Lemma 3.5.5. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $f_\bullet, g_\bullet : C_\bullet \rightarrow D_\bullet$ (resp. $f^\bullet, g^\bullet : C^\bullet \rightarrow D^\bullet$) be a morphism of chain complexes (Definition 3.0.1) (resp. of cochain complexes) that are chain homotopic (Definition 3.5.1). The induced maps

$$H_n(f_\bullet) : H_n(C_\bullet) \rightarrow H_n(D_\bullet)$$

and

$$H_n(g_\bullet) : H_n(C_\bullet) \rightarrow H_n(D_\bullet)$$

of homology (Definition 3.2.2) (resp.

$$H^n(f^\bullet) : H^n(C^\bullet) \rightarrow H^n(D^\bullet)$$

and

$$H^n(g^\bullet) : H^n(C^\bullet) \rightarrow H^n(D^\bullet)$$

of cohomology) are equal.

Proof. This follows from Lemma 3.5.4. □

Definition 3.5.6. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). A chain or cochain complex C in $\mathcal{A}$ is said to be *split* if there are maps $s = \{s_n : C_n \rightarrow C_{n+1}\}_n$ (or $s = \{s^n : C_n \rightarrow C^{n-1}\}_n$) such that $d = dsd$. The maps s_n are called *splitting maps*. The complex C is said to be *split exact* if it is split and acyclic (Definition 3.0.2).

Lemma 3.5.7. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). A chain/cochain complex C in $\mathcal{A}$ is split exact (Definition 3.5.6) if and only if the identity map on C is null homotopic (Definition 3.5.1).

3.6. Mapping cones of morphisms between topological spaces.

Definition 3.6.1. 1. Let $f : X \rightarrow Y$ be a continuous map (Definition C.0.6) of topological spaces (Definition C.0.5).

The *mapping cylinder of f* , denoted by M_f , is the quotient space of the disjoint union $(X \times I) \sqcup Y$ (where $I = [0, 1]$ is the unit interval) by the equivalence relation generated by:

$$(x, 1) \sim f(x) \quad \text{for all } x \in X.$$

In other words, M_f is obtained by attaching the cylinder $X \times I$ to Y by identifying the "top" end $X \times \{1\}$ with the image of X in Y under f .

2. Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs. The *relative mapping cylinder of f* (or *mapping cylinder of the map of pairs f*) is the pair of topological spaces $(M_f, M_{f|_A})$, where M_f is the mapping cylinder of the total map and $M_{f|_A} \subseteq M_f$ is the mapping cylinder of the restricted map $f|_A : A \rightarrow B$.

(♠ TODO: Verify quotient topology for mapping cylinder)

Definition 3.6.2. 1. Let $f : X \rightarrow Y$ be a continuous map (Definition C.0.6) of topological spaces (Definition C.0.5).

The *mapping cone of f* , denoted by C_f , is the topological space obtained by taking the disjoint union of the cone on X and the space Y , and identifying the base of the cone with its image in Y under f . Formally:

$$C_f = ((X \times I) \amalg Y) / \sim$$

where the equivalence relation (Definition C.0.43) $\sim$ is generated by: (♠ TODO: quotient of a topological space by an equivalence relation)

- $(x, 0) \sim (x', 0)$ for all $x, x' \in X$ (collapsing one end of the cylinder to a vertex).
- $(x, 1) \sim f(x)$ for all $x \in X$ (attaching the other end to Y via f).

The mapping cone fits into a sequence $X \xrightarrow{f} Y \xrightarrow{i} C_f$, where i is the natural inclusion.

2. Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs.

The *relative mapping cone of f* (or *mapping cone of the map of pairs f*) is the pair of topological spaces $(C_f, C_{f|_A})$, where C_f is the mapping cone of the total map and $C_{f|_A} \subseteq C_f$ is the mapping cone of the restricted map $f|_A : A \rightarrow B$ sharing the same cone vertex.

(♠ TODO: Diagram for mapping cone of pairs)

Proposition 3.6.3. Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs.

1. The pair $(M_f, M_{f|_A})$ deformation retracts onto the subspace (Y, B) via the map induced by $H_t(x, s) = (x, (1-t)s + t)$ and $H_t(y) = y$.
2. Every map of pairs $f : (X, A) \rightarrow (Y, B)$ factors as $f = r \circ i$, where $i : (X, A) \hookrightarrow (M_f, M_{f|_A})$ is a cofibration and $r : (M_f, M_{f|_A}) \rightarrow (Y, B)$ is a homotopy equivalence.

Theorem 3.6.4. Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs and G an abelian group (Definition C.0.3). There exists a long exact sequence of pairs in singular homology (Definition 3.3.6):

$$\cdots \rightarrow H_n(X, A; G) \xrightarrow{f_*} H_n(Y, B; G) \rightarrow H_n(C_f, C_{f|_A}; G) \rightarrow H_{n-1}(X, A; G) \rightarrow \cdots$$

(♠ TODO: singular cohomology version)

Proposition 3.6.5. Let X and Y be topological spaces (Definition C.0.5) and $f : X \rightarrow Y$ a continuous map (Definition C.0.6).

The mapping cylinder M_f satisfies the following fundamental homotopy properties: (♠ TODO: deformation retraction, cofibration)

1. **Deformation Retraction:** The mapping cylinder M_f deformation retracts onto the subspace $Y \times \{1\} \cong Y$ via the map $H_t(x, s) = (x, (1-t)s + t)$ and $H_t(y) = y$. Thus, the inclusion $j : Y \hookrightarrow M_f$ is a homotopy equivalence.
2. **Cofibration:** The natural inclusion $i : X \hookrightarrow M_f$ defined by $x \mapsto (x, 0)$ is a cofibration.
3. **Factorization:** Every continuous map $f : X \rightarrow Y$ factors as $f = r \circ i$, where $i : X \hookrightarrow M_f$ is an inclusion and $r : M_f \rightarrow Y$ is a homotopy equivalence (the retraction).

Proposition 3.6.6. (♠ TODO:) Let $f : (X, A) \rightarrow (Y, B)$ be a continuous map of pairs. Let G be an abelian group. The mapping cone of the pair $(C_f, C_{f|_A})$ relates to the mapping cylinder pair $(M_f, M_{f|_A})$ via a natural isomorphism:

$$H_n(C_f, C_{f|_A} \cup \{pt\}; G) \cong H_n(M_f, M_{f|_A} \cup X; G)$$

Furthermore, if f is an inclusion of CW pairs $(X, A) \hookrightarrow (Y, B)$, then the relative mapping cone $(C_f, C_{f|_A})$ is homotopy equivalent to the pair of quotients:

$$(C_f, C_{f|_A}) \simeq (Y/X, B/A)$$

under the condition that the inclusions are good pairs. (♠ TODO: good pairs)

3.7. Mapping cones of morphisms between chain complexes.

Definition 3.7.1. 1. Let $f : (C_\bullet, d_\bullet^C) \rightarrow (D_\bullet, d_\bullet^D)$ be a morphism of chain complexes (Definition 3.0.1) in an additive category $\mathcal{A}$ (Definition 2.3.4).

The *mapping cylinder of f* , denoted $\text{Cyl}(f)$, is the chain complex defined by:

- Objects: For each n ,

$$\text{Cyl}(f)_n = C_n \oplus C_{n-1} \oplus D_n.$$

- Differential: For each n , define

$$d_n^{\text{Cyl}(f)} : \text{Cyl}(f)_n \rightarrow \text{Cyl}(f)_{n-1}$$

by the matrix morphism

$$d_n^{\text{Cyl}(f)} = \begin{pmatrix} d_n^C & -\text{id}_{C_{n-1}} & 0 \\ 0 & -d_{n-1}^C & 0 \\ 0 & f_{n-1} & d_n^D \end{pmatrix} : \begin{pmatrix} C_n \\ C_{n-1} \\ D_n \end{pmatrix} \rightarrow \begin{pmatrix} C_{n-1} \\ C_{n-2} \\ D_{n-1} \end{pmatrix}.$$

This construction defines a chain complex, i.e., $d_{n-1}^{\text{Cyl}(f)} \circ d_n^{\text{Cyl}(f)} = 0$.

2. Dually, let $g : (C^\bullet, d_C^\bullet) \rightarrow (D^\bullet, d_D^\bullet)$ be a morphism of cochain complexes (Definition 3.0.1) in $\mathcal{A}$.

The *mapping cylinder of g* , denoted $\text{Cyl}(g)$, is the cochain complex (Definition 3.0.1) defined by:

- Objects: For each n ,

$$\text{Cyl}(g)^n = C^n \oplus C^{n+1} \oplus D^n.$$

- Differential: For each n , define

$$d_{\text{Cyl}(g)}^n : \text{Cyl}(g)^n \rightarrow \text{Cyl}(g)^{n+1}$$

by the matrix morphism

$$d_{\text{Cyl}(g)}^n = \begin{pmatrix} d_C^n & -\text{id}_{C^{n+1}} & 0 \\ 0 & -d_C^{n+1} & 0 \\ 0 & g^{n+1} & d_D^n \end{pmatrix} : \begin{pmatrix} C^n \\ C^{n+1} \\ D^n \end{pmatrix} \rightarrow \begin{pmatrix} C^{n+1} \\ C^{n+2} \\ D^{n+1} \end{pmatrix}.$$

This construction defines a cochain complex, i.e., $d_{\text{Cyl}(g)}^{n+1} \circ d_{\text{Cyl}(g)}^n = 0$.

Proposition 3.7.2. Let $\mathcal{A}$ be an additive category (Definition 2.3.4) and let $f : (C_\bullet, d_\bullet^C) \rightarrow (D_\bullet, d_\bullet^D)$ be a morphism of chain complexes (Definition 3.0.1) in $\mathcal{A}$.

1. **Factorization:** The morphism f factors as the composition $p \circ \alpha$, where $\alpha : C_\bullet \rightarrow \text{Cyl}(f)$ is the canonical inclusion $c_n \mapsto (c_n, 0, 0)$ and $p : \text{Cyl}(f) \rightarrow D_\bullet$ is the projection $(c_n, c_{n-1}, d_n) \mapsto f(c_n) + d_n$.
2. **Homotopy Equivalence:** The projection $p : \text{Cyl}(f) \rightarrow D_\bullet$ is a chain homotopy equivalence (Definition 3.5.1). Thus, the mapping cylinder is isomorphic to $D_\bullet$ in the homotopy category (Definition 8.1.1) $K(\mathcal{A})$.
3. **Cofibration Analog:** The inclusion $\alpha : C_\bullet \rightarrow \text{Cyl}(f)$ is a split monomorphism in each degree, representing the algebraic analog of a topological cofibration. (♠ TODO: cofibration)

Definition 3.7.3. Let $\mathcal{A}$ be an additive category (Definition 2.3.4)

1. Let $f : (C_\bullet, d_\bullet^C) \rightarrow (D_\bullet, d_\bullet^D)$ be a morphism of chain complexes (Definition 3.0.1) in $\mathcal{A}$. The *mapping cone of f* , denoted $\text{Cone}(f)$, is the chain complex defined by:
 - Objects: For each n ,

$$\text{Cone}(f)_n = D_n \oplus C_{n-1}.$$

- Differential: For each n , define

$$d_n^{\text{Cone}(f)} : \text{Cone}(f)_n \rightarrow \text{Cone}(f)_{n-1}$$

by the matrix morphism

$$d_n^{\text{Cone}(f)} = \begin{pmatrix} d_n^D & f_{n-1} \\ 0 & -d_{n-1}^C \end{pmatrix} : D_n \oplus C_{n-1} \rightarrow D_{n-1} \oplus C_{n-2}.$$

This construction defines a chain complex, i.e., $d_{n-1}^{\text{Cone}(f)} \circ d_n^{\text{Cone}(f)} = 0$.

2. Dually, let $g : (C^\bullet, d_C^\bullet) \rightarrow (D^\bullet, d_D^\bullet)$ be a morphism of cochain complexes (Definition 3.0.1) in $\mathcal{A}$.

The *mapping cone of g* , denoted $\text{Cone}(g)$, is the cochain complex (Definition 3.0.1) defined by:

- Objects: For each n ,

$$\text{Cone}(g)^n = D^n \oplus C^{n+1}.$$

- Differential: For each n , define

$$d_{\text{Cone}(g)}^n : \text{Cone}(g)^n \rightarrow \text{Cone}(g)^{n+1}$$

by the matrix morphism

$$d_{\text{Cone}(g)}^n = \begin{pmatrix} d_D^n & g^{n+1} \\ 0 & -d_C^{n+1} \end{pmatrix} : D^n \oplus C^{n+1} \rightarrow D^{n+1} \oplus C^{n+2}.$$

This construction defines a cochain complex, i.e., $d_{\text{Cone}(g)}^{n+1} \circ d_{\text{Cone}(g)}^n = 0$.

Given a chain complex $(C_\bullet, d_\bullet)$ (or a cochain complex), its *mapping cone* $\text{Cone}(C_\bullet)$ simply refers to the mapping cone of the identity morphism $\text{id}_{C_\bullet} : C_\bullet \rightarrow C_\bullet$.

Lemma 3.7.4. (🔥 TODO: quotient complex) Let $f : C_\bullet \rightarrow D_\bullet$ be a morphism of chain complexes in an abelian category (Definition 2.3.12). The mapping cone (Definition 3.7.3) $\text{Cone}(f)$ is the quotient complex of the mapping cylinder (Definition 3.7.1) by the subcomplex $C_\bullet \oplus 0 \oplus 0 \cong C_\bullet$.

Specifically, there exists a short exact sequence of chain complexes:

$$0 \rightarrow C_\bullet \xrightarrow{\alpha} \text{Cyl}(f) \xrightarrow{q} \text{Cone}(f) \rightarrow 0$$

where q is the natural quotient map. This sequence expresses the mapping cone as the homotopy cokernel of f .

Lemma 3.7.5. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. Let $f : (C_\bullet, d_\bullet^C) \rightarrow (D_\bullet, d_\bullet^D)$ be a morphism of chain complexes (Definition 3.0.1) in the additive category $\mathcal{A}$.

Then there is a natural short exact sequence (Definition 2.4.4) of chain complexes

$$0 \longrightarrow (D_\bullet, d_\bullet^D) \xrightarrow{i} (\text{Cone}(f)_\bullet, d_\bullet^{\text{Cone}(f)}) \xrightarrow{p} (C_\bullet[-1], d_\bullet^{C[-1]}) \longrightarrow 0,$$

where $C_\bullet[-1]$ denotes the shifted chain complex (Definition 3.0.1) with

$$(C_\bullet[-1])_n = C_{n-1}, \quad d_n^{C[-1]} := -d_{n-1}^C.$$

More explicitly:

- At the level of objects, by Definition 3.7.3 we have

$$\text{Cone}(f)_n = D_n \oplus C_{n-1}.$$

- The differential (Definition 3.0.1) of the cone is

$$d_n^{\text{Cone}(f)} = \begin{pmatrix} d_n^D & f_{n-1} \\ 0 & -d_{n-1}^C \end{pmatrix} : D_n \oplus C_{n-1} \longrightarrow D_{n-1} \oplus C_{n-2}.$$

- The inclusion $i : D_\bullet \rightarrow \text{Cone}(f)_\bullet$ is given in degree n by

$$i_n : D_n \longrightarrow D_n \oplus C_{n-1}, \quad i_n(d) = (d, 0).$$

This is a chain map, since

$$d_n^{\text{Cone}(f)} \circ i_n(d) = d_n^{\text{Cone}(f)}(d, 0) = (d_n^D(d), 0) = i_{n-1}(d_n^D(d)).$$

- The projection $p : \text{Cone}(f)_\bullet \rightarrow C_\bullet[-1]$ is given in degree n by

$$p_n : D_n \oplus C_{n-1} \longrightarrow C_{n-1}, \quad p_n(d, c) = c.$$

This is a chain map when we use the shifted differential on $C_\bullet[-1]$:

$$d_n^{C[-1]} \circ p_n(d, c) = -d_{n-1}^C(c),$$

while

$$p_{n-1} \circ d_n^{\text{Cone}(f)}(d, c) = p_{n-1}(d_n^D(d) + f_{n-1}(c), -d_{n-1}^C(c)) = -d_{n-1}^C(c),$$

so $d_n^{C[-1]} \circ p_n = p_{n-1} \circ d_n^{\text{Cone}(f)}$.

- For each n , the underlying sequence in $\mathcal{A}$

$$0 \longrightarrow D_n \xrightarrow{i_n} D_n \oplus C_{n-1} \xrightarrow{p_n} C_{n-1} \longrightarrow 0$$

is split exact (Definition 2.4.8) (with splitting given by the obvious inclusion and projection), hence exact (Definition 3.0.2) in the additive (Definition 2.3.4) sense.

Thus the cone of f fits into a degreewise split (Definition 2.4.8) short exact sequence of chain complexes

$$0 \longrightarrow D_\bullet \xrightarrow{i} \text{Cone}(f)_\bullet \xrightarrow{p} C_\bullet[-1] \longrightarrow 0.$$

2. Dually, let $g : (C^\bullet, d_C^\bullet) \rightarrow (D^\bullet, d_D^\bullet)$ be a morphism of cochain complexes in $\mathcal{A}$.

Then there is a natural short exact sequence (Definition 2.4.4) of cochain complexes (Definition 3.0.1)

$$0 \longrightarrow (C^\bullet[1], d_{C[1]}^\bullet) \xrightarrow{j} (\text{Cone}(g)^\bullet, d_{\text{Cone}(g)}^\bullet) \xrightarrow{q} (D^\bullet, d_D^\bullet) \longrightarrow 0,$$

where the shifted cochain complex $C^\bullet[1]$ is defined by

$$(C^\bullet[1])^n = C^{n+1}, \quad d_{C[1]}^n := -d_C^{n+1}.$$

More explicitly:

- At the level of objects, by Definition 3.7.3 we have

$$\text{Cone}(g)^n = D^n \oplus C^{n+1}.$$

- The differential of the cone is

$$d_{\text{Cone}(g)}^n = \begin{pmatrix} d_D^n & g^{n+1} \\ 0 & -d_C^{n+1} \end{pmatrix} : D^n \oplus C^{n+1} \longrightarrow D^{n+1} \oplus C^{n+2}.$$

- The inclusion $j : C^\bullet[1] \rightarrow \text{Cone}(g)^\bullet$ is given in degree n by

$$j^n : (C^\bullet[1])^n = C^{n+1} \longrightarrow D^n \oplus C^{n+1}, \quad j^n(c) = (0, c).$$

This is a cochain map, since

$$d_{\text{Cone}(g)}^n \circ j^n(c) = d_{\text{Cone}(g)}^n(0, c) = (g^{n+1}(c), -d_C^{n+1}(c)),$$

while

$$j^{n+1}(d_{C[1]}^n(c)) = j^{n+1}(-d_C^{n+1}(c)) = (0, -d_C^{n+1}(c)),$$

and the equality $d_{\text{Cone}(g)}^n \circ j^n = j^{n+1} \circ d_{C[1]}^n$ follows from the defining relation $g^{n+1} \circ d_C^n = d_D^n \circ g^n$ on the original complexes (after reindexing).

- The projection $q : \text{Cone}(g)^\bullet \rightarrow D^\bullet$ is given in degree n by

$$q^n : D^n \oplus C^{n+1} \longrightarrow D^n, \quad q^n(d, c) = d.$$

This is a cochain map:

$$d_D^n \circ q^n(d, c) = d_D^n(d),$$

while

$$q^{n+1} \circ d_{\text{Cone}(g)}^n(d, c) = q^{n+1}(d_D^n(d) + g^{n+1}(c), -d_C^{n+1}(c)) = d_D^n(d) + g^{n+1}(c),$$

and again the compatibility with the differentials is ensured by the fact that g is a morphism of cochain complexes.

- For each n , the underlying sequence in $\mathcal{A}$

$$0 \longrightarrow C^{n+1} \xrightarrow{j^n} D^n \oplus C^{n+1} \xrightarrow{q^n} D^n \longrightarrow 0$$

is split exact (with splitting given by the obvious inclusion and projection), hence exact in the additive sense.

Thus the cone of g fits into a degreewise split short exact sequence of cochain complexes

$$0 \longrightarrow C^\bullet[1] \xrightarrow{j} \text{Cone}(g)^\bullet \xrightarrow{q} D^\bullet \longrightarrow 0.$$

Lemma 3.7.6 (Long Exact Sequence of the Mapping Cone). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. Let $f : (C_\bullet, d_\bullet^C) \rightarrow (D_\bullet, d_\bullet^D)$ be a morphism of chain complexes (Definition 3.0.1) in $\mathcal{A}$. Then the short exact sequence of chain complexes

$$0 \longrightarrow D_\bullet \xrightarrow{i} \text{Cone}(f)_\bullet \xrightarrow{p} C_\bullet[-1] \longrightarrow 0$$

from Lemma 3.7.5 induces a long exact sequence in homology (Theorem 3.2.10):

(M)

$$\cdots \rightarrow H_n(D_\bullet) \xrightarrow{H_n(i)} H_n(\text{Cone}(f)_\bullet) \xrightarrow{H_n(p)} H_{n-1}(C_\bullet) \xrightarrow{f_{n-1}^*} H_{n-1}(D_\bullet) \xrightarrow{H_{n-1}(i)} H_{n-1}(\text{Cone}(f)_\bullet) \rightarrow \cdots$$

2. Dually, let $g : (C^\bullet, d_C^\bullet) \rightarrow (D^\bullet, d_D^\bullet)$ be a morphism of cochain complexes in $\mathcal{A}$. Then the short exact sequence of cochain complexes

$$0 \longrightarrow C^\bullet[1] \xrightarrow{j} \text{Cone}(g)^\bullet \xrightarrow{q} D^\bullet \longrightarrow 0$$

from Lemma 3.7.5 induces a long exact sequence in cohomology (Theorem 3.2.10):

(N)

$$\cdots \rightarrow H^{n-1}(C^\bullet) \xrightarrow{H^{n-1}(g)} H^n(D^\bullet) \xrightarrow{H^n(q)} H^n(\text{Cone}(g)^\bullet) \xrightarrow{H^n(j)} H^n(C^\bullet[1]) \xrightarrow{H^n(g)} H^{n+1}(D^\bullet) \rightarrow \cdots$$

Note that $H^n(C^\bullet[1]) \cong H^{n+1}(C^\bullet)$ naturally.

Proof. This follows from Lemma 3.7.5 and Theorem 3.2.10. □

In accordance to the following and Lemma 3.7.5, studying whether a morphism of complexes is a quasi-isomorphism is tantamount to studying the split exact sequence of the mapping cone.

Corollary 3.7.7. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $f : C \rightarrow D$ be a morphism of chain or cochain complexes (Definition 3.0.1) in $\mathcal{A}$. f is a quasi-isomorphism (Definition 3.2.8) if and only if the mapping cone (Definition 3.7.3) $\text{cone}(f)$ is acyclic (Definition 3.0.2).

3.7.1. *Mapping cones for morphisms of topological spaces are mapping cones of the chain maps of the singular chain complexes.* Just as morphisms of pairs of topological spaces possess the notion of mapping cylinders (Definition 3.6.1) and (Definition 3.6.2) cones, morphisms of chain complexes also possess analogous notions; while the construction of chain mapping cylinders and cones are convoluted and seem ad hoc, they ultimately come from topology.

Theorem 3.7.8. (♠ TODO:) Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs of topological spaces (Definition C.0.45) and G an abelian group (Definition C.0.3). Let $f_{\#} : C_{\bullet}(X, A; G) \rightarrow C_{\bullet}(Y, B; G)$ be the induced chain map (Proposition 3.3.7) of singular chain complexes (Definition 3.3.5).

The algebraic mapping cylinder (Definition 3.7.1) of the $f_{\#} : C_{\bullet}(X, A; G) \rightarrow C_{\bullet}(Y, B; G)$ arises from the topological mapping cylinder of pairs via the following precise relationship:

1. **Cylindrical Decomposition:** The singular chain complex of the cylinder $C_{\bullet}(X \times I)$ is chain homotopic to the tensor product of complexes $C_{\bullet}(X) \otimes C_{\bullet}(I)$.
2. **The Interval Complex:** The singular complex of the interval $C_{\bullet}(I)$ is homotopy equivalent to the simple complex $K_{\bullet}$:

$$K_1 \xrightarrow{\partial} K_0, \quad \mathbb{Z}[e] \xrightarrow{\partial} \mathbb{Z}[v_1] \oplus \mathbb{Z}[v_0]$$

where $\partial(e) = v_1 - v_0$.

3. **Emergence of the Algebraic Cylinder:** Applying the algebraic pushout to the diagram induced by the topological gluing $(x, 1) \sim f(x)$ yields:

$$C_{\bullet}(M_f; G) \simeq \text{Cyl}(f_{\#}) := C_{\bullet}(X; G) \oplus C_{\bullet-1}(X; G) \oplus C_{\bullet}(Y; G).$$

The "convoluted" differential in $\text{Cyl}(f_{\#})$ is the unique differential making this a homotopy pushout in the category of chain complexes.

Theorem 3.7.9. Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs (Definition C.0.45). The algebraic mapping cone (Definition 3.7.3) $\text{Cone}(f_{\#})$ is the chain complex obtained by "collapsing" the domain component of the mapping cylinder:

1. **Algebraic Quotient:** There is a natural isomorphism of chain complexes:

$$C_{\bullet}(C_f, C_{f|_A} \cup \{v\}; G) \cong \text{Cyl}(f_{\#}) / C_{\bullet}(X, A; G) \cong \text{Cone}(f_{\#}).$$

2. **The Origin of the Matrix Differential:** The matrix differential $d^{\text{Cone}(f)} = \begin{pmatrix} d^D & f \\ 0 & -d^C \end{pmatrix}$ precisely tracks the boundary of the "prisms" created by the cone construction. The term f represents the base of the cone attached to Y , while $-d^C$ represents the lateral boundary of the cone, with the sign change arising from the boundary of a product formula $\partial(c \otimes e) = \partial c \otimes e + (-1)^{\deg c} c \otimes \partial e$.

4. DERIVED FUNCTORS

This section roughly follows [Wei94, Section 2].

4.1. δ -functors. δ -functors are nice enough functors that yield long exact sequences of homology/cohomology from short exact sequences. The main general examples of (universal) δ -functors are left/right derived functors (Definition 4.3.1) of right/left exact functors (Definition 2.4.12) (Theorem 4.3.4). In particular, the homology/cohomology (Definition 3.2.2) functors $\mathbf{Ch}(\mathcal{A}) \rightarrow \mathcal{A}$ are the quintessential examples of (universal) δ -functors.

Definition 4.1.1. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories (Definition 2.3.12).

1. A *homological δ functor from $\mathcal{A}$ to $\mathcal{B}$* is a pair $(T_n, \delta_n)_{n \geq 0}$ consisting of:
 - a sequence of additive functors (Definition 2.4.1) $T_n : \mathcal{A} \rightarrow \mathcal{B}$ for each integer $n \geq 0$, and
 - for every short exact sequence (Definition 2.4.4) $0 \rightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \rightarrow 0$ in $\mathcal{A}$, a *connecting morphism*

$$\delta_n(A', A, A'') : T_n(A'') \rightarrow T_{n-1}(A')$$

in $\mathcal{B}$ for each $n > 0$

that make the induced sequence

$$\cdots \rightarrow T_{n+1}(A'') \xrightarrow{\delta_{n+1}} T_n(A') \rightarrow T_n(A) \rightarrow T_n(A'') \xrightarrow{\delta_n} T_{n-1}(A') \rightarrow \cdots$$

exact (Definition 3.0.2) in $\mathcal{B}$, and are natural in short exact sequences. That is, for any morphism of short exact sequences, the induced morphisms between these long exact sequences commute.

2. A *cohomological δ -functor from $\mathcal{A}$ to $\mathcal{B}$* is defined dually: it consists of additive functors (Definition 2.4.1) $T^n : \mathcal{A} \rightarrow \mathcal{B}$ for $n \geq 0$ and *connecting morphisms*

$$\delta^n(A', A, A'') : T^n(A') \rightarrow T^{n+1}(A'')$$

such that for each short exact sequence (Definition 2.4.4) $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ the resulting sequence

$$0 \rightarrow T^0(A') \rightarrow T^0(A) \rightarrow T^0(A'') \xrightarrow{\delta^0} T^1(A') \rightarrow T^1(A) \rightarrow T^1(A'') \xrightarrow{\delta^1} \cdots$$

is exact (Definition 3.0.2) and the construction is natural with respect to morphisms of short exact sequences.

3. Let (T_n, δ_n) and (S_n, ∂_n) be homological δ -functors from $\mathcal{A}$ to $\mathcal{B}$. A *morphism of (homological) δ -functors*

$$\eta : T_\bullet \rightarrow S_\bullet$$

is a family of natural transformations (Definition 1.3.1) $\eta_n : T_n \Rightarrow S_n$ for each $n \geq 0$ such that for every short exact sequence $0 \rightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \rightarrow 0$, the following diagram in $\mathcal{B}$ commutes for all $n > 0$:

$$\begin{array}{ccc} T_n(A'') & \xrightarrow{\delta_n} & T_{n-1}(A') \\ \downarrow \eta_n & & \downarrow \eta_{n-1} \\ S_n(A'') & \xrightarrow{\partial_n} & S_{n-1}(A') \end{array}$$

The dual notion (for cohomological δ -functors) is defined analogously, reversing the direction of the connecting morphisms.

Definition 4.1.2. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories (Definition 2.3.12). A cohomological (resp. homological) δ -functor (Definition 4.1.1) $T = (T_n : \mathcal{A} \rightarrow \mathcal{B}, \delta_n)_{n \geq 0}$ (resp. $T = (T^n : \mathcal{A} \rightarrow \mathcal{B}, \delta^n)_{n \geq 0}$) is called **universal** if, for any other δ -functor S of the same type and any natural transformation $f : T_0 \rightarrow S_0$ (or $f : T^0 \rightarrow S^0$), there exists a unique morphism of δ -functors (Definition 4.1.1) extending f .

4.2. Abelian categories with enough objects of a class and resolutions. To be able to define and compute left/right derived functors (Definition 4.3.1) of right/left exact functors, the source abelian category needs to “have enough objects (Definition 4.2.1)” of certain classes; in practice, these classes would be the classes of projective (Lemma 4.2.30) (or flat (Definition 4.2.16) objects when the functor considered is a kind of tensor product, e.g. Definition 2.1.9) or injective (Definition 4.2.2) objects. When the abelian category has enough objects, every object has an appropriate resolution (Definition 4.2.29), see Lemma 4.2.31.

Definition 4.2.1. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let $\mathcal{X}$ be a class of objects in $\mathcal{A}$.

We say that $\mathcal{A}$ **has enough objects of class $\mathcal{X}$ on the left (resp. on the right)** if for any object $M \in \mathcal{A}$, there exists an object X of the class $\mathcal{X}$ and an epimorphism (Definition 2.3.8) $X \twoheadrightarrow M$ (resp. a monomorphism $M \hookrightarrow X$).

4.2.1. Injective and projective objects. The most basic classes of objects that we will use to consider resolutions are injective and projective objects

Definition 4.2.2 (Injective and Projective objects in a general category). Let $\mathcal{C}$ be a category (Definition 1.1.1)

- An object $I \in \mathcal{C}$ is called **injective** if it satisfies the following extension property: for every monomorphism (Definition 2.3.8) $m : A \rightarrow B$ in $\mathcal{C}$ and every morphism $f : A \rightarrow I$, there exists a morphism $\tilde{f} : B \rightarrow I$ such that the diagram

$$\begin{array}{ccc} & I & \\ & \uparrow f & \\ A & \xrightarrow{m} & B \end{array} \quad \begin{array}{c} \nwarrow \tilde{f} \\ \end{array}$$

commutes, i.e., $\tilde{f} \circ m = f$.

- Dually, an object $P \in \mathcal{C}$ is called **projective** if it satisfies the following lifting property: for every epimorphism (Definition 2.3.8) $e : X \twoheadrightarrow Y$ in $\mathcal{C}$ and every morphism $g : P \rightarrow Y$, there exists a morphism $\tilde{g} : P \rightarrow X$ such that the diagram

$$\begin{array}{ccc} & P & \\ & \downarrow g & \\ X & \xrightarrow{e} & Y \end{array} \quad \begin{array}{c} \nwarrow \tilde{g} \\ \end{array}$$

commutes, i.e., $e \circ \tilde{g} = g$.

Lemma 4.2.3. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $f : A_1 \rightarrow B$, $g : A_2 \rightarrow B$ be morphisms in $\mathcal{A}$. If f is a monomorphism (Definition 2.3.8) (resp. epimorphism (Definition 2.3.8)), then the pullback (Definition C.0.24) morphism $A_1 \times_B A_2 \rightarrow A_2$ of f by g is a monomorphism (resp. epimorphism).

Proof. We prove the monomorphism case; the epimorphism case is similar. In $\mathcal{A}$, a morphism is a monomorphism if and only if its kernel is zero by Lemma 2.3.17. Let $p_2 : A_1 \times_B A_2 \rightarrow A_2$ be the pullback of f along g . By Proposition 2.3.16, kernels in $\mathcal{A}$ commute with pullbacks, which are limits. In this case, this tells us that $\ker(p_2)$ fits in the following cartesian diagram:

$$\begin{array}{ccc} \ker(p_2) & \longrightarrow & \ker(f) \\ \downarrow & \lrcorner & \downarrow \\ A_1 \times_B A_2 & \xrightarrow{p_1} & A_1 \\ \downarrow p_2 & & \downarrow f \\ A_2 & \xrightarrow{g} & B \end{array}$$

Since $\ker(f) = 0$, $\ker(p_2) \cong 0$.

□

Proposition 4.2.4. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let $P, I \in \mathcal{A}$.

The following are equivalent (Definition 1.3.4) for an object I (Injective conditions):

1. I is injective (Definition 4.2.2): for every monomorphism (Definition 2.3.8) $f : A \rightarrow B$, any morphism $g : A \rightarrow I$ factors through f .
2. The contravariant functor (Definition 1.2.2) $\text{Hom}_{\mathcal{A}}(-, I)$ is exact (Definition 2.4.12).
3. Every short exact sequence (Definition 2.4.4) $0 \rightarrow I \rightarrow B \rightarrow C \rightarrow 0$ splits (Definition 2.4.8).

Dually, the following are equivalent for an object P :

1. P is projective (Definition 4.2.2): for every epimorphism (Definition 2.3.8) $f : M \rightarrow N$, any morphism $g : P \rightarrow N$ factors through f .
2. The functor (Definition 1.2.2) $\text{Hom}_{\mathcal{A}}(P, -)$ is exact (Definition 2.4.12).
3. Every short exact sequence (Definition 2.4.4) $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ splits (Definition 2.4.8).

Proof. We prove the projective case; the injective case follows by duality in the opposite category $\mathcal{A}^{\text{op}}$.

(1 $\implies$ 2): The functor $\text{Hom}_{\mathcal{A}}(P, -)$ is always left-exact (Definition 2.4.12). To show it is exact, we show it preserves epimorphisms (Definition 2.3.8). Let $f : M \rightarrow N$ be an epimorphism. By the lifting property of P , any g factors through f as seen in the following

diagram:

$$\begin{array}{ccccc} & & P & & \\ & \swarrow h & \downarrow g & & \\ M & \xrightarrow{f} & N & \longrightarrow & 0 \end{array}$$

The existence of h such that $f \circ h = g$ implies that $f_* : \text{Hom}(P, M) \rightarrow \text{Hom}(P, N)$ is surjective, hence the functor is exact.

(2 $\implies$ 3): Consider the short exact sequence (Definition 2.4.4) $0 \rightarrow A \rightarrow B \xrightarrow{f} P \rightarrow 0$. Applying the exact functor $\text{Hom}_{\mathcal{A}}(P, -)$ yields:

$$\text{Hom}_{\mathcal{A}}(P, B) \xrightarrow{f_*} \text{Hom}_{\mathcal{A}}(P, P) \rightarrow 0$$

Since f_* is surjective, id_P has a preimage $s : P \rightarrow B$ such that $f \circ s = \text{id}_P$. This s is a section, so the sequence splits (Definition 2.4.8):

$$0 \longrightarrow A \longrightarrow B \xleftarrow{s} P \xrightarrow{f} 0$$

(3 $\implies$ 1): Given an epimorphism $f : M \rightarrow N$ and map $g : P \rightarrow N$, form the pullback (Definition C.0.24) $M \times_N P$; note that $\mathcal{A}$ admits finite limits by Lemma 2.3.14. Since f is an epimorphism, the projection π_P is an epimorphism by Lemma 4.2.3. The resulting SES splits (Definition 2.4.8) by assumption:

$$\begin{array}{ccccc} M \times_N P & \xrightarrow{\pi_P} & P & & \\ \downarrow \pi_M & \swarrow s & \downarrow g & & \\ M & \xrightarrow{f} & N & & \end{array}$$

The composition $h = \pi_M \circ s$ provides the required lift since $f \circ (\pi_M \circ s) = g \circ \pi_P \circ s = g \circ \text{id}_P = g$. $\square$

Definition 4.2.5. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. $\mathcal{A}$ is said to *have enough injectives* if for every object A in $\mathcal{A}$, there is an monomorphism (Definition 2.3.8) $A \rightarrow I$ with I an injective object (Definition 4.2.2) of $\mathcal{A}$. Equivalently, $\mathcal{A}$ has enough injectives if it has enough objects of the class of injectives on the right (Definition 4.2.1)
2. $\mathcal{A}$ is said to *have enough projectives* if for every object A in $\mathcal{A}$, there is a epimorphism (Definition 2.3.8) $P \rightarrow A$ with P a projective object (Definition 4.2.2) of $\mathcal{A}$. Equivalently, $\mathcal{A}$ has enough projectives if it has enough objects of the class of projectives on the left (Definition 4.2.1)

Theorem 4.2.6. 1. Examples of abelian categories with enough injectives include:

- The category of abelian groups.
- Any category of modules over a ring; equivalently, any category of bimodules over two rings.
- The category of sheaves of abelian groups on a ringed space or more generally on an essentially small site.

2. Examples of abelian categories with enough projectives include:
 - Any category of modules over a ring; equivalent, any category of bimodules over two rings.
 - The category of finitely generated modules over a semisimple ring.

However, not all common abelian categories possess enough injectives or projectives:

Proposition 4.2.7. (♠ TODO: finitely generated abelian groups) The category of finitely generated abelian groups does not have enough injectives (Definition 4.2.5).

Categories of sheaves (on topological spaces) of abelian groups (or more generally of modules over sheaves of rings) do not generally have enough projectives.

Baer's criterion describes injective objects in a category of modules.

Theorem 4.2.8 (Baer's Criterion). Let R be a (not necessarily commutative) ring (Definition C.0.7) with unit and let I be an R -module (Definition 2.1.1). Then I is an injective (Definition 4.2.2) R -module if and only if for every left ideal $J \subseteq R$, every R -module homomorphism (Definition 2.1.2)

$$f : J \rightarrow I$$

extends to an R -module homomorphism

$$\tilde{f} : R \rightarrow I.$$

There are multiple equivalent ways to describe projective objects in a category of modules.

Definition 4.2.9. Let R and S be (not necessarily commutative) rings (Definition C.0.7). A *projective R - S -bimodule* is an (R, S) -bimodule (Definition 2.1.1) P that is a projective object (Definition 4.2.2) in the category ${}_R\mathbf{Mod}_S$. That is, for every surjective homomorphism of (R, S) -bimodules $f : M \rightarrow N$ and every homomorphism $g : P \rightarrow N$, there exists a homomorphism $h : P \rightarrow M$ such that $f \circ h = g$.

Note that P is a projective R - S -bimodule if and only if it is a projective left module over the ring $R \otimes_{\mathbb{Z}} S^{\text{op}}$ (Definition 2.1.10) (Definition C.0.38) (see ??).

Being a projective bimodule is a strictly stronger condition than being projective as a left or right module.

- A bimodule ${}_R P_S$ may be projective as a left R -module (i.e., projective in ${}_R\mathbf{Mod}$) without being a projective bimodule.
- Similarly, it may be projective as a right S -module (i.e., projective in $\mathbf{Mod}_S$) without being a projective bimodule.
- A bimodule that is projective on both sides is sometimes called *biprojective*, but this does not imply it is a projective object in ${}_R\mathbf{Mod}_S$. For example, let $R = S = \mathbb{Q}[x]$. The bimodule $P = \mathbb{Q}[x]$ (with the natural action on both sides) is free (hence projective) as both a left R -module and a right R -module. However, it is *not* a projective (R, R) -bimodule because it is not a direct summand of the free bimodule $R \otimes_{\mathbb{Z}} R^{\text{op}}$.

Definition 4.2.10. Let R and S be rings (Definition C.0.7). An *injective R - S -bimodule* is an (R, S) -bimodule (Definition 2.1.1) E that satisfies any of the following equivalent conditions:

1. The functor (Definition 1.2.2)

$$\mathrm{Hom}_{R\mathrm{Mod}_S}(-, E) : R\mathrm{Mod}_S \rightarrow \mathbf{Ab}$$

(Definition 2.1.8) (Definition 2.1.3) (Definition 1.1.8) is an exact functor (Definition 2.4.12) from the abelian category (Definition 2.3.12) of (R, S) -bimodules (Definition 2.1.1) to the category of abelian groups (Definition 1.1.8).

2. E is an injective object (Definition 4.2.2) in the category (Definition 1.1.1) $R\mathrm{Mod}_S$ (Definition 2.1.3). That is, for every monomorphism of (R, S) -bimodules $f : M \rightarrow N$ and every homomorphism $g : M \rightarrow E$, there exists a homomorphism $h : N \rightarrow E$ such that $h \circ f = g$.

Proposition 4.2.11. Let R and S be (not necessarily commutative) rings (Definition C.0.7). Let P be an (R, S) -bimodule (Definition 2.1.1). The following are equivalent:

1. The functor

$$\mathrm{Hom}_{R\mathrm{Mod}_S}(P, -) : R\mathrm{Mod}_S \rightarrow \mathbf{Ab}$$

is an exact functor (Definition 2.4.12) between the abelian categories $R\mathrm{Mod}_S$ (Definition 2.1.3) and $\mathbf{Ab}$ (Definition 1.1.8).

2. P is a projective left module over the ring $R \otimes_{\mathbb{Z}} S^{\mathrm{op}}$ (Definition 2.1.10) (Definition C.0.38).
3. P is a direct summand (Definition 4.2.13) of a free (R, S) -bimodule. (A free (R, S) -bimodule is a direct sum of copies of the tensor product $R \otimes_{\mathbb{Z}} S^{\mathrm{op}}$ (Definition 2.1.10), equipped with the natural left R -action and right S -action).
4. P is a projective object (Definition 4.2.2) in the category $R\mathrm{Mod}_S$. That is, for every surjective homomorphism of (R, S) -bimodules $f : M \rightarrow N$ and every homomorphism $g : P \rightarrow N$, there exists a homomorphism $h : P \rightarrow M$ such that $f \circ h = g$.

Proof. (♠ TODO:)

□

Definition 4.2.12. Let R and S be (not necessarily commutative) rings (Definition C.0.7). Let I be a set.

The *free $R-S$ -bimodule indexed by I* is the direct sum (Definition 2.2.5) $(R \otimes_{\mathbb{Z}} S^{\mathrm{op}})^{\oplus I}$ where $R \otimes_{\mathbb{Z}} S^{\mathrm{op}}$ (Definition 2.1.10) (Definition C.0.38) is equipped with the natural R - S -bimodules structure.

An R - S -bimodule is said to be *free* if it is isomorphic to such a bimodule.

Equivalently, a free R - S -bimodule can be described by a *basis* $B = \{b_i\}_{i \in I}$ — the *free R - S -bimodule generated by B* is the abelian group of formal finite sums

$$(O) \quad \sum_{j=1}^n r_{i_j} b_{i_j} s_{i_j}$$

where $n \in \mathbb{N}$, $i_j \in I$, $r_{i_j} \in R$, and $s_{i_j} \in S$, subject only to the relations required by the R - S -bimodule axioms. More explicitly, the relations are, for all $i \in I$:

- **Distributivity (Left):** $(r_1 + r_2)b_i s = r_1 b_i s + r_2 b_i s$
- **Distributivity (Right):** $r b_i (s_1 + s_2) = r b_i s_1 + r b_i s_2$

- **Scalar Interaction:** $rb_i(s_1s_2) = (rb_is_1)s_2$ and $(r_1r_2)b_is = r_1(r_2b_is)$
- **Centrality of $\mathbb{Z}$:** For any $n \in \mathbb{Z}$, $n(rb_is) = (nr)b_is = rb_i(ns)$

Furthermore, an element of the form (O) is zero if and only if for each $i \in I$, the term $\sum_{j:i_j=i} r_{i_j} \otimes s_{i_j}$ is zero in the tensor product $R \otimes_{\mathbb{Z}} S^{\text{op}}$.

A *free left R -module* can thus simply be defined/identified as a free $R - \mathbb{Z}$ -bimodule and a *free right R -module* can thus simply be defined/identified as a free $\mathbb{Z} - R$ -bimodule.

Definition 4.2.13. Let $\mathcal{C}$ be an additive category (Definition 2.3.4) and $X \in \text{Ob}(\mathcal{C})$. A subobject (Definition 1.3.6) $i : Y \hookrightarrow X$ is called a *direct summand of X* if there exists a morphism $p : X \rightarrow Y$ such that $p \circ i = \text{id}_Y$. In this case, X is isomorphic to the direct sum (Definition 2.3.4) $Y \oplus Z$ for some object Z , with i the canonical inclusion.

Proposition 4.2.14. Let R, S be (not necessarily commutative) rings (Definition C.0.7). An R - S bimodule (Definition 2.1.1) is projective (Definition 4.2.9) if and only if it is a direct summand (Definition 4.2.13) of a free R - S bimodule (Definition 4.2.12).

Proof. It suffices to (♠ TODO:) □

Proposition 4.2.15. Let R be a local ring or a PID (Definition C.0.41). All projective (Definition 4.2.9) modules (Definition 2.1.1) over R are free (Definition 4.2.12).

4.2.2. *Flat modules.* When a tensor product is being considered, “flat” objects are used to construct resolutions; in particular, in a category of modules, the notion of flatness is as follows:

Definition 4.2.16 (Flat module over a ring). Let R be a (not necessarily commutative) ring (Definition C.0.7).

1. Let M be a left R -module. The module M is said to be *flat (with respect to the left R -module structure)* if the functor

$$- \otimes_R M : \text{Mod}_R \rightarrow \mathbf{Ab}$$

(♠ TODO: comment on how I might as well be talking about other more specific kinds of modules) (Definition 2.1.9) from the category of right R -modules to abelian groups is exact; that is, for every exact sequence of right R -modules

$$0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0,$$

the induced sequence

$$0 \rightarrow N' \otimes_R M \rightarrow N \otimes_R M \rightarrow N'' \otimes_R M \rightarrow 0$$

is exact.

Equivalently, M is flat if $\text{Tor}_1^R(-, M) = 0$ (Definition 4.5.1).

2. Let M be a right R -module. The module M is said to be *flat (with respect to the right R -module structure)* if the functor

$$M \otimes_R - : {}_R\text{Mod} \rightarrow \mathbf{Ab}$$

from the category of left R -modules to abelian groups is exact; that is, for every exact sequence of right R -modules

$$0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0,$$

the induced sequence

$$0 \rightarrow M \otimes_R N' \rightarrow M \otimes_R N \rightarrow M \otimes_R N'' \rightarrow 0$$

is exact.

Equivalently, M is flat if $\mathrm{Tor}_1^R(M, -) = 0$ (Definition 4.5.1).

Equivalently in either case, M is flat if it is flat with respect to the biadditive functor

$$- \otimes_R - : \mathbf{Mod}_R \times {}_R\mathbf{Mod} \rightarrow \mathbf{Ab}.$$

(Definition 4.6.9)

(♠ TODO: forgetful functor from category of modules to category of abelian groups is faithful and exact.)

Proposition 4.2.17. Let R, S be not necessarily commutative rings (Definition C.0.7). If P is a projective (Definition 4.2.9) R - S -bimodule (Definition 2.1.1), then P is both flat (Definition 4.2.16) with respect to the left R -module (Definition 2.1.1) structure and flat with respect to the right S -module (Definition 2.1.1) structure. In other words, the functors (Definition 1.2.2)

$$- \otimes_R P : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$$

and

$$P \otimes_S - : {}_S\mathbf{Mod} \rightarrow {}_R\mathbf{Mod}$$

(Definition 2.1.9) are exact (Definition 3.0.2).

Proof. Recall that tensor products are right exact by Proposition 2.5.5. It thus suffices to show that tensor products are left exact. By Theorem 2.5.11, the category of R - S -bimodules is Grothendieck (Definition 2.5.7) and hence AB4 (Definition 2.5.7) (♠ TODO:) $\square$

Definition 4.2.18. A ring (Definition C.0.7) R is

1. *left perfect* if every flat (Definition 4.2.16) left R -module (Definition 2.1.1) is projective (Definition 4.2.9).
2. *right perfect* if every flat right R -module (Definition 2.1.1) is projective.

Definition 4.2.19. Let R be a ring (Definition C.0.7), not assumed commutative. The *Jacobson radical of R* , denoted by $J(R)$, is the intersection of all maximal left ideals (Definition C.0.22) of R :

$$J(R) = \bigcap \{M \subseteq R : M \text{ is a maximal left ideal of } R\}.$$

Equivalently, $J(R)$ is also the intersection of all maximal right ideals of R . Thus, $J(R)$ is a two-sided ideal of R .

Definition 4.2.20. Let R be a ring (Definition C.0.7), not assumed commutative. A subset $I \subseteq R$ is

1. **left T-nilpotent** if for every sequence $\{a_1, a_2, a_3, \dots\}$ of elements in I , there exists an integer $n \geq 1$ such that the product $a_1 a_2 \dots a_n = 0$.
2. **right T-nilpotent** if for every sequence $\{a_1, a_2, a_3, \dots\}$ of elements in I , there exists an integer $n \geq 1$ such that the product $a_n a_{n-1} \dots a_1 = 0$.

Definition 4.2.21. Let R be a ring (Definition C.0.7), not assumed commutative. Let M be an (left or right) R -module. A **projective cover of M** is a pair (P, π) where P is a projective R -module (Definition 4.2.9) and $\pi : P \rightarrow M$ is a superfluous epimorphism (Definition 2.3.8), meaning that for any submodule (Definition 2.1.4) $K \subseteq P$, if $\pi(K) = M$, then $K = P$. (♠
TODO: superfluous epimorphism)

Theorem 4.2.22. Let R be a ring (Definition C.0.7). The following conditions are equivalent:

1. R is left perfect (Definition 4.2.18), i.e. every flat (Definition 4.2.16) left R -module (Definition 2.1.1) is projective (Definition 4.2.9).
2. Every left R -module has a projective cover.
3. $R/J(R)$ is a semisimple ring and $J(R)$ is left T-nilpotent.
4. R satisfies the descending chain condition on principal right ideals.
5. Every descending chain of principal right ideals $a_1 R \supseteq a_1 a_2 R \supseteq a_1 a_2 a_3 R \supseteq \dots$ terminates.

Proposition 4.2.23. If R is a left perfect (Definition 4.2.18) ring (Definition C.0.7) and $C_\bullet$ is a chain complex (Definition 3.0.1) of flat (Definition 4.2.16) left R -modules (Definition 2.1.1), then $C_\bullet$ is a complex (Definition 3.0.1) of projective (Definition 4.2.9) modules (Definition 2.1.1). Furthermore, if $f : C_\bullet \rightarrow D_\bullet$ is a quasi-isomorphism (Definition 3.2.8) between such complexes, the mapping cone (Definition 3.7.3) $C(f)_\bullet$ is a split exact (Definition 3.5.6) complex if it is acyclic (Definition 3.0.2).

Corollary 4.2.24. Every left Artinian ring (Definition C.0.7) is left perfect (Definition 4.2.18). However, a left perfect ring need not be left Artinian, as the DCC is only required on principal right ideals.

A related notion to flatness is torsion-freeness:

Definition 4.2.25. Let $(R, +, \cdot)$ be a not-necessarily commutative ring (Definition C.0.7).

1. An element $a \in R$ is a **left zero-divisor** if there exists a nonzero $x \in R$ such that $ax = 0$. Otherwise, a is called **left regular** or **left cancellable**.
2. An element $a \in R$ is a **right zero-divisor** if there exists a nonzero $x \in R$ such that $xa = 0$. Otherwise, a is called **right regular** or **right cancellable**.
3. An element $a \in R$ is a **zero-divisor** if it is a left zero-divisor or a right zero-divisor.
4. An element $a \in R$ is a **two-sided zero-divisor** if it is both a left zero-divisor and a right zero-divisor.
5. An element $a \in R$ is **regular**, **cancellable**, or a **non-zero-divisor** if it is both left and right regular.

A zero-divisor of any kind that is not itself 0 is said to be a **nonzero zero divisor** or a **nontrivial zero divisor** of its kind.

Definition 4.2.26 (Zero-divisor and Non-zero-divisor on a Module). Let R be a ring and let $r \in R$ be an element.

1. Let M be a left R -module.
 - The element r is a **(left) zero-divisor on M** if there exists a non-zero element $m \in M$ such that $rm = 0$. Equivalently, r is a zero-divisor on M if the left-multiplication map $\lambda_r: M \rightarrow M$ defined by $\lambda_r(m) = rm$ is not injective.
 - Otherwise, the element r is a **(left) non-zero-divisor on M** (or is **M -regular**), meaning λ_r is injective.
2. Let M be a right R -module.
 - The element r is a **(right) zero-divisor on M** if there exists a non-zero element $m \in M$ such that $mr = 0$. Equivalently, the right-multiplication map $\rho_r: M \rightarrow M$ defined by $\rho_r(m) = mr$ is not injective.
 - Otherwise, the element r is a **(right) non-zero-divisor on M** , meaning ρ_r is injective.

Definition 4.2.27. Let R be a ring (Definition C.0.7).

1. A left R -module (Definition 2.1.1) M is said to be **(left) R -torsion-free** if for every regular (Definition 4.2.25) element $r \in R$ and every $m \in M$, $rm = 0$ implies $m = 0$.
2. A right R -module (Definition 2.1.1) M is said to be **(right) R -torsion-free** if for every regular element $r \in R$ and every $m \in M$, $mr = 0$ implies $m = 0$.

Proposition 4.2.28. Let R be a Dedekind ring. A torsion free module (Definition 4.2.27) over R is flat (Definition 4.2.16).

The relationships among free, projective, flat, and torsion-free modules can be summarized in the following diagram found in Wikipedia's page on projective modules:

$$\begin{array}{ccccccc}
 \text{free} & \xRightarrow{\quad} & \text{projective} & \xRightarrow{\quad} & \text{flat} & \xRightarrow{\quad} & \text{torsion-free} \\
 \uparrow \text{local ring or PID} & & \uparrow \text{perfect ring} & & \uparrow \text{Dedekind ring} & &
 \end{array}$$

4.2.3. *Resolutions.* Intuitively speaking, resolutions are approximations of an object in an abelian category by acyclic sequences.

Definition 4.2.29. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let $\mathcal{X}$ be a class of objects in $\mathcal{A}$. Let M be an object of $\mathcal{A}$.

1. A **right resolution of M** is a cochain complex (Definition 3.0.1) $I^\bullet$ with $I^i = 0$ for $i < 0$ and a map $M \rightarrow I^0$ such that the augmented complex

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

is exact (Definition 3.0.2).

2. A **left resolution of M** is a chain complex (Definition 3.0.1) $P_\bullet$ with $P_i = 0$ for $i < 0$ and a map $P_0 \rightarrow M$ such that the augmented complex

$$\dots P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

is exact (Definition 3.0.2).

3. An **$\mathcal{X}$ -left resolution** of an object $M \in \mathcal{A}$ a left resolution (Definition 4.2.29) by objects of $\mathcal{X}$, i.e. an exact complex

$$\cdots \rightarrow X_2 \rightarrow X_1 \rightarrow X_0 \rightarrow M \rightarrow 0$$

with each $X_i \in \mathcal{X}$.

4. An **$\mathcal{X}$ -right resolution** of an object $M \in \mathcal{A}$ a right resolution (Definition 4.2.29) by objects of $\mathcal{X}$, i.e. an exact complex

$$0 \rightarrow M \rightarrow X^0 \rightarrow X^1 \rightarrow X^2 \rightarrow \cdots$$

with each $X_i \in \mathcal{X}$.

5. A **projective resolution of M** is a left resolution $P^\bullet$ for which the objects P^i are all projective (Definition 4.2.2).
6. An **injective resolution of M** is a right resolution $I^\bullet$ for which the objects I^i are all injective (Definition 4.2.2).

The notion of right/left resolutions above may be regarded as special cases of the notions of right/left resolutions of Definition 8.6.1 where $\mathcal{A}$ is embedded into the category $\mathbf{Ch}(\mathcal{A})$ by identifying an object A of $\mathcal{A}$ with the (co)chain complex concentrated in degree 0 associated to A — the right/left resolutions as defined above are then right/left resolutions of $A^\bullet$ by cochain/chain complexes $K^\bullet$ whose negative degree components are 0.

Lemma 4.2.30. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. A projective object (Definition 4.2.2) $\mathcal{P}$ always has a projective resolution (Definition 4.2.29) given by

$$\cdots \rightarrow 0 \rightarrow \mathcal{P} \xrightarrow{\text{id}} \mathcal{P} \rightarrow 0.$$

2. A injective object (Definition 4.2.2) $\mathcal{I}$ always has a injective resolution (Definition 4.2.29) given by

$$0 \rightarrow \mathcal{I} \xrightarrow{\text{id}} \mathcal{I} \rightarrow 0 \rightarrow \cdots.$$

Proof. This is clear. □

Lemma 4.2.31 (cf. [Wei94, Lemma 2.2.5, Lemma 2.3.6]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let $\mathcal{X}$ be a class of objects in $\mathcal{A}$.

1. If $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the right (Definition 4.2.1), then for every object $A \in \mathcal{A}$ there exists an $\mathcal{X}$ -right resolution of A (Definition 4.2.29).
2. If $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the left (Definition 4.2.1), then for every object $A \in \mathcal{A}$ there exists an $\mathcal{X}$ -left resolution of A (Definition 4.2.29).

Note that this is a special case of Proposition 4.2.33 obtained by letting the complex $M^\bullet$ be the complex such that

$$M^i = \begin{cases} A & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}.$$

In particular,

- If $\mathcal{A}$ has enough injective objects (Definition 4.2.5), then for every object $A \in \mathcal{A}$ there exists an injective resolution of A (Definition 4.2.29).
- If $\mathcal{A}$ has enough projective objects (Definition 4.2.5), then for every object $A \in \mathcal{A}$ there exists a projective resolution of A (Definition 4.2.29).
- If $F : \mathcal{A} \rightarrow \mathcal{B}$ is a left (resp. right) exact functor (Definition 2.4.12) between abelian categories and $\mathcal{A}$ has enough F -acyclic objects on the right (resp. left), then for every object $A \in \mathcal{A}$, there exists an right (resp. left) F -acyclic resolution (Definition 4.4.2) of A .

Proof. 1. Let $A \in \mathcal{A}$ be an object. Since $\mathcal{A}$ has enough objects of class $\mathcal{X}$ of the right, there is an object X_0 of $\mathcal{X}$ and a monomorphism $\varepsilon_0 : A \rightarrow X_0$. Let $A_0 = \text{coker } \varepsilon_0$ (Definition 2.3.6). Inductively, given an object A_{n-1} of $\mathcal{A}$, choose an object X_n of $\mathcal{X}$ and a monomorphism $\varepsilon_n : A_{n-1} \hookrightarrow X_n$. Let $A_n = \text{coker } \varepsilon_n$. In particular, there is a surjection $X_n \twoheadrightarrow A_n$. Let d_n be the composition

$$X_{n-1} \twoheadrightarrow A_{n-1} \xrightarrow{\varepsilon_n} X_n.$$

The chain complex

$$0 \rightarrow A \xrightarrow{\varepsilon_0} X_0 \xrightarrow{d_0} X_1 \xrightarrow{d_1} \dots$$

is thus an $\mathcal{X}$ -right resolution of A .

2. This is simply the dual statement of the next statement.

□

In fact, a generalization is possible: if the abelian category $\mathcal{A}$ has enough objects in a class $\mathcal{X}$ (on the right/left) *complex*, then the complex has a “resolution” by objects of $\mathcal{X}$.

Lemma 4.2.32. Abelian categories (Definition 2.3.12) are finitely complete and finitely cocomplete (Definition 2.2.11).

Proof. Abelian categories have finite products and finite coproducts (in the form of direct sums), empty products and coproducts (in the form of the zero object), and equalizers and coequalizers (Definition 2.2.14) (in the form of kernels and cokernels of morphisms), so Theorem 2.2.15 applies. □

Proposition 4.2.33. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let $\mathcal{X}$ be a class of objects in $\mathcal{A}$.

1. If $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the right (Definition 4.2.1), then for every bounded below (cohomological) complex $M^\bullet$ of objects in $\mathcal{A}$, there exists a bounded below complex $I^\bullet$ of objects in $\mathcal{X}$ and a quasi-isomorphism (Definition 3.2.8) $M^\bullet \rightarrow I^\bullet$.
2. If $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the left (Definition 4.2.1), then for every bounded above (cohomological) complex $M^\bullet$ of objects in $\mathcal{A}$, there exists a bounded above complex $P^\bullet$ of objects in $\mathcal{X}$ and a quasi-isomorphism (Definition 3.2.8) $P^\bullet \rightarrow M^\bullet$.

Proof. We prove that if $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the left, then there exists a complex $P^\bullet$ of objects in $\mathcal{X}$ and a quasi-isomorphism $P^\bullet \rightarrow M^\bullet$. The other statement can be proven basically symmetrically.

First suppose that $M^\bullet$ is bounded above; say that $M^i = 0$ for all $i > n$. We inductively construct $P^\bullet$ and the quasi-isomorphism to $M^\bullet$. Choose an object P^n from $\mathcal{X}$ and a surjective morphism $\epsilon_n : P^n \twoheadrightarrow M^n$, and let $d : P^n \rightarrow P^{n+1}$ be the zero map. Assume inductively that we have constructed the complex $P^\bullet$ and maps $\epsilon_i : P^i \rightarrow M^i$ for $i = k+1, k+2, \dots, n$. We want to construct P^k , the differential $d : P^k \rightarrow P^{k+1}$ and the map $\epsilon_k : P^k \rightarrow M^k$.

Let $L_k = Z^{k+1}(P) \times_{Z^{k+1}(M)} M^k$

$$\begin{array}{ccc} L_k & \longrightarrow & Z^{k+1}(P) \\ \downarrow & & \downarrow \epsilon_{k+1} \\ M^k & \xrightarrow{d} & Z^{k+1}(M) \end{array}$$

where Z^i denotes the i th cycle (Definition 3.2.1) of a complex (Definition 3.2.2); recall that abelian categories have finite limits by Lemma 4.2.32, so fiber products (Definition C.0.24) exist. Choose an object P^k from $\mathcal{X}$ and a surjective morphism $\pi : P^k \twoheadrightarrow L_k$. Set the differential $d : P^k \rightarrow P^{k+1}$ to be $\text{proj}_{Z^{k+1}(P)} \circ \pi$ and the map $\epsilon_k : P^k \rightarrow M^k$ to be $\text{proj}_{M^k} \circ \pi$.

We verify that the square

$$\begin{array}{ccc} P^k & \xrightarrow{d} & P^{k+1} \\ \downarrow \epsilon_k & & \downarrow \epsilon_{k+1} \\ M^k & \xrightarrow{d} & M^{k+1} \end{array}$$

commutes, i.e. that $\epsilon_{k+1} \circ d = d \circ \epsilon_k$. The left is $\epsilon_{k+1} \circ \text{proj}_{Z^{k+1}(P)} \circ \pi$ and the right is $d \circ \text{proj}_{M^k} \circ \pi$, which do indeed coincide.

We verify that the chain map ϵ induces isomorphisms $H^*(P) \xrightarrow{\sim} H^*(M)$ on cohomology objects. We first show that the induced maps on cohomology are epimorphisms. Let $u : Z^k(M) \rightarrow L_k = Z^{k+1}(P) \times_{Z^{k+1}(M)} M^k$ be the unique morphism corresponding to the morphisms $0 : Z^k(M) \rightarrow Z^{k+1}(P)$ and $Z^k(M) \hookrightarrow M^k$. Let Y be the pullback of π along u :

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{\pi}} & Z^k(M) \\ \downarrow \tilde{u} & & \downarrow u \\ P^k & \xrightarrow{\pi} & L_k. \end{array}$$

Note that $\text{im } \tilde{u}$ is a subobject of $Z^k(P) = \ker(d : P^k \rightarrow P^{k+1})$ because

$$d \circ \tilde{u} = \text{proj}_{Z^{k+1}(P)} \circ \pi \circ \tilde{u} = \text{proj}_{Z^{k+1}(P)} \circ u \circ \tilde{\pi} = 0.$$

Therefore, $\tilde{u}$ factors through $Z^k(P)$. Writing $[\tilde{u}]$ for the composition $Y \xrightarrow{\tilde{u}} Z^k(P) \twoheadrightarrow H^k(P)$, note that

$$H^k(\epsilon) \circ [\tilde{u}] = [\epsilon_k \circ \tilde{u}] = [\text{proj}_{M^k} \circ \pi \circ \tilde{u}] = [\text{proj}_{M^k} \circ u \circ \tilde{\pi}] = [(\text{id} : Z^k(M) \rightarrow Z^k(M)) \circ \tilde{\pi}].$$

The right most expression is the composition

$$Y \xrightarrow{\tilde{\pi}} Z^k(M) \twoheadrightarrow H^k(M).$$

Since π is an epimorphism, $\tilde{\pi}$ is an epimorphism, so the above composition is an epimorphism. We have thus shown that $H^k(\epsilon) \circ [\tilde{u}]$ is an epimorphism, so $H^k(\epsilon)$ is an epimorphism.

We now show that $H^{k+1}(\epsilon) : H^{k+1}(P) \rightarrow H^{k+1}(M)$ is a monomorphism. Let K be the kernel of $Z^{k+1}(P) \xrightarrow{\epsilon_{k+1}} Z^{k+1}(M) \twoheadrightarrow H^{k+1}(M)$; this kernel coincides with “the $(k+1)$ -cycles of P mapping to $(k+1)$ -boundaries of M ”. More precisely, K can be regarded as the fiber product

$$\begin{array}{ccc} K & \longrightarrow & Z^{k+1}(P) \\ \downarrow & & \downarrow \epsilon_{k+1} \\ B^{k+1}(M) & \hookrightarrow & Z^{k+1}(M), \end{array}$$

and note that this Cartesian diagram displays K as a subobject of $Z^{k+1}(P)$. Further note that the morphism $d : M^k \rightarrow B^{k+1}(M)$ naturally induces a morphism $L_k \rightarrow K$; in fact, K is then the image of the projection map $\text{proj}_{Z^{k+1}(P)} : L_k \rightarrow Z^{k+1}(P)$. On the other hand, by definition,

$$B^{k+1}(P) = \text{im}(d : P^k \rightarrow Z^{k+1}(P)) = \text{im}(\text{proj}_{Z^{k+1}(P)} \circ \pi).$$

Since π is an epimorphism, this image in turn equals $\text{im}(\text{proj}_{Z^{k+1}(P)})$, which equals K as we have seen. Therefore, K coincides with $B^{k+1}(P)$, which means that the map $Z^{k+1}(P) \rightarrow H^{k+1}(M)$, whose kernel is K by definition, naturally induces a monomorphism $H^{k+1}(P) \rightarrow H^{k+1}(M)$ as desired.

□

Lemma 4.2.34 (cf. [Wei94, Porism 2.2.7, Comparison Theorem 2.2.6, Comparison Theorem 2.3.7]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. Let

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

be a chain complex (Definition 3.0.1) with P_i projective (Definition 4.2.2). For every left resolution (Definition 4.2.29) $Q_\bullet \rightarrow N$ of an object N , every map $M \rightarrow N$ lifts to a complex map (Definition 3.0.1) $P_\bullet \rightarrow Q_\bullet$ unique up to chain homotopy (Definition 3.5.1).

2. Let

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \cdots$$

be a (co)chain complex (Definition 3.0.1) with I^i injective (Definition 4.2.2). For every right resolution (Definition 4.2.29) $N \rightarrow Q^\bullet$ of an object N , every map $N \rightarrow M$ lifts to a complex map (Definition 3.0.1) $Q^\bullet \rightarrow I^\bullet$ unique up to chain homotopy (Definition 3.5.1).

Proof. 1. The map $P_0 \rightarrow M \rightarrow N$ lifts to a map $P_0 \rightarrow Q_0$ because P_0 is projective and $Q_0 \rightarrow N$ is an epimorphism. Inductively suppose that there are morphisms $P_i \rightarrow Q_i$ for $0 \leq i \leq n$, where $n \geq 0$ that make

$$\begin{array}{ccccccccc} P_n & \longrightarrow & P_{n-1} & \longrightarrow & \cdots & \longrightarrow & P_0 & \longrightarrow & M & \longrightarrow & 0 \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \\ Q_n & \longrightarrow & Q_{n-1} & \longrightarrow & \cdots & \longrightarrow & Q_0 & \longrightarrow & N & \longrightarrow & 0 \end{array}$$

into a commuting diagram are established. The morphism $Q_n \rightarrow Q_{n-1}$ (where we let $Q_{-1} = N$ and $P_{-1} = M$ here in case that $n = 0$) acts as 0 when restricted to $\mathfrak{I} := \text{im}(P_{n+1} \rightarrow P_n \rightarrow Q_n)$ (Definition 2.3.10) because the composition

$$P_{n+1} \rightarrow P_n \rightarrow Q_n \rightarrow Q_{n-1}$$

equals the composition

$$P_{n+1} \rightarrow P_n \rightarrow P_{n-1} \rightarrow Q_{n-1}.$$

In other words, $\mathfrak{I}$ is a subobject of $\ker(Q_n \rightarrow Q_{n-1})$ (Definition 2.3.6), which is isomorphic to $\text{im}(Q_{n+1} \rightarrow Q_n)$ by the acyclicity of the sequence of the Q_i 's. Therefore, we have a map $P_{n+1} \rightarrow \mathfrak{I} \hookrightarrow \text{im}(Q_{n+1} \rightarrow Q_n)$ along with an epimorphism $Q_{n+1} \twoheadrightarrow \text{im}(Q_{n+1} \rightarrow Q_n)$. Since P_{n+1} is projective, the former map lifts to a map $P_{n+1} \rightarrow Q_{n+1}$ in a way that is compatible with the latter, i.e. the following commutes:

$$\begin{array}{ccc} P_{n+1} & & \\ \downarrow \text{dotted} & \searrow & \\ Q_{n+1} & \longrightarrow & \text{im}(Q_{n+1} \rightarrow Q_n). \end{array}$$

By induction, this shows that $M \rightarrow N$ lifts to a morphism $P_\bullet \rightarrow Q_\bullet$ of complexes.

We show that the morphism of complexes is unique up to chain homotopy, i.e. if $f_1, f_2 : P_\bullet \rightarrow Q_\bullet$ are two morphisms of complexes, then $h := f_1 - f_2$ is null homotopic. We construct a chain contraction (Definition 3.5.1) $\{s_n : P_n \rightarrow Q_{n+1}\}$ of h by induction on n . If $n < 0$, then set $s_n = 0$. If $n = 0$, note that the composition $P_0 \xrightarrow{h_0} Q_0 \rightarrow N$ equals the composition $P_0 \rightarrow M \xrightarrow{0} N$, so $\text{im}(h_0)$ is a subobject of $\ker(Q_0 \rightarrow N) \cong \text{im}(Q_1 \rightarrow Q_0)$. The projectivity of P_0 thus yields a lift $s_0 : P_0 \rightarrow Q_1$ such that h_0 equals the composition $P_0 \xrightarrow{s_0} Q_1 \xrightarrow{d} Q_0$:

$$\begin{array}{ccc} & P_0 & \\ & \downarrow h_0 & \\ Q_1 & \xrightarrow{d} & Q_0 \end{array} \quad \begin{array}{c} \text{dotted arrow } s_0 \\ \text{from } P_0 \text{ to } Q_1 \end{array}$$

Note moreover that $h_0 = ds_0 + s_{-1}d$ because $s_{-1} = 0$. Inductively suppose that we have maps s_i for $i \leq n$ such that $h_n = ds_n + s_{n-1}d$ or equivalently that $ds_n = h_n - s_{n-1}d$. Consider the map $h_{n+1} - s_n d : P_{n+1} \rightarrow Q_{n+1}$. Compute

$$d(h_{n+1} - s_n d) = dh_{n+1} - ds_n d = dh_{n+1} - (h_n - s_{n-1}d)d = (dh_{n+1} - h_n d) + s_{n-1}dd = 0$$

Therefore, $\text{im}(h_{n+1} - s_n d)$ is a subobject of $\ker(Q_{n+1} \rightarrow Q_n) \cong \text{im}(Q_{n+2} \rightarrow Q_{n+1})$, which is in turn a quotient of Q_{n+2} . Since P_{n+1} is projective, there is a morphism $s_{n+1} : P_{n+1} \rightarrow Q_{n+2}$ such that $ds_{n+1} = h_{n+1} - s_n d$.

$$\begin{array}{ccc} & & P_{n+1} \\ & \nearrow s_{n+1} & \downarrow h_n - s_{n-1}d = ds_n \\ Q_{n+2} & \xrightarrow{d} & \text{im}(Q_{n+2} \rightarrow Q_{n+1}) \cong \ker(Q_{n+1} \rightarrow Q_n) \end{array}$$

The s_n thus form a chain contraction as needed.

2. This is simply dual to the previous part.

□

4.3. Derived functors of right or left exact functors between abelian categories where the source category has enough projectives or injectives.

Definition 4.3.1. (♠ TODO: I think that the definition of derived categories might be doable for more general kinds of resolutions? Perhaps it is that if I have a right exact functor F , then $L^i F$ can be computed with resolutions of F -acyclic objects? (Definition 4.4.1)) (♠ TODO: Apparently, left/right derived functors may be defined for functors that are additive and preserve finite coproducts, and not necessarily right/left exact; the exactness condition ensures that the zeroth derived functor agrees with F .) Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories (Definition 2.3.12), and let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 2.4.1).

1. Suppose that the functor F is right exact (Definition 2.4.12) and suppose that $A \in \mathcal{A}$ is an object for which a projective resolution (Definition 4.2.29)

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$$

exists in $\mathcal{A}$. We define the **left derived object** $L_n F A \in \mathcal{B}$ by applying F to obtain a complex

$$\cdots \rightarrow F(P_2) \rightarrow F(P_1) \rightarrow F(P_0) \rightarrow 0$$

and letting $L_n F(A)$ be the n -th homology object (Definition 3.2.2) of this complex in $\mathcal{B}$:

$$L_n F(A) := H_n(F(P_\bullet)).$$

The object $L_n F(A)$ is independent of the choice of projective resolution up to natural isomorphism (Proposition 4.3.3).

By convention, set $L_n F = 0$ for $n < 0$.

The **higher left derived objects** refer to the object $L_n F(A)$ for $n > 0$.

2. Suppose that the functor F is right exact (Definition 2.4.12) and that $\mathcal{A}$ has enough projectives (Definition 4.2.5). The **left derived functors** refer to the family of functors

$$L_n F : \mathcal{A} \rightarrow \mathcal{B}, \quad A \mapsto L_n F(A).$$

The **higher left derived functors** refer to the functors $L_n F$ for $n > 0$.

3. Suppose that the functor F is right exact (Definition 2.4.12) and suppose that $A \in \mathcal{A}$ is an object for which a injective resolution (Definition 4.2.29)

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

exists in $\mathcal{A}$. We define the *right derived object* $R_n F A \in \mathcal{B}$, also often denoted by $R^n F A$, by applying F to obtain a complex

$$0 \rightarrow F(I^0) \rightarrow F(I^1) \rightarrow F(I^2) \rightarrow \dots$$

and letting $R_n F(A)$ be the n -th cohomology object (Definition 3.2.2) of this complex in $\mathcal{B}$:

$$R_n F(A) := H^n(F(I_\bullet)).$$

The object $R_n F(A)$ is independent of the choice of injective resolution up to natural isomorphism (Proposition 4.3.3).

By convention, set $R_n F = 0$ for $n < 0$.

The *higher right derived objects* refer to the object $R_n F(A)$ for $n > 0$.

4. Suppose that the functor F is right exact (Definition 2.4.12) and that $\mathcal{A}$ has enough injectives (Definition 4.2.5). The *right derived functors* refer to the family of functors

$$R_n F : \mathcal{A} \rightarrow \mathcal{B}, \quad A \mapsto R_n F(A).$$

The right derived functors are also often denoted by $R^n F$. The *higher right derived functors* refer to the functors $R_n F$ for $n > 0$.

Lemma 4.3.2 (Horseshoe lemma, cf. [Wei94, Horseshoe Lemma 2.2.8]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. Suppose that

$$0 \rightarrow A' \xrightarrow{i_A} A \xrightarrow{\pi_A} A'' \rightarrow 0$$

is a short exact sequence in $\mathcal{A}$, and that $\varepsilon' : P'_\bullet \rightarrow A'$ and $\varepsilon'' : P''_\bullet \rightarrow A''$ are respectively projective resolutions (Definition 4.2.29).

$$\begin{array}{ccccccc} & & & & 0 & & \\ & & & & \downarrow & & \\ \cdots P'_2 & \longrightarrow & P'_1 & \longrightarrow & P'_0 & \xrightarrow{\varepsilon'} & A' \longrightarrow 0 \\ & & & & \downarrow i_A & & \\ & & & & A & & \\ & & & & \downarrow \pi_A & & \\ \cdots P''_2 & \longrightarrow & P''_1 & \longrightarrow & P''_0 & \xrightarrow{\varepsilon''} & A'' \longrightarrow 0 \\ & & & & \downarrow & & \\ & & & & 0 & & \end{array}$$

Let $P_\bullet = P'_\bullet \oplus P''_\bullet$. The complex $P_\bullet$ is a projective resolution of A , and the short exact sequence lifts to an exact esquence of complexes

$$0 \rightarrow P' \xrightarrow{i} P \xrightarrow{\pi} P'' \rightarrow 0$$

where $i_n : P'_n \rightarrow P_n$ and $\pi_n : P_n \rightarrow P''_n$ are the natural inclusion and projection respectively.

2. Suppose that

$$0 \rightarrow A' \xrightarrow{i_A} A \xrightarrow{\pi_A} A'' \rightarrow 0$$

is a short exact sequence in $\mathcal{A}$, and that $\eta' : A' \rightarrow I'^\bullet$ and $\eta'' : A'' \rightarrow I''^\bullet$ are respectively injective resolutions (Definition 4.2.29).

$$\begin{array}{ccccccc} & & 0 & & & & \\ & & \downarrow & & & & \\ 0 & \longrightarrow & A' & \xrightarrow{\eta'} & I'^0 & \longrightarrow & I'^1 \longrightarrow I'^2 \dots \\ & & \downarrow i_A & & & & \\ & & A & & & & \\ & & \downarrow \pi_A & & & & \\ 0 & \longrightarrow & A' & \xrightarrow{\eta''} & I''^0 & \longrightarrow & I''^1 \longrightarrow I''^2 \dots \\ & & \downarrow & & & & \\ & & 0 & & & & \end{array}$$

Let $I^\bullet = I'^\bullet \oplus I''^\bullet$. The complex $I^\bullet$ is an injective resolution of A , and the short exact sequence lifts to a short exact sequence of complexes

$$0 \rightarrow I'^\bullet \xrightarrow{i} I^\bullet \xrightarrow{\pi} I''^\bullet \rightarrow 0,$$

where $i^n : I'^n \rightarrow I^n$ and $\pi^n : I^n \rightarrow I''^n$ are the natural inclusion and projection at each degree n .

Proof. (♠ TODO:)

□

Proposition 4.3.3 (cf.[Wei94, Lemma 2.4.1]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12). Let A be an object of $\mathcal{A}$.

1. Suppose that F is right exact (Definition 2.4.12), and suppose that a projective resolution (Definition 4.2.29)

$$\dots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$$

of A exists in $\mathcal{A}$. Let

$$\dots \rightarrow Q_2 \rightarrow Q_1 \rightarrow Q_0 \rightarrow A \rightarrow 0$$

be any projective resolution of A in $\mathcal{A}$. For all n , there are natural isomorphisms

$$H_n(F(P_\bullet)) \cong H_n(F(Q_\bullet)).$$

In other words, the left derived objects $L_n F(A)$ (Definition 4.3.1) is well defined.

2. Suppose that F is left exact (Definition 2.4.12), and suppose that a injective resolution (Definition 4.2.29)

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

of A exists in $\mathcal{A}$. Let

$$0 \rightarrow A \rightarrow Q^0 \rightarrow Q^1 \rightarrow Q^2 \rightarrow \dots$$

be any injective resolution of A in $\mathcal{A}$. For all n , there are natural isomorphisms

$$H_n(F(I^\bullet)) \cong H_n(F(Q^\bullet)).$$

In other words, the right derived objects $R_n F(A)$ (Definition 4.3.1) is well defined.

Proof. 1. By Lemma 4.2.34, there is a lift $f : P_\bullet \rightarrow Q_\bullet$ of the identity map $A \rightarrow A$ unique up to chain homotopy. There are then induced natural maps $H_n(F(f)) : H_n(F(P_\bullet)) \rightarrow H_n(F(Q_\bullet))$ (Proposition 3.2.7). There is also a lift $f' : Q_\bullet \rightarrow P_\bullet$ of the identity map $A \rightarrow A$ unique up to chain homotopy, and this also induces natural maps $H_n(F(f')) : H_n(F(Q_\bullet)) \rightarrow H_n(F(P_\bullet))$. The chain maps f and f' are in fact chain homotopy inverses (Definition 3.5.1) because Lemma 4.2.34 also implies that any lifts $P_\bullet \rightarrow P_\bullet$ and $Q_\bullet \rightarrow Q_\bullet$ of the identity map $A \rightarrow A$ are chain homotopic to the identity chain maps. Therefore, $H_n(F(f))$ and $H_n(F(f'))$ are inverses of each other as morphisms in $\mathcal{B}$ by Lemma 3.5.5.

2. This is dual to the previous part. □

Theorem 4.3.4 (cf. [Wei94, Theorem 2.4.6, Theorem 2.4.7, 2.5.1]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

1. Suppose that F is right exact (Definition 2.4.12).

(a) Let

$$0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$$

be a short exact sequence (Definition 2.4.4) in $\mathcal{A}$. Suppose that there exist projective resolutions (Definition 4.2.29) $P' \rightarrow A'$ and $P'' \rightarrow A''$. There exists a long exact sequence

$$\dots \xrightarrow{\partial} L_i F(A') \rightarrow L_i F(A) \rightarrow L_i F(A'') \xrightarrow{\partial} L_{i-1} F(A') \rightarrow L_{i-1} F(A) \rightarrow L_{i-1} F(A'') \xrightarrow{\partial} \dots$$

of derived objects (Definition 4.3.1) and this long exact sequence is natural.

(b) If $\mathcal{A}$ has enough projectives (Definition 4.2.5), then the derived functor $L_* F$ form a universal homological δ -functor (Definition 4.1.2).

2. Suppose that G is left exact (Definition 2.4.12).

(a) Let

$$0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$$

be a short exact sequence (Definition 2.4.4) in $\mathcal{A}$. Suppose that there exist injective resolutions (Definition 4.2.29) $I' \rightarrow A'$, $I \rightarrow A$, and $I'' \rightarrow A''$. There exists a long exact sequence

$$\dots \xrightarrow{\partial} R^i G(A') \rightarrow R^i G(A) \rightarrow R^i G(A'') \xrightarrow{\partial} R^{i+1} G(A') \rightarrow R^{i+1} G(A) \rightarrow R^{i+1} G(A'') \xrightarrow{\partial} \dots$$

of derived objects (Definition 4.3.1), and this long exact sequence is natural.

(b) If $\mathcal{A}$ has enough injectives (Definition 4.2.5), then the derived functors $R^* G$ form a universal cohomological δ -functor (Definition 4.1.2).

Proof. 1. (a) By the Horseshoe lemma (Lemma 4.3.2), there is a projective resolution $P \rightarrow A$ fitting into a short exact sequence $0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0$. Since the P_n'' are projective, each sequence $0 \rightarrow P_n' \rightarrow P_n \rightarrow P_n'' \rightarrow 0$ is split exact (Definition 2.4.8) by Proposition 4.2.4. Since F is additive and since $P_n \cong P_n' \oplus P_n''$, we have isomorphisms $F(P_n) \cong F(P_n') \oplus F(P_n'')$ making each sequence

$$0 \rightarrow F(P_n') \rightarrow F(P_n) \rightarrow F(P_n'') \rightarrow 0$$

is split exact in $\mathcal{B}$. Therefore,

$$0 \rightarrow F(P') \rightarrow F(P) \rightarrow F(P'') \rightarrow 0$$

is a short exact sequence of chain complexes in $\mathcal{B}$. Its associated long exact sequence in homology (Theorem 4.3.4) is the desired long exact sequence.

We now show that the long exact sequence is natural. In other words, given a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' \longrightarrow 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' \\ 0 & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & B'' \longrightarrow 0 \end{array}$$

in $\mathcal{A}$ such that A' , A'' , B' , and B'' have projective resolutions, we want to show that the morphisms

$$L_i F(f') : L_i F(A') \rightarrow L_i F(B'), \quad L_i F(f) : L_i F(A) \rightarrow L_i F(B), \quad L_i F(f'') : L_i F(A'') \rightarrow L_i F(B'')$$

arising from the functoriality of the $L_i F$ yield a morphism

$$\begin{array}{ccccccc} \cdots & \longrightarrow & L_i F(A') & \longrightarrow & L_i F(A) & \longrightarrow & L_i F(A'') \xrightarrow{\partial} L_{i-1} F(A') \longrightarrow \cdots \\ & & \downarrow L_i F(f') & & \downarrow L_i F(f) & & \downarrow L_i F(f'') \\ \cdots & \longrightarrow & L_i F(B') & \longrightarrow & L_i F(B) & \longrightarrow & L_i F(B'') \xrightarrow{\partial} L_{i-1} F(B') \longrightarrow \cdots \end{array}$$

of the long exact sequences as chain complexes (Definition 3.0.1)

Say that we given projective resolutions (Definition 4.2.29) $\epsilon' : P' \rightarrow A'$, $\epsilon'' : P'' \rightarrow A''$, $\eta' : Q' \rightarrow B'$, and $\eta'' : Q'' \rightarrow B''$. Obtain projective resolutions $\epsilon : P \rightarrow A$ and $\eta : Q \rightarrow B$ via the Horseshoe Lemma (Lemma 4.3.2). Obtain chain maps $F' : P' \rightarrow Q'$ and $F'' : P'' \rightarrow Q''$ lifting f' and f'' respectively via Lemma 4.2.34.

We show that there is also a chain map $F : P \rightarrow Q$ lifting f that also yields the following commutative diagram of maps in $\mathbf{Ch}(\mathcal{A})$ with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & P' & \longrightarrow & P & \longrightarrow & P'' \longrightarrow 0 \\ & & \downarrow F' & & \downarrow F & & \downarrow F'' \\ 0 & \longrightarrow & Q' & \longrightarrow & Q & \longrightarrow & Q'' \longrightarrow 0 \end{array}$$

The Horseshoe Lemma construction provides the following split-exact sequences in each degree n :

$$0 \longrightarrow P'_n \xrightarrow{i_n} P_n \xrightarrow{p_n} P''_n \longrightarrow 0$$

$\xleftarrow{s_n} \quad \quad \quad \xleftarrow{t_n}$

$$0 \longrightarrow Q'_n \xrightarrow{j_n} Q_n \xrightarrow{q_n} Q''_n \longrightarrow 0$$

$\xleftarrow{u_n} \quad \quad \quad \xleftarrow{v_n}$

where $P_n \cong P'_n \oplus P''_n$ and $Q_n \cong Q'_n \oplus Q''_n$.

We define the chain map $F_n : P_n \rightarrow Q_n$ in terms of the isomorphisms $P_n \cong P'_n \oplus P''_n$ and $Q_n \cong Q'_n \oplus Q''_n$ as the block matrix:

$$F_n = \begin{pmatrix} F'_n & \gamma_n \\ 0 & F''_n \end{pmatrix} : P'_n \oplus P''_n \rightarrow Q'_n \oplus Q''_n$$

where $\gamma_n : P''_n \rightarrow Q'_n$ is a morphism chosen such that F commutes with the differentials d^P and d^Q , i.e. $d^Q \circ F = F \circ d^P$. In other words, the following diagrams need to commute:

$$\begin{array}{ccc} P'_n \oplus P''_n & \xrightarrow{\begin{pmatrix} F'_n & \gamma_n \\ 0 & F''_n \end{pmatrix}} & Q'_n \oplus Q''_n \\ \left(\begin{array}{cc} d_n^{P'} & \delta'_n \\ 0 & d_n^{P''} \end{array} \right) \downarrow & & \downarrow \left(\begin{array}{cc} d_n^{Q'} & \delta_n \\ 0 & d_n^{Q''} \end{array} \right) \\ P'_{n-1} \oplus P''_{n-1} & \xrightarrow{\begin{pmatrix} F'_{n-1} & \gamma_{n-1} \\ 0 & F''_{n-1} \end{pmatrix}} & Q'_{n-1} \oplus Q''_{n-1} \end{array}$$

Expanding the matrix multiplication $d_n^Q F_n = F_{n-1} d_n^P$, the diagonal entries commute by the definition of F' and F'' . The upper-right entry yields the condition:

$$(P) \quad d_n^{Q'} \gamma_n + \delta_n F''_n = F'_{n-1} \delta'_n + \gamma_{n-1} d_n^{P''}$$

We construct γ_n by induction on n :

- **Base Case** ($n = 0$): Since $P_{-1} = 0$, the condition simplifies to the lifting property. Using the projectivity of P''_0 , we lift the map through the resolution to obtain γ_0 .
- **Inductive Step**: Suppose γ_{n-1} is defined. Let $\Psi_n = F'_{n-1} \delta'_n + \gamma_{n-1} d_n^{P''} - \delta_n F''_n$. We want to find γ_n such that $d_n^{Q'} \gamma_n = \Psi_n$.
- **Verification**: Applying $d_{n-1}^{Q'}$ to Ψ_n yields zero:

$$d_{n-1}^{Q'} \Psi_n = d_{n-1}^{Q'} F'_{n-1} \delta'_n + d_{n-1}^{Q'} \gamma_{n-1} d_n^{P''} - d_{n-1}^{Q'} \delta_n F''_n$$

By substituting the inductive hypothesis for $d_{n-1}^{Q'} \gamma_{n-1}$ and using the fact that differentials squares are zero, the terms cancel out. Since P''_n is projective, there exists a lift $\gamma_n : P''_n \rightarrow Q'_n$ satisfying (P).

Applying F to the resulting morphism of SESs of complexes and invoking the naturality of the homology long exact sequence completes the proof.

- (b) That $L_* F$ form a homological δ -functor is essentially the content of the above. We refer to [Wei94, Theorem 2.4.7] for the proof that the homological δ -functor is in fact universal.

2. This is dual to the previous part.

□

4.4. **F -acyclic objects.** More generally, resolutions by F -acyclic objects, instead of projectives or injectives, may be used to compute derived functors.

Definition 4.4.1 (F -acyclic object for a functor of categories). Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories (Definition 2.3.12).

1. Let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be a right exact functor (Definition 2.4.12).

An object $A \in \mathcal{A}$ for which a projective resolution (Definition 4.2.29) exists is called **F -acyclic** if for all integers $n > 0$, its higher left derived functors (Definition 4.3.1) vanish:

$$L_n F(A) = 0 \text{ for all } n > 0$$

2. Let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be a left exact functor (Definition 2.4.12).

An object $A \in \mathcal{A}$ for which an injective resolution (Definition 4.2.29) exists is called **F -acyclic** if for all integers $n > 0$, its higher right derived functors (Definition 4.3.1) vanish:

$$R_n F(A) = 0 \text{ for all } n > 0$$

Definition 4.4.2 (F -acyclic resolution for a right/left exact functor). Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories (Definition 2.3.12) and let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 2.4.1).

1. Suppose that F is a right exact functor (Definition 2.4.12). **An (left) F -acyclic resolution** of an object $A \in \mathcal{A}$ is a left resolution of A for (Definition 4.2.29) the class of F -acyclic objects (Definition 4.4.1).
2. Suppose that F is a left exact functor (Definition 2.4.12). **An (right) F -acyclic resolution** of an object $A \in \mathcal{A}$ is a right resolution of A for (Definition 4.2.29) the class of F -acyclic objects (Definition 4.4.1).

Lemma 4.4.3. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

1. If F is right exact (Definition 2.4.12), then any projective object (Definition 4.2.2) $\mathcal{P}$ is F -acyclic (Definition 4.4.1).
2. If F is left exact (Definition 2.4.12), then any injective object (Definition 4.2.2) $\mathcal{I}$ is F -acyclic (Definition 4.4.1).

Proof. Note that any projective/injective object has a projective/injective resolution (Definition 4.2.29) (Lemma 4.2.30), so we may talk about the derived objects $L_n F(\mathcal{P})$ and $R^n F(\mathcal{I})$ (Definition 4.3.1) of projective objects $\mathcal{P}$ and injective objects $\mathcal{I}$ of $\mathcal{A}$. In fact, the projective/injective resolution is simply given by

$$\cdots \rightarrow 0 \rightarrow \mathcal{P} \xrightarrow{\text{id}} \mathcal{P} \rightarrow 0$$

and

$$0 \rightarrow \mathcal{I} \xrightarrow{\text{id}} \mathcal{I} \rightarrow 0 \rightarrow \cdots$$

so we compute $L_n F(\mathcal{P}) = 0$ and $R^n F(\mathcal{I}) = 0$. $\square$

Lemma 4.4.4 (Dimension shifting, cf. [Wei94, Exercise 2.4.3]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12). Let A be an object of $\mathcal{A}$.

1. Suppose that F is right exact (Definition 2.4.12).
 - (a) Suppose that $0 \rightarrow M \rightarrow C \rightarrow A \rightarrow 0$ is an exact sequence in $\mathcal{A}$ where C is F -acyclic, and that A and M have projective resolutions (Definition 4.2.29). We have $L_i F(A) \cong L_{i-1} F(M)$ for $i \geq 2$ and $L_1 F(A) \cong \ker(F(M) \rightarrow F(C))$.
 - (b) Suppose that $\mathcal{A}$ has enough projectives (Definition 4.2.5). Let

$$0 \rightarrow M_m \rightarrow C_m \rightarrow C_{m-1} \rightarrow \cdots \rightarrow C_0 \rightarrow A \rightarrow 0$$

be an acyclic complex (Definition 3.0.2) with the C_i F -acyclic (Definition 4.4.1). We have

- (i) $L_i F(A) \cong L_{i-m-1} F(M_m)$ for $i \geq m+2$ and
- (ii) $L_{m+1} F(A) \cong \ker(F(M_m) \rightarrow F(C_m))$.

2. Suppose that $G : \mathcal{A} \rightarrow \mathcal{B}$ is left exact (Definition 2.4.12).
 - (a) Suppose that $0 \rightarrow A \rightarrow I \rightarrow N \rightarrow 0$ is a short exact sequence in $\mathcal{A}$ where I is G -acyclic, and that A and N have injective resolutions (Definition 4.2.29). We have $R^i G(A) \cong R^{i-1} G(N)$ for $i \geq 2$ and $R^1 G(A) \cong \text{coker}(G(I) \rightarrow G(N))$.
 - (b) Suppose that $\mathcal{A}$ has enough injectives (Definition 4.2.5). Let

$$0 \rightarrow A \rightarrow I_0 \rightarrow I_1 \rightarrow \cdots \rightarrow I_m \rightarrow N_m \rightarrow 0$$

be an acyclic complex (Definition 3.0.2) with the I_j G -acyclic (Definition 4.4.1). We have

- (i) $R^i G(A) \cong R^{i-m-1} G(N_m)$ for $i \geq m+2$ and
- (ii) $R^{m+1} G(A) \cong \text{coker}(G(I_m) \rightarrow G(N_m))$.

Proof. 1. Suppose that F is right exact

- (a) Since C is F -acyclic, the long exact sequence of derived functors associated to (Theorem 4.3.4) $0 \rightarrow M \rightarrow C \rightarrow A \rightarrow 0$ yields an exact sequence

$$0 \rightarrow L_1 F(A) \rightarrow F(M) \rightarrow F(C) \rightarrow F(A) \rightarrow 0$$

along with isomorphisms $L_i F(A) \cong L_{i-1} F(M)$ for $i \geq 2$. In particular, $L_1 F(A) = \ker(F(M) \rightarrow F(C))$.

- (b) We now proceed by induction. The above proves the base case of $m = 0$. Associated to the acyclic complex

$$0 \rightarrow M_m \rightarrow C_m \rightarrow C_{m-1} \rightarrow \cdots \rightarrow C_0 \rightarrow A \rightarrow 0$$

is a short exact sequence

$$0 \rightarrow M_m \rightarrow C_m \rightarrow M_{m-1} \rightarrow 0$$

where $M_{m-1} = \text{coker}(M_m \rightarrow C_m)$ and an acyclic complex

$$0 \rightarrow M_{m-1} \rightarrow C_{m-1} \rightarrow \cdots \rightarrow C_0 \rightarrow A \rightarrow 0.$$

The derived functor long exact sequence of the short exact sequence yields isomorphisms

$$L_i F(M_{m-1}) \cong L_{i-1} F(M_m)$$

for all $i \geq 1$. By induction, we also have $L_i F(A) \cong L_{i-(m-1)-1} F(M_{m-1}) = L_{i-m} F(M_{m-1})$ for $i \geq m+1$, so we in fact have

$$L_i F(A) \cong L_{i-m} F(M_{m-1}) \cong L_{i-m-1} F(M_m)$$

for all $i \geq m+1$. Moreover, the derived functor long exact sequence also includes

$$0 \rightarrow L_1 F(M_{m-1}) \rightarrow F(M_m) \rightarrow F(C_m) \rightarrow F(M_{m-1}) \rightarrow 0,$$

so $L_1 F(M_{m-1}) \cong \ker(F(M_m) \rightarrow F(C_m))$. We found that $L_1 F(M_{m-1}) \cong L_{m+1} F(A)$, so $L_{m+1} F(A)$ is the kernel of $F(M_m) \rightarrow F(C_m)$ as desired.

2. These hold dually.

□

Proposition 4.4.5 (cf.[Wei94, Exercise 2.4.3]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12). Let A be an object of $\mathcal{A}$.

1. Suppose that F is right exact (Definition 2.4.12) and suppose that $\mathcal{A}$ has enough projectives (Definition 4.2.5). Let

$$\cdots \rightarrow Q_2 \rightarrow Q_1 \rightarrow Q_0 \rightarrow A \rightarrow 0$$

be any F -acyclic (left) resolution (Definition 4.4.2) of A in $\mathcal{A}$. For all n , there are natural isomorphisms

$$L_n F(A) \cong H_n(F(Q_\bullet)).$$

In particular, the left derived objects $L_n F(A)$ may be computed using any F -acyclic (left) resolution of A and are independent of the choice of F -acyclic resolution. In particular, since all projective objects are F -acyclic (Lemma 4.4.3), $L_n F(A)$ is well defined.

2. Dually, suppose that F is left exact (Definition 2.4.12), and suppose that $\mathcal{A}$ has enough injectives (Definition 4.2.5).

Let

$$0 \rightarrow A \rightarrow Q^0 \rightarrow Q^1 \rightarrow Q^2 \rightarrow \cdots$$

be any F -acyclic (right) resolution (Definition 4.4.2) of A in $\mathcal{A}$. For all n , there are natural isomorphisms

$$(R^n F)(A) \cong H^n(F(Q^\bullet)).$$

In particular, the right derived objects $R_n F(A)$ may be computed using any F -acyclic (right) resolution of A and are independent of the choice of F -acyclic. In particular, since all injectives objects are F -acyclic (Lemma 4.4.3), $R_n F(A)$ is well defined.

Proof. (♠ TODO: There is a “dimension shifting” fact that goes into proving this kind of thing.) □

4.5. Tor and Ext. Recall by Proposition 2.5.4 that left/right adjoint functors between abelian categories are right/left exact. Also recall the tensor-hom adjunction (Theorem 2.5.2) of modules. We can thus define derived functors as follows:

Definition 4.5.1. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7), let M be an R - S bimodule (Definition 2.1.1), and let N be an S - T bimodule. Let $n \geq 0$ be an integer. The *Tor of M and N* , denoted by $\text{Tor}_n(M, N)$ or $\text{Tor}_n^S(M, N)$ (or other notations such as $\text{Tor}_n^S({}_R M_S, {}_S N_T)$ or ${}_R(\text{Tor}_n^S(M, N))_T$ to further emphasize the module structures), is defined in the following ways:

1. as the n th left derived functor (Definition 4.3.1) $L_n(M \otimes_S -)(N)$ of the right exact functor (Definition 2.4.12) $M \otimes_S - : {}_S \mathbf{Mod}_T \rightarrow {}_R \mathbf{Mod}_T$ (Definition 2.1.3) (Definition 2.1.9) (Theorem 2.5.2) (Proposition 2.5.4)
2. as the n th left derived functor (Definition 4.3.1) $L_n(- \otimes_S N)(M)$ of the right exact functor (Definition 2.4.12) $- \otimes_S N : {}_R \mathbf{Mod}_S \rightarrow {}_R \mathbf{Mod}_T$ (Definition 2.1.3) (Definition 2.1.9) (Theorem 2.5.2) (Proposition 2.5.4).

The two definitions turn out to coincide up to natural isomorphism; moreover, the Tor objects may be obtained by using flat resolutions instead of projective resolutions (Corollary 4.7.10).

Definition 4.5.2. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7), let M be an R - S bimodule (Definition 2.1.1), and let N be an R - T bimodule. Let $n \geq 0$ be an integer. The *Ext of M and N* , denoted by $\text{Ext}^n(M, N)$ or $\text{Ext}_R^n(M, N)$ (or other notations such as $\text{Ext}_{R-\mathbb{Z}}^n(M, N)$, $\text{Ext}_R^n({}_R M_S, {}_R N_T)$ or ${}_S(\text{Ext}_R^n(M, N))_T$ to further emphasize the module structures), is defined in the following ways:

1. as the n th right derived functor (Definition 4.3.1) $R^n \text{Hom}_R(M, -)(N)$ of the left exact functor (Definition 2.4.12) $\text{Hom}_R(M, -) = \text{Hom}_{R-\mathbb{Z}}(M, -) : {}_R \mathbf{Mod}_T \rightarrow {}_S \mathbf{Mod}_T$ (Definition 2.1.3) (Definition 2.1.8) (Theorem 2.5.2) (Proposition 2.5.4)
2. as the n th right derived functor (Definition 4.3.1) $R^n \text{Hom}_R(-, N)(M)$ of the left exact functor (Definition 2.4.12) $\text{Hom}_R(-, N) = \text{Hom}_{R-\mathbb{Z}}(-, N) : ({}_R \mathbf{Mod}_S)^{\text{op}} \rightarrow {}_S \mathbf{Mod}_T$ (Definition 2.1.3) (Definition 1.2.1) (Definition 2.1.8) (Theorem 2.5.2) (Proposition 2.5.4).

The two definitions turn out to coincide up to natural isomorphism, see Corollary 4.7.6.

Note that the computation of $R^n \text{Hom}_R(M, -)(N)$ uses injective resolutions (Definition 4.2.29) of N in the category ${}_R \mathbf{Mod}_T \rightarrow {}_S$ and the computation of $R^n \text{Hom}_R(-, N)(M)$ uses injective resolutions of N in the category $({}_R \mathbf{Mod}_S)^{\text{op}}$, i.e. projective resolutions (Definition 4.2.29) of N in the category ${}_R \mathbf{Mod}_S$.

Dually, one can define the notion of $\text{Ext}(M, N)$ when M and N are right R -modules instead as derived functors of $\text{Hom}_R = \text{Hom}_{\mathbb{Z}-R}$.

The above definitions are essentially specializations of those of Definition 4.7.2.

4.6. **Balancing Ext and Tor.** The goal is to show that the two definitions of Ext (Definition 4.5.2) agree with each other (Corollary 4.7.6) and similarly for Tor (Definition 4.5.1) (Corollary 4.7.10). Double complexes (Definition 4.6.1) are useful to do so.

4.6.1. Double complexes.

Definition 4.6.1. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. A **(cohomological) double complex** (also called a **bicomplex**) in $\mathcal{A}$ is a collection of objects

$$\{A^{p,q}\}_{(p,q) \in \mathbb{Z}^2}$$

together with morphisms

$$d'_A{}^{p,q} : A^{p,q} \rightarrow A^{p+1,q}, \quad d''_A{}^{p,q} : A^{p,q} \rightarrow A^{p,q+1},$$

satisfying the identities

$$(d'_A)^{p+1,q} \circ (d'_A)^{p,q} = 0, \quad (d''_A)^{p,q+1} \circ (d''_A)^{p,q} = 0,$$

and the **anti-commutativity relation**

$$(d''_A)^{p+1,q} \circ (d'_A)^{p,q} + (d'_A)^{p,q+1} \circ (d''_A)^{p,q} = 0.$$

$$\begin{array}{ccccccc} & & \vdots & & \vdots & & \\ & & \uparrow d'' & & \uparrow d'' & & \\ \dots & \longrightarrow & A^{p,q+1} & \xrightarrow{d'} & A^{p+1,q+1} & \longrightarrow & \dots \\ & & \uparrow d'' & & \uparrow d'' & & \\ \dots & \longrightarrow & A^{p,q} & \xrightarrow{d'} & A^{p+1,q} & \longrightarrow & \dots \\ & & \uparrow & & \uparrow & & \\ & & \vdots & & \vdots & & \end{array}$$

An alternative (equivalent) convention defines a double complex $(A^{p,q}, d'_A, d''_A)$ so that

$$(d''_A)^{p+1,q} \circ (d'_A)^{p,q} - (d'_A)^{p,q+1} \circ (d''_A)^{p,q} = 0.$$

This convention differs by a sign and corresponds to replacing d''_A by $-d''_A$. Thus, both conventions are equivalent up to the natural isomorphism $A^{p,q} \mapsto A^{p,q}$, $d'_A \mapsto d'_A$, $d''_A \mapsto -d''_A$.

2. A **(homological) double complex** (also called a **bicomplex**) in $\mathcal{A}$ is a collection of objects $\{A_{p,q}\}_{(p,q) \in \mathbb{Z}^2}$ together with morphisms

$$d'_{Ap,q} : A_{p,q} \rightarrow A_{p-1,q}, \quad d''_{Ap,q} : A_{p,q} \rightarrow A_{p,q-1},$$

satisfying the identities

$$(d'_{Ap-1,q} \circ (d'_{Ap,q}) = 0, \quad (d''_{Ap,q-1} \circ (d''_{Ap,q}) = 0,$$

and the **anti-commutativity relation**

$$(d''_{Ap-1,q} \circ (d'_{Ap,q}) + (d'_{Ap,q-1} \circ (d''_{Ap,q}) = 0.$$

$$\begin{array}{ccccc}
& \vdots & & \vdots & \\
& \downarrow & & \downarrow & \\
\cdots & \longleftarrow & A_{p,q} & \xleftarrow{d'} & A_{p+1,q} & \longrightarrow \cdots \\
& & \downarrow d'' & & \downarrow d'' & \\
\cdots & \longleftarrow & A_{p,q-1} & \xleftarrow{d'} & A_{p+1,q-1} & \longrightarrow \cdots \\
& & \downarrow & & \downarrow & \\
& \vdots & & \vdots &
\end{array}$$

An alternative (equivalent) convention defines a homological double complex $(A_{p,q}, d'_A, d''_A)$ such that the horizontal and vertical maps satisfy the *commutativity relation*

$$(d''_A)_{p-1,q} \circ (d'_A)_{p,q} = (d'_A)_{p,q-1} \circ (d''_A)_{p,q}.$$

The two conventions are equivalent under the assignment $(d'_A, d''_A) \mapsto (d'_A, (-1)^p d''_A)$ or simply by replacing d''_A with $-d''_A$.

The choice to speak of cohomological double complexes or homological ones is merely a choice of convention; the theories are equivalent. Any homological double complex $A_{\bullet,\bullet}$ can be viewed as a cohomological double complex $A^{\bullet,\bullet}$ by setting $A^{p,q} = A_{-p,-q}$. Under this re-indexing, the decreasing differentials $d_{p,q} : A_{p,q} \rightarrow A_{p-1,q}$ correspond to increasing differentials $d^{-p,-q} : A^{-p,-q} \rightarrow A^{-p+1,-q}$.

In practice, the homological notation is standard in the study of derived functors of right-exact functors (such as the Tor functor), while the cohomological notation is preferred for left-exact functors (such as the Ext functor and sheaf cohomology). This symmetry ensures that any result proved for one type of double complex—such as the existence of a spectral sequence or the properties of the total complex—has a mirror image in the other.

Remark 4.6.2. The homological/cohomological conventions for the theory of double complexes may not be kept consistent in this document.

Definition 4.6.3. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. Let

$$(A^{p,q}, d'_A, d''_A)_{(p,q) \in \mathbb{Z}^2}, \quad (B^{p,q}, d'_B, d''_B)_{(p,q) \in \mathbb{Z}^2}$$

be cohomological double complexes (Definition 4.6.1) in $\mathcal{A}$.

A *morphism of double complexes* $f : A \rightarrow B$ is a collection of morphisms

$$f^{p,q} : A^{p,q} \rightarrow B^{p,q},$$

for all $(p,q) \in \mathbb{Z}^2$, such that the following diagrams commute:

$$\begin{aligned}
d'_B{}^{p,q} \circ f^{p,q} &= f^{p+1,q} \circ d'_A{}^{p,q}, \\
d''_B{}^{p,q} \circ f^{p,q} &= f^{p,q+1} \circ d''_A{}^{p,q}.
\end{aligned}$$

2. Let

$$(A_{p,q}, d'_A, d''_A)_{(p,q) \in \mathbb{Z}^2}, \quad (B_{p,q}, d'_B, d''_B)_{(p,q) \in \mathbb{Z}^2}$$

be homological double complexes (Definition 4.6.1) in $\mathcal{A}$.

A **morphism of double complexes** $f : A \rightarrow B$ is a collection of morphisms

$$f_{p,q} : A_{p,q} \rightarrow B_{p,q},$$

for all $(p, q) \in \mathbb{Z}^2$, such that the following diagrams commute:

$$\begin{aligned} d'_{Bp,q} \circ f_{p,q} &= f_{p-1,q} \circ d'_{Ap,q}, \\ d''_{Bp,q} \circ f_{p,q} &= f_{p,q-1} \circ d''_{Ap,q}. \end{aligned}$$

In other words, f respects both horizontal and vertical differentials of the double complexes.

The double complexes and their morphisms form a category, sometimes denoted by **DC**($\mathcal{A}$).

Definition 4.6.4. Let $(A^{p,q}, d'_A, d''_A)$ be a double complex in an additive category $\mathcal{A}$.

- The double complex is called **bounded above** if there exist integers p_0, q_0 such that $A^{p,q} = 0$ whenever $p > p_0$ or $q > q_0$.
- The double complex is called **bounded below** if there exist integers p_0, q_0 such that $A^{p,q} = 0$ whenever $p < p_0$ or $q < q_0$.
- The double complex is called **bounded** if it is both bounded above and below.
- The double complex is said to be in the **first quadrant** (also called *first-quadrant double complex*) if $A^{p,q} = 0$ whenever $p < 0$ or $q < 0$. In particular, any first quadrant double complex is bounded below.
- The double complex is said to be in the **third quadrant** (also called *third-quadrant double complex*) if $A^{p,q} = 0$ whenever $p > 0$ or $q > 0$. In particular, any third quadrant double complex is bounded above.
- Let us say that the double complex is **locally finite along diagonals** or **locally bounded along diagonals**⁶ if for each integer n , there exist at most finitely many pairs (p, q) with $p + q = n$ such that $A^{p,q} \neq 0$.
- Let us say that the double complex is **bounded in total degree**⁷ if there exist integers m and M such that $A^{p,q} = 0$ whenever $m \leq p + q \leq M$.

Definition 4.6.5. Let $(A^{p,q}, d'_A, d''_A)$ be a double complex (Definition 4.6.1) in an additive category (Definition 2.3.4) $\mathcal{A}$. For each integer n , define:

$$\begin{aligned} \text{Tot}_{\oplus}^n(A) &= \bigoplus_{p+q=n} A^{p,q}, \\ \text{Tot}_{\Pi}^n(A) &= \prod_{p+q=n} A^{p,q}. \end{aligned}$$

⁶These do not seem to be standard terminology.

⁷This does not seem to be standard terminology.

assuming that the direct sum $\bigoplus_{p+q=n} A^{p,q}$ and the product $\prod_{p+q=n} A^{p,q}$ respectively exist. These are called the *direct-sum total complex* and the *product total complex*, respectively.

Define the differentials categorically as follows:

- For the direct sum total complex, the differential $d^n : \text{Tot}_{\oplus}^n(A) \rightarrow \text{Tot}_{\oplus}^{n+1}(A)$ is the unique morphism such that, for each (p, q) with $p + q = n$, we have

$$d^n \circ \iota_{p,q} = \iota_{p+1,q} \circ d'_A{}^{p,q} + (-1)^p \iota_{p,q+1} \circ d''_A{}^{p,q},$$

where $\iota_{p,q} : A^{p,q} \rightarrow \text{Tot}_{\oplus}^n(A)$ is the canonical inclusion.

- For the product total complex, the differential $d^n : \text{Tot}_{\Pi}^n(A) \rightarrow \text{Tot}_{\Pi}^{n+1}(A)$ is the unique morphism such that, for each (p, q) with $p + q = n + 1$, we have

$$\pi_{p,q} \circ d^n = d'_A{}^{p-1,q} \circ \pi_{p-1,q} + (-1)^{p-1} d''_A{}^{p,q-1} \circ \pi_{p,q-1},$$

where $\pi_{p,q} : \text{Tot}_{\Pi}^{n+1}(A) \rightarrow A^{p,q}$ is the canonical projection.

Then $(\text{Tot}_{\oplus}^{\bullet}(A), d)$ and $(\text{Tot}_{\Pi}^{\bullet}(A), d)$ are chain complexes (Definition 3.0.1) in $\mathcal{A}$, whenever the corresponding sums or products exist. The complexes $\text{Tot}_{\oplus}^{\bullet}(A)$ and $\text{Tot}_{\Pi}^{\bullet}(A)$ may also be denoted by notations such as $\text{Tot}^{\oplus}(A)$ and $\text{Tot}^{\Pi}(A)$ — note the superscripted direct sum and product symbols — especially when cohomological conventions (Definition 3.0.1) are adopted.

Definition 4.6.6 (Double Complex associated to biadditive functor and chain complexes). Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be additive categories (Definition 2.3.4), and suppose that

$$F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$$

is a biadditive functor (Definition 2.4.2). Let $X_{\bullet}$ and $Y_{\bullet}$ be chain complexes (Definition 3.0.1) of objects in $\mathcal{A}$ and $\mathcal{B}$ respectively.

Construct the double complex (Definition 4.6.1) $Z_{\bullet,\bullet} = (Z_{n,m}, d_{n,m}^h, d_{n,m}^v)$ in $\mathcal{C}$ associated to F , $X_{\bullet}$, and $Y_{\bullet}$ as follows:

$$Z_{n,m} := F(X_n, Y_m),$$

with horizontal differentials

$$d_{n,m}^h := F(d_n^X, \text{id}_{Y_m}) : Z_{n,m} \rightarrow Z_{n-1,m},$$

and vertical differentials

$$d_{n,m}^v := (-1)^n F(\text{id}_{X_n}, d_m^Y) : Z_{n,m} \rightarrow Z_{n,m-1}.$$

These differentials indeed satisfy the double complex conditions:

$$d^h \circ d^h = 0, \quad d^v \circ d^v = 0, \quad \text{and} \quad d^h \circ d^v + d^v \circ d^h = 0.$$

We may often denote the double complex $Z_{\bullet,\bullet}$ by $F(X_{\bullet}, Y_{\bullet})$. In particular, F induces a bifunctor

$$F : \mathbf{Ch}(\mathcal{A}) \times \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$$

(Definition 4.6.3) that is in fact a biadditive functor of additive categories (see Proposition 3.1.8) (♠ TODO: verify that we indeed get a biadditive functor)

In particular, we may speak of the total complexes (Definition 4.6.5) $\text{Tot}^\oplus(F(X_\bullet, Y_\bullet))$ and $\text{Tot}^\Pi(F(X_\bullet, Y_\bullet))$, and these specify biadditive functors

$$\text{Tot}^\oplus(F(-, -)), \text{Tot}^\Pi(F(-, -)) : \mathbf{Ch}(\mathcal{A}) \times \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{Ch}(\mathcal{C}).$$

4.6.2. Acyclic assembly lemma.

Lemma 4.6.7. Let C be a double complex (Definition 4.6.1) of objects in an additive category (Definition 2.3.4) $\mathcal{A}$. If C is locally bounded along diagonals (Definition 4.6.4), then the complexes $\text{Tot}^\oplus(A)$ and $\text{Tot}^\Pi(A)$ (Definition 4.6.5) are naturally isomorphic.

Proof. Since C is locally bounded along diagonals, the degree n components

$$\begin{aligned} (\text{Tot}^\oplus)^n(A) &= \bigoplus_{p+q=n} A^{p,q}, \\ (\text{Tot}^\Pi)^n(A) &= \prod_{p+q=n} A^{p,q}. \end{aligned}$$

are finite direct sums and finite products respectively and hence are naturally isomorphic (Lemma 2.3.5). The differential maps of the two total complexes also naturally coincide. $\square$

Lemma 4.6.8 (cf. [Wei94, Acyclic Assembly Lemma 2.7.3]). (♠ TODO: It may be the case that this is generalizable beyond first quadrant double complexes, but I don't have a slick way to show this. See the commented out code for the statements; also, it may be necessary to assume something like AB4* for such statements) Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) for which (small) filtered colimits (Definition 2.2.13) which exist are exact (e.g. which holds if $\mathcal{A}$ satisfies Ab5 (Definition 2.5.7)). Let C be a double complex (Definition 4.6.1) in $\mathcal{A}$.

If C has exact columns or has exact rows and C is a bounded below or bounded above double complex (Definition 4.6.4), then $\text{Tot}^\Pi(C)$ is an acyclic chain complex (Definition 3.0.2).

Proof. We show that if C has exact columns and C is bounded below, then $\text{Tot}^\Pi(C)$ is an acyclic chain complex (Definition 3.0.2); it can then be argued symmetrically that if C has exact columns and C is bounded above, then $\text{Tot}^\Pi(C)$ is acyclic. Moreover, the case of exact rows can be deduced by reflecting the rows and columns of double complexes.

Note that since C is assumed to be bounded below and hence is locally bounded along diagonals (Definition 4.6.4), $\text{Tot}^\Pi(C)$ and $\text{Tot}^\oplus(C)$ exist, are constructed by finite products (which are also finite coproducts), and are naturally isomorphic by Lemma 4.6.7. Further recall that finite coproducts in an abelian category are exact.

Define the sub-double complexes $F^k C$ of C by

$$(F^k C)^{p,q} = \begin{cases} C^{p,q} & \text{if } p \leq k \\ 0 & \text{otherwise} \end{cases}.$$

This yields a filtration

$$\dots \subseteq F^{k-1} C \subseteq F^k C \subseteq F^{k+1} C \subseteq \dots \subseteq C.$$

Moreover, for each n ,

$$(\mathrm{Tot}^\Pi(F^k C))^n = \prod_{p \leq k, p+q=n} C^{p,q}.$$

For each n , the above stabilizes as $k \rightarrow \infty$ to $\mathrm{Tot}^\Pi(C)^n$. Now let

$$D^k = F^k C / F^{k-1} C = \begin{cases} C^{p,q} & \text{if } p = k \\ 0 & \text{otherwise.} \end{cases}$$

Since each column $C^{k,*}$ is exact by assumption, the total complex $\mathrm{Tot}^\Pi(D^k)$ is acyclic. Note that we have short exact sequences

$$0 \rightarrow F^{k-1} C \rightarrow F^k C \rightarrow D^k \rightarrow 0$$

of double complexes. The totalization functor $\mathrm{Tot}^\Pi(-)$ in this case is exact because all of the double complexes are locally bounded along diagonals (Definition 4.6.4). We hence have a short exact sequence

$$0 \rightarrow \mathrm{Tot}^\Pi(F^{k-1} C) \rightarrow \mathrm{Tot}^\Pi(F^k C) \rightarrow \mathrm{Tot}^\Pi(D^k) \rightarrow 0.$$

Since $\mathrm{Tot}^\Pi(D^k)$ is acyclic (Definition 3.0.2), the long exact cohomology sequences (Theorem 4.3.4) yield isomorphisms

$$H^n(\mathrm{Tot}^\Pi(F^{k-1} C)) \cong H^n(\mathrm{Tot}^\Pi(F^k C)).$$

Since C is assumed to be bounded, the subcomplex $F^k C$ is zero for sufficiently negative k , in which case $\mathrm{Tot}^\Pi(F^k C)$ is acyclic. By induction on k , $\mathrm{Tot}^\Pi(F^k C)$ remains acyclic for all k . Moreover, the filtered colimit $\varinjlim_k \mathrm{Tot}^\Pi(F^k C)$ is $\mathrm{Tot}^\Pi(C)$. The assumed exactness of filtered colimits in $\mathcal{A}$ concludes that $\mathrm{Tot}^\Pi(C)$ is acyclic.

By symmetry, if C instead has exact rows, then $\mathrm{Tot}^\Pi(C)$ is an acyclic chain complex. $\square$

4.6.3. Flat objects with respect to biadditive functors.

Definition 4.6.9 (Flat object with respect to a tensor (monoidal) product). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) (in practice, the following definitions are usually considered when F is some kind of “tensor product” $\otimes$ and is right exact (Definition 2.4.12) in each variable).

1. An object $X \in \mathcal{A}$ is called *flat (with respect to F on the left)* if the functor $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ is exact (Definition 2.4.12).
2. An object $Y \in \mathcal{B}$ is called *flat (with respect to F on the right)* if the functor $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ is exact (Definition 2.4.12).

If $\mathcal{A} = \mathcal{B} = \mathcal{C}$ and F makes $\mathcal{A}$ into a symmetric monoidal category (Definition A.0.2), then certainly $F(X, -)$ is exact if and only if $F(-, Y)$ is exact, i.e. flatness is equivalent on the two sides of $\otimes$.

Remark 4.6.10. The above notion of flatness is non-standard terminology, unless the bifunctor F is some kind of tensor product.

Definition 4.6.11 (Having enough flat objects in an abelian category). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2)

1. We say that $\mathcal{A}$ has enough flat objects with respect to $\otimes$ if for every object $A \in \mathcal{A}$ there exists an epimorphism

$$F \twoheadrightarrow A$$

in $\mathcal{A}$ where F is flat with respect to $\otimes$ (Definition 4.6.9), i.e. $F \otimes - : \mathcal{B} \rightarrow \mathcal{C}$ is exact. Equivalently, $\mathcal{A}$ has enough flat objects with respect to $\otimes$ if it has enough objects of the class of objects of $\mathcal{A}$ which are flat with respect to $\otimes$ (Definition 4.2.1) on the left.

2. Similarly, we say that $\mathcal{B}$ has enough flat objects with respect to $\otimes$ if for every $B \in \mathcal{B}$ there exists an epimorphism

$$G \twoheadrightarrow B$$

in $\mathcal{B}$ with G flat with respect to $\otimes$, i.e. $- \otimes G : \mathcal{A} \rightarrow \mathcal{C}$ is exact. Equivalently, $\mathcal{B}$ has enough flat objects with respect to $\otimes$ if it has enough objects of the class of objects of $\mathcal{B}$ which are flat with respect to $\otimes$ (Definition 4.2.1) on the left.

Definition 4.6.12. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bifunctor that is additive (Definition 2.4.1) and right exact (Definition 2.4.12) in each variable.

1. A *flat (left-)resolution of an object $F \in \mathcal{A}$* is a left resolution of F (Definition 4.2.29) consisting of flat objects (Definition 4.6.9) in $\mathcal{A}$ with respect to $\otimes$.
2. A *flat (left-)resolution of an object $G \in \mathcal{B}$* is a left resolution of G (Definition 4.2.29) consisting of flat objects (Definition 4.6.9) in $\mathcal{B}$ with respect to $\otimes$.

Note that we are reserving these default term “flat resolution” for “left resolutions with flat objects”. We may still use *flat right-resolution* to be a right resolution (Definition 4.2.29) consisting of flat objects (Definition 4.6.9).

4.6.4. Derived functors for biadditive functors computed via resolutions of flat objects.

Definition 4.6.13 (Derived functors for biadditive functors of abelian categories via resolutions by flat objects). Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) of abelian categories (Definition 2.3.12).

1. Fix an object A of $\mathcal{A}$. Let B be an object of $\mathcal{B}$.
 (a) Let $L_n^I F(A, B) = L_n^I(F(A, -))(B)$ be an object of $\mathcal{C}$ obtained as follows: take a left resolution (Definition 4.2.29)

$$P_\bullet \rightarrow B$$

of B of objects P_i for which the functors $F(-, P_i) : \mathcal{A} \rightarrow \mathcal{C}$ are exact (Definition 2.4.12), i.e. $P_\bullet$ is a flat resolution of B (Definition 4.6.12). Let

$$L_n^I(F(A, -))(B) := H_n(F(A, P_\bullet))$$

(Definition 3.2.2).

- (b) Let $R_{II}^n F(A, B) = R_{II}^n(F(A, -))(B)$ be an object of $\mathcal{C}$ obtained as follows: take a right resolution (Definition 4.2.29)

$$B \rightarrow I^\bullet$$

of B of objects I^i for which the functors $F(-, I^i) : \mathcal{A} \rightarrow \mathcal{C}$ are exact (Definition 2.4.12), i.e. $I^\bullet$ is a flat right-resolution of B (Definition 4.6.12). Let

$$R_{II}^n(F(A, -))(B) := H^n(F(A, I^\bullet))$$

(Definition 3.2.2).

2. Fix an object B of $\mathcal{B}$. Let A be an object of $\mathcal{A}$.

- (a) Let $L_n^I F(A, B) = L_n^I(F(-, B))(A)$ be an object of $\mathcal{C}$ obtained as follows: take a left resolution (Definition 4.2.29)

$$P_\bullet \rightarrow A$$

of A of objects P_i for which the functors $F(P_i, -) : \mathcal{B} \rightarrow \mathcal{C}$ are exact (Definition 2.4.12), i.e. $P_\bullet$ is a flat resolution (Definition 4.6.12) of A . Let

$$L_n^I(F(-, B))(A) := H_n(F(P_\bullet, B))$$

(Definition 3.2.2).

- (b) Let $R_I^n F(A, B) = R_I^n(F(-, B))(A)$ be an object of $\mathcal{C}$ obtained as follows: take a right resolution (Definition 4.2.29)

$$A \rightarrow I^\bullet$$

of A of objects I^i for which the functors $F(I^i, -) : \mathcal{B} \rightarrow \mathcal{C}$ are exact (Definition 2.4.12), i.e. $I^\bullet$ is a flat right-resolution of A (Definition 4.6.12). Let

$$R_I^n(F(-, B))(A) := H^n(F(I^\bullet, B))$$

(Definition 3.2.2).

A priori, the objects $L_n^{II} F(A, B)$, $R_{II}^n F(A, B)$, $L_n^I F(A, B)$, and $R_I^n F(A, B)$ are not well defined, i.e. they may possibly depend on the choice of resolution. See Theorem 4.6.15, which asserts that the pair $L_n^{II} F(A, B)$ and $L_n^I F(A, B)$ (resp. $R_{II}^n F(A, B)$ and $R_I^n F(A, B)$) are in agreement and are well defined under mild conditions.

In case that F is a functor thought of as some kind of “tensor product” and denoted by $\otimes$ (and usually, but not necessarily, such that $\otimes$ is right exact in each variable), it is customary to denote $L_n^{II} F(A, B)$ and $L_n^I F(A, B)$ by $\text{Tor}_n^{II}(A, B)$ and $\text{Tor}_n^I(A, B)$ respectively and to call these objects *Tor objects*. In other words, Tor objects are obtained by taking flat (Definition 4.6.9) resolutions with respect to $\otimes$

In case that F is a functor thought of as some kind of “Hom” and denoted as (some variation of) $\text{Hom} : \mathcal{A}^{\text{op}} \times \mathcal{B} \rightarrow \mathcal{C}$ (most usually, $\mathcal{A} = \mathcal{B}$ for such Hom functors), it is customary to denote $R_{II}^n F(A, B)$ and $R_I^n F(A, B)$ by $\text{Ext}_n^{II}(A, B)$ and $\text{Ext}_n^I(A, B)$ respectively and to call these objects *Ext objects*.

Lemma 4.6.14. Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) of abelian categories (Definition 2.3.12). Assume that (small) filtered colimits which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 2.5.7)).

Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$ be objects.

1. Suppose that left resolutions (Definition 4.2.29) $P_{A,\bullet} \rightarrow A$ and $P_{B,\bullet} \rightarrow B$ exist such that $P_{A,i}$ and $P_{B,i}$ are flat (Definition 4.6.9) with respect to F on the left and right respectively, i.e. $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i .

The complexes $F(P_{A,\bullet}, B)$ and $F(A, P_{B,\bullet})$ are quasi-isomorphic (Definition 3.2.8) to the complex $\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))$ (Definition 4.6.5) (Definition 4.6.6).

2. Suppose that right resolutions (Definition 4.2.29) $A \rightarrow I^{A,\bullet}$ and $B \rightarrow I^{B,\bullet}$ exist such that $I^{A,i}$ and $I^{B,i}$ are flat (Definition 4.6.9) with respect to F on the left and right respectively, $F(I^{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, I^{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i .

The complexes $F(I^{A,\bullet}, B)$ and $F(A, I^{B,\bullet})$ are quasi-isomorphic (Definition 3.2.8) to the complex $\text{Tot}(F(I^{A,\bullet}, I^{B,\bullet}))$ (Definition 4.6.5) (Definition 4.6.6).

Proof. We prove 1. The other part is the dual statement.

Choose resolutions $P_{A,\bullet} \xrightarrow{\varepsilon} A$ and $P_{B,\bullet} \xrightarrow{\eta} B$ such that $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i . Identifying A and B with complexes concentrated in degree 0, we can form (Definition 4.6.6) the three double complexes (Definition 4.6.1) $F(P_{A,\bullet}, P_{B,\bullet})$, $F(A, P_{B,\bullet})$, and $F(P_{A,\bullet}, B)$. Note that the augmentation morphisms ε and η induce morphisms $\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet})) \rightarrow \text{Tot}(F(A \otimes P_{B,\bullet}))$, $\text{Tot}(F(P_{A,\bullet}, B))$.

Let C be the double complex of objects in $\mathcal{C}$ obtained from $F(P_{A,\bullet}, P_{B,\bullet})$ by adding $F(A, P_{B,\bullet}[-1])$ in the column $p = -1$. One can show that the translate $\text{Tot}(C)[1]$ is the mapping cone (Definition 3.7.3) of the map

$$\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet})) \xrightarrow{\varepsilon \otimes \text{id}} \text{Tot}(F(A, P_{B,\bullet})) = F(A, P_{B,\bullet}).$$

Moreover, since each $F(-, P_{B,i})$ is an exact functor, every row of C is exact, so $\text{Tot}(C)$ is exact by Lemma 4.6.8. Therefore, $F(\varepsilon, \text{id})$ is a quasi-isomorphism and hence

$$H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \xrightarrow{H_*(F(\varepsilon, P_{B,\bullet}))} H_*(F(A, P_{B,\bullet}))$$

is a natural isomorphism.

By symmetry, there is a natural isomorphism $H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \rightarrow H_*(F(P_{A,\bullet}, B))$. $\square$

Theorem 4.6.15 (Balancing generalized derived functors of a biadditive functor of abelian categories computed via flat resolutions). Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) of abelian categories (Definition 2.3.12). Assume that (small) filtered colimits which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 2.5.7)).

Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$ be objects.

1. Suppose that left resolutions (Definition 4.2.29) $P_{A,\bullet} \rightarrow A$ and $P_{B,\bullet} \rightarrow B$ exist such that $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i , i.e. $P_{A,\bullet}$ and $P_{B,\bullet}$ are flat resolutions (Definition 4.6.12) of A and B respectively.
 - (a) The objects $L_n^I F(A, B)$ (Definition 4.6.13) and $L_n^{II} F(A, B)$ are naturally isomorphic.
 - (b) The objects $L_n^I F(A, B)$ and $L_n^{II} F(A, B)$ are well defined (up to natural isomorphism), i.e. do not depend on the choice of left resolutions of A and B respectively.

2. Suppose that right resolutions (Definition 4.2.29) $A \rightarrow I^{A,\bullet}$ and $B \rightarrow I^{B,\bullet}$ exist such that $F(I^{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, I^{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i .
 - (a) The objects $R_I^n F(A, B)$ (Definition 4.6.13) and $R_{II}^n F(A, B)$ are naturally isomorphic.
 - (b) The objects $R_I^n F(A, B)$ and $R_{II}^n F(A, B)$ are well defined (up to natural isomorphism), i.e. do not depend on the choice of left resolutions of A and B respectively.

Proof. We prove 1. The other part is the dual statement.

Choose resolutions $P_{A,\bullet} \xrightarrow{\varepsilon} A$ and $P_{B,\bullet} \xrightarrow{\eta} B$ such that $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i . As per Lemma 4.6.14, $F(\varepsilon, \text{id})$ is a quasi-isomorphism and hence

$$H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \xrightarrow{H_*(F(\varepsilon, P_{B,\bullet}))} H_*(F(A, P_{B,\bullet}))$$

is a natural isomorphism. By symmetry, there is a natural isomorphism $H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \rightarrow H_*(F(P_{A,\bullet}, B))$. Therefore, $L_n^I(A, B)$ and $L_n^{II}(A, B)$ are naturally isomorphic as claimed. In particular, $L_n^I(A, B)$ and $L_n^{II}(A, B)$ are independent of the choice of resolution of A and B respectively.

□

4.7. Tor/derived functor of a (right exact) biadditive functor via projective resolutions or flat resolutions. Recall that the derived functor (Definition 4.6.13) of a biadditive functor (Definition 2.4.2) may be computed via flat resolutions (Definition 4.6.12); this seems to be a departure from the notion of a left derived functor (Definition 4.3.1) of a single-variate right exact (Definition 2.4.12) additive functor, which a priori uses projective (Definition 4.2.2) resolutions (Definition 4.2.29). The motivation for this is that certain common abelian categories, such as those of sheaves (Definition B.0.8) of abelian groups or modules (Definition B.0.18), may have enough flats (Definition 4.6.11) (with respect to the bi-additive functor of interest, which is a right exact “tensor product” $\otimes$ in common contexts) but not enough projectives (Definition 4.2.5), so projective resolutions have not be available to compute the derived functors. Nevertheless, in the case that the abelian categories involved may both have enough flats and have enough projectives (e.g. when the abelian categories are the categories of abelian groups or modules over rings), then using either flat resolutions or projective resolutions to compute the derived functors of the bi-additive functor (which again is usually some kind of right exact “tensor product” $\otimes$) will yield the same result, see Corollary 4.7.9.

Definition 4.7.1 (General circumstances for the existence of Tor functors). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) that is right exact (Definition 2.4.12) in each variable.

(♠ TODO: Do notation for tor in two categories, one category, modules over rings)

For objects $A \in \mathcal{A}$ and $B \in \mathcal{B}$, the *Tor objects*

$$\text{Tor}_n(A, B) = \text{Tor}_n^{\mathcal{A}, \mathcal{B}}(A, B)$$

may be defined in one of several not-necessarily-equivalent ways:

1. as the left derived functors (Definition 4.3.1) of $A \otimes -$ computed via projective resolutions (Definition 4.2.29) of B in $\mathcal{B}$, assuming that such a projective resolution exists.
2. as the left derived functors (Definition 4.3.1) of $- \otimes B$ computed via projective resolutions (Definition 4.2.29) of B in $\mathcal{A}$, assuming that such a projective resolution exists.
3. as the “left derived functors” of $A \otimes -$ computed via flat resolutions (Definition 4.6.12) of B in $\mathcal{B}$, assuming that such a flat resolution exists. More precisely, $\mathrm{Tor}_n(A, B) = H_n(A \otimes F_\bullet)$ in this case where $F_\bullet \rightarrow B$ is a flat resolution. Equivalently, $\mathrm{Tor}_n(A, B) = \mathrm{Tor}_n^{II}(A, B)$ as in Definition 4.6.13. A priori, this might depend on the choice of flat resolution.
4. as the “left derived functors” of $- \otimes B$ computed via flat resolutions (Definition 4.6.12) of A in $\mathcal{A}$. More precisely, $\mathrm{Tor}_n(A, B) = H_n(F_\bullet \otimes B)$ in this case where $F_\bullet \rightarrow A$ is a flat resolution. Equivalently, $\mathrm{Tor}_n(A, B) = \mathrm{Tor}_n^I(A, B)$ as in Definition 4.6.13. A priori, this might depend on the choice of flat resolution.

See Theorem 4.7.4 and Lemma 4.7.8, which describe sufficient conditions under which the Tor functors defined via flat resolutions are independent of the choice of flat resolution. In particular Corollary 4.7.9 shows that all of the above notions are in natural agreement if $\mathcal{A}$ and $\mathcal{B}$ have enough projectives (Definition 4.2.5) and have enough flats (Definition 4.6.11) and filtered direct limits that exist in $\mathcal{C}$ are exact.

For any of the above notions of Tor_n , note that

1. for fixed $A \in \mathcal{A}$, if $\mathrm{Tor}_n(A, B)$ exists and is well defined for any $B \in \mathcal{B}$, then $\mathrm{Tor}_n(A, -)$ is an additive functor $\mathcal{B} \rightarrow \mathcal{C}$.
2. for fixed $B \in \mathcal{B}$, if $\mathrm{Tor}_n(A, B)$ exists and is well defined for any $A \in \mathcal{A}$, then $\mathrm{Tor}_n(-, B)$ is an additive functor $\mathcal{A} \rightarrow \mathcal{C}$.
3. if $\mathrm{Tor}_n(A, B)$ exists and is well defined for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$, then $\mathrm{Tor}_n(-, -)$ is an biadditive $\mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$.

In the case that $\mathcal{A}$ is the category of $R - S$ -modules, $\mathcal{B}$ is the category of $S - T$ -modules, $\mathcal{C}$ is the category of $R - T$ -modules for some (not necessarily commutative) rings R, S, T (Definition C.0.7), and $\otimes$ is the usual tensor product between $R - S$ -modules and $S - T$ -modules producing $R - T$ -modules, then the Tor functors may be denoted by

$$\mathrm{Tor}_n^S(A, B)$$

for $A \in \mathcal{A}$ and $B \in \mathcal{B}$.

Definition 4.7.2. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). For objects $A_1, A_2 \in \mathcal{A}$, the *Ext objects*

$$\mathrm{Ext}^n(A, B) = \mathrm{Ext}_{\mathcal{A}}^n(A, B)$$

(♠ TODO: I think they would be equivalent when defined) may be defined in one of the several not-necessarily-equivalent ways:

1. As the right derived functors (Definition 4.3.1) of $\mathrm{Hom}_{\mathcal{A}}(A_1, -) : \mathcal{A} \rightarrow \mathrm{Ab}$ computed via injective resolutions (Definition 4.2.29) of A_2 in $\mathcal{A}$, assuming that such an injective resolution (Definition 4.2.29) exists.

2. As the right derived functors of $\text{Hom}_{\mathcal{A}}(-, A_2) : \mathcal{A}^{\text{op}} \rightarrow \text{Ab}$ computed by injective resolutions of A_1 in $\mathcal{A}^{\text{op}}$ (i.e. projective resolutions (Definition 4.2.29) of A_1 in $\mathcal{A}$), assuming that such a resolution (Definition 4.2.29) exists.

In particular, by identifying the biadditive functor (Definition 2.4.2)

$$\text{Hom}_{\mathcal{A}} : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \text{Ab},$$

which is left exact (Definition 2.4.12) in each variable, with the biadditive functor

$$\text{Hom}_{\mathcal{A}}^{\text{op}} : \mathcal{A} \times \mathcal{A}^{\text{op}} \rightarrow \text{Ab}^{\text{op}}$$

(♠ TODO: talk about how functors $\mathcal{A} \rightarrow \mathcal{B}$ are equivalent to functors $\text{Aop} \rightarrow \text{Bop}$, say $\text{Fun}(\mathcal{A}, \mathcal{B}) \cong \text{Fun}(\text{Aop}, \text{Bop})$ is an isomorphism of categories) that is right exact (Definition 2.4.12) in each variable, one can see that the above definitions are special cases of the definitions in Definition 4.7.1 that involve projective resolutions — the Ext objects are left derived objects (Definition 4.3.1) of $\text{Hom}_{\mathcal{A}}^{\text{op}}$ computed via projective resolutions in each variable. The above definitions are furthermore special cases of the definitions in Definition 4.7.1 that involve flat resolutions (Definition 4.6.12) because projective and injective objects in an abelian category respectively make the covariant and contravariant Hom functors exact, see Proposition 4.2.4 and because flatness (Definition 4.6.9) in this general context is defined in terms of the exactness of the biadditive functor.

Remark 4.7.3. Some common abelian categories do not have enough projectives (Definition 4.2.5), but nevertheless have enough flats (Definition 4.6.11) and hence Tor objects are well defined as per Theorem 4.7.4. One such abelian category is the category of torsion abelian groups. Similarly, some categories of sheaves of torsion abelian groups on some sites may also have enough flats, but not enough projectives.

Theorem 4.7.4 (Balancing of Tor, cf. [Wei94, Theorem 2.7.2]). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) that is right exact (Definition 2.4.12) in each variable. Assume that (small) filtered colimits which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 2.5.7)).

Given $A \in \mathcal{A}$ and $B \in \mathcal{B}$ for which flat resolutions exist (Definition 4.6.12), let $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ respectively be the Tor objects $\text{Tor}_n^{\mathcal{A}}(A, B)$ (Definition 4.7.1) (Definition 4.6.13) computed via flat resolutions (Definition 4.6.12) of A in $\mathcal{A}$ and of B in $\mathcal{B}$.

1. $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ are naturally isomorphic.
2. $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ are independent of the choice of flat resolution of A and B respectively.

In particular, we may identify the objects $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ and simply write $\text{Tor}_n(A, B)$ for either.

Proof. This follows from Theorem 4.6.15.

□

Theorem 4.7.5. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $A_1, A_2 \in \mathcal{A}$ be objects.

1. The objects $\text{Ext}_{\mathcal{A}}^n(A_1, A_2)$ computed via injective resolutions (Definition 4.2.29) of A_2 are well defined. In other words, if $A_2 \rightarrow I^{i,\bullet}$ are injective resolutions of A_2 for $i = 1, 2$, then there are natural isomorphisms (Definition 1.3.1)

$$H^n(\text{Hom}_{\mathcal{A}}(A_1, I^{1,\bullet})) \cong H^n(\text{Hom}_{\mathcal{A}}(A_1, I^{2,\bullet}))$$

for all $n \geq 0$.

2. Dually, the objects $\text{Ext}_{\mathcal{A}}^n(A_1, A_2)$ computed via projective resolutions (Definition 4.2.29) of A_1 are well defined. In other words, if $P_{i,\bullet} \rightarrow A_1$ are projective resolutions of A_1 for $i = 1, 2$, then there are natural isomorphisms

$$H^n(\text{Hom}_{\mathcal{A}}(P_{1,\bullet}, A_2)) \cong H^n(\text{Hom}_{\mathcal{A}}(P_{2,\bullet}, A_2))$$

for all $n \geq 0$.

3. Suppose that an injective resolution $A_2 \rightarrow I^\bullet$ of A_2 and a projective resolution $P_\bullet \rightarrow A_1$ exist. The two objects $\text{Ext}_{\mathcal{A}}^n(A_1, A_2)$ computed by each of these two resolutions are naturally isomorphic. In other words, there are natural isomorphisms

$$H^n(\text{Hom}_{\mathcal{A}}(P_\bullet, A_2)) \cong H^n(\text{Hom}_{\mathcal{A}}(A_1, I^\bullet))$$

for all $n \geq 0$.

Proof. 1. This follows from Proposition 4.3.3.

2. This follows from Proposition 4.3.3.

3. This follows from Theorem 4.7.4; note that projective objects and injective objects in $\mathcal{A}$ are flat (Definition 4.6.9)/exact for the first and second inputs of $\text{Hom}_{\mathcal{A}}$ respectively by Proposition 4.2.4.

□

Corollary 4.7.6. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7), let M be an R - S bimodule (Definition 2.1.1), and let N be an R - T bimodule. Let $n \geq 0$ be an integer.

The Ext (Definition 4.5.2) objects $\text{Ext}^n(M, N)$ are well defined up to natural isomorphism (Definition 1.3.1), independent of

- Whether a projective resolution (Definition 4.2.29) of M or an injective resolution (Definition 4.2.29) of N is used to compute the right derived functor (Definition 8.4.1),
- the choice of projective resolution of M and,
- the choice of injective resolution of N .

Proof. Note that categories of bimodules over rings have enough projectives and enough injectives (Theorem 4.2.6), so projective and injective resolutions of M and N exist by Proposition 4.2.33. The statement follow from Theorem 4.7.5. □

Lemma 4.7.7. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) that is right exact (Definition 2.4.12) in each variable.

1. Say that $\mathcal{B}$ has enough projectives (Definition 4.2.5) and let $\text{Tor}_n(A, B)$ be the Tor object computed as the left derived functor (Definition 4.3.1) of $A \otimes -$ applied to B . If A is flat (Definition 4.6.9), then $\text{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all B .

2. Say that $\mathcal{A}$ has enough projectives (Definition 4.2.5) and let $\mathrm{Tor}_n(A, B)$ be the Tor object computed as the left derived functor of $- \otimes B$ applied to A . If B is flat (Definition 4.6.9), then $\mathrm{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all A .
3. Say that $\mathcal{B}$ has enough flats (Definition 4.6.11) and let $\mathrm{Tor}_n(A, B)$ be the Tor object computed as $H_n(A \otimes F_\bullet)$ for a flat resolution $F_\bullet \rightarrow B$. If A is flat (Definition 4.6.9), then $\mathrm{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all B .
4. Say that $\mathcal{A}$ has enough flats (Definition 4.6.11) and let $\mathrm{Tor}_n(A, B)$ be the Tor object computed as $H_n(F_\bullet \otimes B)$ for a flat resolution $F_\bullet \rightarrow A$. If B is flat (Definition 4.6.9), then $\mathrm{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all A .

Proof. We prove the first part. The other parts hold similarly.

If A is flat, then for any $B \in \mathcal{B}$, letting $P_\bullet \rightarrow B$ be some projective resolution (Definition 4.2.29), the exactness of $A \otimes -$ implies that $\mathrm{Tor}_n(A, B) = H_n(A \otimes P_\bullet) = 0$ for all $n \neq 0$. $\square$

Lemma 4.7.8 (cf. [Wei94, Flat Resolution Lemma 3.2.8]). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) that is right exact (Definition 2.4.12) in each variable. Suppose that $\mathcal{A}$ and $\mathcal{B}$ have enough projective objects (Definition 4.2.5).

Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$ be objects.

1. Suppose that A has some flat resolution (Definition 4.6.12). The Tor objects (Definition 4.7.1)

$$\mathrm{Tor}_n(A, B) \in \mathcal{C}$$

obtained via a projective resolution of A (Definition 4.2.29) and via a flat resolution of A (Definition 4.6.12) are naturally isomorphic. In particular, either notion of Tor_n is well defined.

2. Suppose that B has some flat resolution (Definition 4.6.12). The Tor objects

$$\mathrm{Tor}_n(A, B) \in \mathcal{C}$$

obtained via a projective resolution of B (Definition 4.2.29) and via a flat resolution of B (Definition 4.6.12) are naturally isomorphic. In particular, either notion of Tor_n is well defined.

Proof. We show 1. Write $\mathrm{Tor}_n(A, B)$ for the tor object obtained via a projective resolution of A . Write $H_n(F_\bullet \otimes B)$ for the tor object obtained via a flat resolution $F_\bullet \rightarrow A$ of A . We argue by induction. For $n = 0$, we have $\mathrm{Tor}_0(A, B) \cong H_0(F_\bullet \otimes B)$ because $- \otimes B$ is right exact by assumption.

Let K be such that $0 \rightarrow K \rightarrow F_0 \rightarrow A \rightarrow 0$ is exact. write $E_\bullet = (\cdots \rightarrow F_2 \rightarrow F_1)$ so that $E_\bullet \rightarrow K$ is a resolution of K by flat objects (Lemma 4.2.31). Since $\mathcal{B}$ has enough projectives, further note that all flat objects of $\mathcal{A}$ are acyclic (Definition 4.4.1) for the functor $- \otimes B$ by Lemma 4.7.7.

Since $\mathcal{A}$ has enough projectives, Lemma 4.4.4 thus applies to yield

$$\mathrm{Tor}_1(A, B) = \ker(K \otimes B \rightarrow F_0 \otimes B) = \ker \left\{ \frac{F_1 \otimes B}{\mathrm{im}(F_2 \otimes B) \rightarrow F_0 \otimes B} = H_1(F \otimes B) \right\},$$

thus establishing the desired isomorphism for $n = 1$. For $n \geq 2$, use induction to see that

$$\mathrm{Tor}_n(A, B) \cong \mathrm{Tor}_{n-1}(K, B) \cong H_{n-1}(E_\bullet \otimes B) = H_n(F \otimes B).$$

□

Corollary 4.7.9. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition 2.4.2) that is right exact (Definition 2.4.12) in each variable. Suppose that $\mathcal{A}$ and $\mathcal{B}$ have enough projectives (Definition 4.2.5) and flats (Definition 4.6.11). Further suppose that small (Definition 2.2.10) filtered colimits (Definition 2.2.13) which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 2.5.7)).

For any objects $A \in \mathcal{A}$ and $B \in \mathcal{B}$, all notions of $\mathrm{Tor}_n(A, B)$ as defined in Definition 4.7.1 naturally agree with one another.

Proof. This follows from Theorem 4.7.4 and Lemma 4.7.8. □

Corollary 4.7.10. Let R, S, T be (not necessarily commutative) rings (Definition C.0.7), let M be an R - S bimodule (Definition 2.1.1), and let N be an R - T bimodule. Let $n \geq 0$ be an integer.

The Tor objects (Definition 4.7.1) $\mathrm{Tor}_n(M, N)$ are well defined up to natural isomorphism (Definition 1.3.1), independent of

- Whether a projective resolution (Definition 4.2.29) of M , a flat (Definition 4.2.16) resolution (Definition 4.6.12) of M , an projective resolution of N , or a flat resolution of N is used to compute the derived functor (Definition 4.6.13), and
- the choice of projective (Definition 4.2.2) or flat resolution of M or N .

Proof. Note that a category of bimodules has enough (Definition 4.6.12) free modules (Definition 4.2.12) on the left. Moreover, all free modules are projective (Proposition 4.2.14) and thus flat (Proposition 4.2.17), so the categories of bimodules have enough projectives and have enough flats. Moreover, categories of bimodules are Grothendieck (Definition 2.5.7) (Theorem 2.5.11) and hence Ab5 (Definition 2.5.7), so the desired result follows from Corollary 4.7.9. □

5. EXT AND TOR

5.1. Tor for abelian groups.

Proposition 5.1.1 ([Wei94, Proposition 3.1.2]). Let A and B be abelian groups (Definition C.0.3).

1. $\mathrm{Tor}_1^{\mathbb{Z}}(A, B)$ (Definition 4.5.1) is a torsion abelian group. (♠ TODO: torsion abelian group)

2. $\text{Tor}_n^{\mathbb{Z}}(A, B) = 0$ for $n \geq 2$.

Proposition 5.1.2 ([Wei94, Proposition 3.1.3]). $\text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B)$ (Definition 4.5.1) is the torsion subgroup of B for every abelian group B . (♠ TODO: torsion subgroup)

Proposition 5.1.3 ([Wei94, Proposition 3.1.4]). If A is a torsion-free abelian group, (♠ TODO: torsion-free) then $\text{Tor}_n^{\mathbb{Z}}(A, B) = 0$ (Definition 4.5.1) for all $n \geq 1$ and all abelian groups B .

5.2. Tor and Flatness.

Theorem 5.2.1 ([Wei94, Theorem 3.2.2]). Let R be a ring (Definition C.0.7). Let $S \subseteq Z(R)$ be a multiplicative subset of R contained in its center. (♠ TODO: localization of a general ring) The localization $S^{-1}R$ is a flat R -module (Definition 4.2.16).

Proposition 5.2.2 ([Wei94, Exercise 3.2.1]). Let R be a ring (Definition C.0.7).

1. Given a right R -module (Definition 2.1.1) A , the following are equivalent:
 - (a) A is flat (Definition 4.2.16).
 - (b) $\text{Tor}_n^R(A, B) = 0$ (Definition 4.5.1) for all $n \neq 0$ and all left R -modules (Definition 2.1.1) B .
 - (c) $\text{Tor}_1^R(A, B) = 0$ for all left R -modules B .
2. Dually, Given a left R -module (Definition 2.1.1) B , the following are equivalent:
 - (a) B is flat.
 - (b) $\text{Tor}_n^R(A, B) = 0$ for all $n \neq 0$ and all right R -modules (Definition 2.1.1) A .
 - (c) $\text{Tor}_1^R(A, B) = 0$ for all right R -modules A .

(♠ TODO: flat base change 3.2.9, 3.2.10) (♠ TODO: localization for Tor 3.2.13)

5.3. Ext and extensions.

Definition 5.3.1. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). Let $A, B \in \mathcal{A}$.

An *extension of A by B* refers to a short exact sequence (Definition 2.4.4) of the form

$$0 \rightarrow B \rightarrow X \rightarrow A \rightarrow 0.$$

Two extensions

$$\begin{aligned} \xi : 0 &\rightarrow B \rightarrow X \rightarrow A \rightarrow 0 \text{ and} \\ \xi' : 0 &\rightarrow B \rightarrow X' \rightarrow A \rightarrow 0 \text{ and} \end{aligned}$$

of A by B are said to be *equivalent* if there is a commutative diagram

$$\begin{array}{ccccccccc} \xi : & 0 & \longrightarrow & B & \longrightarrow & X & \longrightarrow & A & \longrightarrow & 0 \\ & & & \parallel & & \downarrow \cong & & \parallel & & \\ \xi' : & 0 & \longrightarrow & B & \longrightarrow & X' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

An extension is *split* if it is equivalent to a split exact sequence (Definition 2.4.8)

$$0 \rightarrow B \rightarrow A \oplus B \rightarrow A \rightarrow 0.$$

Lemma 5.3.2 (cf. [Wei94, Lemma 3.4.1] for a discussion for modules). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) with enough injectives (Definition 4.2.5) or with enough projectives (Definition 4.2.5). Let $A, B \in \mathcal{A}$.

If $\text{Ext}^1(A, B) = 0$ (Definition 4.7.2), then every extension (Definition 5.3.1) of A by B is split (Definition 5.3.1).

Proof. First suppose that $\mathcal{A}$ has enough projectives so that $\text{Ext}^1(A, -) : \mathcal{A} \rightarrow \text{Ab}$ is defined as a derived functor (Definition 4.3.1). Given an extension $\xi : 0 \rightarrow B \rightarrow X \rightarrow A \rightarrow 0$, the long exact sequence (Theorem 4.3.4) in $\text{Ext}^n(A, -)$ yields an exact sequence

$$\text{Hom}(A, X) \rightarrow \text{Hom}(A, A) \xrightarrow{\partial} \text{Ext}^1(A, B) = 0$$

so the identity map $\text{id}_A \in \text{Hom}(A, A)$ lifts to a map $\sigma : A \rightarrow X$ that is in fact a section (Definition 2.4.8) of $X \rightarrow A$. Thus, ξ is split.

Dually, suppose that $\mathcal{A}$ has enough injectives so that $\text{Ext}^1(-, B) : \mathcal{A}^{\text{op}} \rightarrow \text{Ab}$ is defined as a derived functor. Given an extension $\xi : 0 \rightarrow B \rightarrow X \rightarrow A \rightarrow 0$, the long exact sequence in $\text{Ext}^n(-, B)$ yields the exact sequence

$$\text{Hom}(X, B) \rightarrow \text{Hom}(B, B) \xrightarrow{\partial} \text{Ext}^1(A, B) = 0$$

This time, $\text{id}_B \in \text{Hom}(B, B)$ thus extends to a map $X \rightarrow B$ that is in fact a retract (Definition 2.4.8) of $B \rightarrow X$. Thus, ξ is split. $\square$

Definition 5.3.3. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $A, B \in \mathcal{A}$.

1. Suppose that $\mathcal{A}$ has enough projectives (Definition 4.2.5) so that $\text{Ext}^1(A, -) : \mathcal{A} \rightarrow \text{Ab}$ is defined as a derived functor (Definition 4.3.1).

Given an extension $\xi : 0 \rightarrow B \rightarrow X \rightarrow A \rightarrow 0$, the long exact sequence (Theorem 4.3.4) in $\text{Ext}^n(A, -)$ yields an exact sequence

$$\text{Hom}(A, X) \rightarrow \text{Hom}(A, A) \xrightarrow{\partial} \text{Ext}^1(A, B)$$

Define $\Theta(\xi) = \partial(\text{id}_A)$, called the *characteristic class of ξ* .

2. Suppose that $\mathcal{A}$ has enough injectives (Definition 4.2.5) so that $\text{Ext}^1(-, B) : \mathcal{A}^{\text{op}} \rightarrow \text{Ab}$ is defined as a derived functor (Definition 4.3.1).

Given an extension $\xi : 0 \rightarrow B \rightarrow X \rightarrow A \rightarrow 0$, the long exact sequence (Theorem 4.3.4) in $\text{Ext}^n(-, B)$ yields an exact sequence

$$\text{Hom}(X, B) \rightarrow \text{Hom}(B, B) \xrightarrow{\partial} \text{Ext}^1(A, B)$$

Define $\Theta(\xi) = \partial(\text{id}_B)$ called the *characteristic class of ξ* .

(♠ TODO: show these notions agree)

Lemma 5.3.4. (♠ TODO: show these notions agree)

Theorem 5.3.5 (cf. [Wei94, Theorem 3.4.3] for a discussion for modules). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) with enough injectives (Definition 4.2.5) or with enough projectives (Definition 4.2.5). Let $A, B \in \mathcal{A}$.

The assignment Θ establishes a bijective correspondence

$$\{\text{equivalence classes of extensions of } A \text{ by } B\} \leftrightarrow \cong \text{Ext}^1(A, B)$$

in which the equivalence class of the split extension corresponds to $0 \in \text{Ext}^1(A, B)$.

5.4. Universal Coefficient Theorems and Künneth Formulae. The universal coefficient theorems and Künneth formulae quantify the extent to which $\otimes$ and Hom fail to commute with homology/cohomology.

Definition 5.4.1 (Hereditary Ring). A ring (Definition C.0.7) R is called **right (resp. left) hereditary** if it satisfies any of the following equivalent conditions:

1. Every right (resp. left) ideal of R is a projective R -module (Definition 4.2.9).
2. Every submodule (Definition 2.1.4) of a projective (Definition 4.2.9) right (resp. left) R -module (Definition 2.1.1) is projective.
3. Every submodule of a free right (resp. left) R -module is projective.
4. The right (resp. left) global dimension (Definition 6.0.5) of R is at most 1 ($\text{gl dim } R \leq 1$).

Lemma 5.4.2. Any PID (Definition C.0.41) is a hereditary ring (Definition 5.4.1) (both on the left and the right).

Lemma 5.4.3. Any commutative (Definition C.0.9) Dedekind domain is a hereditary ring (both on the left and the right).

Theorem 5.4.4 (Homological Künneth Formula for Complexes, see [Wei94, Theorem 3.6.3]). Let R be a ring (Definition C.0.7). Let P be a chain complex (Definition 3.0.1) of flat (Definition 4.2.16) right R -modules (Definition 2.1.1) such that each submodule (Definition 2.1.4) of boundaries (Definition 3.2.1) $d(P_n) \subseteq P_{n-1}$ is also flat. Let Q be any chain complex of left R -modules (Definition 2.1.1). Then there is a natural exact sequence (Definition 2.4.4) for every n :

$$0 \rightarrow \bigoplus_{i+j=n} H_i(P) \otimes_R H_j(Q) \rightarrow H_n(\text{Tot}^\oplus(P \otimes_R Q)) \rightarrow \bigoplus_{i+j=n-1} \text{Tor}_1^R(H_i(P), H_j(Q)) \rightarrow 0.$$

(Definition 4.6.5) (Definition 4.6.6) (Definition 3.2.2) (Definition 4.5.1)

In the case that R is additionally a right hereditary ring (Definition 5.4.1) and P is any complex (Definition 3.0.1) of projective (Definition 4.2.9) right R -modules, the modules $d(P_n)$ are automatically projective and hence the above holds — In fact, the above short exact sequence splits (Definition 2.4.8) (non-canonically) in this case. See Theorem 5.4.5 for a dual statement.

Theorem 5.4.5 (Cohomological Künneth Formula over General Rings). Let R be any ring (Definition C.0.7). Let P be a cochain complex (Definition 3.0.1) of flat (Definition 4.2.16)

right R -modules (Definition 2.1.1) such that each submodule (Definition 2.1.4) of coboundaries (Definition 3.2.1) $d(P^n) \subseteq P^{n+1}$ is also flat. Let Q be a cochain complex of left R -modules.

There is a natural short exact sequence (Definition 2.4.4):

$$0 \rightarrow \bigoplus_{i+j=n} H^i(P) \otimes_R H^j(Q) \rightarrow H^n(\text{Tot}^\oplus(P \otimes_R Q)) \rightarrow \bigoplus_{i+j=n+1} \text{Tor}_1^R(H^i(P), H^j(Q)) \rightarrow 0.$$

(Definition 4.6.5) (Definition 4.6.6) (Definition 3.2.2) (Definition 4.5.1)

In the case that R is additionally a right hereditary ring (Definition 5.4.1) and P is any complex (Definition 3.0.1) of projective (Definition 4.2.9) right R -modules, the modules $d(P^n)$ are automatically projective and hence the above holds — In fact, the above short exact sequence splits (Definition 2.4.8) (non-canonically) in this case.

See Theorem 5.4.4 for a dual statement.

Theorem 5.4.6 (Universal coefficient theorem for homology, see e.g. [Wei94, Theorem 3.6.1, 3.6.2, Exercise 3.6.2]). Let R be a (not necessarily commutative) ring (Definition C.0.7). Let P be a chain complex (Definition 3.0.1) of flat right R modules (Definition 4.2.16) such that each submodule (Definition 2.1.4) $d(P_n)$ of P_{n-1} is also flat. Then for every n and every left R -module (Definition 2.1.1) M , there is an exact sequence (Definition 2.4.4)

$$0 \rightarrow H_n(P) \otimes_R M \rightarrow H_n(P \otimes_R M) \rightarrow \text{Tor}_1^R(H_{n-1}(P), M) \rightarrow 0$$

(Definition 4.5.1) (Definition 3.2.2) When R is a hereditary ring (Definition 5.4.1) (e.g. R is a PID (Definition C.0.41)), this exact sequence splits (Definition 2.4.8) non-canonically.

Theorem 5.4.7 (Universal coefficient theorem for cohomology, see e.g. [Wei94, Theorem 3.6.5, Exercise 3.6.2]). Let R be a (not necessarily commutative) ring (Definition C.0.7). Let P be a chain complex (Definition 3.0.1) of projective right R modules such that each submodule (Definition 2.1.4) $d(P_n)$ of P_{n-1} is also projective. Then for every n and every right R -module (Definition 2.1.1) M , there is an exact sequence

$$0 \rightarrow \text{Ext}_R^1(H_{n-1}(P), M) \rightarrow H^n(\text{Hom}_R(P, M)) \rightarrow \text{Hom}_R(H_n(P), M) \rightarrow 0$$

(Definition 4.5.2) (Definition 3.2.2) When R is a hereditary ring (Definition 5.4.1) (e.g. R is a PID (Definition C.0.41)), this exact sequence splits (Definition 2.4.8) non-canonically.

6. HOMOLOGICAL DIMENSION

Definition 6.0.1. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and $\mathcal{X} \subseteq \mathcal{A}$ a class of objects. For an object $M \in \mathcal{A}$:

1. The *left $\mathcal{X}$ -dimension of M* , denoted $\text{res}_{\mathcal{X}}(M)$, is the infimum of the set of integers $n \geq 0$ such that there exists an exact sequence (Definition 3.0.2):

$$0 \rightarrow X_n \rightarrow X_{n-1} \rightarrow \cdots \rightarrow X_0 \rightarrow M \rightarrow 0$$

with $X_i \in \mathcal{X}$ for all $0 \leq i \leq n$. If no such finite sequence exists, the dimension is ∞ .

2. The *right $\mathcal{X}$ -dimension of M* , denoted $\text{cores}_{\mathcal{X}}(M)$, is the infimum of the set of integers $n \geq 0$ such that there exists an exact sequence:

$$0 \rightarrow M \rightarrow X^0 \rightarrow X^1 \rightarrow \cdots \rightarrow X^n \rightarrow 0$$

with $X^i \in \mathcal{X}$ for all $0 \leq i \leq n$. If no such finite sequence exists, the dimension is ∞ .

Definition 6.0.2. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

1. Suppose that F is left exact (Definition 2.4.12) and that $\mathcal{A}$ has enough objects (Definition 4.2.1) that are F -acyclic (Definition 4.4.1) (e.g. $\mathcal{A}$ has enough injectives (Definition 4.2.5)). The *cohomological dimension of F* , denoted $\text{cd}(F)$, is the supremum of the set of integers n such that the n -th right derived functor (Definition 4.3.1) $R^n F$ is non-zero:

$$\text{cd}(F) = \sup\{n \in \mathbb{Z} \mid R^n F \neq 0\}$$

2. Suppose that F is right exact (Definition 2.4.12) and that $\mathcal{A}$ has enough objects (Definition 4.2.1) that are F -acyclic (Definition 4.4.1) (e.g. $\mathcal{A}$ has enough projectives (Definition 4.2.5)). The *homological dimension of F* , denoted $\text{hd}(F)$, is the supremum of the set of integers n such that the n -th left derived functor (Definition 4.3.1) $L_n F$ is non-zero:

$$\text{hd}(F) = \sup\{n \in \mathbb{Z} \mid L_n F \neq 0\}$$

If the respective derived functors (Definition 4.3.1) are zero for all $n > N$, the dimension is N ; otherwise, it is ∞ .

Proposition 6.0.3. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor and $\mathcal{I}$ the class of injective objects in $\mathcal{A}$. If $\mathcal{A}$ has enough injectives, then for any $M \in \mathcal{A}$, the right $\mathcal{I}$ -dimension (injective dimension) of M provides an upper bound for the cohomological dimension of F at M :

$$\min\{n \mid R^i F(M) = 0 \text{ for all } i > n\} \leq \text{cores}_{\mathcal{I}}(M)$$

Definition 6.0.4. Let R be a not necessarily commutative ring (Definition C.0.7) and M a left R -module (Definition 2.1.1). (♠ TODO: Probably need to recheck whether supremum or infimum is to be here.)

1. The *(left) projective dimension of M* , denoted $\text{pd}_R(M)$, is the supremum of lengths n of finite projective resolutions (Definition 4.2.29):

$$0 \rightarrow P_n \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

If no such finite resolution (Definition 4.2.29) exists, $\text{pd}_R(M) = \infty$.

2. The *(left) injective dimension of M* , denoted $\text{id}_R(M)$, is the supremum of lengths n of finite injective resolutions (Definition 4.2.29):

$$0 \rightarrow M \rightarrow E^0 \rightarrow E^1 \rightarrow \cdots \rightarrow E^n \rightarrow 0$$

If no such finite resolution exists, $\text{id}_R(M) = \infty$.

3. The *(left) flat dimension of M* , denoted $\text{fd}_R(M)$, is the supremum of lengths n of finite flat resolutions (Definition 4.6.12):

$$0 \rightarrow F_n \rightarrow \cdots \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$$

In other words,

- the projective dimension of M is the left $\mathcal{X}$ dimension in the sense of Definition 6.0.1 where $\mathcal{X}$ is the class of (left) projective R -modules,
- the injective dimension of M is the right $\mathcal{X}$ dimension in the sense of Definition 6.0.1 where $\mathcal{X}$ is the class of (left) projective R -modules, and
- the flat dimension of M is the left $\mathcal{X}$ dimension in the sense of Definition 6.0.1 where $\mathcal{X}$ is the class of (left) projective R -modules.

Right-module projective/injective/flat (Definition 2.1.1) dimensions are defined analogously using resolutions of right R -module (Definition 2.1.1).

Definition 6.0.5. Let R be a not necessarily commutative ring (Definition C.0.7). The *left global dimension of R* , denoted $\text{lgldim}(R)$, is defined as:

$$\text{lgldim}(R) = \sup\{\text{pd}_R(M) \mid M \in {}_R\mathbf{Mod}\}$$

(Definition 6.0.4) The *right global dimension* $\text{rgldim}(R)$ is defined similarly using right R -modules.

When it is clear which global dimension is discussed, one may simply write $\text{gldim}(R)$.

Definition 6.0.6. Let R be a not necessarily commutative ring (Definition C.0.7). The *Tor-dimension* (or *weak global dimension*) of a ring R , denoted $\text{Tor-dim}(R)$ or $\text{wdim}(R)$, is defined as:

$$\text{wdim}(R) = \sup\{\text{fd}_R(M) \mid M \in {}_R\mathbf{Mod}\}$$

Theorem 6.0.7. (Global Dimension Theorem) For any not necessarily commutative ring (Definition C.0.7) R , the following values are equal:

1. $\text{lgldim}(R)$, i.e. $\sup\{\text{pd}_R(M) \mid M \in {}_R\mathbf{Mod}\}$. (Definition 6.0.5) (Definition 6.0.4)
2. $\sup\{\text{id}_R(M) \mid M \in {}_R\mathbf{Mod}\}$.
3. $\sup\{\text{pd}_R(R/I) \mid I \text{ is a left ideal of } R\}$.

If R is Noetherian, then $\text{lgldim}(R) = \text{rgldim}(R)$. In general, these may differ.

Theorem 6.0.8. (Tor Dimension Theorem) For any not necessarily commutative ring (Definition C.0.7) R , the Tor-dimension is symmetric and relates to global dimension as follows:

1. $\text{wdim}(R) = \sup\{\text{fd}_R(M) \mid M \in {}_R\mathbf{Mod}\} = \sup\{\text{fd}_R(N) \mid N \in \mathbf{Mod}_R\}$. (Definition 6.0.6)
2. $\text{wdim}(R) \leq \min(\text{lgldim}(R), \text{rgldim}(R))$. (Definition 6.0.5)
3. If R is left and right Noetherian, then $\text{wdim}(R) = \text{lgldim}(R) = \text{rgldim}(R)$.

7. SHEAVES ON TOPOLOGICAL SPACES

Sheaves (say on a topological space) of abelian groups form an abelian category with a rich theory.

7.1. Presheaves and sheaves on topological spaces.

Definition 7.1.1 (Presheaf on a category). Let C and $\mathcal{A}$ be (large) categories (Definition 1.1.1).

1. A *presheaf* $\mathcal{F}$ on C with values in $\mathcal{A}$ is a functor

$$\mathcal{F} : C^{\text{op}} \rightarrow \mathcal{A}.$$

In other words, a presheaf $\mathcal{F}$ on C with values in $\mathcal{A}$ is simply a contravariant functor (Definition 1.2.2) from C to $\mathcal{A}$. Explicitly, for every object U in C , one has an object $\mathcal{F}(U)$ in $\mathcal{A}$ (called the *U -valued sections/sections evaluated at U of $\mathcal{F}$* , cf. Definition 7.1.5), and for every morphism $f : V \rightarrow U$ in C , one has a morphism (called the *restriction map*)

$$\mathcal{F}(f) : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

in $\mathcal{A}$, such that for all composable morphisms $W \xrightarrow{g} V \xrightarrow{f} U$ in C , the following diagram in $\mathcal{A}$ commutes:

$$\begin{array}{ccccc} & & \mathcal{F}(f \circ g) & & \\ & \searrow & \text{arc} & \searrow & \\ \mathcal{F}(U) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(V) & \xrightarrow{\mathcal{F}(g)} & \mathcal{F}(W) \end{array}$$

That is,

$$\mathcal{F}(g) \circ \mathcal{F}(f) = \mathcal{F}(f \circ g),$$

and for every object U in C , $\mathcal{F}(\text{id}_U) = \text{id}_{\mathcal{F}(U)}$.

2. Let $\mathcal{F}, \mathcal{G} : C^{\text{op}} \rightarrow \mathcal{A}$ be two presheaves on C with values in $\mathcal{A}$. A *morphism of presheaves*

$$\varphi : \mathcal{F} \rightarrow \mathcal{G}$$

is a natural transformation of functors (Definition 1.3.1): for each object U of C , one has a morphism

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

in $\mathcal{A}$, such that for every morphism $f : V \rightarrow U$ in C , the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(V) \\ \varphi_U \downarrow & & \downarrow \varphi_V \\ \mathcal{G}(U) & \xrightarrow{\mathcal{G}(f)} & \mathcal{G}(V) \end{array}$$

commutes, i.e.,

$$\varphi_V \circ \mathcal{F}(f) = \mathcal{G}(f) \circ \varphi_U$$

for all objects and morphisms in C .

3. Given a universe (Definition A.0.3) U , a *U -presheaf on C* typically refers to a presheaf of U -sets on C .

4. The *presheaf category/category of $\mathcal{A}$ -valued presheaves on $\mathcal{C}$* is the (large) category whose objects are the presheaves on $\mathcal{C}$ with values in $\mathcal{A}$ and whose morphisms are the presheaf morphisms. Common notations for the presheaf category include, but are not limited to: $\mathcal{A}^{\mathcal{C}^{\text{op}}}$, $\text{PreShv}(\mathcal{C}, \mathcal{A})$, $[\mathcal{C}^{\text{op}}, \mathcal{A}]$. If the value category $\mathcal{A}$ is clear from context, then notations such as $\text{PreShv}(\mathcal{C})$ are also common. Note that the presheaf category $\text{PreShv}(\mathcal{C}, \mathcal{A})$ is equivalent to the category of functors (Definition 2.2.6) $\mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ and hence notations for the functor categories are applicable as notations for presheaf categories.

Definition 7.1.2 (Category of opens of a topological space). Let X be a topological space (Definition C.0.5). The *category of opens of X* , sometimes denoted **Open**(X) (or $\text{Open}(X)$) or **Ouv**(X) (for the French word “ouvert”, meaning open), etc.), is the small (Definition 1.1.2) category (Definition 1.1.1) defined as follows:

- The objects are the open subsets $U \subseteq X$.
- For two open sets $U, V \subseteq X$, the morphism set is

$$\text{Hom}_{\text{Open}(X)}(U, V) = \begin{cases} \{\iota_{U,V}\}, & \text{if } U \subseteq V, \\ \emptyset, & \text{otherwise,} \end{cases}$$

where $\iota_{U,V}$ denotes the inclusion morphism $U \hookrightarrow V$.

- Composition of morphisms is given by composition of set-theoretic inclusions, i.e.

$$\iota_{V,W} \circ \iota_{U,V} = \iota_{U,W} \quad \text{whenever } U \subseteq V \subseteq W.$$

- The identity morphism on an object U is the inclusion $\iota_{U,U} = \text{id}_U$.

See also Definition B.0.7.

Definition 7.1.3. Let (X, τ_X) and (Y, τ_Y) be topological spaces (Definition C.0.5), and let $f : X \rightarrow Y$ be a continuous map (Definition C.0.6). Let $\text{Open}(X)$ and $\text{Open}(Y)$ be their respective categories of open sets (Definition 7.1.2) with inclusion morphisms, equipped with the canonical (Definition B.0.7) Grothendieck topologies (Definition B.0.5) given by open coverings.

Define the functor

$$f^{-1} : \text{Open}(Y) \rightarrow \text{Open}(X), \quad U \mapsto f^{-1}(U).$$

It is a continuous functor of sites from $\text{Open}(Y)$ to $\text{Open}(X)$ which induces a site morphism

$$f : (\text{Open}(X), \text{can}) \rightarrow (\text{Open}(Y), \text{can})$$

.

Definition 7.1.4 (Presheaf on a topological space). Let X be a topological space (Definition C.0.5). Let $\mathcal{D}$ be a category.

A *presheaf (of objects of $\mathcal{D}$ /valued in $\mathcal{D}$) on X* is a rule $\mathcal{F}$ that assigns:

- to each open set $U \subseteq X$, an object $\mathcal{F}(U) \in \text{Ob } \mathcal{D}$, called the *sections of $\mathcal{F}$ over U* ,
- to each inclusion of open sets $V \subseteq U$, a morphism

$$\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V), \quad s \mapsto s|_V,$$

in the category $\mathcal{D}$ called the *restriction map* such that the following conditions hold:

- (Identity) For every open set $U \subseteq X$, the restriction map ρ_U^U is the identity on $\mathcal{F}(U)$.
- (Transitivity) For inclusions $W \subseteq V \subseteq U$ of open sets, one has

$$\rho_W^U = \rho_W^V \circ \rho_V^U.$$

For instance, we may speak of a *presheaf of sets/groups/rings/etc. on the topological space X* .

Equivalently, a presheaf on X (of objects in a category $\mathcal{D}$) is a functor (Definition 1.2.2)

$$\mathbf{Open}(X)^{\text{op}} \rightarrow \mathcal{D}$$

from the opposite of the category $\mathbf{Open}(X)$ (Definition 7.1.2) of open subsets of X (see also Definition 7.1.1) .

Equivalently, a presheaf on X is a presheaf on the category $\mathbf{Open}(X)$ in the sense of Definition 7.1.1 .

The sections object $\mathcal{F}(U)$ is also denoted by $\Gamma(U, \mathcal{F})$ (see Definition 7.1.5) . Moreover, the object $\mathcal{F}(X) = \Gamma(X, \mathcal{F})$ is called the *global sections object of $\mathcal{F}$* . This agrees with the notion of global sections as discussed in Definition 7.1.5.

Definition 7.1.5. Let $\mathcal{C}$ be a (large) category (Definition 1.1.1), and let $\mathcal{D}$ be a (large) category (Definition 1.1.1). Let $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ be a presheaf valued in $\mathcal{D}$ (Definition 7.1.1).

1. For an object $U \in \mathcal{C}$, the *sections functor evaluated at U* is the functor

$$\Gamma(U, -) : \mathbf{PSh}(\mathcal{C}, \mathcal{D}) \rightarrow \mathcal{D}$$

defined by

$$\Gamma(U, \mathcal{F}) := \mathcal{F}(U),$$

i.e., the value of the presheaf $\mathcal{F}$ at the object U .

2. The *global sections of $\mathcal{F}$* is the object $\Gamma(\mathcal{F})$ of $\mathcal{D}$ defined as the limit (Definition 2.2.8)

$$\Gamma(\mathcal{F}) = \varprojlim_{U \in \mathcal{C}^{\text{op}}} \mathcal{F}(U)$$

assuming that such a limit exists, where the limit is taken over objects $U \in \mathcal{C}$ and the restriction morphisms $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$ in $\mathcal{D}$ for morphisms $U \rightarrow V$ in $\mathcal{C}$.

If a final object (Definition 2.3.1) $*$ $\in \mathcal{C}$ exists, then $\Gamma(\mathcal{F})$ exists and coincides with $\Gamma(*, \mathcal{F}) = \mathcal{F}(*)$. The construction $\Gamma(\mathcal{F})$ is functorial; in particular, if $\Gamma(\mathcal{F})$ exists for all $\mathcal{F}$ in $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$, e.g. if limits of (Definition 2.2.8) diagrams in $\mathcal{D}$ indexed by $\mathcal{C}$ exist, then Γ is a functor

$$\Gamma : \mathbf{PSh}(\mathcal{C}, \mathcal{D}) \rightarrow \mathcal{D}$$

called the *global sections functor on $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$* .

Definition 7.1.6 (Sheaf on a topological space). Let X be a topological space (Definition C.0.5), let $\mathcal{D}$ be a category (Definition 1.1.1); a posteriori, it is necessary for $\mathcal{D}$ to have a terminal object for the following definition to be non-void. Let $\mathcal{F}$ be a presheaf valued in

$\mathcal{D}$ on X (Definition 7.1.4). Then $\mathcal{F}$ is a *sheaf* if it satisfies the following additional condition (known as the *sheaf axioms*):

For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U , let $\mathcal{J}$ be the diagram (Definition 2.2.6) in the category of opens (Definition 7.1.2) of U consisting of the inclusions $U_i \cap U_j \hookrightarrow U_i$ for all $i, j \in I$. Then $\mathcal{F}$ is a sheaf if the limit (Definition 2.2.8) of the diagram $\mathcal{F} \circ \mathcal{J}$ exists in $\mathcal{D}$ and the natural morphism

$$\mathcal{F}(U) \rightarrow \lim_{j \in \mathcal{J}} \mathcal{F}(j)$$

is an isomorphism. More precisely, $\mathcal{J} : J \rightarrow \text{Open}(U)$ should be the functor whose index category J consists of

1. An object i for every $i \in I$ and an object (i, j) for every pair $i, j \in I$,
2. Morphisms $p_1 : (i, j) \rightarrow i$ and $p_2 : (i, j) \rightarrow j$ for every pair $i, j \in I$

and which sends the objects and morphisms as follows:

1. $\mathcal{J}(i) = U_i$
2. $\mathcal{J}(i, j) = U_i \cap U_j$
3. $\mathcal{J}(p_1) : U_i \cap U_j \hookrightarrow U_i$
4. $\mathcal{J}(p_2) : U_i \cap U_j \hookrightarrow U_j$.

In particular, taking $U = \emptyset$ and taking the empty open cover of the empty set, $\mathcal{F}(\emptyset)$ must be the terminal object of $\mathcal{D}$

In the case that $\mathcal{D}$ admits all small limits (Definition 2.2.10), the sheaf condition is equivalent to the following: For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U , the following equalizer diagram is exact (Definition 2.2.14):

$$\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j).$$

Here, the morphism $\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i)$ and the two morphisms $\prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j)$ are induced by the restriction maps (Definition 7.1.4) $\mathcal{F}(U) \rightarrow \mathcal{F}(U_i)$ and $\mathcal{F}(U_i) \rightarrow \mathcal{F}(U_i \cap U_j)$.

In the case that $\mathcal{D}$ is some subcategory (Definition 1.3.6) of the category of sets (Definition 1.1.7), the sheaf condition is equivalent to the following: For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U ,

- (Locality) If $s, t \in \mathcal{F}(U)$ are such that $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.
- (Gluing) If for each i there is $s_i \in \mathcal{F}(U_i)$ such that for all i, j one has $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there exists a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

Equivalently, a sheaf on a topological space X may be defined as a sheaf on (Definition B.0.8) the site (Definition B.0.5) of opens on X (Definition B.0.7).

Definition 7.1.7 (Morphism of presheaves on a topological space). Let X be a topological space (Definition C.0.5), let $\mathcal{D}$ be a category (Definition 1.1.1) and let $\mathcal{F}$, and $\mathcal{G}$ be presheaves valued in $\mathcal{D}$ on X (Definition 7.1.4).

A **morphism of presheaves** $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a collection of maps

$$\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

in $\mathcal{D}$ defined for every open set $U \subseteq X$, such that the maps are compatible with restriction: for every inclusion $V \subseteq U$ of open sets, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow \rho_V^U & & \downarrow \rho_V^U \\ \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \end{array}$$

commutes.

Equivalently, φ is a natural transformation (Definition 1.3.1) of $\mathcal{F}, \mathcal{G}$, regarded as (Definition 7.1.4) functors

$$\mathcal{F}, \mathcal{G} : \mathbf{Open}(X)^{\text{op}} \rightarrow \mathcal{D}.$$

If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then a **morphism of sheaves** is a morphism between $\mathcal{F}$ and $\mathcal{G}$ as presheaves.

(♠ TODO: talk about how γ -objects in categories of sheaves valued in sets are equivalent to categories valued in γ -sets)

7.1.1. Examples of sheaves.

Example 7.1.8. The following are examples of sheaves on topological spaces (Definition 7.1.6):

1. Given topological spaces (Definition C.0.5) X and Y , let $\underline{Y} : \mathbf{Open}(X)^{\text{op}} \rightarrow \mathbf{Sets}$ (Definition 7.1.2) be given by $\underline{Y}(U) = C(U, Y)$, the set of continuous maps (Definition C.0.6) $U \rightarrow Y$.

The sheaf condition here expresses the topological fact that ‘continuous maps that are pairwise compatible on intersections glue uniquely to continuous maps’.

2. In particular, given sets X and Y , one can consider them with the discrete topology, in which case the above sheaf $\underline{Y}$ assigns to a general subset U of X the set of functions $U \rightarrow Y$.

This captures the intuition that ‘functions that are pairwise compatible on intersections glue uniquely to functions’.

3. Similarly, for $k \in \mathbb{Z}^{\geq 0} \cup \{\infty\}$, and given C^k -differentiable manifolds (Definition C.0.37) M and N with or without boundary, let $\underline{N}^k : \mathbf{Open}(M)^{\text{op}} \rightarrow \mathbf{Sets}$ be given by $\underline{N}^k(U) = C^k(U, N)$, the set of C^k -maps (Definition C.0.35) $U \rightarrow N$.
4. Similarly, and given complex analytic manifolds M and N with or without boundary, let $\underline{N} : \mathbf{Open}(M)^{\text{op}} \rightarrow \mathbf{Sets}$ be given by $\underline{N}(U) = C^k(U, N)$, the set of complex analytic maps $U \rightarrow N$. (♠ TODO: complex analytic maps)

5. In particular, given an open subset $D \subseteq \mathbb{C}$, the functor $\mathbf{Open}(D)^{\text{op}} \rightarrow \mathbf{Sets}$ given by $U \mapsto \{f : U \rightarrow \mathbb{C} : f \text{ holomorphic}\}$ is a sheaf.

The sheaf condition here expresses the complex analytic fact that ‘holomorphic functions that are pairwise compatible on intersections glue to uniquely to holomorphic functions’.

Definition 7.1.9 (Constant sheaf on a topological space). Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be a category (Definition 1.1.1) with a terminal object (Definition 2.3.1) denoted 1 , and let A be an object of $\mathcal{A}$.

1. The *constant presheaf on X with value A* is the presheaf (Definition 7.1.4) P_A defined by

$$P_A(U) = \begin{cases} A & \text{if } U \neq \emptyset \\ 1 & \text{if } U = \emptyset \end{cases}$$

where for any inclusion of non-empty open sets $V \subseteq U$, the restriction map is the identity id_A , and for any inclusion $\emptyset \subseteq U$, the restriction map is the unique morphism $P_A(U) \rightarrow 1$.

2. Assume that a sheafification functor (Definition 7.3.2)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists. The *constant sheaf on X associated to A* , denoted $\underline{A}$ or sometimes just A , is the sheaf associated to (Definition 7.3.2) the constant presheaf P with value A defined above.

Lemma 7.1.10. Let $\mathcal{A}$ be a category (Definition 1.1.1) with a final object (Definition 2.3.1). Let $\{x\}$ be a singleton topological space (Definition C.0.5). The following categories are equivalent (Definition 1.3.4):

1. $\mathcal{A}$
2. $\text{PreShv}(\{x\}, \mathcal{A})$ (Definition 7.1.4)
3. $\text{Shv}(\{x\}, \mathcal{A})$ (Definition 7.1.6)

Explicitly, an equivalence $\mathcal{A} \rightarrow \text{Shv}(\{x\}, \mathcal{A})$ sends $A \in \mathcal{A}$ to the constant sheaf (Definition 7.1.9) on $\{x\}$ valued in A , and sends $\mathcal{P} \in \text{Shv}(\{x\}, \mathcal{A})$ to $\mathcal{P}(\{x\})$; these assignments also describe an equivalence between $\mathcal{A}$ and $\text{PreShv}(\{x\}, \mathcal{A})$.

Moreover, a sheafification functor (Definition 7.3.2)

$$\text{PreShv}(\{x\}, \mathcal{A}) \rightarrow \text{Shv}(\{x\}, \mathcal{A})$$

can simply be given by the identity assignment.

(♠ TODO: describe constant sheaves on sets by the connectivity of X .)

Definition 7.1.11 (Skyscraper sheaf). Let X be a topological space (Definition C.0.5), and let $x \in X$ be a point. Let $\mathcal{A}$ be a category (Definition 1.1.1) with a terminal object (Definition 2.3.1) 1 . Given an object $E \in \text{ob}(\mathcal{A})$, the *skyscraper sheaf at x with value E* ,

denoted $i_{x,*}(E)$, is the sheaf defined for each open set $U \subseteq X$ by

$$(i_{x,*}(E))(U) = \begin{cases} E & \text{if } x \in U \\ 1 & \text{if } x \notin U \end{cases}$$

The restriction maps $\rho_{UV} : (i_{x,*}(E))(U) \rightarrow (i_{x,*}(E))(V)$ for an inclusion $V \subseteq U$ are defined as:

- The identity map id_E if $x \in V$ (which implies $x \in U$).
- The unique morphism to the terminal object $E \rightarrow 1$ if $x \in U$ but $x \notin V$.
- The identity map id_1 if $x \notin U$ (which implies $x \notin V$).

Equivalently, this skyscraper sheaf is the direct image (Definition 7.7.2) $(i_x)_*(\underline{E})$ of constant sheaf (Definition 7.1.9) on $\{x\}$ with value E . As such, one might also denote the skyscraper sheaf as $(i_x)_*(E)$.

(♠ TODO: define additive group, multiplicative group, roots of unity, etc.)

Definition 7.1.12. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1).

The *additive group* $\mathbb{G}_a = \mathbb{G}_{a,(X,\mathcal{O}_X)}$ is simply the sheaf $\mathcal{O}_X$ regarded as a sheaf (Definition 7.1.6) of abelian groups on X .

Definition 7.1.13. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1).

The *multiplicative group* $\mathbb{G}_m = \mathbb{G}_{m,(X,\mathcal{O}_X)}$, also written $\mathcal{O}_X^\times$, is the sheaf $\mathbf{Open}(X)^{\text{op}} \rightarrow \mathbf{Grp}$ given as follows:

$$\mathcal{O}_X^\times(U) = \{s \in \mathcal{O}_X(U) : \text{there exists some } t \in \mathcal{O}_X(U) \text{ such that } st = 1\}.$$

Definition 7.1.14. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1).

Given an integer $n \geq 1$, the *sheaf of n th roots of unity* is the sheaf $\mu_n = \mu_{n,(X,\mathcal{O}_X)} : \mathbf{Open}(X)^{\text{op}} \rightarrow \mathbf{Grp}$ given as follows:

$$\mu_n(U) = \{s \in \mathcal{O}_X^\times(U) : s^n = 1\}.$$

Another particularly interesting example of a sheaf is that of the structure sheaf of an affine scheme (Definition C.0.23).

Lemma 7.1.15. Let X be a topological space (Definition C.0.5) and let $\mathcal{A}$ be a locally small category (Definition 1.1.2).

1. The category (Definition 1.1.1) $\mathbf{PreShv}(X, \mathcal{A})$ is locally small (Definition 1.1.2).
2. The category $\mathbf{Sh}(X, \mathcal{A})$ is locally small.

Proof. Note that $\mathbf{Open}(X)$ (Definition 7.1.2) is a small category (Definition 1.1.2), so $\mathbf{PreShv}(X, \mathcal{A})$ is locally small by Lemma 2.2.7. The category $\mathbf{Sh}(X, \mathcal{A})$ is necessarily a (full) subcategory (Definition 1.3.7) of $\mathbf{PreShv}(X, \mathcal{A})$ and is thus locally small. $\square$

7.2. Sheaves on more general sites. Note that on a topological space X , it is natural to talk about open sets $\{U_i\}$ which cover an open set U of X . One can express more general notions of “coverings” of objects in a category through the notion of a “site” (Definition B.0.5), which consists of a category and a “Grothendieck topology” (Definition B.0.5) on the category. The notion of “sheaf on a topological space” (Definition 7.1.6) above is a specialization of this more general notion.

A site (Definition B.0.5) is a category (Definition 1.1.1) equipped with a Grothendieck topology (Definition B.0.5), which is a notion that generalizes the idea for an object to have “covers”. One then has a general notion of sheaves (Definition B.0.8) on a site; a sheaf on a topological space X (Definition 7.1.6) is then simply a sheaf on the category $\mathbf{Open}(X)$ (Definition 7.1.2) equipped with its canonical Grothendieck topology (Definition B.0.7). It can therefore be fruitful to gain intuition about sheaves in general by studying sheaves on topological spaces and to have the understanding that many notions discussed for sheaves on topological spaces are special cases of those for sheaves on general sites. We may at times state definitions and concepts in terms of sheaves on somewhat general sites (e.g. sites that are small (Definition B.0.6)/are essentially small/have a small topologically generating family (Definition B.0.5)), but it should be understood that they specialize to sheaves on topological spaces (valued in sets).

A (Grothendieck) topos (Definition B.0.9) is a category that is equivalent to (Definition 1.3.4) a category of sheaves valued in sets on some small site, or equivalently a category that satisfies Giraud’s axioms. The theory of topoi is quite robust and appears in many contexts. Moreover, all locally small categories (Definition 1.1.2) are basically full subcategories (Definition 1.3.7) of topoi (Definition B.0.9) essentially due to Yoneda embedding (Corollary 1.3.14).

See Section B.

7.3. Sheafification of presheaves on topological spaces.

Definition 7.3.1. (♠ TODO: Move these notations to the definitions of presheaves and sheaves on topological spaces) Let X be a topological space (Definition C.0.5), and let $\mathcal{D}$ be a category (Definition 1.1.1) with a terminal object (Definition 2.3.1).

The presheaves on X valued in $\mathcal{D}$ (Definition 7.1.4), along with the morphisms (Definition 7.1.7) thereof, form a (in general large) category (Definition 1.1.1) often denoted by notations such as $\mathbf{PreShv}(X, \mathcal{D})$ (♠ TODO: include more notations) (or $\mathbf{PreShv}(X)$ if the category $\mathcal{D}$ is clear). If $\mathcal{D}$ is locally small (Definition 1.1.2), then so is $\mathbf{PreShv}(X, \mathcal{D})$.

Similarly, the sheaves on X valued in $\mathcal{D}$ (Definition 7.1.6), along with the morphisms (Definition 7.1.7) thereof, form a (in general large) category (Definition 1.1.1) often denoted by notations such as $\mathbf{Shv}(X, \mathcal{D})$ (♠ TODO: include more notations) (or $\mathbf{Shv}(X)$ if the category $\mathcal{D}$ is clear). The category $\mathbf{Shv}(X, \mathcal{D})$ is a full subcategory (Definition 1.3.7) of $\mathbf{PreShv}(X, \mathcal{D})$.

Equivalently, the categories of presheaves and sheaves are the categories $\mathbf{PreShv}(\mathbf{Open}(X), \mathcal{D})$ and $\mathbf{Shv}(\mathbf{Open}(X), \mathcal{D})$ of presheaves (Definition 7.1.1) and sheaves (Definition B.0.8) where

$\mathbf{Open}(X)$ (Definition 7.1.2) is the category of open subsets of X equipped with its usual (Definition B.0.7) Grothendieck pretopology (Definition B.0.4).

Definition 7.3.2. Let $\mathcal{A}$ be a category (Definition 1.1.1) with a terminal object. Let X be a topological space (Definition C.0.5).

1. Let $\mathcal{F}$ be a presheaf (Definition 7.1.4) on X valued in $\mathcal{A}$.

A **sheafification** of $\mathcal{F}$ is a pair $(\mathcal{F}^+, \theta)$, where $\mathcal{F}^+$ is a sheaf on X and $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$ is a morphism of presheaves, satisfying the following universal property:

For any sheaf $\mathcal{G}$ on X and any morphism of presheaves $\phi : \mathcal{F} \rightarrow \mathcal{G}$, there exists a unique morphism of sheaves $\psi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\psi \circ \theta = \phi$.

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\theta} & \mathcal{F}^+ \\ & \searrow \phi & \downarrow \psi \\ & & \mathcal{G} \end{array}$$

2. A **sheafification functor** refers to a functor

$$a : \mathbf{PreShv}(X, \mathcal{A}) \rightarrow \mathbf{Shv}(X, \mathcal{A})$$

that is left adjoint (Definition 2.5.1) to the inclusion functor

$$i : \mathbf{Shv}(X, \mathcal{A}) \hookrightarrow \mathbf{PreShv}(X, \mathcal{A}).$$

If such a sheafification functor exists, then it is unique up to unique natural isomorphism. Given a presheaf P , the sheafification $a(P)$ is also sometimes called the **sheaf associated to P** . The sheafification functor on $\mathbf{PreShv}(X, \mathcal{A})$ is a special case of the general definition of sheafification functor (Definition B.0.14), applied to the site (Definition B.0.5) whose underlying category is $\mathbf{Open}(X)$ (Definition 7.1.2) and whose Grothendieck topology is the canonical one (Definition B.0.7).

If $\mathcal{A}$ is a category admitting direct colimits (Definition 2.2.13) (e.g. the category of sets, groups, abelian groups, modules over rings, or vector spaces over fields), then the sheaf $\mathcal{P}^+$ associated to a presheaf $\mathcal{P}$ valued in $\mathcal{A}$ may be constructed by assigning to each open set $U \subseteq X$ the set (or object in $\mathcal{D}$)

$$\mathcal{P}^+(U) := \left\{ s = (s_x)_{x \in U} \in \prod_{x \in U} \mathcal{P}_x \left| \begin{array}{l} \forall x \in U, \\ \exists \text{ an open } V \subseteq U \text{ with } x \in V, \\ \exists t \in \mathcal{P}(V) \text{ such that} \\ \forall y \in V, s_y = t_y \end{array} \right. \right\}.$$

where $\mathcal{P}_x$ is the stalk (Definition 7.5.1) of $\mathcal{P}$ at x , and t_y is the germ (Definition 7.5.1) of t at y . In particular, $\mathcal{P}^+$ exists, and $\mathcal{P} \mapsto \mathcal{P}^+$ is a sheafification functor.

Equivalently, the assignment $\mathcal{P} \mapsto \mathcal{P}^+$ is the sheafification functor as defined in Definition B.0.14 in the case that the site is the site of opens (Definition B.0.7) on the topological space X . See also Theorem B.0.15.

Definition 7.3.3 ((Co)limits in Presheaf Categories). (♠ **TODO: verify**) Let $\mathcal{C}$ be a category and $\mathcal{A}$ a category. Let I be an index category.

1. If $\mathcal{A}$ admits colimits of shape I , the **presheaf colimit** of a diagram $F : I \rightarrow \text{PreShv}(\mathcal{C}, \mathcal{A})$ is the presheaf defined pointwise by

$$(\varinjlim_{i \in I} F(i))(U) := \varinjlim_{i \in I} (F(i)(U))$$

for every $U \in \text{ob}(\mathcal{C})$.

2. If $\mathcal{A}$ admits limits of shape I , the **presheaf limit** of a diagram $F : I \rightarrow \text{PreShv}(\mathcal{C}, \mathcal{A})$ is the presheaf defined pointwise by

$$(\varprojlim_{i \in I} F(i))(U) := \varprojlim_{i \in I} (F(i)(U))$$

for every $U \in \text{ob}(\mathcal{C})$.

Definition 7.3.4 (Limits in Sheaf Categories). (♠ TODO: verify) Let $(\mathcal{C}, J)$ be a site and $\mathcal{A}$ a category admitting limits of shape I . The category $\text{Shv}(\mathcal{C}, J, \mathcal{A})$ is closed under I -indexed limits in $\text{PreShv}(\mathcal{C}, \mathcal{A})$. Specifically, for any diagram $F : I \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$, the limit exists in the category of sheaves and is preserved by the natural inclusion $i : \text{Shv}(\mathcal{C}, J, \mathcal{A}) \hookrightarrow \text{PreShv}(\mathcal{C}, \mathcal{A})$:

$$\varprojlim_{\text{Shv}} F \cong \varprojlim_{\text{PreShv}} i \circ F$$

Definition 7.3.5 (Colimits in Sheaf Categories). (♠ TODO: verify)

Let $(\mathcal{C}, J)$ be a site and $\mathcal{A}$ a category admitting colimits of shape I . Suppose there exists a **sheafification functor** (a reflector) $a : \text{PreShv}(\mathcal{C}, \mathcal{A}) \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$ left adjoint to the inclusion i . For any diagram $F : I \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$, the colimit in the category of sheaves is constructed by:

$$\varinjlim_{\text{Shv}} F := a \left(\varinjlim_{\text{PreShv}} i \circ F \right)$$

Definition 7.3.6 (Plus-Function and Exactness). (♠ TODO: verify) Let $(\mathcal{C}, J)$ be a site. For a presheaf $\mathcal{P} \in \text{PreShv}(\mathcal{C}, \mathcal{A})$, the **plus-functor** is defined by

$$\mathcal{P}^+(U) := \varinjlim_{S \in J(U)} \text{Hom}_{\text{PreShv}}(S, \mathcal{P})$$

where $J(U)$ is the filtered system of sieves covering U . If $\mathcal{A}$ is AB5 and admits colimits indexed by $J(U)$, the functor $(-)^+ : \text{PreShv} \rightarrow \text{PreShv}$ is exact. Consequently, if the sheafification functor $a \cong (+)^2$ exists, it is an exact functor.

7.4. Sections functors. Sections functors are left exact.

Proposition 7.4.1. Let $\mathcal{C}$ be an essentially small category and let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. Let $U \in \text{Ob}(\mathcal{C})$ be some fixed object. The sections functor (Definition 7.1.5)

$$\Gamma(U, -) : \text{PSh}(\mathcal{C}, \mathcal{A}) \rightarrow \mathcal{A}$$

is left exact (Definition 2.4.12). (♠ TODO: state that presheaves and sheaves valued in an abelian category form abelian categories)

2. Assume that $\Gamma(\mathcal{F})$ exists for all $\mathcal{F}$ in $\text{PSh}(\mathcal{C}, \mathcal{A})$ so that Γ is a functor

$$\Gamma : \text{PSh}(\mathcal{C}, \mathcal{A}) \rightarrow \mathcal{A}.$$

The functor Γ is left exact.

Proof. Recall that $\text{PSh}(\mathcal{C}, \mathcal{D})$ is an abelian category (??) since $\mathcal{C}$ is essentially small. (♠
 TODO: Talk about how limits are left exact and) $\square$

Proposition 7.4.2. Let X be a topological space (Definition C.0.5) and let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. The functor (Definition 1.2.2)

$$\Gamma : \text{PSh}(\mathcal{C}, \mathcal{A}) \rightarrow \mathcal{A}.$$

(Definition 7.1.4) (Definition 7.1.5) is exact (Definition 2.4.12).

2. Assume that a sheafification functor (Definition 7.3.2)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists; in particular, $\text{Shv}(X, \mathcal{A})$ is an abelian category (Proposition 7.5.3).

The functor

$$\Gamma : \text{Sh}(\mathcal{C}, \mathcal{A}) \rightarrow \mathcal{A}.$$

(Definition 7.1.6) is left exact (Definition 2.4.12).

7.5. Stalks of presheaves and sheaves on topological spaces.

Definition 7.5.1 (Stalk of a sheaf). Let X be a topological space, and let $\mathcal{D}$ be a category (Definition 1.1.1) Let $\mathcal{F}$ be a presheaf on X valued in $\mathcal{D}$ (Definition 7.1.4). For a point $x \in X$, the *stalk of $\mathcal{F}$ at x* , denoted $\mathcal{F}_x$, is defined as the filtered direct limit (Definition 2.2.13)

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U),$$

where the limit ranges over all open neighborhoods U of x in X ordered by inclusion, assuming that such a direct limit exists.

If $\mathcal{D}$ is some kind of category of sets (e.g. $\mathcal{D} = \mathbf{Sets}, \mathbf{Grps}, \mathbf{Rings}$), then an element of $\mathcal{F}_x$ is called a *germ of a section at x* . Concretely, a germ at x is given by a pair (U, s) with U an open neighborhood of x and $s \in \mathcal{F}(U)$, modulo the equivalence relation: $(U, s) \sim (V, t)$ if there exists an open neighborhood $W \subseteq U \cap V$ of x such that $s|_W = t|_W$, see also ??.

If $\mathcal{F}$ is a presheaf of groups, rings, or modules, then each stalk $\mathcal{F}_x$ inherits some corresponding algebraic structure. More generally, if $\mathcal{D}$ is so that $\mathcal{F}_x$ exists for every presheaf $\mathcal{F} \in \text{PreShv}(X, \mathcal{D})$ (resp. every sheaf $\mathcal{F} \in \text{Sh}(X, \mathcal{D})$), then $\mathcal{F} \mapsto \mathcal{F}_x$ is a functor $\text{PreShv}(X, \mathcal{D}) \rightarrow \mathcal{D}$ (resp. $\text{Sh}(X, \mathcal{D}) \rightarrow \mathcal{D}$).

This notion of stalk is a specialization of the more general notion of a stalk of an object of a topos (Definition B.0.9) at a point of said topos (Definition B.0.19)

Proposition 7.5.2. (♠ TODO: prove that if $\mathcal{A}$ is AB5 and sheafification exists, then the sheafification is exact; apparently this should be true on a general small site)

Proposition 7.5.3. (♠ TODO: verify this for general $\mathcal{A}$; this is certainly true for modules.)
 (♠ TODO: prove that if $\mathcal{A}$ is AB5 and sheafification exists, then the sheafification is exact)
 Let X be a topological space (Definition C.0.5). (♠ TODO: verify these)

1. If $\mathcal{A}$ is a pre-additive category (Definition 2.3.4), then $\text{PreShv}(X, \mathcal{A})$ (Definition 7.1.4) and $\text{Shv}(X, \mathcal{A})$ (Definition 7.1.6) are pre-additive (Definition 2.4.1).
2. If $\mathcal{A}$ is an additive category (Definition 2.3.4), then $\text{PreShv}(X, \mathcal{A})$ and $\text{Shv}(X, \mathcal{A})$ are additive.
3. Suppose that $\mathcal{A}$ is an abelian category (Definition 2.3.12).
 - (a) The category $\text{PreShv}(X, \mathcal{A})$ is abelian (Definition 2.3.12).
 - (b) Assuming that a sheafification functor (Definition B.0.14)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists and is exact (Definition 2.4.12), $\text{Shv}(X, \mathcal{A})$ is abelian.

4. Suppose that $\mathcal{A}$ is an $\text{AB}n$ category for $n \in \{3, 4, 5\}$.
 - (a) $\text{PreShv}(X, \mathcal{A})$ is $\text{AB}n$.
 - (b) Assume that a sheafification functor (Definition B.0.14)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists. Further suppose that a is exact for $n \in \{4, 5\}$, $\text{Shv}(X, \mathcal{A})$ is $\text{AB}n$. In particular, the exactness condition is not needed for $n = 3$. Moreover, the exactness condition is always satisfied for $n = 5$.

5. Suppose that $\mathcal{A}$ is AB5 and has enough injectives (Definition 4.2.5).
 - (a) $\text{PreShv}(X, \mathcal{A})$ has enough injectives.
 - (b) Assuming that a sheafification functor (Definition B.0.14)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists, $\text{Shv}(X, \mathcal{A})$ has enough injectives.

6. If $\mathcal{A}$ is a Grothendieck category (Definition 2.5.7), then so are $\text{PreShv}(X, \mathcal{A})$ and $\text{Shv}(X, \mathcal{A})$.

Proof. 1. (♠ TODO:)

□

Proposition 7.5.4. (♠ TODO: verify this for general $\mathcal{A}$; this is certainly true for modules.)
 Let $\mathcal{A}$ be an AB5 category (Definition 2.5.7). Let X be a topological space (Definition C.0.5), and let $x \in X$ be a point.

1. The stalk (Definition 7.5.1) functor $\mathcal{F} \mapsto \mathcal{F}_x$ is an exact functor (Definition 2.4.12) $\text{PreShv}(X, \mathcal{A}) \rightarrow \mathcal{A}$ (Definition 7.1.4).
2. Assume that a sheafification (Definition 7.3.2) functor

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists. The stalk functor $\mathcal{F} \mapsto \mathcal{F}_x$ is an exact functor $\text{Shv}(X, \mathcal{A}) \rightarrow \mathcal{A}$.

3. Assume that a sheafification (Definition 7.3.2) functor

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists. A sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$$

is exact (Definition 2.4.6) if and only if the induced sequences

$$0 \rightarrow \mathcal{F}_x \rightarrow \mathcal{G}_x \rightarrow \mathcal{H}_x \rightarrow 0$$

on stalks are exact for all $x \in X$.

Proof. (♠ TODO:)

□

(♠ TODO: define surjection, injection of sheaves of sets)

Proposition 7.5.5. (♠ TODO: verify this for general $\mathcal{A}$; it is true for $\mathcal{A} = \mathbf{Sets}$) Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be a category (Definition 1.1.1) with a final object (Definition 2.3.1). Assume that for all $x \in X$ and all $\mathcal{F} \in \mathbf{Sh}(X, \mathcal{A})$, the stalks (Definition 7.5.1) $\mathcal{F}_x$ exist in $\mathcal{A}$.

A sheaf morphism (Definition 7.1.6) $f : \mathcal{F} \rightarrow \mathcal{G}$ in $\mathbf{Sh}(X, \mathcal{A})$ is an isomorphism if and only if all of the stalk morphisms $f : \mathcal{F}_x \rightarrow \mathcal{G}_x$ are isomorphisms.

Proof. (♠ TODO:)

□

Proposition 7.5.6. Let X be a topological space (Definition C.0.5). A morphism $f : \mathcal{F} \rightarrow \mathcal{G}$ in $\mathbf{Sh}(X, \mathbf{Sets})$ is an isomorphism (resp. monomorphism (Definition 2.3.8), resp. epimorphism (Definition 2.3.8)) if and only if all of the stalk (Definition 7.5.1) morphisms $f : \mathcal{F}_x \rightarrow \mathcal{G}_x$ are bijections (resp. injections, resp. surjections). In particular, $\mathbf{Open}(X)$ (Definition 7.1.2) is a site (Definition B.0.5) with enough points.

7.5.1. Sheaf theoretic statements of analysis and manifolds.

Example 7.5.7. One function studied in complex analysis is the complex logarithm function $\log(z)$. Consider the following (true) statements:

- $\log(z)$ can be locally defined as a holomorphic function at any point in $X = \mathbb{C} \setminus \{0\}$.
- $\log(z)$ cannot be globally defined as a holomorphic function on X .

Let $\mathcal{O}$ be the sheaf on X of holomorphic functions (i.e. $\mathcal{O}(U) = \{f : U \rightarrow \mathbb{C} \text{ holomorphic}\}$ for any $U \subseteq X$ open). There is a sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{n \mapsto 2\pi i n} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \rightarrow 1$$

(♠ TODO: ref) of sheaves of abelian groups on X . One can verify that it is a short exact sequence by checking exactness at all stalks (♠ TODO:). The first statement is tantamount to saying that $\mathcal{O} \xrightarrow{\exp} \mathcal{O}^*$ is a surjection of sheaves (♠ TODO:) — one can verify this at the level of stalks. The second statement is tantamount to saying that the map $\Gamma(X, \mathcal{O}) \xrightarrow{\exp} \Gamma(X, \mathcal{O}^*)$ on global sections is not a surjection.

(♠ TODO:) Once we study sheaf cohomology, we obtain the cohomological long exact sequence

$$\begin{aligned} 0 \rightarrow \Gamma(X, \underline{\mathbb{Z}}) \rightarrow \Gamma(X, \mathcal{O}) \xrightarrow{\exp} \Gamma(X, \mathcal{O}^*) \\ \xrightarrow{\delta} H^1(X, \underline{\mathbb{Z}}) \rightarrow \dots \end{aligned}$$

To show that $\Gamma(X, \mathcal{O}) \xrightarrow{\exp} \Gamma(X, \mathcal{O}^*)$ is not a surjection, one can then show that the image of the connecting morphism $\delta : \Gamma(X, \mathcal{O}^*) \rightarrow H^1(X, \underline{\mathbb{Z}})$ is not zero.

7.6. Sheaf of rings, ringed spaces, and sheaves of modules.

Definition 7.6.1 (Ringed space). A **ringed space** is a pair $(X, \mathcal{O}_X)$ where

- X is a topological space (Definition C.0.5), and
- $\mathcal{O}_X$ is a sheaf of (Definition 7.1.6) commutative rings (Definition C.0.9) on X .

Equivalently, a ringed space is a ringed site (Definition B.0.17) where the site is the site of opens (Definition B.0.7) of the topological space X . The sheaf $\mathcal{O}_X$ may be suppressed from the notation and only X may be used to denote a ringed space. The sheaf $\mathcal{O}_X$, also commonly denoted by $\mathcal{O}_X$, is called the **structure sheaf of X** .

Definition 7.6.2. 1. Let $\mathcal{C}$ be a site (Definition B.0.5), and let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition B.0.8) of (not necessarily commutative) rings (Definition C.0.7) on $\mathcal{C}$.

- (a) An **$(\mathcal{A}, \mathcal{B})$ -bimodule** (or a **bimodule over $(\mathcal{A}, \mathcal{B})$**) is a sheaf (Definition B.0.8) $\mathcal{M}$ of abelian groups on $\mathcal{C}$ equipped with a left $\mathcal{A}$ -module structure given by a morphism of sheaves (Definition B.0.8) of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$ -module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every object U in $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$\lambda_U(a, \rho_U(m, b)) = \rho_U(\lambda_U(a, m), b)$$

holds in $\mathcal{M}(U)$. In standard multiplicative notation where $\lambda(a, m)$ is denoted $a \cdot m$ and $\rho(m, b)$ is denoted $m \cdot b$, this condition is the associativity axiom

$$(a \cdot m) \cdot b = a \cdot (m \cdot b).$$

In particular, for every object $U \in \mathcal{C}$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$ -bimodule (Definition 2.1.1).

- (b) Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$ -bimodules. A **homomorphism of $(\mathcal{A}, \mathcal{B})$ -bimodules** (or an **$(\mathcal{A}, \mathcal{B})$ -linear morphism**) is a morphism of sheaves of abelian groups $f : \mathcal{M} \rightarrow \mathcal{N}$ such that for every object U of $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$ -bimodules, with morphisms being morphisms of sheaves of abelian groups compatible with both the left $\mathcal{A}$ -action and the right $\mathcal{B}$ -action, by $\mathcal{A}\text{-}\mathcal{B}\text{-Mod}$ or sometimes by ${}_{\mathcal{A}}\text{Mod}_{\mathcal{B}}$ (♠ TODO: talk about how bimodules can be identifies with left/right modules)

2. Let $(\mathcal{C}, J)$ be a site (Definition B.0.5). Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on $(\mathcal{C}, J)$ (Definition B.0.8), i.e. $((\mathcal{C}, J), \mathcal{O})$ is a ringed site (Definition B.0.17).

(a) An *(left/right/two-sided) $\mathcal{O}$ -module* consists of the following data:

- A sheaf $\mathcal{F}$ of abelian groups on $(\mathcal{C}, J)$,
- for every object $U \in \mathcal{C}$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$ -module on $\mathcal{F}(U)$,

such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

is $\mathcal{O}(U)$ -linear when the $\mathcal{O}(U)$ -action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \rightarrow \mathcal{O}(V)$$

induced by f .

- (b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$ -modules (Definition B.0.18).

A *morphism of $\mathcal{O}$ -modules* $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves (Definition B.0.8) of abelian groups such that, for every object $U \in \mathcal{C}$, the component map

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is $\mathcal{O}(U)$ -linear, i.e. it satisfies

$$\varphi_U(r \cdot s) = r \cdot \varphi_U(s) \quad \text{for all } r \in \mathcal{O}(U), s \in \mathcal{F}(U).$$

The collection of all $\mathcal{O}$ -modules together with their morphisms of $\mathcal{O}$ -modules forms the *category of $\mathcal{O}$ -modules*, denoted $\mathbf{Mod}(\mathcal{O})$. The Hom functors $\text{Hom}_{\mathbf{Mod}(\mathcal{O})}$ of this category may be denoted by $\mathbf{Hom}_{\mathcal{O}}$.

In case that a sheafification functor (Definition B.0.14)

$$\text{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \text{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists, a left, right, two-sided $\mathcal{O}$ -module (and morphisms thereof) is equivalent to a $(\mathcal{O}, \mathbb{Z})$ -bimodule, $(\mathbb{Z}, \mathcal{O})$ -bimodule, and $(\mathcal{O}, \mathcal{O})$ -bimodule (and morphisms thereof) respectively, where $\mathbb{Z}$ is the constant sheaf of the integer ring $\mathbb{Z}$. In particular, the category of $(\mathbb{Z}, \mathbb{Z})$ -bimodules is equivalent to the category of sheaves of abelian groups on $\mathcal{C}$.

Definition 7.6.3. Let X be a topological space (Definition C.0.5). Let $\mathcal{A}, \mathcal{B}$ be sheaves (Definition 7.1.6) of rings on X . (♠ TODO: continue) These notions are merely special cases of those of Definition B.0.18

Theorem 7.6.4 (e.g. see [Sta25, Tag 01DU]). For any site (Definition B.0.5) $(\mathcal{C}, J)$ on an essentially small category $\mathcal{C}$ and a sheaf of rings (Definition B.0.8) $\mathcal{O}$ on $\mathcal{C}$, the category $\mathbf{Mod}(\mathcal{O})$ of $\mathcal{O}$ -modules (Definition B.0.18) is an abelian category (Definition 2.3.12) that has enough injectives (Definition 4.2.5). In fact, there is a functorial injective embedding (♠ TODO: what does this mean?)

(♠ TODO: talk about how O-modules form a grothendieck abelian category)

(♠ TODO: do definitions of sheaf hom for topological spaces)

Definition 7.6.5 (Sheafified Hom). Let $(\mathcal{D}, K)$ be a site (Definition B.0.5) that admits a small topologically generating family (Definition B.0.5) (e.g. by being essentially small) and let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves of rings on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf (Definition B.0.8) of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition B.0.18) and $\mathcal{N}$ be a sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules.

The *sheafified Hom*, denoted by notations such as $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, or $\underline{\text{Hom}}_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, is the sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules defined section-wise by:

$$\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})(U) = \text{Hom}_{\text{Sh}(\mathcal{D}/U, K|_U; \text{Mod}(\mathcal{R}|_U))}(\mathcal{M}|_U, \mathcal{N}|_U).$$

(??) (??) (??)

Equivalently, it is the unique sheaf which satisfies the right adjoint property relative to the sheafified tensor product (Theorem 7.6.7).

Definition 7.6.6 (Sheafified Tensor Product). (♠ TODO: Consolidate with) Let $(\mathcal{D}, K)$ be a site (Definition B.0.5) that admits a small topologically generating family (Definition B.0.5) (e.g. by being essentially small) and let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves of rings (Definition B.0.8) on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf (Definition B.0.8) of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition B.0.18) and $\mathcal{N}$ be a sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules. The *sheafified tensor product*, denoted $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$, is the sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules defined as the sheafification (Definition B.0.14) of the section-wise tensor product presheaf:

$$U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U).$$

(Definition 2.1.9) Explicitly, $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N} = a(U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U))$, where a is the sheafification functor, which exists by Theorem B.0.15. (♠ TODO: to apply the theorem, it seems necessary to show that the category of sheaves of $\mathcal{R}$ - $\mathcal{T}$ bimodules is complete, cocomplete (probably need sheafification of pointwise construction), has exact filtered colimits, and the IPC property)

In particular, we have a bi-additive functor (Definition 2.4.2)

$$\otimes : {}_{\mathcal{R}}\text{Mod}_{\mathcal{S}} \times {}_{\mathcal{S}}\text{Mod}_{\mathcal{T}} \rightarrow {}_{\mathcal{R}}\text{Mod}_{\mathcal{T}}$$

Theorem 7.6.7 (Tensor-Hom Adjunction for Sheaves of Bimodules). Let $(\mathcal{D}, K)$ be a site (Definition B.0.5) that admits a small topologically generating family (Definition B.0.5) (e.g. by being essentially small). Let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves (Definition B.0.8) of rings on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition B.0.18), $\mathcal{N}$ be a sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules, and $\mathcal{P}$ be a sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules.

Then there is a natural isomorphism:

$$\text{Hom}_{\text{Sh}(\mathcal{R}-\mathcal{T})}(\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}, \mathcal{P}) \cong \text{Hom}_{\text{Sh}(\mathcal{S}-\mathcal{T})}(\mathcal{N}, \mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{P}))$$

where $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$ is the sheafified tensor product and $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{P})$ is the sheaf of local $\mathcal{R}$ -linear homomorphisms.

Proof. (♠ TODO: Verify this proof We apply ?? by specifying the categories $\mathcal{A}, \mathcal{B}, \mathcal{C}$ and the functors F, G as follows:

1. **Categories:** Let $\mathcal{A}$ be the category whose objects are triples (R, S, M) where M is an R - S bimodule. Define $\mathcal{B}$ and $\mathcal{C}$ similarly for S - T and R - T bimodules.
2. **Base Functors:** Define $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ by $(M, N) \mapsto M \otimes_S N$ and $G : \mathcal{A}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{B}$ by $(M, P) \mapsto \text{Hom}_R(M, P)$. At each component, $F(A, -) \dashv G(A, -)$ is the standard algebraic tensor-hom adjunction for bimodules.
3. **Sheaf Interpretation:** The sheaves $\mathcal{M}, \mathcal{N}, \mathcal{P}$ are viewed as objects in $\text{Sh}(\mathcal{D}, K; \mathcal{A})$, $\text{Sh}(\mathcal{D}, K; \mathcal{B})$, and $\text{Sh}(\mathcal{D}, K; \mathcal{C})$ respectively, where the ring components of these sheaves are fixed to be the sheaves $\mathcal{R}, \mathcal{S}, \mathcal{T}$.

By ??, the sheafy functor $\mathcal{F}$ associated to the tensor product is:

$$\mathcal{F}(\mathcal{M}, \mathcal{N}) = a(U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U))$$

which is precisely the definition of the sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$.

The right adjoint functor $\mathcal{G}_{\mathcal{M}}$ is defined section-wise by:

$$\mathcal{G}_{\mathcal{M}}(\mathcal{P}) : U \mapsto \text{Hom}_{\mathcal{R}(U)}(\mathcal{M}(U), \mathcal{P}(U))$$

Per the theorem, since G is a right adjoint, $\mathcal{G}_{\mathcal{M}}(\mathcal{P})$ is a sheaf. In the category of modules, this section-wise Hom sheaf is naturally isomorphic to the internal sheaf $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{P})$ because any morphism of sheaves $\mathcal{M}|_U \rightarrow \mathcal{P}|_U$ is uniquely determined by its action on sections when the sheafification of the section-wise Hom agrees with the internal Hom.

Thus, the general adjunction $\mathcal{F}(\mathcal{M}, -) \dashv \mathcal{G}_{\mathcal{M}}(-)$ specializes to the required isomorphism of Hom-sets in the category of sheaves of bimodules.) $\square$

7.7. Pullbacks and pushforwards.

Definition 7.7.1 (Pullback (inverse image) of a sheaf). Let $f : X \rightarrow Y$ be a continuous map between topological spaces (Definition C.0.5). Let $\mathcal{D}$ be a category with a terminal object (Definition 2.3.1).

1. Let $\mathcal{G}$ be a presheaf on Y valued in a $\mathcal{D}$ (Definition 7.1.4). The **pullback** or **inverse image presheaf** $f^{-1}\mathcal{G}$ on X is defined as the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V)$$

provided these colimits (Definition 2.2.8) exist for all U , where U ranges over open subsets of X and the colimit is taken over all open subsets $V \subseteq Y$ containing $f(U)$. This construction admits a natural isomorphism

$$(f^{-1}\mathcal{G})_x \rightarrow \mathcal{G}_{f(x)}$$

of stalks (Definition 7.5.1) for every $x \in X$.

2. If $\mathcal{G}$ is a sheaf (Definition 7.1.6) valued in $\mathcal{D}$, then we can define the **pullback** or **inverse image sheaf** $f^*\mathcal{G}$ on X as the sheaf associated to the presheaf (Definition B.0.14) $f^{-1}\mathcal{G}$, assuming it exists.

If $\mathcal{D}$ is the category of sets (Definition 1.1.7), of groups, of abelian groups, of modules over rings, for example, then the relevant colimits (Definition 2.2.8) needed to define the inverse image presheaves and the sheafification needed to define the inverse image sheaves exist.

Definition 7.7.2 (Pushforward (direct image) of a sheaf). Let $f : X \rightarrow Y$ be a continuous map between topological spaces (Definition C.0.5), and let $\mathcal{F}$ be a presheaf on X valued in a category $\mathcal{D}$ (Definition 7.1.4) with a terminal object. The *pushforward* or *direct image presheaf* $f_*\mathcal{F}$ on Y is the presheaf valued in $\mathcal{D}$ on Y (Definition 7.1.1) defined as follows: For every open set $V \subseteq Y$, the value of the pushforward is given by

$$f_*\mathcal{F}(V) := \mathcal{F}(f^{-1}(V)).$$

For an inclusion of open sets $V' \subseteq V$ in Y , the restriction morphism

$$\text{res}_{V,V'}^{f_*\mathcal{F}} : f_*\mathcal{F}(V) \rightarrow f_*\mathcal{F}(V')$$

is defined as the restriction morphism of $\mathcal{F}$ associated with the inclusion of preimages $f^{-1}(V') \subseteq f^{-1}(V)$ in X :

$$\text{res}_{f^{-1}(V), f^{-1}(V')}^{\mathcal{F}} : \mathcal{F}(f^{-1}(V)) \rightarrow \mathcal{F}(f^{-1}(V')).$$

If $\mathcal{F}$ is a sheaf (Definition 7.1.6), then so is $f_*\mathcal{F}$.

(♠ TODO: comment on what is happening on the stalks)

Definition 7.7.3. Let $i : X \hookrightarrow Y$ be the embedding of a subspace (Definition C.0.39) X into a topological space Y . Let $\mathcal{D}$ be a category with a terminal object (Definition 2.3.1).

1. Let $\mathcal{G}$ be a presheaf on Y valued in $\mathcal{D}$. The *restriction of $\mathcal{G}$ to X* is defined as the inverse image presheaf (Definition 7.7.1) $i^{-1}\mathcal{G}$, assuming it exists. It is often denoted by $\mathcal{G}|_X$.
2. Let $\mathcal{G}$ be a sheaf on Y valued in $\mathcal{D}$. The *restriction of $\mathcal{G}$ to X* is defined as the inverse image sheaf (Definition 7.7.1) $i^*\mathcal{G}$, assuming it exists. It is often denoted by $\mathcal{G}|_X$.

Note, in particular, that if $i : X \hookrightarrow Y$ is the embedding of an open subspace, then $\mathcal{G}|_X$ is the functor $\mathbf{Open}(X)^{\text{op}} \rightarrow \mathcal{D}$ given by $U \mapsto \mathcal{G}(i(U))$.

If $\mathcal{D}$ is the category of sets (Definition 1.1.7), of groups, of abelian groups, of modules over rings, for example, then the relevant colimits (Definition 2.2.8) needed to define the restriction presheaves and the sheafification needed to define the restriction sheaves exist.

Lemma 7.7.4. (♠ TODO: check for general categories) Let X be a topological space (Definition C.0.5). Let $i_x : \{x\} \hookrightarrow X$ be the inclusion of a point. Let $\mathcal{A}$ be a category (Definition 1.1.1). For any presheaf $\mathcal{F}$ on X valued in $\mathcal{A}$,

1. The restriction (Definition 7.7.3) $\mathcal{F}|_{\{x\}} = i_x^*\mathcal{F}$ (Definition 7.7.1) exists if and only if the stalk (Definition 7.5.1) $\mathcal{F}_x$ does.
2. $\mathcal{F}|_{\{x\}}$ is the presheaf on $\{x\}$ corresponding to (Lemma 7.1.10) $\mathcal{F}_x$.

Theorem 7.7.5. Let $f : X \rightarrow Y$ be a continuous map between topological spaces (Definition C.0.5). Let $\mathcal{A}$ be a locally small category (Definition 1.1.2) such that $\mathcal{A}$ is cocomplete (Definition 2.2.11), i.e. admits small colimits (Definition 2.2.10), and such that $\text{PreShv}(Y, \mathcal{A})$

admits sheafification (Definition 7.3.2); for example, the categories of sets, of groups, of abelian groups, of rings, of modules over a ring, etc. satisfy these hypotheses.

The inverse image functor (Definition 7.7.1)

$$f^* : \mathrm{Sh}(Y, \mathcal{A}) \rightarrow \mathrm{Sh}(X, \mathcal{A})$$

and the direct image functor (Definition 7.7.2)

$$f_* : \mathrm{Sh}(X, \mathcal{A}) \rightarrow \mathrm{Sh}(Y, \mathcal{A})$$

form an adjoint pair (Definition 2.5.1) $f^* \dashv f_*$, i.e., for any sheaves $\mathcal{F} \in \mathrm{Sh}(Y, \mathcal{A})$ and $\mathcal{G} \in \mathrm{Sh}(X, \mathcal{A})$, there is a natural isomorphism

$$\mathrm{Hom}_{\mathrm{Sh}(X, \mathcal{A})}(f^* \mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{Sh}(Y, \mathcal{A})}(\mathcal{F}, f_* \mathcal{G}).$$

Theorem 7.7.8 is an application of the properties of inverse images (Definition 7.7.1) and direct images (Definition 7.7.2); we can show that the category of $\mathcal{O}_X$ -modules has enough injectives (Definition 4.2.5) (cf. Theorem 7.6.4)

Lemma 7.7.6. Let $\mathcal{C}$ be a category (Definition 1.1.1).

1. If $\{I_\alpha\}_\alpha$ is a collection of injective objects (Definition 4.2.2) in $\mathcal{C}$ such that $\prod_\alpha I_\alpha$ (Definition 2.2.1) exists in $\mathcal{C}$, then $\prod_\alpha I_\alpha$ is injective.
2. Dually, if $\{I_\alpha\}_\alpha$ is a collection of projective objects (Definition 4.2.2) in $\mathcal{C}$ such that $\prod_\alpha I_\alpha$ (Definition 2.2.1) exists in $\mathcal{C}$, then $\prod_\alpha I_\alpha$ is projective.

Proof. We prove the statement about injective objects. Suppose that $m : A \rightarrow B$ is a monomorphism in $\mathcal{C}$, and let $f : A \rightarrow \prod_\alpha I_\alpha$ be a morphism. Let $f_\alpha = \mathrm{proj}_\alpha \circ f : A \rightarrow I_\alpha$ where $\mathrm{proj}_\alpha : \prod_\alpha I_\alpha \rightarrow I_\alpha$ is the projection. Since I_α is injective, there exist morphisms $\tilde{f}_\alpha : B \rightarrow I_\alpha$ such that $\tilde{f}_\alpha \circ m = f_\alpha$. Let $\tilde{f} : B \rightarrow \prod_\alpha I_\alpha$ be the canonical morphism induced by the $\tilde{f}_\alpha$ by the universal property of the product. To show that $\tilde{f} \circ m = f$, it is tantamount to show that post-composing these maps with the projections proj_α results in the same maps for all α — doing so on $\tilde{f} \circ m$ results in $\mathrm{proj}_\alpha \circ (\tilde{f} \circ m) = \tilde{f}_\alpha \circ m = f_\alpha$ and doing so on f results in f_α . $\square$

Lemma 7.7.7. Let $(X, \mathcal{O}_X)$ be a ringed space. Let $i_x : \{x\} \hookrightarrow X$ be the embedding of a point x of X . Let I_x be an injective (Definition 4.2.10) $\mathcal{O}_{X,x}$ (Definition 7.5.1)-module.

The skyscraper sheaf (Definition 7.1.11) $(i_x)_* \underline{I_x}$ (Definition 7.7.2) of the (constant sheaf (Definition 7.1.9) whose value is the) stalk (Definition 7.5.1) is an injective object (Definition 4.2.2) of $\mathrm{Mod}(\mathcal{O}_X)$ (Definition 7.6.3).

Proof. By Proposition 4.2.4, it is tantamount to show that the Hom-functor $\mathrm{Hom}_{\mathrm{Mod}(\mathcal{O}_X)}(-, (i_x)_* \underline{I_x}) : \mathrm{Mod}(\mathcal{O}_X) \rightarrow \mathrm{Ab}$ (Definition 1.1.8) is exact (Definition 2.4.12). By abuse of notation, write $\mathcal{O}_{X,x}$ for inverse image $i_x^*(\mathcal{O}_X)$, which is a sheaf of rings on $\{x\}$; by Lemma 7.7.4, it can be identified with (Lemma 7.1.10) the stalk $\mathcal{O}_{X,x}$.

For all $\mathcal{A} \in \mathrm{Mod}(\mathcal{O}_X)$, we have

$$\mathrm{Hom}_{\mathrm{Mod}(\mathcal{O}_X)}(\mathcal{A}, (i_x)_* \underline{I_x}) \cong \mathrm{Hom}_{\mathrm{Mod}(\mathcal{O}_{X,x})}(i_x^* \mathcal{A}, \underline{I_x}) \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, I_x)$$

where the rightmost Hom is that of modules over the stalk $\mathcal{O}_{X,x}$. The stalk functor $\mathcal{A} \mapsto \mathcal{A}_x$ is exact (Proposition 7.5.4) and $\text{Hom}_{\mathcal{O}_{X,x}}(-, I_x)$ is also exact (Proposition 4.2.4), so

$$\mathcal{A} \mapsto \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, I_x) \cong \text{Hom}_{\text{Mod}(\mathcal{O}_X)}(\mathcal{A}, (i_x)_* \underline{I_x})$$

is exact as desired. $\square$

Theorem 7.7.8 (e.g. see [Sta25, Tag 01DU]). Let $(X, \mathcal{O}_X)$ be a ringed space. The category $\mathbf{Mod}(\mathcal{O}_X)$ of $\mathcal{O}_X$ -modules is an abelian category (Definition 2.3.12) that has enough injectives (Definition 4.2.5).

This fact is merely a special case of Theorem 7.6.4.

Proof. Let $\mathcal{F} \in \mathbf{Mod}(\mathcal{O}_X)$. For every point $x \in X$, the stalk $\mathcal{F}_x$ is a $\mathcal{O}_{X,x}$ -module (Definition 2.1.1). The category of $\mathcal{O}_{X,x}$ -modules has enough injectives (Theorem 4.2.6), so there exists an injective object I_x and a monomorphism $\mathcal{F}_x \hookrightarrow I_x$ of $\mathcal{O}_{X,x}$ -modules.

Define the following sheaf:

$$\mathcal{I} = \prod_{x \in X} (i_x)_* I_x$$

(Definition 7.3.4) (Definition 2.2.1) (Definition 7.1.11) (Definition 7.7.2) where $(i_x)_* I_x$ is the skyscraper sheaf at x with value I_x .

We produce a natural morphism $\mathcal{F} \rightarrow \mathcal{I}$ — note that products for sheaves of sets are constructible as products of presheaves (without needing to apply sheafification), so this is tantamount to producing morphisms $\mathcal{F} \rightarrow (i_x)_* \underline{I_x}$ for every x . By the adjunction (Definition 2.5.1) of invers images (Definition 7.7.1) and direct images (Definition 7.7.2) (Theorem 7.7.5), such a morphism corresponds to a morphism $(i_x)^* \mathcal{F} \rightarrow \underline{I_x}$, which is equivalent to (Lemma 7.7.4) a morphism $\mathcal{F}_x \rightarrow I_x$ — we already have such a morphism and in fact an injection.

(♠ TODO: injection of sheaves) The morphism $\mathcal{F} \rightarrow \mathcal{I}$ is a monomorphism, i.e. an injection of sheaves, because (Proposition 7.5.5) the induced morphism $\mathcal{F}_x \rightarrow \mathcal{I}_x$ on stalks is simply $\mathcal{F}_x \rightarrow I_x$ (♠ TODO: justify), which is injective. Moreover, $\mathcal{I}$ is an injective object by Lemma 7.7.7. Therefore, $\mathbf{Mod}(\mathcal{O}_X)$ indeed has enough injectives. $\square$

Definition 7.7.9 (Godement Sheaf). Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1) and $\mathcal{F} \in \text{Mod}(\mathcal{O}_X)$. The *Godement sheaf associated to $\mathcal{F}$* , denoted by $\mathcal{C}^0(\mathcal{F})$, is defined as the product of the skyscraper sheaves (Definition 7.1.11) of its stalks (Definition 7.5.1):

$$\mathcal{C}^0(\mathcal{F}) := \prod_{x \in X} (i_x)_* \underline{\mathcal{F}_x}$$

(Definition 7.3.4) (Definition 2.2.1) where $i_x : \{x\} \hookrightarrow X$ is the inclusion of the point x and $\underline{\mathcal{F}_x}$ denotes the constant sheaf (Definition 7.1.9) on $\{x\}$ with value the stalk (Definition 7.5.1) $\mathcal{F}_x$.

There is a canonical monomorphism of $\mathcal{O}_X$ -modules:

$$d^{-1} : \mathcal{F} \hookrightarrow \mathcal{C}^0(\mathcal{F})$$

which, by the adjunction (Definition 2.5.1) $(i_x^*, (i_x)_*)$, corresponds at each factor to the identity map on stalks $\text{id} : \mathcal{F}_x \rightarrow \mathcal{F}_x$.

Definition 7.7.10. Let $\mathcal{F}$ be a sheaf of sets (Definition 7.1.6) on a topological space (Definition C.0.5) X . The sheaf $\mathcal{F}$ is called *flasque* or *flabby* if for every open $U \subseteq X$, the restriction map (Definition 7.7.3)

$$\rho_{X,U} : \mathcal{F}(X) \rightarrow \mathcal{F}(U)$$

is surjective.

Lemma 7.7.11. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1). For any $\mathcal{F} \in \text{Mod}(\mathcal{O}_X)$, the Godement sheaf (Definition 7.7.9) $\mathcal{C}^0(\mathcal{F})$ is flasque (flabby).

Proof. (♠ TODO: verify) A section of $\mathcal{C}^0(\mathcal{F})$ over an open set $U \subseteq X$ is an element of $\prod_{x \in U} \mathcal{F}_x$. Given such a section $s = (s_x)_{x \in U}$, we can define a global section $\tilde{s} = (\tilde{s}_x)_{x \in X}$ by setting $\tilde{s}_x = s_x$ if $x \in U$ and $\tilde{s}_x = 0$ if $x \notin U$. The restriction of $\tilde{s}$ to U is clearly s , hence the restriction map $\Gamma(X, \mathcal{C}^0(\mathcal{F})) \rightarrow \Gamma(U, \mathcal{C}^0(\mathcal{F}))$ is surjective. $\square$

Theorem 7.7.12. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1). The category (Definition 1.1.1) $\text{Mod}(\mathcal{O}_X)$ (Definition 7.6.3) of $\mathcal{O}_X$ -modules has enough (Definition 4.2.1) flasque sheaves (Definition 7.7.10) (on the right).

See also Theorem 7.7.13.

Theorem 7.7.13 (Godement Resolution). Let $(X, \mathcal{O}_X)$ be a ringed space. Every $\mathcal{O}_X$ -module $\mathcal{F}$ admits a canonical resolution by flasque sheaves (Definition 7.7.10):

$$0 \rightarrow \mathcal{F} \xrightarrow{d^{-1}} \mathcal{C}^0(\mathcal{F}) \xrightarrow{d^0} \mathcal{C}^1(\mathcal{F}) \xrightarrow{d^1} \mathcal{C}^2(\mathcal{F}) \rightarrow \dots$$

where the sheaves $\mathcal{C}^n(\mathcal{F})$ are defined inductively as follows:

1. Let $\mathcal{C}^0(\mathcal{F})$ be the Godement sheaf (Definition 7.7.9) of $\mathcal{F}$.
2. Let $\mathcal{Z}^1 = \text{coker}(d^{-1} : \mathcal{F} \rightarrow \mathcal{C}^0(\mathcal{F}))$. Define $\mathcal{C}^1(\mathcal{F}) := \mathcal{C}^0(\mathcal{Z}^1)$, and let d^0 be the composition $\mathcal{C}^0(\mathcal{F}) \twoheadrightarrow \mathcal{Z}^1 \hookrightarrow \mathcal{C}^1(\mathcal{F})$.
3. Inductively, let $\mathcal{Z}^{n+1} = \text{coker}(\mathcal{C}^{n-1}(\mathcal{F}) \rightarrow \mathcal{C}^n(\mathcal{F}))$ and $\mathcal{C}^{n+1}(\mathcal{F}) := \mathcal{C}^0(\mathcal{Z}^{n+1})$.

The resulting complex $\mathcal{C}^\bullet(\mathcal{F})$ is an acyclic resolution for the global sections functor $\Gamma(X, -)$.

Proof. (♠ TODO: verify) By construction, each $\mathcal{C}^n(\mathcal{F})$ is a Godement sheaf of some $\mathcal{O}_X$ -module, and thus is flasque by ???. Exactness at each $\mathcal{C}^n(\mathcal{F})$ follows from the local exactness of the kernel-cokernel sequences in the abelian category $\text{Mod}(\mathcal{O}_X)$. Since flasque sheaves are $\Gamma(X, -)$ -acyclic (???), $\mathcal{C}^\bullet(\mathcal{F})$ can be used to compute the sheaf cohomology $H^i(X, \mathcal{F})$. $\square$

Additionally, flasque sheaves are acyclic for direct image functors and hence right resolutions by flasque sheaves may be used to compute left derived direct image functors.

Proposition 7.7.14. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1).

1. Flasque sheaves (Definition 7.7.10) of $\mathcal{O}_X$ -modules (Definition 7.6.3) are acyclic for (Definition 4.4.1) the global sections functor (Definition 7.1.5) $\Gamma : \text{Mod}(\mathcal{O}_X) \rightarrow \text{Ab}$, which is left exact (Definition 2.4.12).

2. More generally, Flasque sheaves (Definition 7.7.10) of $\mathcal{O}_X$ -modules (Definition 7.6.3) are acyclic for (Definition 4.4.1) the direct image functor (Definition 7.7.2) $f_* : \text{Mod}(\mathcal{O}_X) \rightarrow \text{Sh}(Y, \text{Ab})$, which is left exact, for any continuous map $f : X \rightarrow Y$.

Proposition 7.7.15. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1). If $\mathcal{F}$ is a sheaf on X (Definition 7.1.6) of $\mathcal{O}_X$ -modules (Definition 7.6.3) that is injective (Definition 4.2.2) in the category of $\mathcal{O}_X$ -modules, then $\mathcal{F}$ is flasque.

7.7.1. Locally constant sheaves.

Definition 7.7.16 (Locally Constant Sheaf). Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be a category (Definition 1.1.1) with a terminal object (Definition 2.3.1) such that a sheafification functor (Definition 7.3.2)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists.

A sheaf (Definition 7.1.6) $\mathcal{F}$ on X valued in $\mathcal{A}$ is said to be *locally constant* if for every point $x \in X$, there exists an open neighborhood $U \subseteq X$ of x such that the restriction (Definition 7.7.3) $\mathcal{F}|_U$ is isomorphic to a constant sheaf (Definition 7.1.9) on U valued in $\mathcal{A}$.

In other words, the sheaf $\mathcal{F}$ is locally constant if it is if it is locally constant in the sense of Definition B.0.20 with respect to the site associated to the X (Definition B.0.7).

Accordingly, $\mathcal{F}$ is *of finite rank* if for each such open neighborhood U , the constant sheaf $\mathcal{F}|_U$ is isomorphic to the constant sheaf associated to a finite-dimensional vector space (or finitely generated free abelian group) over the base ring. Equivalently, a locally constant sheaf on a topological space is a locally constant sheaf for the natural site of the topological space.

(♠ TODO: talk about examples of locally constant sheaves, how they are equivalent to representations or to bundles, how they are locally isomorphic to constant sheaves and yet are not isomorphic to constant sheaves)

(♠ TODO: talk about how sheaves of sets on topological spaces are equivalent to étale spaces over the topological spaces)

The following is an essential idea; it has analogues for more sheaves of more general sites such as the étale site of a scheme.

Theorem 7.7.17. Let X Be a connected, locally path connected, and semi-locally simply connected (Definition C.0.56). Let $x \in X$. Let R be a ring. There is an equivalence (Definition 1.3.4)

$$\mathbf{LCS}(X, R) \simeq \mathbf{Rep}_R(\pi_1(X, x))$$

(♠ TODO: $\mathbf{LCS}(X, R)$) of categories where $\mathbf{LCS}(X, R)$ is the category of locally constant sheaves (Definition B.0.20) of R -modules, and $\mathbf{Rep}_R(\pi_1(X, x))$ is the category of R -linear representations of the fundamental group $\pi_1(X, x)$.

(♠ TODO: fiber,) (♠ TODO: associated bundle) Natural inverse equivalences (Definition 1.3.4) are given as follows: given a locally constant sheaf $\mathcal{F}$, the fiber at the basepoint

x is the module $M = \mathcal{F}_x$ (Definition 7.5.1), and for any loop γ in X based at x , “lifting” sections along the loop defines an automorphism of M , which in turn defines a representation $\rho : \pi_1(X, x) \rightarrow \text{Aut}_R(M)$. Given a representation $\rho : \pi_1(X, x) \rightarrow \text{Aut}_R(M)$, and letting $\tilde{X}$ be the universal cover of X , form the associated bundle of the locally constant sheaf as the quotient

$$\mathcal{F} = (\tilde{X} \times M) / \pi_1(X, x)$$

where π_1 acts on $\tilde{X}$ via deck transformations and on M via the representation ρ .

7.7.2. Vector bundles.

7.7.3. The definition of schemes. Now that we have defined the notion of sheaves on topological spaces, the definition of schemes is not too far off.

Definition 7.7.18 (Affine scheme). Let A be a commutative ring with unity (Definition C.0.9). Define the set $\text{Spec}(A)$ to be the set of all prime ideals (Definition C.0.22) of A . Equip it with the *Zariski topology*, which is the topology (Definition C.0.5) whose closed sets are given by *vanishing loci*

$$V(I) = \{\mathfrak{p} \in \text{Spec}(A) : I \subseteq \mathfrak{p}\}$$

for ideals $I \subseteq A$. Define the sheaf $\mathcal{O}_{\text{Spec}(A)}$, called the *structure sheaf of $\text{Spec } A$* , by

$$\mathcal{O}_{\text{Spec}(A)}(U) = \{ \text{locally defined fractions of elements of } A \text{ on } U \},$$

for each open set $U \subseteq \text{Spec}(A)$. More precisely, $\mathcal{O}_{\text{Spec}(A)}(U)$ can be described as the set of functions $s : U \rightarrow \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ such that

1. For each $\mathfrak{p} \in U$, the value $s(\mathfrak{p})$ lies in the localization $A_{\mathfrak{p}}$
2. For each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ containing $\mathfrak{p}$ and elements $a, f \in A$ such that for all $\mathfrak{q} \in V$,
 - $f \notin \mathfrak{q}$, so that the fraction $\frac{a}{f}$ is well-defined in $A_{\mathfrak{q}}$.
 - $s(\mathfrak{q}) = \frac{a}{f}$ in $A_{\mathfrak{q}}$.

It is the case that the stalk (Definition 7.5.1) at $\mathfrak{p} \in \text{Spec}(A)$ is canonically the localization $A_{\mathfrak{p}}$. Then $(\text{Spec}(A), \mathcal{O}_{\text{Spec}(A)})$ is a locally ringed space (Definition 7.7.19), called the *affine scheme associated to A* .

Moreover, given $f \in A$, we define the locus $D(f)$ by

$$D(f) = \text{Spec } A \setminus V((f)) = \{\mathfrak{p} \in \text{Spec } A : f \notin \mathfrak{p}\}$$

Definition 7.7.19 (Locally ringed space). A *locally ringed space* is a ringed space (Definition 7.6.1) $(X, \mathcal{O}_X)$ such that for every point $x \in X$, the stalk $\mathcal{O}_{X,x}$ (Definition 7.5.1) is a local ring. The unique maximal ideal (Definition C.0.22) of $\mathcal{O}_{X,x}$ is often denoted $\mathfrak{m}_x$ and is called the *maximal ideal at x* .

Definition 7.7.20 (Morphism of ringed spaces). Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be ringed spaces (Definition 7.7.19). A *morphism of ringed spaces*

$$(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$$

consists of:

- a continuous map $f : X \rightarrow Y$, and
- a morphism of (Definition 7.1.7) sheaves of rings (Definition 7.1.6)

$$f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X,$$

on Y where $f_* \mathcal{O}_X$ is the direct image sheaf (Definition 7.7.2) of $\mathcal{O}_X$ on Y . Equivalently, due to the adjunction (Definition 2.5.1) between direct image (Definition 7.7.2) and inverse image (Definition 7.7.1) (Theorem 7.7.5), such a morphism can alternatively be given by a morphism of sheaves of rings

$$f^* \mathcal{O}_Y \rightarrow \mathcal{O}_X$$

on X .

Equivalently, for each open set $V \subseteq Y$, there is a ring homomorphism

$$f^\#(V) : \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V)),$$

compatible with restriction maps of sections.

Definition 7.7.21 (Morphism of locally ringed spaces). Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be locally ringed spaces (Definition 7.7.19). A **morphism of locally ringed spaces** is a morphism of ringed spaces (Definition 7.7.20) $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ such that for each $x \in X$, the induced ring homomorphism of stalks (Definition 7.5.1)

$$f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$$

is a local homomorphism of local rings; that is, $(f_x^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{m}_{f(x)}$ where $\mathfrak{m}_x \subseteq \mathcal{O}_{X, x}$ and $\mathfrak{m}_{f(x)} \subseteq \mathcal{O}_{Y, f(x)}$ are the maximal ideals.

Definition 7.7.22 (Scheme). A **scheme** is a locally ringed space (Definition 7.7.19) $(X, \mathcal{O}_X)$ that admits an open cover $\{U_i\}_{i \in I}$ such that each $(U_i, \mathcal{O}_X|_{U_i})$ (Definition 7.7.3) is isomorphic (as a locally ringed space) (Definition 7.7.21) to an affine scheme $(\text{Spec}(A_i), \mathcal{O}_{\text{Spec}(A_i)})$ (Definition C.0.23) for some commutative ring (Definition C.0.7) A_i . In other words, a scheme is a locally ringed space locally isomorphic to affine schemes.

7.7.4. Sheaf cohomology.

Definition 7.7.23. (♠ TODO: generalize to more general abelian value categories)

Let $(\mathcal{C}, J)$ be a site (Definition B.0.5) on a locally small category (Definition 1.1.2) or a U -site for some universe (Definition A.0.3) U . Let $\mathcal{O}$ be a sheaf of rings (Definition B.0.8) on $\mathcal{C}$, so that $(\mathcal{C}, J, \mathcal{O})$ is a ringed site (Definition B.0.17). Recall that the category $\mathbf{Mod}(\mathcal{O})$ (Definition B.0.18) of $\mathcal{O}$ -modules is abelian and has enough injectives (Definition 4.2.5) (Theorem 7.6.4).

Assume that global sections objects $\Gamma(\mathcal{G})$ (Definition 7.1.5) exist for all objects $\mathcal{G}$ of $\text{Sh}(\mathcal{C}, \mathbf{Ab})$ ⁸ so that Γ is a functor

$$\text{Sh}(\mathcal{C}, \mathbf{Ab}) \rightarrow \mathbf{Ab},$$

⁸for example, this occurs when $\mathcal{C}$ is essentially small

which is a left exact functor (Definition 2.4.12) (Proposition 7.4.1). Note that Γ restricts to a left exact functor

$$\mathbf{Mod}(\mathcal{O}) \rightarrow \mathbf{Ab}.$$

If $\mathcal{C}$ has a final object (Definition 2.3.1) $*$ as well, then recall that $\Gamma(\mathcal{F}) = \mathcal{F}(*)$.

Let $\mathcal{F}$ be an object of $\mathbf{Mod}(\mathcal{O})$.

1. For each integer $n \geq 0$, the *n -th (abelian) (global) sheaf cohomology group of $\mathcal{F}$* is

$$H^n(\mathcal{C}, J; \mathcal{F}) := R^n\Gamma(\mathcal{F}),$$

where $R^n\Gamma$ is the n -th right derived functor (Definition 4.3.1) of the global sections functor Γ (Definition 7.1.5).

In particular, each H^n is a functor

$$H^n : \mathbf{Mod}(\mathcal{O}) \rightarrow \mathbf{Ab}.$$

2. Given an object $U \in \mathcal{C}$ and for each integer $n \geq 0$, the *n -th (abelian) sheaf cohomology group of $\mathcal{F}$ of sections at U* is

$$H^n(U, \mathcal{F}) := (R^n\Gamma(U, -))(\mathcal{F}),$$

where $R^n\Gamma(U, -)$ is the n -th right derived functor (Definition 4.3.1) of the sections functor $\Gamma(U, -)$ evaluated at U (Definition 7.1.5).

In particular, $H^n(U, \mathcal{F})$ can be regarded as the n th global sheaf cohomology group of the restriction $\mathcal{F}|_U$ of $\mathcal{F}$ to U .

Definition 7.7.24. Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) with enough injectives (Definition 4.2.5) and such that a sheafification functor (Definition 7.3.2)

$$a : \mathbf{PreShv}(X, \mathcal{A}) \rightarrow \mathbf{Shv}(X, \mathcal{A})$$

exists. In particular, $\mathbf{Shv}(X, \mathcal{A})$ (Definition 7.1.6) is (Proposition 7.5.3) an abelian category and $\Gamma(U, -) : \mathbf{Shv}(X, \mathcal{A}) \rightarrow \mathcal{A}$ is (Proposition 7.4.2) a left exact functor (Definition 2.4.12) for all open (Definition C.0.5) subsets $U \subseteq X$. In particular, these conditions are satisfied for $\mathcal{A} = \mathbf{Ab}$ (Theorem 4.2.6).

Let $\mathcal{F} \in \mathbf{Shv}(X, \mathcal{A})$ and let $U \subseteq X$ be an open subset. For each integer $n \geq 0$, the *n th (abelian) sheaf cohomology object of $\mathcal{F}$ of sections over U* is

$$H^n(U, \mathcal{F}) := (R^n\Gamma(U, -))(\mathcal{F})$$

where $R^n\Gamma(U, -)$ is the n th right derived functor (Definition 4.3.1) of the sections functor $\Gamma(U, -)$. In the case that $U = X$, this sheaf cohomology object may be called the *global sheaf cohomology object*. The sheaf cohomology objects $H^n(U, \mathcal{F})$ for $n \geq 1$ are often called the *higher sheaf cohomology objects of $\mathcal{F}$ with sections over U* .

In the case that X is equipped with a sheaf (Definition B.0.8) $\mathcal{O}_X$ of rings (Definition C.0.7), i.e. $(X, \mathcal{O}_X)$ is a ringed space (Definition 7.6.1) and $\mathcal{F} \in \mathbf{Mod}(\mathcal{O}_X)$ (Definition 7.6.3), we might define $H^n(U, \mathcal{F})$ in either of the following ways:

1. as the n th right derived functor of $\Gamma(U, -) : \text{Mod}(\mathcal{O}_X) \rightarrow \text{Ab}$ via (right) resolutions (Definition 4.2.29) of $\mathcal{F}$ by injective (Definition 4.2.2) $\text{Mod}(\mathcal{O}_X)$ (Definition 7.6.3)-modules (Definition B.0.18).
2. as the n th right derived functor of $\Gamma(U, -) : \text{Shv}(X, \text{Ab}) \rightarrow \text{Ab}$ via (right) resolutions of $\mathcal{F}$ by injective objects (Definition 4.2.2) of $\text{Shv}(X, \text{Ab})$.

It turns out the corresponding sheaf cohomology (Definition 7.7.23) groups (Definition C.0.3) defined in these two ways are naturally isomorphic (Definition 1.3.1). (♠ TODO: prove)

The above notion of sheaf cohomology of sheaves of abelian groups/modules over a sheaf on a topological space is a special case of the more general notion of sheaf cohomology (Definition 7.7.23) of sheaves of abelian groups/modules over a sheaf on a site (Definition B.0.5) in the case that the site is that on **Open**(X) (Definition B.0.7).

While the above notion of sheaf cohomology is for sheaves of abelian groups, there are sensible notions of zeroth and first sheaf cohomology for sheaves of not-necessarily abelian groups

Definition 7.7.25 (Torsors under a Sheaf of Groups). (♠ TODO: group object, action by a group object) Let $(\mathcal{C}, J)$ be a site (Definition B.0.5) and $\mathcal{G}$ a sheaf of groups on it (Definition B.0.8). A **$\mathcal{G}$ -torsor** (or synonymously a **principal $\mathcal{G}$ -bundle** or **principal $\mathcal{G}$ -homogeneous space**, etc.) is a sheaf $\mathcal{P}$ of sets on $(\mathcal{C}, J)$ equipped with a (right) action of $\mathcal{G}U$ such that (♠ TODO: This needs some genuine fixing to be definable on a general site, not just one given by a pretopology)

1. For each object U of $\mathcal{C}$, the action of the group $G(U)$ on $\mathcal{P}(U)$ is free and transitive whenever $\mathcal{P}(U)$ is nonempty.
2. $\mathcal{P}$ is locally trivial. More explicitly, for each object U of $\mathcal{C}$, there exists some covering $\{V_i \rightarrow U\}_{i \in I}$ in J such that there is an isomorphism

$$\mathcal{P}|_{U_i} \cong \mathcal{G}|_{U_i}$$

as a sheaves with $\mathcal{G}|_{U_i}$ actions for each i .

A $\mathcal{G}$ -torsor is called **trivial on U** if it is isomorphic (as a sheaf with a right action of $\mathcal{G}$) to $\mathcal{G}|_U$ acting on itself by right multiplication, or equivalently, if $\mathcal{G}(U) \neq \emptyset$. In particular, any $\mathcal{G}$ -torsor is *locally* trivial.

Definition 7.7.26 (Nonabelian Sheaf Cohomology: H^0 and H^1). Let $(\mathcal{C}, J)$ be a site (Definition B.0.5) and $\mathcal{G}$ a sheaf (Definition B.0.8) of groups on $(\mathcal{C}, J)$. For $U \in \mathcal{C}$, we define the **0th and 1st (nonabelian) sheaf cohomology sets** as follows:

- The zeroth sheaf cohomology group **$H^0(\mathcal{C}, J, \mathcal{G}) := \Gamma(\mathcal{G})$** is the group of global sections (Definition 7.1.5) of $\mathcal{G}$, assuming that it exists (which is always the case when $\mathcal{C}$ is essentially small for example since the category of groups is closed under small projective limits (♠ TODO:)).
- The first sheaf cohomology set **$H^1(\mathcal{C}, J, \mathcal{G})$** is the pointed set of isomorphism classes of $\mathcal{G}$ -torsors (Definition 7.7.25) on the site $(\mathcal{C}, J)$.

Proposition 7.7.27. See [Sta25, Tag 03AJ] for a statement⁹ Let $(\mathcal{C}, J)$ be a site (Definition B.0.5). Assume that global sections objects $\Gamma(\mathcal{G})$ (Definition 7.1.5) exist for all objects $\mathcal{G}$ of $\mathrm{Sh}(\mathcal{C}, \mathbf{Ab})$ ¹⁰ so that Γ is a functor

$$\mathrm{Sh}(\mathcal{C}, \mathbf{Ab}) \rightarrow \mathbf{Ab}.$$

Let $\mathcal{G}$ be a sheaf (Definition B.0.8) of abelian groups on $(\mathcal{C}, J)$. For $i = 0, 1$, there is a canonical bijection between the set of isomorphism classes of $\mathcal{G}$ -torsors (Definition 7.7.25) and the (abelian) sheaf cohomology group $H^i(\mathcal{C}, J; \mathcal{G})$ (Definition 7.7.23). In other words, for $i = 0, 1$, there is a canonical bijection between the i th nonabelian sheaf cohomology (Definition 7.7.26) and the i th abelian sheaf cohomology (Definition 7.7.23) of $\mathcal{G}$.

Proof. In the case of $i = 0$, both are identifiable with the groups of global sections. In the case of $i = 1$, see [Sta25, Tag 03AJ]. $\square$

7.8. Čech cohomology. (♠ TODO: Čech cohomology for more general sites)

Definition 7.8.1. Let X be a topological space (Definition C.0.5), let $\mathcal{A}$ be a $\mathrm{AB3}^*$ category (Definition 2.5.7), let $\mathcal{F}$ be a presheaf on X valued in (Definition 7.1.4) $\mathcal{A}$, and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . Impose a total ordering on the index set I .

1. For $i_0, i_1, \dots, i_p \in I$, write

$$U_{i_0, \dots, i_p} := U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_p}.$$

The *object of p -cochains of $\mathcal{U}$ on $\mathcal{F}$* is the product

$$C^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0, \dots, i_p}).$$

In particular, an element $\alpha \in C^p(\mathcal{U}, \mathcal{F})$ precisely corresponds to a tuple of elements of $\mathcal{F}(U_{i_0, \dots, i_p})$; write $\alpha_{i_0, \dots, i_p}$ for the element of $\mathcal{F}(U_{i_0, \dots, i_p})$ corresponding α .

2. Define the *coboundary operator*

$$d^p : C^p(\mathcal{U}, \mathcal{F}) \rightarrow C^{p+1}(\mathcal{U}, \mathcal{F})$$

so that

$$\pi_{i_0, \dots, i_{p+1}} \circ d^p = \sum_{j=0}^{p+1} (-1)^j \mathrm{res}_{U_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}}, U_{i_0, \dots, i_{p+1}}} \circ \pi_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}}$$

(Definition 7.1.4) where $\pi_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}} : C^p(\mathcal{U}, \mathcal{F}) \rightarrow \mathcal{F}(U_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}})$ is the projection map from the product onto the factor indexed by the $(p+1)$ -tuple $(i_0, \dots, \hat{i}_j, \dots, i_{p+1})$.

⁹While the Stacks Project defines a site as a category equipped with a Grothendieck *pretopology*, its proof of this statement should be applicable for more general sites

¹⁰for example, this occurs when $\mathcal{C}$ is essentially small

At the level of generalized elements, for any object $A \in \mathcal{A}$ and any morphism $\alpha \in \text{Hom}_{\mathcal{A}}(A \rightarrow C^p(\mathcal{U}, \mathcal{F}))$, the coboundary $d^p \circ \alpha$ is determined by its components:

$$(d^p \alpha)_{i_0, \dots, i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j \alpha_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}} \big|_{U_{i_0, \dots, i_{p+1}}}$$

where $\alpha_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}} = \pi_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}} \circ \alpha$ is the component of α in the factor $\mathcal{F}(U_{i_0, \dots, \hat{i}_j, \dots, i_{p+1}})$.

One can verify that $d^{p+1} \circ d^p = 0$ (for instance, by checking $d^{p+1} \circ d^p \circ \alpha = 0$ for all generalized elements α) and hence that $(C^\bullet(\mathcal{U}, \mathcal{F}), d^\bullet)$ is a cochain complex (Definition 3.0.1) of objects in $\mathcal{A}$.

3. The *p th Čech cohomology object of $\mathcal{F}$ with respect to the cover $\mathcal{U}$* is the cohomology (Definition 3.2.2) of the cochain complex $(C^\bullet(\mathcal{U}, \mathcal{F}), d^\bullet)$:

$$\check{H}^p(\mathcal{U}, \mathcal{F}) := H^p(C^\bullet(\mathcal{U}, \mathcal{F})) = \ker(d^p) / \text{im}(d^{p-1})$$

4. Whenever an open cover $\mathcal{V}$ of X is a refinement of $\mathcal{U}$, (i.e. every $V_j \in \mathcal{V}$ is a subset of some $U_i \in \mathcal{U}$), there is an induced map

$$\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow \check{H}^p(\mathcal{V}, \mathcal{F}).$$

on Čech cohomology groups. The collection of open covers of X ordered by refinement specifies a filtered category (Definition 2.2.12) $\mathcal{C}$, and the above maps on Čech cohomology groups yields a diagram (Definition 2.2.6) $\mathcal{C} \rightarrow \mathcal{A}$. Assuming that $\mathcal{A}$ is additionally an *AB4* (Definition 2.5.7) category (in practice, the theory works better for *AB5* categories), define the *p th Čech cohomology object of X with coefficients in $\mathcal{F}$* as

$$\check{H}^p(X, \mathcal{F}) := \text{colim}_{\mathcal{U} \in \mathcal{C}} \check{H}^p(\mathcal{U}, \mathcal{F}).$$

One often considers Čech cohomology for $\mathcal{A} = \text{Ab}$ (Definition 1.1.8).

Remark 7.8.2. Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be an *AB3* category (Definition 2.5.7). Assume that a sheafification functor (Definition 7.3.2)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists so that $\text{Sh}(X, \mathcal{A})$ is an abelian caetgory (Proposition 7.5.3).

Given an arbitrary open cover $\mathcal{U}$, the functors (Definition 1.2.2)

$$\{\check{H}^p(\mathcal{U}, -) : \text{Sh}(X, \mathcal{A}) \rightarrow \mathcal{A}\}_{p \geq 0}$$

(Definition 7.1.6) (Definition 1.1.8) (Definition 7.8.1) may not form a cohomological δ -functor (Definition 4.1.1). In other words, short exact sequences (Definition 2.4.4)

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

in $\text{Sh}(X, \mathcal{A})$ may not yield long exact sequences (Theorem 3.2.10) of the form

$$\begin{aligned} 0 \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}') &\rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}) \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}'') \\ &\rightarrow \check{H}^1(\mathcal{U}, \mathcal{F}') \rightarrow \check{H}^1(\mathcal{U}, \mathcal{F}) \rightarrow \check{H}^1(\mathcal{U}, \mathcal{F}'') \\ &\rightarrow \cdots \end{aligned}$$

This is because, for general $\mathcal{U}$, the sequence

$$0 \rightarrow C^\bullet(\mathcal{U}, \mathcal{F}') \rightarrow C^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow C^\bullet(\mathcal{U}, \mathcal{F}'') \rightarrow 0$$

is generally only left exact (Definition 2.4.12) because sections are left exact (Proposition 7.4.2).

On the other hand, at least in the case that $\mathcal{A} = \text{Ab}$ and given any open cover $\mathcal{U}$ of X , there is a long exact sequence

$$\begin{aligned} 0 \rightarrow \check{H}^0(X, \mathcal{F}') \rightarrow \check{H}^0(X, \mathcal{F}) \rightarrow \check{H}^0(X, \mathcal{F}'') \\ \rightarrow \check{H}^1(X, \mathcal{F}') \rightarrow \check{H}^1(X, \mathcal{F}) \rightarrow \check{H}^1(X, \mathcal{F}'') \\ \rightarrow \dots \end{aligned}$$

Lemma 7.8.3. Let X be a topological space (Definition C.0.5). Let

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

be a short exact sequence (Definition 2.4.4) in $\text{Sh}(X, \text{Ab})$ (♠ TODO: generalize to an AB3star + AB5 category (Definition 2.5.7)). There is a short exact sequence

$$0 \rightarrow \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}') \rightarrow \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}'') \rightarrow 0.$$

where the $\mathcal{U}$ run over the open (Definition C.0.5) covers of X .

Proof. (♠ TODO:) Take an element $\alpha \in \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}'')$ □

Proposition 7.8.4. Let X be a topological space (Definition C.0.5). The functors (Definition 1.2.2)

$$\{\check{H}^p(X, -) : \text{Sh}(X, \text{Ab}) \rightarrow \text{Ab}\}_{p \geq 0}$$

(Definition 7.1.6) (Definition 1.1.8) (Definition 7.8.1) form a cohomological δ -functor (Definition 4.1.1).

Proof. Let

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

be a short exact sequence (Definition 2.4.4) in $\text{Sh}(X, \text{Ab})$. By Lemma 7.8.3, there is a short exact sequence

$$0 \rightarrow \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}') \rightarrow \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}'') \rightarrow 0.$$

where the $\mathcal{U}$ run over the open (Definition C.0.5) covers of X . Take the long exact sequence in cohomology (Theorem 3.2.10):

$$\begin{aligned} 0 \rightarrow \check{H}^0(\varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}')) \rightarrow \check{H}^0(\varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F})) \rightarrow \check{H}^0(\varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}'')) \\ \rightarrow \check{H}^1(\varinjlim_{\mathcal{U}} C^\bullet(\mathcal{U}, \mathcal{F}')) \end{aligned}$$

Noting that $\mathbf{Ab}$ is an $AB5$ category (Theorem 2.5.9) and hence filtered colimits are exact, the cohomology functors and the filtered colimits commute, yielding a long exact sequence

$$\begin{aligned} 0 \rightarrow \varinjlim_{\mathcal{U}} \check{H}^0(C^\bullet(\mathcal{U}, \mathcal{F}')) &\rightarrow \varinjlim_{\mathcal{U}} \check{H}^0(C^\bullet(\mathcal{U}, \mathcal{F})) \rightarrow \varinjlim_{\mathcal{U}} \check{H}^0(C^\bullet(\mathcal{U}, \mathcal{F}'')) \\ &\rightarrow \varinjlim_{\mathcal{U}} \check{H}^1(C^\bullet(\mathcal{U}, \mathcal{F}')). \end{aligned}$$

This is precisely a long exact sequence in Čech cohomology (Definition 3.2.2):

$$\begin{aligned} 0 \rightarrow \check{H}^0(X, \mathcal{F}') &\rightarrow \check{H}^0(X, \mathcal{F}) \rightarrow \check{H}^0(X, \mathcal{F}'') \\ &\rightarrow \check{H}^1(X, \mathcal{F}') \end{aligned}$$

□

Leray's theorem Theorem 7.8.12 states that Čech cohomology $\check{H}(\mathcal{U}, \mathcal{F})$ with respect to an acyclic cover naturally agrees with sheaf cohomology $H^p(X, \mathcal{F})$. Moreover, Theorem 7.8.13 states that Čech cohomology $\check{H}(X, \mathcal{F})$ agrees with sheaf cohomology $H^p(X, \mathcal{F})$ under nice enough circumstances.

Definition 7.8.5. Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) with enough injectives (Definition 4.2.5) and such that a sheafification functor (Definition 7.3.2)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists. In particular, the notion of sheaf cohomology objects (Definition 7.7.24) is well defined. Moreover, these conditions are satisfied for $\mathcal{A} = \mathbf{Ab}$ (Theorem 4.2.6). Let $\mathcal{F}$ be a sheaf of (Definition 7.1.6) abelian groups (Definition C.0.3) on X , and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X .

We say that the open cover $\mathcal{U}$ is **acyclic for $\mathcal{F}$** or a **Leray cover for $\mathcal{F}$** if for every finite nonempty intersection $U_{i_0, \dots, i_n} = U_{i_0} \cap \dots \cap U_{i_n}$ and all $q > 0$, the sheaf cohomology (Definition 7.7.24) $H^q(U_{i_0, \dots, i_n}, \mathcal{F})$ equals 0.

Definition 7.8.6. Let X be a topological space (Definition C.0.5). An open cover $\{U_i\}_{i \in I}$ is said to be a **good cover** if every finite intersection $\bigcap_{j=1}^n U_{i_j}$ is empty or contractible (Definition C.0.54).

Definition 7.8.7. Let X be a topological space (Definition C.0.5). We will say that an open cover $\{U_i\}_{i \in I}$ is a **semi-good open cover** if every finite intersection $\bigcap_{j=1}^n U_{i_j}$ is empty or is a disjoint union of contractible (Definition C.0.54) open subsets.

Remark 7.8.8. The terminology of “semi-good cover” is probably not standard.

Theorem 7.8.9. Let X be a topological space (Definition C.0.5). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) with enough injectives (Definition 4.2.5) and such that a sheafification functor (Definition 7.3.2)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists. In particular, the notion of sheaf cohomology objects (Definition 7.7.24) is well defined. Moreover, these conditions are satisfied for $\mathcal{A} = \mathbf{Ab}$ (Theorem 4.2.6). Let $\mathcal{F}$ be a sheaf of (Definition 7.1.6) abelian groups (Definition C.0.3) on X .

Assuming that X is a contractible and locally contractible topological space and $\mathcal{F}$ is a constant sheaf (Definition 7.1.9) on X valued in $\mathcal{A}$, the sheaf cohomology objects $H^p(X, \mathcal{F})$ (Definition 7.7.24) all vanish for $p > 0$.

Proposition 7.8.10. Let X be a topological space (Definition C.0.5), let $\mathcal{F}$ be a sheaf of (Definition 7.1.6) abelian groups (Definition C.0.3) on X , and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X .

Under the following circumstances, $\mathcal{U}$ is acyclic for $\mathcal{F}$ (Definition 7.8.5):

1. X is a separated scheme (Definition C.0.53), $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -module, and every $U_i \in \mathcal{U}$ is an affine open subset of X .
2. X is locally contractible (Definition C.0.55), $\mathcal{F}$ is a constant sheaf (Definition 7.1.9), and $\mathcal{U}$ is a semi-good cover (Definition 7.8.7) of X .
3. (♠ TODO: something about complex manifolds, holomorphic sheaves, and each U_i being a Stein manifold)
4. $\mathcal{F}$ is a flasque sheaf (Definition 7.7.10).

Proposition 7.8.11. Let M be a smooth manifold (Definition C.0.37). There exists a good cover of M (Definition 7.8.6).

Theorem 7.8.12 (Leray's theorem). Let X be a topological space (Definition C.0.5), let $\mathcal{F}$ be a sheaf of (Definition 7.1.6) abelian groups (Definition C.0.3) on X , and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . If $\mathcal{U}$ is acyclic for $\mathcal{F}$ (Definition 7.8.5), then there is a natural isomorphism (Definition 1.3.1)

$$\check{H}^p(\mathcal{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$$

for all $p \geq 0$ where $\check{H}^p(\mathcal{U}, \mathcal{F})$ is the p th Čech cohomology group of $\mathcal{F}$ with respect to $\mathcal{U}$ (Definition 7.8.1) and $H^p(X, \mathcal{F})$ is the p th sheaf cohomology group of $\mathcal{F}$ (Definition 7.7.24).

Theorem 7.8.13. (♠ TODO: there might be more circumstances under which these are true.) Let X be a topological space (Definition C.0.5). There is a natural isomorphism (Definition 1.3.1)

$$\check{H}^p(X, \mathcal{F}) \cong H^p(X, \mathcal{F})$$

for all $p \geq 0$ where $\check{H}^p(X, \mathcal{F})$ is the p th Čech cohomology group of $\mathcal{F}$ (Definition 7.8.1) and $H^p(X, \mathcal{F})$ is the p th sheaf cohomology group of $\mathcal{F}$ (Definition 7.7.24) under any of the following circumstances:

1. X is a paracompact (Definition C.0.49) Hausdorff space (Definition C.0.51)
2. X is a quasi-compact (Definition C.0.52) and quasi-separated (Definition C.0.53) scheme (Definition 7.7.22) and $\mathcal{F}$ is a quasi-coherent sheaf.
3. X is a locally contractible (Definition C.0.55) and semi-locally simply connected (Definition C.0.56) topological space (Definition C.0.5) and $\mathcal{F}$ is a locally constant sheaf (Definition 7.7.16).

Let us compute some examples of Čech cohomology groups (which may coincide with sheaf cohomology as per Theorem 7.8.13).

Example 7.8.14. Let us compute $\check{H}^p(S^1, \underline{\mathbb{Z}})$ (Definition 7.8.1) (Definition 7.1.9). Take $\mathcal{U} = \{U_0, U_1, U_2\}$ to consist of three arcs each covering about 120° and overlapping slightly — note that this open cover is good (Definition 7.8.6) and that S^1 is a smooth manifold (Definition C.0.37), so Proposition 7.8.10 states that $\mathcal{U}$ is acyclic (Definition 7.8.5) for $\underline{\mathbb{Z}}$ and hence Leray’s theorem (Theorem 7.8.12) and Theorem 7.8.13 together show that $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong \check{H}^p(S^1, \mathcal{F})$ (and that these are isomorphic to sheaf cohomology (Definition 7.7.24) $H^p(S^1, \mathcal{F})$).

The complex is given by

$$\begin{aligned} C^0(\mathcal{U}, \underline{\mathbb{Z}}) &= \Gamma(U_0, \underline{\mathbb{Z}}) \oplus \Gamma(U_1, \underline{\mathbb{Z}}) \oplus \Gamma(U_2, \underline{\mathbb{Z}}) = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \\ C^1(\mathcal{U}, \underline{\mathbb{Z}}) &= \Gamma(U_0 \cap U_1, \underline{\mathbb{Z}}) \oplus \Gamma(U_0 \cap U_2, \underline{\mathbb{Z}}) \oplus \Gamma(U_1 \cap U_2, \underline{\mathbb{Z}}) = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \\ C^2(\mathcal{U}, \underline{\mathbb{Z}}) &= \Gamma(U_0 \cap U_1 \cap U_2, \underline{\mathbb{Z}}) = \Gamma(\emptyset, \underline{\mathbb{Z}}) = 0 \end{aligned}$$

The differential map

$$d^0 : C^0(\mathcal{U}, \underline{\mathbb{Z}}) \rightarrow C^1(\mathcal{U}, \underline{\mathbb{Z}})$$

is given by

$$d^0(s_0, s_1, s_2) = (s_1 - s_0, s_2 - s_1, s_0 - s_2)$$

computing the cohomology of the cochain complex yields

$$\begin{aligned} \check{H}^0(S^1, \underline{\mathbb{Z}}) &\cong \mathbb{Z} \\ \check{H}^1(S^1, \underline{\mathbb{Z}}) &\cong \mathbb{Z}. \end{aligned}$$

Example 7.8.15. Let $r \in \mathbb{Q}^\times$ and let $\mathcal{F}_r$ be the locally constant sheaf of $\mathbb{Q}$ -vector spaces on S^1 with “monodromy” r — by this, we mean that $\mathcal{F}_r$ is the (isomorphism class of) sheaf on S^1 corresponding to (Theorem 7.7.17) the homomorphism $\pi_1(S^1, x) \rightarrow \text{Aut}(\mathbb{Q}) \cong \mathbb{Q}^\times$ given by sending a generator of $\pi_1(S^1, x)$ to $r \in \mathbb{Q}^\times$.

Consider the open cover $\mathcal{U} = \{U_0, U_1\}$ consisting of two overlapping semicircles. The intersection $U_0 \cap U_1 = V_a \sqcup V_b$ is a disjoint union of two contractible (Definition C.0.54) arcs, making $\mathcal{U}$ a semi-good cover (Definition 7.8.7).

As the specification of $\mathcal{F}_r$ by monodromy only determines its isomorphism class, we provide an explicit description of a representative element. Since each U_i is contractible (Definition C.0.54) and locally contractible, the restriction $\mathcal{F}_r|_{U_i}$ is a constant sheaf (Definition 7.1.9) with value $\mathbb{Q}$. Thus, we have isomorphisms $\phi_i : \Gamma(U_i, \mathcal{F}_r) \xrightarrow{\sim} \mathbb{Q}_i$, where $\mathbb{Q}_0, \mathbb{Q}_1$ are two copies of $\mathbb{Q}$.

Note that $\Gamma(U_0 \cap U_1, \mathcal{F}_r) \cong \Gamma(V_a, \mathcal{F}_r) \oplus \Gamma(V_b, \mathcal{F}_r)$. Since $V_a, V_b \subset U_i$ are inclusions of contractible sets into a larger contractible set, the restriction maps $\Gamma(U_i, \mathcal{F}_r) \rightarrow \Gamma(V_j, \mathcal{F}_r)$ are isomorphisms. By identifying each $\Gamma(V_j, \mathcal{F}_r)$ with $\Gamma(U_i, \mathcal{F}_r)$ via these internal restrictions, the map $\Gamma(U_i, \mathcal{F}_r) \rightarrow \Gamma(U_0 \cap U_1, \mathcal{F}_r)$ is identified with the diagonal:

$$s \mapsto (s, s) \in \Gamma(V_a, \mathcal{F}_r) \oplus \Gamma(V_b, \mathcal{F}_r).$$

The monodromy r is encoded in the transition between the choices of basis $1 \in \mathbb{Q}_0$ and $1 \in \mathbb{Q}_1$. We construct $\mathcal{F}_r$ such that:

1. On the overlap V_a , the identification $\mathbb{Q}_0 \cong \Gamma(V_a, \mathcal{F}_r) \cong \mathbb{Q}_1$ is the identity ($1 \mapsto 1$).
2. On the overlap V_b , the identification $\mathbb{Q}_0 \cong \Gamma(V_b, \mathcal{F}_r) \cong \mathbb{Q}_1$ is multiplication by r ($1 \mapsto r$).

With these identifications, an element $(c_0, c_1) \in \mathbb{Q}_0 \oplus \mathbb{Q}_1$ maps via the Čech differential d^0 to:

$$d^0(c_0, c_1) = (c_1 - c_0, c_1 - rc_0) \in \mathbb{Q} \oplus \mathbb{Q}.$$

For $r \neq 1$, the only solution to $d^0(c_0, c_1) = (0, 0)$ is $c_0 = c_1 = 0$, implying $\check{H}^0(S^1, \mathcal{F}_r) = 0$.

To compute $\check{H}^1(S^1, \mathcal{F}_r) = \text{coker } d^0$, we observe that for any target vector $(a, b) \in \mathbb{Q} \oplus \mathbb{Q}$, we seek (c_0, c_1) such that:

$$\begin{cases} c_1 - c_0 = a \\ c_1 - rc_0 = b \end{cases}$$

Subtracting the first equation from the second yields $(1 - r)c_0 = b - a$. Since $r \neq 1$, we can solve for $c_0 = \frac{b-a}{1-r}$ and subsequently $c_1 = a + c_0$. Thus, d^0 is surjective, and the quotient group $\check{H}^1(S^1, \mathcal{F}_r)$ is zero.

Example 7.8.16. As per the situation in Example 7.5.7, let $X = \mathbb{C} \setminus \{0\}$, let $\mathcal{O}_X$ be the sheaf of holomorphic functions on X , and let $\mathcal{O}_X^*$ be the sheaf of multiplicatively invertible holomorphic functions on X . There is a short exact sequence

$$0 \rightarrow \underline{\mathbb{Z}} \xrightarrow{n \mapsto 2\pi i n} \mathcal{O}_X \xrightarrow{f \mapsto \exp(f)} \mathcal{O}_X^* \rightarrow 1$$

of sheaves of abelian groups on X .

We show that the global section $z \in \Gamma(X, \mathcal{O}_X^*)$ is not in the image of $\exp : \Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X, \mathcal{O}_X^*)$.

By Proposition 7.8.4, there is an exact sequence

$$(Q) \quad 0 \rightarrow \check{H}^0(X, \underline{\mathbb{Z}}) \rightarrow \check{H}^0(X, \mathcal{O}_X) \rightarrow \check{H}^0(X, \mathcal{O}_X^*) \xrightarrow{\delta} \check{H}^1(X, \underline{\mathbb{Z}})$$

The degree 0 cohomology terms are identifiable with the global sections $\Gamma(X, -)$. To show that z is not in the image of $\exp : \Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X, \mathcal{O}_X^*)$ is equivalent to show that $\delta(z) \in \check{H}^1(X, \underline{\mathbb{Z}})$ is nonzero. It is the case that $\check{H}^0(\mathcal{U}, -)$ is naturally isomorphic to $\Gamma(X, -)$ for any open cover $\mathcal{U}$ of X , so $\check{H}^0(X, -)$ is also naturally isomorphic to $\Gamma(X, -)$. Moreover, $\check{H}^1(X, \underline{\mathbb{Z}})$ is naturally isomorphic to $H^1(X, \underline{\mathbb{Z}})$ by Theorem 7.8.13 since X is a manifold and hence is (Theorem C.0.50) a paracompact Hausdorff space. If $\mathcal{U}$ is an open cover that is acyclic for $\underline{\mathbb{Z}}$ then Leray's theorem (Theorem 7.8.12) implies that $\check{H}^1(X, \underline{\mathbb{Z}})$ is naturally isomorphic to $\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}})$. Therefore, as long as $\mathcal{U}$ is an acyclic open cover for $\underline{\mathbb{Z}}$, the exact sequence (Q) is identifiable as

$$(R) \quad 0 \rightarrow \check{H}^0(\mathcal{U}, \underline{\mathbb{Z}}) \rightarrow \check{H}^0(\mathcal{U}, \mathcal{O}_X) \rightarrow \check{H}^0(\mathcal{U}, \mathcal{O}_X^*) \xrightarrow{\delta} \check{H}^1(\mathcal{U}, \underline{\mathbb{Z}})$$

.

We also choose $\mathcal{U}$ to be nice enough so that

$$(S) \quad 0 \rightarrow C^\bullet(\mathcal{U}, \underline{\mathbb{Z}}) \rightarrow C^\bullet(\mathcal{U}, \mathcal{O}_X) \rightarrow C^\bullet(\mathcal{U}, \mathcal{O}_X^*),$$

which is a priori only left exact, is also a short exact sequence; this allows us to compute $\delta(z)$ using the diagram chasing logic of the Snake lemma.

We let $\mathcal{U} = \{U_0, U_1\}$ be given by $U_0 = \mathbb{C} \setminus (-\infty, 0]$ and $U_1 = \mathbb{C} \setminus [0, \infty)$. Note that the intersection $U_{01} = U_0 \cap U_1$ has two disjoint connected components, which we describe as V_{up} and V_{low} , given as the upper half-plane and lower half-plane respectively. To say that (S) is exact is tantamount to saying that $\Gamma(U, \mathcal{O}_X) \xrightarrow{\text{exp}} \Gamma(U, \mathcal{O}_X^*)$ is surjective for all $U \in \{U_0, U_1, U_{01}\}$, which is true due by complex analysis.

To have the exact sequence Equation (R), recall that we need to justify that $\mathcal{U}$ is acyclic for $\underline{\mathbb{Z}}$, i.e. that

$$H^q(U_0, \underline{\mathbb{Z}}), H^q(U_1, \underline{\mathbb{Z}}), H^q(U_0 \cap U_1, \underline{\mathbb{Z}}) = 0.$$

The first two vanish due to Theorem 7.8.9 because U_0 and U_1 are contractible and $\underline{\mathbb{Z}}$ is constant. The third is identifiable with $H^q(V_{\text{up}}, \underline{\mathbb{Z}}) \oplus H^q(V_{\text{low}}, \underline{\mathbb{Z}})$, which is also 0.

We now compute $\delta(z)$. The diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^0(\mathcal{U}, \underline{\mathbb{Z}}) & \longrightarrow & C^0(\mathcal{U}, \mathcal{O}_X) & \longrightarrow & C^0(\mathcal{U}, \mathcal{O}_X^*) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C^1(\mathcal{U}, \underline{\mathbb{Z}}) & \longrightarrow & C^1(\mathcal{U}, \mathcal{O}_X) & \longrightarrow & C^1(\mathcal{U}, \mathcal{O}_X^*) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \vdots & & \vdots & & \vdots \end{array}$$

is identified with

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma(U_0, \underline{\mathbb{Z}}) \oplus \Gamma(U_1, \underline{\mathbb{Z}}) & \longrightarrow & \Gamma(U_0, \mathcal{O}_X) \oplus \Gamma(U_1, \mathcal{O}_X) & \longrightarrow & \Gamma(U_0, \mathcal{O}_X^*) \oplus \Gamma(U_1, \mathcal{O}_X^*) \longrightarrow 0 \\ & & \downarrow d^0 & & \downarrow d^0 & & \downarrow d^0 \\ 0 & \longrightarrow & \Gamma(U_0 \cap U_1, \underline{\mathbb{Z}}) & \longrightarrow & \Gamma(U_0 \cap U_1, \mathcal{O}_X) & \longrightarrow & \Gamma(U_0 \cap U_1, \mathcal{O}_X^*) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

In the nonzero upper right corner, $z \in \check{H}^0(X, \mathcal{O}_X^*) \cong \check{H}^0(\mathcal{U}, \mathcal{O}_X^*)$ is represented by $(z, z) \in \Gamma(U_0, \mathcal{O}_X^*) \oplus \Gamma(U_1, \mathcal{O}_X^*)$. Lift (z, z) to $f \in \Gamma(U_0, \mathcal{O}_X) \oplus \Gamma(U_1, \mathcal{O}_X)$ by letting $f = (f_0, f_1)$ for local lifts $f_i \in \Gamma(U_i, \mathcal{O}_X)$ of $z \in \Gamma(U_i, \mathcal{O}_X^*)$; in other words, f_i is a local logarithm $\log z$ on U_i . The differential $d(f) \in C^1(\mathcal{U}, \mathcal{O}_X) = \Gamma(U_0 \cap U_1, \mathcal{O}_X)$ is $f_1 - f_0$, as a function on $U_0 \cap U_1$. Note that the image of this in $\Gamma(U_0 \cap U_1, \mathcal{O}_X^*)$ is 0, so there exists a unique element $c_{01} \in \Gamma(U_0 \cap U_1, \underline{\mathbb{Z}})$ whose image in $\Gamma(U_0 \cap U_1, \mathcal{O}_X)$ is $f_1 - f_0$. By the logic of the Snake lemma, this c_{01} is well defined, no matter how f was chosen.

Explicitly, let $f_0(z) = \log |z| + i \arg(z)$ with $\arg(z) \in (-\pi, \pi)$ and let $f_1(z) = \log |z| + i \arg(z)$ with $\arg(z) \in (0, 2\pi)$. We note that

$$f_1(z) - f_0(z) = \begin{cases} 0 & \text{if } z \in V_{\text{up}} \\ 2\pi i & \text{if } z \in V_{\text{low}} \end{cases}.$$

Therefore, $c_{01} \in \Gamma(U_0 \cap U_1, \underline{\mathbb{Z}})$ is the section on $U_0 \cap U_1$ given by 0 on V_{up} and 1 on V_{low} .

Now the differential map

$$d^0 : \Gamma(U_0, \underline{\mathbb{Z}}) \oplus \Gamma(U_1, \underline{\mathbb{Z}}) \rightarrow \Gamma(U_0 \cap U_1, \underline{\mathbb{Z}})$$

is given by

$$\mathbb{Z} \oplus \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}, \quad (a, b) \mapsto (a - b, a - b).$$

The cokernel of this map, which is exactly $\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}})$ by definition, is thus the quotient of $\mathbb{Z} \oplus \mathbb{Z}$ by pairs of the form $(a - b, a - b)$ for $a, b \in \mathbb{Z}$. This quotient is isomorphic to $\mathbb{Z}$ and the element $c_{01} = (0, 1)$ can be identified with 1 in this $\mathbb{Z}$. Therefore, $\delta(z)$ is a generator of $\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) \cong \check{H}^1(X, \underline{\mathbb{Z}}) \cong \mathbb{Z}$.

8. LEFT/RIGHT DERIVED FUNCTORS OF RIGHT/LEFT EXACT FUNCTORS

8.1. Homotopy category of chain complexes of an additive category. We will define homotopy categories and derived categories using the cohomological convention (Definition 3.0.1)

Definition 8.1.1 (Homotopy category $K(\mathcal{A})$). Let $\mathcal{A}$ be an additive category (Definition 2.3.4). (♠ TODO: describe the distinguished triangles of this category)

1. The *homotopy category* $K(\mathcal{A})$ has as objects the (cochain) complexes over $\mathcal{A}$, and as morphisms the cochain maps (Definition 3.0.1) modulo cochain homotopy equivalence (Definition 3.5.1). More explicitly,

$$\text{Hom}_{K(\mathcal{A})}(C^\bullet, D^\bullet) := \frac{\{\text{cochain maps } C^\bullet \rightarrow D^\bullet\}}{\{\text{cochain homotopies}\}},$$

where morphisms differing by a cochain homotopy are identified.

2. Let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 2.4.1) of additive categories.

The functor F naturally induces a functor between the homotopy categories of complexes,

$$K(F) : K(\mathcal{A}) \rightarrow K(\mathcal{B}),$$

by applying F degreewise to complexes and cochain maps. That is, for a complex $(C^\bullet, d^\bullet)$ in $\mathcal{A}$, define

$$K(F)(C^\bullet) := (F(C^\bullet), F(d^\bullet)),$$

and for a cochain map $f^\bullet : C^\bullet \rightarrow D^\bullet$, define

$$K(F)(f^\bullet) := (F(f^\bullet)).$$

The functor $K(F)$ respects cochain homotopies, thus it is well-defined on the homotopy category.

3. (See [DBG⁺77, Catégories dérivées Chapitre 1 2-3] for notation) We write $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, and $K^b(\mathcal{A})$ for the full subcategories of bounded below, bounded above, and bounded complexes respectively.
4. (See [DBG⁺77, Catégories dérivées Chapitre 1 2-3] for notation) If $\mathcal{A}$ is abelian, we may also write $K^{\infty,+}(\mathcal{A})$, $K^{\infty,-}(\mathcal{A})$, $K^{\infty,b}(\mathcal{A})$, and $K^{\infty,\emptyset}(\mathcal{A})$ for the full subcategories of cohomologically bounded below, cohomologically bounded above, cohomologically bounded, and acyclic complexes (Definition 3.0.2) respectively. We may further write $K^{?,??}(\mathcal{A})$ for the full subcategory where $? \in \{+, -, b\}$ indicates the boundedness of the complex and $?? \in \{+, -, b, (\text{blank})\}$ indicates the cohomological boundedness of the complex.

The various categories here are (Proposition 8.1.6) triangulated categories.

Definition 8.1.2 (Triangulated category). A *triangulated category* is a triple $(\mathcal{T}, [1], \Delta)$ where:

- $\mathcal{T}$ is an additive category,
- $[1] : \mathcal{T} \rightarrow \mathcal{T}$ (also denoted by notations such as Σ or T) is an additive auto-equivalence called the *shift* or *suspension functor*,
- Δ is a class of *distinguished triangles* of the form

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

with objects $X, Y, Z \in \mathcal{T}$ and morphisms f, g, h in $\mathcal{T}$,

satisfying the following axioms:

- (TR1) The class Δ is closed under isomorphisms of triangles and contains the triangles isomorphic to

$$A \xrightarrow{\text{id}} A \rightarrow 0 \rightarrow A[1].$$

Moreover, for every morphism $f : X \rightarrow Y$ in $\mathcal{T}$, there exists a distinguished triangle

$$X \xrightarrow{f} Y \rightarrow Z \rightarrow X[1].$$

- (TR2) (Rotation) If

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is a distinguished triangle, then so are

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$$

and

$$Z[-1] \xrightarrow{-h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z.$$

- (TR3) (Octahedral axiom) Given morphisms $X \xrightarrow{f} Y$ and $Y \xrightarrow{g} Z$ in $\mathcal{T}$, there exist distinguished triangles

$$\begin{aligned} X &\xrightarrow{f} Y \xrightarrow{u} C(f) \xrightarrow{v} X[1], \\ Y &\xrightarrow{g} Z \xrightarrow{w} C(g) \xrightarrow{x} Y[1], \end{aligned}$$

$$X \xrightarrow{g \circ f} Z \xrightarrow{y} C(g \circ f) \xrightarrow{z} X[1],$$

and morphisms

$$s : C(f) \rightarrow C(g \circ f), \quad t : C(g \circ f) \rightarrow C(g)$$

such that the following diagram commutes and its rows and columns are distinguished triangles:

$$\begin{array}{ccccccc}
X & \xrightarrow{f} & Y & \xrightarrow{u} & C(f) & \xrightarrow{v} & X[1] \\
\parallel & & \downarrow g & & \downarrow s & & \parallel \\
X & \xrightarrow{g \circ f} & Z & \xrightarrow{y} & C(g \circ f) & \xrightarrow{z} & X[1] \\
& & \downarrow w & & \downarrow t & & \\
& & C(g) & \xlongequal{\quad} & C(g) & & \\
& & \downarrow x & & \downarrow & & \\
& & Y[1] & \xrightarrow{f[1]} & X[1][1] & &
\end{array}$$

In particular,

- The compositions $s \circ u = y$, $t \circ s = w$, and $x \circ t = v[1]$ hold,
- All rows and columns form distinguished triangles.

(TR4) The class Δ is closed under the shift functor $[1]$: if

$$X \rightarrow Y \rightarrow Z \rightarrow X[1]$$

is distinguished, then so is

$$X[1] \rightarrow Y[1] \rightarrow Z[1] \rightarrow X[2].$$

Definition 8.1.3. A functor $T : D_1 \rightarrow D_2$ between triangulated categories (Definition 8.1.2) is called *exact* if it is additive, is translation preserving, and transforms distinguished triangles (Definition 8.1.2) to distinguished triangles. Oftentimes, a *morphism between triangulated categories* or a *triangulated functor* refers to an exact functor between triangulated categories.

Definition 8.1.4. In the context of a triangulated category (Definition 8.1.2) $\mathcal{D}$, a multiplicative system (Definition 8.2.1) S is *compatible with the triangulation* if:

1. For any $s \in S$, $T(s) \in S$ where T is the translation functor.
2. For any commutative square between the first two steps of two distinguished triangles where the vertical arrows are in S , there exists a morphism in S between the third objects that makes the whole diagram of triangles commute.

Definition 8.1.5. Let I be a finite set and $\varepsilon : I \rightarrow \{1, -1\}$ a variance function. Let $(\mathcal{A}_i)_{i \in I}$ and $\mathcal{A}$ be triangulated categories (Definition 8.1.2). A multifunctor $F : \prod_{i \in I} \mathcal{A}_i \rightarrow \mathcal{A}$ is an *exact multifunctor of variance ε* if for every index $k \in I$, the partial functor

$$F_k : \mathcal{A}_k \rightarrow \mathcal{A}$$

(obtained by fixing all objects A_j for $j \neq k$) satisfies the following:

1. If $\varepsilon(k) = 1$, F_k is a covariant exact functor: it maps any distinguished triangle $X \rightarrow Y \rightarrow Z \rightarrow T_k X$ in $\mathcal{A}_k$ to a distinguished triangle

$$F_k(X) \rightarrow F_k(Y) \rightarrow F_k(Z) \rightarrow TF_k(X)$$

in $\mathcal{A}$.

2. If $\varepsilon(k) = -1$, F_k is a contravariant exact functor: it maps any distinguished triangle $X \rightarrow Y \rightarrow Z \rightarrow T_k X$ in $\mathcal{A}_k$ to a distinguished triangle

$$F_k(Z) \rightarrow F_k(Y) \rightarrow F_k(X) \rightarrow F_k(T_k^{-1}X)$$

in $\mathcal{A}$, using the isomorphism $F_k(T_k^{-1}X) \cong TF_k(X)$.

Furthermore, F must be a graded functor equipped with natural isomorphisms $\varphi_i : T \circ F \xrightarrow{\sim} F \circ T_i^{\varepsilon_i}$ for each $i \in I$. These must satisfy the anticommutativity condition: for any pair of distinct indices $i, j \in I$ ($i \neq j$), the following diagram of natural transformations commutes with a factor of -1 :

$$(T) \quad \begin{array}{ccc} T^2 F & \xrightarrow{T\varphi_i} & TF T_i^{\varepsilon_i} \\ \downarrow T\varphi_j & & \downarrow \varphi_j T_i^{\varepsilon_i} \\ TF T_j^{\varepsilon_j} & \xrightarrow{-\varphi_i T_j^{\varepsilon_j}} & FT_i^{\varepsilon_i} T_j^{\varepsilon_j} \end{array}$$

Equivalently, the following square is anticommutative:

$$(U) \quad (\varphi_j T_i^{\varepsilon_i}) \circ (T\varphi_i) = -(\varphi_i T_j^{\varepsilon_j}) \circ (T\varphi_j)$$

as morphisms of functors $T^2 F \rightarrow FT_i^{\varepsilon_i} T_j^{\varepsilon_j}$.

In particular in the case that $\varepsilon(i) = 1$ for all i , we are reduced to the following definition: Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $\mathcal{A}$ be triangulated categories (Definition 8.1.2). A multifunctor $F : \prod \mathcal{A}_i \rightarrow \mathcal{A}$ is called an *exact covariant multifunctor* if it is covariant in each argument and if for every index $i \in \{1, \dots, n\}$ and every fixed collection of objects $(A_j)_{j \neq i}$, the partial functor $F(\dots, A_{i-1}, -, A_{i+1}, \dots) : \mathcal{A}_i \rightarrow \mathcal{A}$ is an exact functor (Definition 8.1.3) of triangulated categories.

Proposition 8.1.6. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

1. The homotopy category $K(\mathcal{A})$ (Definition 8.1.1) is a triangulated category (Definition 8.1.2). (♠ TODO: shift of chain complexes, describe what the distinguished triangles are)
2. The subcategory $K^?(\mathcal{A})$ is a triangulated subcategory of $K(\mathcal{A})$ for $? \in \{+, -, b\}$.
3. Let $\mathcal{A}$ be an abelian category. The subcategory $K^{?,??}(\mathcal{A})$ is a triangulated subcategory of $K(\mathcal{A})$ for $? \in \{\infty, +, -, b\}$ and $?? \in \{+, -, b, (\text{blank})\}$.

(♠ TODO: How homotopy categories of chain complexes relate to topological homotopy categories)

Theorem 8.1.7 (Exact functor induces a functor on homotopy categories, cf. [DA73, Exposé XVII, Proposition 1.1.7.]). Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor (Definition 2.4.2) of abelian categories (Definition 2.3.12). F induces exact multifunctors (Definition 8.1.5)¹¹ between the corresponding homotopy categories (Definition 8.1.1):

$$\begin{aligned} K(\mathcal{A}) &\rightarrow K(\mathcal{B}) \\ K^?(\prod \mathcal{A}_i) &\rightarrow K^?(\mathcal{B}) \\ K^{?,??}(\prod \mathcal{A}_i) &\rightarrow K^{?,??}(\mathcal{B}) \end{aligned}$$

(Definition 8.1.1). We write these functors by $K(F)$.

8.2. Multiplicative systems in categories.

Definition 8.2.1 (Multiplicative system). (♠ TODO: Write a precise definition) Let $\mathcal{C}$ be a category. A *multiplicative system S in $\mathcal{C}$* consists of a collection of morphisms in $\mathcal{C}$ satisfying:

- S contains all identity morphisms,
- S is closed under composition,
- (Ore conditions) For any morphism f in $\mathcal{C}$ and any morphism $s \in S$ with suitable domain and codomain, there exist morphisms to form commutative squares allowing localization.

The precise Ore condition varies depending on context but guarantees the localization exists.

Definition 8.2.2 (Localization by a multiplicative system). Given a category $\mathcal{C}$ and a multiplicative system (Definition 8.2.1) $S \subseteq \text{Mor}(\mathcal{C})$, the *localization $\mathcal{C}[S^{-1}]$* is a category equipped with a functor

$$Q : \mathcal{C} \rightarrow \mathcal{C}[S^{-1}]$$

that sends every morphism in S to an isomorphism satisfying the universal property: any functor from $\mathcal{C}$ sending the morphisms in S to isomorphisms factors uniquely through Q .

Definition 8.2.3. A multiplicative system (Definition 8.2.1) S in a category (Definition 1.1.1) $\mathcal{C}$ is called *saturated* if it satisfies the following condition: any morphism f in $\mathcal{C}$ whose image $Q(f)$ in the localized category (Definition 8.2.2) $\mathcal{C}(S^{-1})$ is an isomorphism belongs to S .

Proposition 8.2.4. Let $\mathcal{C}$ be a category (Definition 1.1.1) and S a multiplicative system (Definition 8.2.1). There exists a unique saturated multiplicative system (Definition 8.2.3) $\bar{S}$ containing S such that the canonical functors $\mathcal{C}(S^{-1}) \rightarrow \mathcal{C}(\bar{S}^{-1})$ define an equivalence of categories (Definition 1.3.4). $\bar{S}$ consists of all morphisms f such that there exist morphisms g, h with $g \circ f \in S$ and $f \circ h \in S$.

¹¹wlog covariant in each variable, but this can be made multi-variant by taking opposite categories (Definition 1.2.1)

If F is contravariant in the j -th variable, the definitions are adjusted by replacing $\mathcal{A}_j$ with its opposite category $\mathcal{A}_j^{\text{op}}$, thereby interchanging K^+ with K^- and RF with LF for that specific variable.

8.3. Derived categories of abelian categories. (♠ TODO: describe the objects and morphisms in a derived category)

Definition 8.3.1 (Derived category). Let $\mathcal{A}$ be an abelian category. Consider the class S of quasi-isomorphisms (Definition 3.2.8) in the homotopy category $K(\mathcal{A})$ (Definition 8.1.1) (Definition 8.1.1).

The *derived category* $D(\mathcal{A})$ is the localization (Definition 8.2.2) of the homotopy category at the multiplicative system (Definition 8.2.1) S :

$$D(\mathcal{A}) := K(\mathcal{A})[S^{-1}],$$

We often write the localization functor (Definition 8.2.2) $K(\mathcal{A}) \rightarrow D(\mathcal{A})$ by q or Q .

- We often write $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, and $D^b(\mathcal{A})$ for the full subcategories of cohomologically bounded below, cohomologically bounded above, and cohomologically bounded complexes respectively.

In general, $D(\mathcal{A})$ (and the aforementioned subcategories) may only exist as a large category (Definition 1.1.1), rather than a locally small category. When $\mathcal{A}$ is a Grothendieck abelian category (Definition 2.5.7), however, $D(\mathcal{A})$ is a locally small category (see [Sta25, Tag 09PA]).

Convention 8.3.2. Given an abelian category (Definition 8.3.1) $\mathcal{A}$, any object M of $\mathcal{A}$ may be identified with an object of the homotopy category $K(\mathcal{A})$ (Definition 8.1.1) and the derived category $D(\mathcal{A})$ (Definition 8.3.1) — simply take the chain complex concentrated at degree 0 with the object M . In fact, this derived object can often be regarded as an object of “nice” enough subcategories of $D(\mathcal{A})$, such as the bounded derived categories $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, $D^b(\mathcal{A})$.

(♠ TODO: State how a derived category is a triangulated category)

Definition 8.3.3. Let $\mathcal{A}$ be an additive category (Definition 2.3.4). For $? \in \{+, -, b, (\text{blank})\}$, write $K^?(\mathcal{I})$ (resp. $K^?(\mathcal{P})$) for the full subcategory of $K^?(\mathcal{A})$ (Definition 8.1.1) whose objects are appropriately bounded complexes of injectives (resp. projectives) (Definition 4.2.2).

Even though derived categories of abelian categories may not be guaranteed to be locally small, the following gives sufficient conditions for the appropriately (cohomologically) bounded derived categories to be locally small. The following also shows that appropriately bounded complexes under these circumstances can be represented in the derived category by a complex of injective or projective objects; we might think of such complexes of injectives or projectives as acting like an injective/projective resolution.

Theorem 8.3.4 (see e.g. [Wei94, Theorem 10.4.8]). Let $\mathcal{A}$ be an abelian category.

1. If $\mathcal{A}$ has enough injectives (Definition 4.2.5), then the category $D^+(\mathcal{A})$ (Definition 8.3.1) is equivalent to the category $K^+(\mathcal{I})$ (Definition 8.3.3), which is locally small (Definition 1.1.2). In fact, an equivalence is given by

$$K^+(\mathcal{I}) \xrightarrow{q} D^+(\mathcal{I}) \hookrightarrow D^+(\mathcal{A}).$$

2. If $\mathcal{A}$ has enough projectives (Definition 4.2.5), then the category $D^-(\mathcal{A})$ (Definition 8.3.1) is equivalent to the category $K^-(\mathcal{P})$ (Definition 8.3.3), which is locally small (Definition 1.1.2).

$$K^-(\mathcal{P}) \xrightarrow{q} D^-(\mathcal{P}) \hookrightarrow D^-(\mathcal{A}).$$

8.4. Total derived functors. The following subsection mostly takes the point of view of [Wei94, Chapter 10]

Definition 8.4.1 (cf. [Wei94, Definition 10.5.1]). Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : K(\mathcal{A}) \rightarrow K(\mathcal{B})$ (Definition 8.1.1) be an exact functor (Definition 8.1.3) of triangulated categories. Write $q_{\mathcal{A}} : K(\mathcal{A}) \rightarrow D(\mathcal{A})$ and $q_{\mathcal{B}} : K(\mathcal{B}) \rightarrow D(\mathcal{B})$ for the localization functors (Definition 8.3.1),

1. A *(total) right derived functor on F on $K(\mathcal{A})$* is an exact functor $RF : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ together with a natural transformation

$$\xi : q_{\mathcal{B}}F \Rightarrow (RF)q_{\mathcal{A}},$$

which is universal in the sense that if $G : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ is another morphism equipped with a natural transformation $\zeta : q_{\mathcal{B}}F \Rightarrow Gq_{\mathcal{A}}$, then there exists a unique natural transformation $\eta : RF \Rightarrow G$ so that $\zeta_A = \eta_{q_{\mathcal{A}}A} \circ \xi_A$ for all A in $K(\mathcal{A})$.

2. Dually, a *(total) left derived functor on F on $K(\mathcal{A})$* is an exact functor $LF : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ together with a natural transformation

$$\xi : (LF)q_{\mathcal{A}} \Rightarrow q_{\mathcal{B}}F,$$

which is universal in the sense that if $G : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ is another exact functor equipped with a natural transformation $\zeta : Gq_{\mathcal{A}} \Rightarrow q_{\mathcal{B}}F$, then there exists a unique natural transformation $\eta : G \Rightarrow LF$ such that

$$\zeta_A = \xi_A \circ \eta_{q_{\mathcal{A}}A}$$

for all A in $K(\mathcal{A})$.

The universal properties guarantee that if the total derived functors exist, then they are unique up to natural isomorphism, and that

These definitions/ideas may be extended to exact functors $K^?(\mathcal{A}) \rightarrow K(\mathcal{B})$ for $? \in \{+, -, b\}$ and to exact functors $K^{?,??}(\mathcal{A}) \rightarrow K(\mathcal{B})$ for $? \in \{\infty, +, -, b\}$ and $?? \in \{+, -, b, (\text{blank})\}$ (Definition 8.1.1)

See Definition 8.5.4 for a more general treatment of derived functors

In particular, the above definitions are specializations of those of Definition 8.5.8, which fit into the framework of Definition 8.5.4.

Proposition 8.4.2. See e.g. [Wei94, 10.5.2] Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor (Definition 2.4.12) of abelian categories (Definition 2.3.12). There is an induced functor

$$D(\mathcal{A}) \rightarrow D(\mathcal{B})$$

(Definition 8.3.1) also denoted by $\mathbf{F}$ by abuse of notation.

This functor is constructible in the “simple” way —

- For an object M of $D(\mathcal{A})$, take a chain complex K representing it. The object FM of $D(\mathcal{B})$ is the object represented by the chain complex FK .
- For a morphism $f : M \rightarrow N$ in $D(\mathcal{A})$, take a morphism $\tilde{f} : K \rightarrow L$ representing it. The morphism $Ff : FM \rightarrow FN$ is the morphism represented by $F\tilde{f} : FK \rightarrow FL$.

Due to the exactness of F , these do not depend on the choices of representatives.

The below shows that right and left total derived functors exist for nice enough abelian categories on sufficiently (cohomologically) bounded objects and can be computed using injective/projective resolutions.

Theorem 8.4.3 (See e.g. [Wei94, 10.5.6]). Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : K^+(\mathcal{A}) \rightarrow K(\mathcal{B})$ (Definition 8.1.1) be a morphism of triangulated categories (Definition 8.1.3).

1. If $\mathcal{A}$ has enough injectives (Theorem 8.1.7), then the right derived functor RF (Definition 8.4.1) exists (as a functor $D^+(\mathcal{A}) \rightarrow D(\mathcal{B})$ (Definition 8.3.1)), and if I is a bounded below complex of injectives (Definition 4.2.2), then

$$RF(I) \cong qF(I)$$

where $q : K(\mathcal{B}) \rightarrow D(\mathcal{B})$ is the localization functor (Definition 8.3.1).

2. Dually, if $\mathcal{A}$ has enough projectives, then the left derived functor LF exists (as a functor $D^-(\mathcal{A}) \rightarrow D(\mathcal{B})$), and if I is a bounded above complex of projectives, then

$$LF(P) \cong qF(P)$$

Corollary 8.4.4. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12). Write $q : K(\mathcal{B}) \rightarrow D(\mathcal{B})$ (Definition 8.1.1 Definition 8.3.1) for the localization functor (Definition 8.2.2).

1. Assuming that $\mathcal{A}$ has enough injectives (Definition 4.2.5), there exists a right derived functor (Definition 8.4.1) $RF : D^+(\mathcal{A}) \rightarrow D(\mathcal{B})$ and if $I \in K^+(\mathcal{I})$ (Definition 8.3.3) is a bounded below complex of injectives (Definition 4.2.2), then

$$RF(I) \cong q(K(F))(I).$$

where $K(F) : K^+(\mathcal{A}) \rightarrow K(\mathcal{B})$ is the exact functor (Definition 8.1.3) induced by F (Theorem 8.1.7).

If F is additionally left exact (Definition 2.4.12), then for all objects A of $\mathcal{A}$ we have

$$H^i(RF(A)) \cong R^iF(A)$$

(Definition 4.3.1) for all integers i where we regard A as the chain complex (Definition 3.0.1) object of $D^+(\mathcal{A})$ with A concentrated in degree 0 (Convention 8.3.2).

2. Assuming that $\mathcal{A}$ has enough projectives (Definition 4.2.5), there exists a left derived functor (Definition 8.4.1) $LF : D^-(\mathcal{A}) \rightarrow D(\mathcal{B})$ and if $P \in K^-(\mathcal{P})$ (Definition 8.3.3) is a bounded above complex of projectives (Definition 4.2.2), then

$$LF(P) \cong q(K(F))(P).$$

where $K(F) : K^-(\mathcal{A}) \rightarrow K(\mathcal{B})$ is the exact functor (Definition 8.1.3) induced by F (Theorem 8.1.7).

If F is additionally right exact (Definition 2.4.12), then for all objects A of $\mathcal{A}$ we have

$$H^{-i}(LF(A)) \cong L_i F(A)$$

(Definition 4.3.1) for all integers i where we regard A as the chain complex (Definition 3.0.1) object of $D^+(\mathcal{A})$ with A concentrated in degree 0 (Convention 8.3.2).

Proof. Apply Theorem 8.4.3 to the exact functor $K(F)$ to obtain the isomorphisms

$$RF(I) \cong q(K(F))(I)$$

$$LF(P) \cong q(K(F))(P).$$

If F is left exact and A is an object of $\mathcal{A}$, then the right derived functors $R^i F(A)$ (Definition 4.3.1) are defined. Recall that they are defined first by taking an injective resolution (Definition 4.2.29)

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

Since the injective resolution is acyclic (Definition 3.0.2), The object A and the complex $0 \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$ are equivalent as objects of $D^+(\mathcal{A})$ as an object of $D^+(\mathcal{A})$ are equivalent objects. Moreover, we have the isomorphism

$$RF(I) \cong q(K(F))(I).$$

In fact, the i th cohomology of the right hand side object precisely describes $R^i F(A)$ by definition. Therefore,

$$H^i(RF(A)) \cong R^i F(A)$$

as desired.

The analogous statement for $L_i F(A)$ holds similarly. □

(♠ TODO: define hyper-derived functors)

Corollary 8.4.5 (See e.g. [Wei94, Corollary 10.5.7, Remark 10.5.8]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

1. If $\mathcal{A}$ has enough injectives (Definition 4.2.5), the hyper-derived functors $\mathbb{R}^i F(X)$ are the cohomology of $\mathbf{R}F(X) : \mathbb{R}^i F(X) \cong H^i \mathbf{R}F(X)$ for all i .
2. If $\mathcal{A}$ has enough projectives (Definition 4.2.5), the hyper-derived functors $\mathbb{L}_i F(X)$ are the cohomology of $\mathbf{L}F(X) : \mathbb{L}_i F(X) \cong H^{-i} \mathbf{L}F(X)$ for all i .

8.5. General derived functors for exact multifunctors between triangulated categories. The following subsection mostly takes the point of view of [DA73, Exposé XVII, 1.2]

Definition 8.5.1 (Ind-category). Let $\mathcal{C}$ be a locally small category (Definition 1.1.2).

1. The *Ind-category of $\mathcal{C}$* , denoted $\mathbf{Ind}(\mathcal{C})$, is defined as follows:

- Objects of $\text{Ind}(\mathcal{C})$ are formal filtered colimits (Definition 2.2.13) of objects in $\mathcal{C}$. More precisely, an object is given by a filtered (Definition 2.2.12) small category I and a functor

$$X : I \rightarrow \mathcal{C}.$$

- Morphisms between objects $X : I \rightarrow \mathcal{C}$ and $Y : J \rightarrow \mathcal{C}$ are defined by

$$\text{Hom}_{\text{Ind}(\mathcal{C})}(X, Y) := \varinjlim_{i \in I} \varprojlim_{j \in J} \text{Hom}_{\mathcal{C}}(X_i, Y_j),$$

(Definition 2.2.13) where X_i and Y_j denote the images of $i \in I$ and $j \in J$ under X and Y , respectively.

The composition of morphisms is induced naturally from composition in $\mathcal{C}$. Hence, $\text{Ind}(\mathcal{C})$ is the completion of $\mathcal{C}$ under filtered colimits. Objects of $\text{Ind}(\mathcal{C})$ are called *Ind-objects of $\mathcal{C}$* .

2. The *Pro-category of $\mathcal{C}$* , denoted $\text{Pro}(\mathcal{C})$, is defined as follows:

- Objects of $\text{Pro}(\mathcal{C})$ are formal cofiltered limits (Definition 2.2.13) of objects in $\mathcal{C}$. More precisely, an object is given by a cofiltered small category I and a functor

$$X : I \rightarrow \mathcal{C}.$$

- Morphisms between objects $X : I \rightarrow \mathcal{C}$ and $Y : J \rightarrow \mathcal{C}$ are defined by

$$\text{Hom}_{\text{Pro}(\mathcal{C})}(X, Y) := \varprojlim_{j \in J} \varinjlim_{i \in I} \text{Hom}_{\mathcal{C}}(X_i, Y_j),$$

where X_i and Y_j denote the images of $i \in I$ and $j \in J$ under X and Y , respectively.

The composition of morphisms is induced naturally from composition in $\mathcal{C}$.

Hence, $\text{Pro}(\mathcal{C})$ is the completion of $\mathcal{C}$ under cofiltered limits. Objects of $\text{Pro}(\mathcal{C})$ are called *Pro-objects of $\mathcal{C}$* .

Since **Sets** has all limits and colimits (♠ TODO:) and hence has all projective and inductive limits and since $\mathcal{C}$ is locally small, $\text{Ind}(\mathcal{C})$ and $\text{Pro}(\mathcal{C})$ are locally small.

The *canonical embedding of a locally small category $\mathcal{C}$ into its Ind-category or Pro-category* is defined by treating each object as a "constant" diagram. Let $\mathbf{1}$ be the category with a single object $*$ and its identity morphism.

1. The Embedding $\iota_{\text{Ind}} : \mathcal{C} \rightarrow \text{Ind}(\mathcal{C})$:

- Objects: An object $C \in \mathcal{C}$ is mapped to the functor $X : \mathbf{1} \rightarrow \mathcal{C}$ sending $*$ to C . This represents C as a formal filtered colimit over the trivial index category.
- Morphisms: Given objects X (representing C) and Y (representing D), the Hom-set in the Ind-category simplifies:

$$\text{Hom}_{\text{Ind}(\mathcal{C})}(X, Y) \cong \varinjlim_{i \in \mathbf{1}} \varprojlim_{j \in \mathbf{1}} \text{Hom}_{\mathcal{C}}(X_i, Y_j) \cong \text{Hom}_{\mathcal{C}}(C, D).$$

2. The Embedding $\iota_{\text{Pro}} : \mathcal{C} \rightarrow \text{Pro}(\mathcal{C})$:

- Objects: An object $C \in \mathcal{C}$ is mapped to the functor $X : \mathbf{1} \rightarrow \mathcal{C}$ sending $*$ to C , representing C as a formal cofiltered limit.
- Morphisms: Similarly, the Pro-morphism definition reduces to:

$$\text{Hom}_{\text{Pro}(\mathcal{C})}(X, Y) \cong \varprojlim_{j \in \mathbf{1}} \varinjlim_{i \in \mathbf{1}} \text{Hom}_{\mathcal{C}}(X_i, Y_j) \cong \text{Hom}_{\mathcal{C}}(C, D).$$

In both cases, the functor ι is fully faithful, allowing us to identify $\mathcal{C}$ with a full subcategory of its respective completion. Given an object C of $\mathcal{C}$, the pro/ind-object $\iota(C)$ may be called the *constant pro/ind-object of C* .

Definition 8.5.2. Let $\mathcal{C}$ be a locally small category (Lemma 2.2.7).

1. An Ind-object (Definition 8.5.1) $X = \varinjlim X_i$ is said to be *essentially constant* if it is isomorphic in $\text{Ind}(\mathcal{C})$ to an object in the image of the canonical fully faithful functor $h : \mathcal{C} \rightarrow \text{Ind}(\mathcal{C})$, which maps an object $A \in \text{ob}(\mathcal{C})$ to the Ind-object represented by the constant diagram.
2. A Pro-object (Definition 8.5.1) $X = \varprojlim X_i$ is said to be *essentially constant* if it is isomorphic in $\text{Pro}(\mathcal{C})$ to an object in the image of the canonical fully faithful functor $h : \mathcal{C} \rightarrow \text{Pro}(\mathcal{C})$, which maps an object $A \in \text{ob}(\mathcal{C})$ to the Pro-object represented by the constant diagram.

In either definition, the object A may be referred to as the *value/limit/constant value of the Ind/Pro object* or the *object representing the Ind/Pro object*.

Proposition 8.5.3. Let $\mathcal{C}$ be a locally small category (Lemma 2.2.7).

1. An Ind-object (Definition 8.5.1) $X = \varinjlim_{i \in I} X_i$ (where I is a filtered (Definition 2.2.12) index category) is essentially constant (Definition 8.5.2) and isomorphic to $A \in \text{ob}(\mathcal{C})$ if and only if there exists a family of morphisms $(f_i : X_i \rightarrow A)_{i \in I}$ such that for every morphism $u : i \rightarrow j$ in I , the diagram

$$\begin{array}{ccc} X_i & \xrightarrow{X(u)} & X_j \\ & \searrow f_i & \downarrow f_j \\ & & A \end{array}$$

commutes, and the induced morphism $\varinjlim X_i \rightarrow A$ (Definition 2.2.13) is an isomorphism in $\text{Ind}(\mathcal{C})$.

2. A Pro-object (Definition 8.5.1) $X = \varprojlim_{i \in I} X_i$ (where I is a filtered (Definition 2.2.12) index category) is essentially constant (Definition 8.5.2) and isomorphic to $A \in \text{ob}(\mathcal{C})$ if and only if there exists a family of morphisms $(g_i : A \rightarrow X_i)_{i \in I}$ such that for every morphism $u : i \rightarrow j$ in I , the diagram

$$\begin{array}{ccc} A & \xrightarrow{g_i} & X_i \\ & \searrow g_j & \downarrow X(u) \\ & & X_j \end{array}$$

commutes, and the induced morphism $A \rightarrow \varprojlim X_i$ (Definition 2.2.13) is an isomorphism in $\text{Pro}(\mathcal{C})$.

Definition 8.5.4 (Mixed-Variance Derived Multifunctors, cf. [DA73, Exposé XVII, Définition 1.2.1., 1.2.5.]). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of triangulated categories (Definition 8.1.2), and let $\mathcal{C}$ be a triangulated category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an exact multifunctor (Definition 8.1.5), covariant in $\mathcal{A}_i$ and contravariant in $\mathcal{B}_j$. Let S_i, T_j, U be saturated multiplicative systems (Definition 8.2.3) in $\mathcal{A}_i, \mathcal{B}_j, \mathcal{C}$ respectively assumed compatible with the triangulated structure (Definition 8.1.4) of its respective category. Write

$$\begin{aligned} Q_{A_i} &: \mathcal{A}_i \rightarrow \mathcal{A}_i(S_i^{-1}) \\ Q_{B_j} &: \mathcal{B}_j \rightarrow \mathcal{B}_j(T_j^{-1}) \\ Q_C &: \mathcal{C} \rightarrow \mathcal{C}(U^{-1}) \end{aligned}$$

for the localization functors (Definition 8.2.2).

1. The *right derived functor RF of F relative to (S_i, T_j) and U* is the functor

$$RF : \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) \rightarrow \text{Ind}(\mathcal{C}(U^{-1}))$$

which makes the following diagram commutative

$$\begin{array}{ccc} \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j & \xrightarrow{F} & \mathcal{C} \\ \downarrow \prod Q_{A_i} \times \prod Q_{B_j} & & \downarrow \overline{Q}_C \\ \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) & \xrightarrow{RF} & \text{Ind}(\mathcal{C}(U^{-1})) \end{array}$$

where $\overline{Q}_C$ is the composition

$$\mathcal{C} \xrightarrow{Q_C} \mathcal{C}(U^{-1}) \xrightarrow{\iota_{\text{Ind}}} \text{Ind}(\mathcal{C}(U^{-1}))$$

of Q_C and the canonical embedding (Definition 8.5.1) ι_{Ind} .

2. The *left derived functor LF of F relative to (S_i, T_j) and U* is the functor

$$LF : \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) \rightarrow \text{Pro}(\mathcal{C}(U^{-1}))$$

which makes the following diagram commutative:

$$\begin{array}{ccc} \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j & \xrightarrow{F} & \mathcal{C} \\ \downarrow \prod Q_{A_i} \times \prod Q_{B_j} & & \downarrow \overline{Q}_C \\ \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) & \xrightarrow{LF} & \text{Pro}(\mathcal{C}(U^{-1})) \end{array}$$

where $\overline{Q}_C$ is the composition

$$\mathcal{C} \xrightarrow{Q_C} \mathcal{C}(U^{-1}) \xrightarrow{\iota_{\text{Pro}}} \text{Pro}(\mathcal{C}(U^{-1}))$$

of Q_C and the canonical embedding (Definition 8.5.1) ι_{Pro} .

Definition 8.5.5 (Derivability and Adapted Families). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of triangulated categories (Definition 8.1.2), and let $\mathcal{C}$ be a triangulated category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an exact multifunctor (Definition 8.1.5), covariant in $\mathcal{A}_i$ and contravariant in $\mathcal{B}_j$. Let S_i, T_j, U be saturated multiplicative systems (Definition 8.2.3) in $\mathcal{A}_i, \mathcal{B}_j, \mathcal{C}$ respectively assumed compatible with the triangulated structure (Definition 8.1.4) of its respective category. Write

$$\begin{aligned} Q_{A_i} : \mathcal{A}_i &\rightarrow \mathcal{A}_i(S_i^{-1}) \\ Q_{B_j} : \mathcal{B}_j &\rightarrow \mathcal{B}_j(T_j^{-1}) \\ Q_C : \mathcal{C} &\rightarrow \mathcal{C}(U^{-1}) \end{aligned}$$

for the localization functors (Definition 8.2.2). Write ι for the appropriate canonical embedding functors of the localized categories into their Ind or Pro categories (Definition 8.5.1).

1. Let RF be the right derived functor (Definition 8.5.4) of F relative to the multiplicative systems.
 - (a) RF is *defined at a family* (A_i, B_j) if the Ind-object (Definition 8.5.1) $RF(A_i, B_j)$ is essentially constant (Definition 8.5.2), i.e. isomorphic to an object of $\mathcal{C}(U^{-1})$. This object of $\mathcal{C}(U^{-1})$ is the *value of RF at (A_i, B_j)* .
 - (b) F is *right derivable* if RF is defined at all objects. In this case, we view RF as a functor into $\mathcal{C}(U^{-1})$.
 - (c) A family (A_i, B_j) is *adapted for RF* (or *deployed for RF* (*déployée pour RF* in French) or *unfolded for RF* , cf. [Spa88]) if the natural morphism $(\iota \circ Q_C \circ F)(A_i, B_j) \rightarrow RF(Q_{A_i}A_i, Q_{B_j}B_j)$ is an isomorphism in $\text{Ind}(\mathcal{C}(U^{-1}))$, i.e. the Ind-object $RF(A_i)$ of $A(S^{-1})$ is essentially constant (Definition 8.5.2) with value $F((A_i), (B_j))$.
2. Let LF be the left derived functor (Definition 8.5.4) of F relative to the multiplicative systems.
 - (a) LF is *defined at a family* (A_i, B_j) if the Pro-object (Definition 8.5.1) $LF(A_i, B_j)$ is essentially constant (Definition 8.5.2), i.e. isomorphic to an object of $\mathcal{C}(U^{-1})$. This object of $\mathcal{C}(U^{-1})$ is the *value of LF at (A_i, B_j)* .
 - (b) F is *left derivable* if LF is defined at all objects. In this case, we view LF as a functor into $\mathcal{C}(U^{-1})$.
 - (c) A family (A_i, B_j) is *adapted for LF* (or *deployed for LF* (*déployée pour LF* in French) or *unfolded for LF* , cf. [Spa88]) if the natural morphism $LF(Q_{A_i}A_i, Q_{B_j}B_j) \rightarrow (\iota \circ Q_C \circ F)(A_i, B_j)$ is an isomorphism in $\text{Pro}(\mathcal{C}(U^{-1}))$, i.e. the Pro-object $LF(A_i)$ of $A(S^{-1})$ is essentially constant (Definition 8.5.2) with value $F((A_i), (B_j))$.

In particular, note that if a family (A_i, B_j) is adapted for RF resp. LF , then RF resp. LF is defined at (A_i, B_j) .

Proposition 8.5.6 (cf. [Spa88]). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of triangulated categories (Definition 8.1.2), and let $\mathcal{C}$ be a triangulated category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an exact multifunctor (Definition 8.1.5), covariant in $\mathcal{A}_i$ and contravariant in $\mathcal{B}_j$. Let S_i, T_j, U be saturated multiplicative systems (Definition 8.2.3) in $\mathcal{A}_i, \mathcal{B}_j, \mathcal{C}$ respectively assumed compatible with the triangulated structure (Definition 8.1.4) of its respective category.

A family (A_i, B_j) is adapted (or deployed/unfolded) (Definition 8.5.5) for RF (Definition 8.5.4) if and only if the following holds: For every family of morphisms $s_i : A_i \rightarrow X_i$ in S_i and $t_j : Y_j \rightarrow B_j$ in T_j , there exist morphisms $\sigma_i : X_i \rightarrow \mathcal{X}_i$ in S_i and $\tau_j : \mathcal{Y}_j \rightarrow Y_j$ in T_j such that the induced morphism in $\mathcal{C}$

$$F(\sigma_i \circ s_i, t_j \circ \tau_j) : F(A_i, B_j) \rightarrow F(\mathcal{X}_i, \mathcal{Y}_j)$$

is an isomorphism in the localized category $\mathcal{C}(U^{-1})$ (i.e., its image under Q_C is an isomorphism, or equivalently, it is an element of U).

See also Proposition 8.7.1

Proof. (♠ TODO: verify this proof) Let $\mathcal{D} = \prod_i \mathcal{A}_i \times \prod_j \mathcal{B}_j^{op}$ and let S be the product multiplicative system $\prod_i S_i \times \prod_j T_j^{op}$. The right derived functor RF evaluated at an object $\mathbf{A} = (A_i, B_j)$ is defined as the Ind-object in $\mathcal{C}(U^{-1})$ represented by the diagram:

$$RF(\mathbf{A}) = \varinjlim_{\mathbf{A} \rightarrow \mathbf{X} \in S} Q_C(F(\mathbf{X}))$$

By definition, $\mathbf{A}$ is *adapted* (deployed) for RF if the natural morphism from the constant Ind-object to the system is an isomorphism:

$$\eta : \iota(Q_C(F(\mathbf{A}))) \xrightarrow{\cong} RF(Q_{\mathcal{D}}(\mathbf{A})) \quad \text{in } \text{Ind}(\mathcal{C}(U^{-1}))$$

($\implies$) Suppose $\mathbf{A}$ is adapted. An isomorphism in an Ind-category $\text{Ind}(\mathcal{M})$ between a constant object M and a filtered system $\{N_k\}_{k \in K}$ implies that for every index $k \in K$ (representing a resolution $\mathbf{A} \rightarrow \mathbf{X}$), there exists a morphism $k \rightarrow \ell$ in the filtering category (representing a further resolution $\mathbf{X} \rightarrow \mathbf{X}'$) such that the composite map $M \rightarrow N_\ell$ is an isomorphism in $\mathcal{M}$.

Translating this to our categories: for any family of morphisms $s_i : A_i \rightarrow X_i$ and $t_j : Y_j \rightarrow B_j$ (defining the resolution $\mathbf{A} \rightarrow \mathbf{X}$), there must exist a further resolution $\mathbf{X} \rightarrow \mathbf{X}'$ (defined by $\sigma_i : X_i \rightarrow \mathcal{X}_i$ and $\tau_j : \mathcal{Y}_j \rightarrow Y_j$) such that the induced morphism $Q_C(F(\mathbf{A})) \rightarrow Q_C(F(\mathbf{X}'))$ is an isomorphism in $\mathcal{C}(U^{-1})$. This is exactly the condition stated.

($\impliedby$) Conversely, assume the condition holds. Let J be the category of resolutions of $\mathbf{A}$. The condition implies that the constant object $Q_C(F(\mathbf{A}))$ is *cofinal* in the system defining $RF(\mathbf{A})$.

Specifically, the assumption guarantees that for every $\mathbf{X} \in J$, there is a $\mathbf{X}' \in J$ further in the diagram such that $Q_C(F(\mathbf{A}))$ and $Q_C(F(\mathbf{X}'))$ are isomorphic via the transition maps of the system. In the category of Ind-objects, if a filtered system contains a cofinal subsystem of objects isomorphic to a constant object M , then the Ind-limit of that system is isomorphic to M . Thus, $RF(\mathbf{A}) \cong \iota(Q_C(F(\mathbf{A})))$, which means $\mathbf{A}$ is adapted. $\square$

Proposition 8.5.7 (cf. [Spa88]). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of abelian categories, and let $\mathcal{C}$ be an abelian category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an additive multifunctor, covariant in $K(\mathcal{A}_i)$ and contravariant in $K(\mathcal{B}_j)$.

A family of complexes $(A_i^\bullet, B_j^\bullet)$ is adapted (or deployed/unfolded) (Definition 8.5.5) for the right derived functor RF if and only if the following holds: For every family of quasi-isomorphisms (Definition 3.2.8) $s_i : A_i^\bullet \rightarrow X_i^\bullet$ and $t_j : Y_j^\bullet \rightarrow B_j^\bullet$, there exist quasi-isomorphisms $\sigma_i : X_i^\bullet \rightarrow \mathcal{X}_i^\bullet$ and $\tau_j : \mathcal{Y}_j^\bullet \rightarrow Y_j^\bullet$ such that the induced morphism in $K(\mathcal{C})$

$$F(\sigma_i \circ s_i, t_j \circ \tau_j) : F(A_i^\bullet, B_j^\bullet) \rightarrow F(\mathcal{X}_i^\bullet, \mathcal{Y}_j^\bullet)$$

is a quasi-isomorphism (i.e., an isomorphism in the derived category $D(\mathcal{C})$).

Proof. This follows from Proposition 8.5.6. □

8.5.1. Derived multifunctors of additive multifunctors between abelian categories.

Definition 8.5.8. Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor (Definition 2.4.2) of abelian categories (Definition 2.3.12).

The *right (resp. left) derived multifunctors of F* are the derived functors (Definition 8.5.4) RF (resp. LF) of the induced multifunctors (Theorem 8.1.7)

- $K(\prod \mathcal{A}_i) \rightarrow K(\mathcal{B})$
- $K^+(\prod \mathcal{A}_i) \rightarrow K^+(\mathcal{B})$
- $K^-(\prod \mathcal{A}_i) \rightarrow K^-(\mathcal{B})$

(Recall Proposition 8.1.6) relative to the systems (Definition 8.2.3) of quasi-isomorphisms (Definition 3.2.8) in the respective homotopy categories (Definition 8.1.1). In particular, the functor RF a priori may refer to one of the functors

$$D^+(\prod \mathcal{A}_i) \rightarrow \text{Ind } D^+(\mathcal{B})$$

$$D^-(\prod \mathcal{A}_i) \rightarrow \text{Ind } D^-(\mathcal{B})$$

$$D(\prod \mathcal{A}_i) \rightarrow \text{Ind } D(\mathcal{B})$$

and the functor LF refers to one of the functors

$$D^+(\prod \mathcal{A}_i) \rightarrow \text{Pro } D^+(\mathcal{B})$$

$$D^-(\prod \mathcal{A}_i) \rightarrow \text{Pro } D^-(\mathcal{B})$$

$$D(\prod \mathcal{A}_i) \rightarrow \text{Pro } D(\mathcal{B})$$

(Definition 8.3.1) (Definition 8.5.1).

Definition 8.5.9 ([DA73, Definition 1.2.3]). Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor (Definition 2.4.2) of abelian categories (Definition 2.3.12).

1. An object $(A_1, \dots, A_n)$ with $A_i \in \text{ob}(\mathcal{A}_i)$ is **right acyclic for F** or **acyclic for RF** if the complex concentrated in degree 0 associated to (A_i) is adapted for (Definition 8.5.5) RF .
2. Similarly, (A_i) is **left acyclic for F** or **acyclic for LF** if the complex concentrated in degree 0 associated to (A_i) is adapted for (Definition 8.5.5) LF .

In the case that $n = 1$ and $A_1 \in \mathcal{A}_1$ has a projective (Definition 4.2.2) resolution (Definition 4.2.29) (resp. injective resolution), the notion of left acyclic for F (resp. right acyclic for F) is equivalent to the notion of F -acyclicity (Definition 4.4.1). (♠ TODO: justify)

Proposition 8.5.10 (cf. [DA73, Exposé XVII, Proposition 1.2.4.]). Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor (Definition 2.4.2) of abelian categories (Definition 2.3.12).

1. The following diagram of categories is commutative for the right derived multifunctor:

$$\begin{array}{ccc} D^+(\prod \mathcal{A}_i) & \longrightarrow & D(\prod \mathcal{A}_i) \\ \downarrow RF & & \downarrow RF \\ \text{Ind } D^+(\mathcal{B}) & \longrightarrow & \text{Ind } D(\mathcal{B}) \end{array}$$

(Definition 8.3.1) (Definition 4.6.9) (Definition 8.5.8) where the horizontal arrows are the natural functors induced by inclusions. If RF is everywhere defined on $D(\prod \mathcal{A}_i)$, its restriction to $D^+(\prod \mathcal{A}_i)$ factors through $D^+(\mathcal{B})$ [DA73, Proposition 1.2.4].

2. Dually, the following diagram is commutative for the left derived multifunctor:

$$\begin{array}{ccc} D^-(\prod \mathcal{A}_i) & \longrightarrow & D(\prod \mathcal{A}_i) \\ \downarrow LF & & \downarrow LF \\ \text{Pro } D^-(\mathcal{B}) & \longrightarrow & \text{Pro } D(\mathcal{B}) \end{array}$$

If LF is everywhere defined on $D(\prod \mathcal{A}_i)$, its restriction to $D^-(\prod \mathcal{A}_i)$ factors through $D^-(\mathcal{B})$.

Theorem 8.5.11 ([DA73, Exposé XVII, Proposition 1.2.7.]). (♠ TODO:) Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 2.4.1) between abelian categories (Definition 2.3.12).

If $\mathcal{A}$ has enough objects (Definition 4.2.1) that are acyclic for (Definition 8.5.9) LF (resp. for RF), the derived functor $LF : D^-(\mathcal{A}) \rightarrow D^-(\mathcal{B})$ (resp. $RF : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$) is left (resp. right) derivable (Definition 8.5.5) and every homologically upper bounded (resp. homologically lower bounded) complex of acyclics for LF (resp. for RF) is adapted for LF (resp. for RF) (Definition 8.5.5).

(♠ TODO:

- **Address the Biadditive/Multifunctor Gap**

- Add a Lemma or Remark following Definition 5.5.5 clarifying that for a biadditive functor $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$, the condition of a pair of complexes $(M^\bullet, N^\bullet)$ being adapted (deployed) is what allows the representation of LF via the Total Complex construction.
- Explicitly define the Total Complex functor $\text{Tot} : K(\mathcal{A}) \times K(\mathcal{B}) \rightarrow K(\mathcal{C})$ used to induce the multifunctor in Definition 5.5.6. Specify the sign convention for the total differential: $d(x \otimes y) = d_x(x) \otimes y + (-1)^p x \otimes d_y(y)$ for x of degree p .

- **Formulate the Balancing Results**

- Add a Proposition (The Balancing Lemma) stating that if $\mathcal{P} \subset \mathcal{A}$ is a class of objects adapted for $F(-, N)$ and $\mathcal{Q} \subset \mathcal{B}$ is a class adapted for $F(M, -)$, there exist canonical isomorphisms in $D(\mathcal{C})$:

$$\mathrm{Tot}(F(P^\bullet, N^\bullet)) \cong \mathrm{Tot}(F(M^\bullet, Q^\bullet)) \cong \mathrm{Tot}(F(P^\bullet, Q^\bullet))$$

- Cite [SGA 4, XVII, 1.2.7] to justify that these isomorphisms are well-defined because complexes of adapted objects are "deployed".

- **Incorporate the “Everywhere Defined” Logic**

- Include the “Finite Dimension” case from [SGA 4, XVII, Prop 1.2.10] to explain the conditions under which LF and RF can be extended to the unbounded derived category $D(\mathcal{A})$.

- **Technical Refinements**

- Provide a precise proof or reference for the property: “If F is bi-right-exact, then projective objects are left-acyclic (adapted) for F .”
- Define the term “Deployed in the first variable”: A complex $P^\bullet$ is deployed in the first variable if the family $(P^\bullet, N^\bullet)$ is deployed for the multifunctor F for every $N^\bullet \in K(\mathcal{A}_2)$.
- Resolve the ♠ TODO regarding the existence of limits/colimits in **Sets** to ensure Ind/Pro categories are locally small.

- **Categorical Equivalence**

- Add a Remark confirming the equivalence of categories $D(\mathcal{A}_1) \times D(\mathcal{A}_2) \simeq D(\mathcal{A}_1 \times \mathcal{A}_2)$, noting that quasi-isomorphisms are defined component-wise.

- **derived tensor product**

- Do proposition 4.1.7. expose XVII

)

8.6. K -flat resolutions of chain complexes with respect to bi-additive functors.

Definition 8.6.1. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. ([Spa88, After Theorem B]) Let $f : A^\bullet \rightarrow B^\bullet$ be a quasi-isomorphism (Definition 3.2.8) of chain complexes (Definition 3.0.1) of objects in $\mathcal{A}$. We say that $A^\bullet$ is a *left resolution of $B^\bullet$* and that $B^\bullet$ is a *right resolution of $A^\bullet$* .
2. Given a chain complex (Definition 3.0.1) $K^\bullet$ of objects in $\mathcal{A}$ and a class $\mathcal{Y}$ of chain complexes of objects in $\mathcal{A}$, an *$\mathcal{Y}$ -left resolution* may refer to a left resolution $Y^\bullet \rightarrow K^\bullet$ where $Y^\bullet \in \mathcal{Y}$ and an *$\mathcal{Y}$ -right resolution* may refer to a right resolution $K^\bullet \rightarrow Y^\bullet$ where $Y^\bullet \in \mathcal{Y}$.

Definition 8.6.2. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bi-additive functor (Definition 2.4.2).

Let us say that a complex (Definition 3.0.1) $K^\bullet$ of $\mathcal{A}$ (resp. of $\mathcal{B}$) is *K -flat with respect to F on the left (resp. right) for direct sum total complexes* if for every acyclic complex (Definition 3.0.2) $M^\bullet$ of $\mathcal{B}$ (resp. of $\mathcal{A}$), the direct sum total complex (Definition 4.6.5) $\mathrm{Tot}^\oplus(F(K^\bullet, M^\bullet))$ (Definition 4.6.6) (resp. $\mathrm{Tot}^\oplus(F(M^\bullet, K^\bullet))$) is acyclic.

Similarly, let us say that $K^\bullet$ is K -flat with respect to F on the left (resp. right) for product total complexes if for every acyclic complex (Definition 3.0.2) $M^\bullet$ of $\mathcal{B}$ (resp. of $\mathcal{A}$), the product total complex (Definition 4.6.5) $\text{Tot}\Pi(F(K^\bullet, M^\bullet))$ (Definition 4.6.6) (resp. $\text{Tot}\Pi(F(M^\bullet, K^\bullet))$) is acyclic.

Definition 8.6.3. Let $\mathcal{A}$ be an additive category (Definition 2.3.4).

Given (cochain) complexes (Definition 3.0.1) $K^\bullet, L^\bullet$ of objects in $\mathcal{A}$, the *internal Hom complex* $\text{Hom}^\bullet(K^\bullet, L^\bullet)$, also called the *(product) total Hom complex* or the *complex of morphisms from $K^\bullet$ to $L^\bullet$* and also denoted by notations such as $\underline{\text{Hom}}(X, Y)$ depending on the context, is the product total complex (Definition 4.6.5) $\text{Tot}\Pi(\text{Hom}(K^\bullet, L^\bullet))$ where $\text{Hom}(K^\bullet, L^\bullet)$ is the double complex associated to (Definition 4.6.6) the biadditive functor $\text{Hom} : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \text{Ab}$. More explicitly, $\text{Hom}^\bullet(K^\bullet, L^\bullet)$ is the complex of objects in Ab given by

- Objects: In each degree $n \in \mathbb{Z}$, the object is the product

$$\text{Hom}^n(K^\bullet, L^\bullet) = \prod_{p \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(K^p, L^{p+n}).$$

An element $f \in \text{Hom}^n(K^\bullet, L^\bullet)$ is thus a family of morphisms $(f_p)_{p \in \mathbb{Z}}$ where $f_p : K^p \rightarrow L^{p+n}$.

- Differential: The differential $d^n : \text{Hom}^n(K^\bullet, L^\bullet) \rightarrow \text{Hom}^{n+1}(K^\bullet, L^\bullet)$ is defined for each component $f = (f_p)$ by

$$(d^n f)_p = d_L^{p+n} \circ f_p - (-1)^n f_{p+1} \circ d_K^p,$$

where d_K and d_L are the differentials of the complexes $K^\bullet$ and $L^\bullet$ respectively.

Under this construction, the 0-cocycles (Definition 3.2.1) of $\text{Hom}^\bullet(K^\bullet, L^\bullet)$ correspond precisely to the chain maps from $K^\bullet$ to $L^\bullet$, and the 0-coboundaries (Definition 3.2.1) correspond to those chain maps which are null-homotopic (Definition 3.5.1).

If $K_\bullet$ and $L_\bullet$ are chain complexes written with homological conventions instead, then the internal Hom complex should be denoted by $\underline{\text{Hom}}_\bullet(K_\bullet, L_\bullet)$ instead; explicitly, $\text{Hom}_\bullet(K_\bullet, L_\bullet)$ is the complex of objects in Ab given by

- Objects: In each degree $n \in \mathbb{Z}$, the object is the product

$$\text{Hom}_n(K_\bullet, L_\bullet) = \prod_{p \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(K_p, L_{p+n}).$$

An element $f \in \text{Hom}_n(K_\bullet, L_\bullet)$ is a family of morphisms $(f_p)_{p \in \mathbb{Z}}$ such that $f_p : K_p \rightarrow L_{p+n}$.

- Differential: The differential $d_n : \text{Hom}_n(K_\bullet, L_\bullet) \rightarrow \text{Hom}_{n-1}(K_\bullet, L_\bullet)$ is defined for a family $f = (f_p)$ by

$$(d_n f)_p = d_L^{p+n} \circ f_p - (-1)^n f_{p-1} \circ d_K^p,$$

where $d_K^p : K_p \rightarrow K_{p-1}$ and $d_L^q : L_q \rightarrow L_{q-1}$ are the homological differentials.

As in the cohomological case, the 0-cycles of $\text{Hom}_\bullet(K_\bullet, L_\bullet)$ are precisely the chain maps between the complexes, and the 0-boundaries are the null-homotopic chain maps.

Definition 8.6.4. Let $\mathcal{A}$ be an abelian category. A chain complex $A^\bullet$ of objects in $\mathcal{A}$ is called *K -projective* (resp. *K -injective*) if for every acyclic complex (Definition 3.0.2) $X^\bullet$ of objects in $\mathcal{A}$, the complex of abelian groups $\text{Hom}^\bullet(A^\bullet, X^\bullet)$ (Definition 8.6.3) (resp. $\text{Hom}^\bullet(X^\bullet, A^\bullet)$) is acyclic.

Equivalently, $A^\bullet$ is K -injective if and only if it is K -flat with respect to (Definition 8.6.2) Hom on the left for product total complexes and $A^\bullet$ is K -projective if and only if it is K -flat with respect to Hom on the right for product total complexes.

Proposition 8.6.5 ([Spa88, 1.2. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12) and let $A^\bullet \in \mathbf{Ch}(\mathcal{A})$ be a complex (Definition 3.0.1) such that $A^i = 0$ for all $i \neq 0$. The complex (Definition 3.0.1) $A^\bullet$ is K -projective (Definition 4.2.2) (resp. K -injective (Definition 4.2.2)) if and only if A^0 is a projective (resp. injective) object of $\mathcal{A}$.

Proposition 8.6.6 ([Spa88, 1.3. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). If two of the vertices of a distinguished triangle (Definition 8.1.2) of $K(\mathcal{A})$ (Definition 8.1.1) are K -projective (Definition 8.6.4) (resp. K -injective (Definition 8.6.4)), then so is the third.

Proposition 8.6.7 ([Spa88, 1.3. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). A complex $A^\bullet$ of objects in $\mathcal{A}$ is K -projective (resp. K -injective) if and only if $A^\bullet[n]$ (Definition 3.0.6) is as well, where n is any integer.

Proposition 8.6.8 ([Spa88, 1.4. Proposition, 1.5. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12), and let $K(\mathcal{A})$ (Definition 8.1.1) and $D(\mathcal{A})$ (Definition 8.3.1) be its homotopy and derived categories, respectively.

1. For every complex (Definition 3.0.1) $A^\bullet$ in $\mathcal{A}$, the following conditions are equivalent:
 - (a) $A^\bullet$ is K -projective (Definition 8.6.4).
 - (b) For every $S^\bullet \in K(\mathcal{A})$, the natural homomorphism

$$\text{Mor}_{K(\mathcal{A})}(A^\bullet, S^\bullet) \rightarrow \text{Mor}_{D(\mathcal{A})}(A^\bullet, S^\bullet)$$

is an isomorphism.

- (c) For every diagram in $K(\mathcal{A})$

$$\begin{array}{ccc} & & A^\bullet \\ & \swarrow g & \downarrow f \\ X^\bullet & \xrightarrow{s} & S^\bullet \end{array}$$

where s is a quasi-isomorphism (Definition 3.2.8), there exists a unique morphism $g : A^\bullet \rightarrow X^\bullet$ such that $s \circ g = f$ in $K(\mathcal{A})$.

- (d) For every quasi-isomorphism $u : S^\bullet \rightarrow A^\bullet$ in $K(\mathcal{A})$, there exists a morphism $v : A^\bullet \rightarrow S^\bullet$ such that $u \circ v = 1_{A^\bullet}$ in $K(\mathcal{A})$.
2. Similarly, for every $A^\bullet$ in $\mathcal{A}$, the following conditions are equivalent:
 - (a) $A^\bullet$ is K -injective (Definition 8.6.4).
 - (b) For every $S^\bullet \in K(\mathcal{A})$, the natural homomorphism

$$\text{Mor}_{K(\mathcal{A})}(S^\bullet, A^\bullet) \rightarrow \text{Mor}_{D(\mathcal{A})}(S^\bullet, A^\bullet)$$

is an isomorphism.

(c) For every diagram in $K(\mathcal{A})$

$$\begin{array}{ccc} S^\bullet & \xrightarrow{s} & X^\bullet \\ f \downarrow & \swarrow g & \\ A^\bullet & & \end{array}$$

where s is a quasi-isomorphism, there exists a unique morphism $g : X^\bullet \rightarrow A^\bullet$ such that $g \circ s = f$ in $K(\mathcal{A})$.

(d) For every quasi-isomorphism $u : A^\bullet \rightarrow S^\bullet$ in $K(\mathcal{A})$, there exists a morphism $v : S^\bullet \rightarrow A^\bullet$ such that $v \circ u = 1_{A^\bullet}$ in $K(\mathcal{A})$.

Proposition 8.6.9. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12).

1. If $P_\bullet$ is a bounded above homological chain complex (or equivalently a bounded below cohomological chain complex) of projective objects (Definition 4.2.2), then $P_\bullet$ is K -projective (Definition 8.6.4).

$$\cdots \rightarrow P_{i+1} \rightarrow P_i \rightarrow 0 \rightarrow 0 \rightarrow \cdots$$

2. If $I^\bullet$ is a bounded below cohomological chain complex (or equivalently a bounded above homological chain complex) of injective objects (Definition 4.2.2), then $I^\bullet$ is K -injective (Definition 8.6.4).

$$\cdots \rightarrow 0 \rightarrow 0 \rightarrow I^i \rightarrow I^{i+1} \rightarrow \cdots$$

Proof. (♠ TODO: verify the following) We prove (1). By the characterization of K -projectivity, it suffices to show that for any acyclic complex $N^\bullet$, every chain map $f : P^\bullet \rightarrow N^\bullet$ is homotopic to zero ($f \sim 0$).

Assume without loss of generality that $P^n = 0$ for $n > 0$. We construct a homotopy $h^n : P^n \rightarrow N^{n-1}$ such that $f^n = d_N^{n-1}h^n + h^{n+1}d_P^n$ by descending induction on n .

Base Case: For $n = 0$, since $P^1 = 0$, the chain map condition $d_N^0 f^0 = f^1 d_P^0$ implies $d_N^0 f^0 = 0$. Thus $\text{im}(f^0) \subseteq \ker(d_N^0)$. By the acyclicity of $N^\bullet$, $\ker(d_N^0) = \text{im}(d_N^{-1})$. Because P^0 is projective, f^0 factors through the surjection $N^{-1} \rightarrow \text{im}(d_N^{-1})$, providing $h^0 : P^0 \rightarrow N^{-1}$ such that $d_N^{-1}h^0 = f^0$. Setting $h^1 = 0$, we have $f^0 = d_N^{-1}h^0 + h^1 d_P^0$.

Inductive Step: Suppose we have constructed $\{h^k\}_{k \geq n}$ satisfying the homotopy relation. Consider the map $g = f^n - h^{n+1}d_P^n : P^n \rightarrow N^n$. We check its image under d_N^n :

$$\begin{aligned} d_N^n g &= d_N^n f^n - d_N^n h^{n+1} d_P^n \\ &= f^{n+1} d_P^n - (f^{n+1} - h^{n+2} d_P^{n+1}) d_P^n \\ &= f^{n+1} d_P^n - f^{n+1} d_P^n + h^{n+2} d_P^{n+1} d_P^n = 0. \end{aligned}$$

Thus $\text{im}(g) \subseteq \ker(d_N^n)$. By acyclicity, $\ker(d_N^n) = \text{im}(d_N^{n-1})$. Since P^n is projective, g factors through the surjection d_N^{n-1} , yielding $h^n : P^n \rightarrow N^{n-1}$ such that $d_N^{n-1}h^n = g = f^n - h^{n+1}d_P^n$.

This construction provides a well-defined sequence of maps $(h^n)_{n \in \mathbb{Z}}$ defining a homotopy $h : f \sim 0$. Thus $\text{Mor}_{K(\mathcal{A})}(P^\bullet, N^\bullet) = 0$, proving $P^\bullet$ is K -projective. The proof for (2) is dual, using ascending induction starting from the lowest non-zero degree of $I^\bullet$. $\square$

Theorem 8.6.10 ([Spa88, 3.4. Theorem, 3.8. Proposition]). 1. Let $\mathcal{A}$ be an AB5 (Definition 2.5.7) abelian category (Definition 2.3.12). Suppose that $\mathfrak{P}$ is a class of objects in $\mathbf{Ch}(\mathcal{A})$ satisfying the following:

Every (cohomologically) bounded above complex $A^\bullet \in \mathbf{Ch}(\mathcal{A})$ has a left $\mathfrak{P}$ -resolution (Definition 8.6.1).

Then every complex in $\mathbf{Ch}(\mathcal{A})$ has a left $\mathfrak{P}$ -resolution.

2. Dually, let $\mathcal{A}$ be an AB5* (Definition 2.5.7) abelian category (Definition 2.3.12). Suppose that $\mathfrak{I}$ is a class of objects in $\mathbf{Ch}(\mathcal{A})$ satisfying the following:

Every (cohomologically) bounded below complex $A^\bullet \in \mathbf{Ch}(\mathcal{A})$ has a left $\mathfrak{I}$ -resolution (Definition 8.6.1).

Then every complex in $\mathbf{Ch}(\mathcal{A})$ has a right $\mathfrak{I}$ -resolution.

Corollary 8.6.11 ([Spa88, 3.5. Corollary, 3.9. Corollary]). 1. Let $\mathcal{A}$ be an AB5 (Definition 2.5.7) abelian category (Definition 2.3.12). If $\mathcal{A}$ has enough projectives (Definition 4.2.5), then every complex in $\mathcal{C}$ has a left K -projective (Definition 8.6.4) resolution (Definition 8.6.1). (♠ TODO: bounded complex of projectives is K -projective)

2. Dually, let $\mathcal{A}$ be an AB5* (Definition 2.5.7) abelian category (Definition 2.3.12). If $\mathcal{A}$ has enough injectives (Definition 4.2.5), then every complex in $\mathcal{C}$ has a right K -injective (Definition 8.6.4) resolution (Definition 8.6.1).

Proof. 1. Since $\mathcal{A}$ has enough projectives, any bounded above homological complex in $\mathcal{A}$ has a left-resolution Proposition 4.2.33 by a bounded above homological complex of projective objects, which is K -projective by Proposition 8.6.9. Therefore, any complex in $\mathcal{A}$ has a left-resolution by a K -projective object by Theorem 8.6.10

2. This is dual to the previous part.

□

Theorem 8.6.12 ([Ser03, Theorem 3.13.]). Let $\mathcal{A}$ be a Grothendieck category (Definition 2.5.7).

For every chain complex (Definition 3.0.1) $A^\bullet$ in $\mathcal{A}$, there is a right resolution (Definition 8.6.1) $A^\bullet \rightarrow I^\bullet$ such that $I^\bullet$ is K -injective (Definition 8.6.4).

Lemma 8.6.13. Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bi-additive functor (Definition 2.4.2) where $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are additive categories (Definition 2.3.4). Let J be a category (Definition 1.1.1).

1. Assume that $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves colimits (Definition 2.2.8) of diagrams of shape J (Definition 2.2.6) for all objects $X \in \mathcal{A}$. Then the induced functor $F(X^\bullet, -) : \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ (Definition 4.6.6) (Definition 3.0.1) (Definition 4.6.3) also preserves colimits of diagrams of shape J for all objects $X^\bullet \in \mathbf{Ch}(\mathcal{A})$.
2. Dually, assume that $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ preserves colimits (Definition 2.2.8) of diagrams of shape J (Definition 2.2.6) for all objects $Y \in \mathcal{B}$. Then the induced functor $F(-, Y^\bullet) : \mathbf{Ch}(\mathcal{A}) \rightarrow \mathbf{DC}(\mathcal{C})$ (Definition 4.6.6) (Definition 3.0.1) (Definition 4.6.3) also preserves colimits of diagrams of shape J for all objects $Y^\bullet \in \mathbf{Ch}(\mathcal{B})$.
3. Assume that $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves limits (Definition 2.2.8) of diagrams of shape J (Definition 2.2.6) for all objects $X \in \mathcal{A}$. Then the induced functor $F(X^\bullet, -) :$

$\mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ (Definition 4.6.6) (Definition 3.0.1) (Definition 4.6.3) also preserves limits of diagrams of shape J for all objects $X^\bullet \in \mathbf{Ch}(\mathcal{A})$.

4. Dually, assume that $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ preserves limits (Definition 2.2.8) of diagrams of shape J (Definition 2.2.6) for all objects $Y \in \mathcal{B}$. Then the induced functor $F(-, Y^\bullet) : \mathbf{Ch}(\mathcal{A}) \rightarrow \mathbf{DC}(\mathcal{C})$ (Definition 4.6.6) (Definition 3.0.1) (Definition 4.6.3) also preserves limits of diagrams of shape J for all objects $Y^\bullet \in \mathbf{Ch}(\mathcal{B})$.

Proof. We prove that if $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves colimits of diagrams of shape J for all $X \in \mathcal{A}$, then so does $F(X^\bullet, -) : \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ for all $X^\bullet \in \mathbf{Ch}(\mathcal{A})$. Given $X^\bullet \in \mathbf{Ch}(\mathcal{A})$ and $Y^\bullet \in \mathbf{Ch}(\mathcal{B})$, the term of $F(X^\bullet, Y^\bullet)$ at position (p, q) is $F(X^p, Y^q)$. Let $J \rightarrow \mathbf{Ch}(\mathcal{B}) : j \mapsto Y_{(j)}^\bullet$ be some diagram (Definition 2.2.6) whose colimit is $Y^\bullet$. In fact, since colimits of chain complexes are computed termwise (??), there are canonical isomorphisms

$$Y^q \cong \operatorname{colim}_{j \in J} Y_{(j)}^q$$

in $\mathcal{B}$.

We wish to show that $F(X^\bullet, Y^\bullet)$ is the colimit of the diagram $j \mapsto F(X^\bullet, Y_{(j)}^\bullet)$ in $\mathbf{DC}(\mathcal{C})$. The colimit in $\mathbf{DC}(\mathcal{C})$ is also computed termwise (note that $\mathbf{DC}(\mathcal{C})$ is identifiable as $\mathbf{Ch}(\mathbf{Ch}(\mathcal{C}))$), it suffices to verify the isomorphism at each position (p, q) and ensure it respects the differentials.

Consider the term at position (p, q) . The functor $F(X^\bullet, -)$ maps the colimit diagram in $\mathbf{Ch}(\mathcal{B})$ to a cocone (Definition 2.2.8) in $\mathbf{DC}(\mathcal{C})$. This induces a canonical comparison morphism:

$$\phi : \operatorname{colim}_{j \in J} F(X^\bullet, Y_{(j)}^\bullet) \longrightarrow F(X^\bullet, \operatorname{colim}_{j \in J} Y_{(j)}^\bullet) = F(X^\bullet, Y^\bullet).$$

At the position (p, q) , this morphism is the map:

$$\phi^{p,q} : \operatorname{colim}_{j \in J} F(X^p, Y_{(j)}^q) \longrightarrow F(X^p, Y^q).$$

By hypothesis, $F(X^p, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves colimits of shape J . Therefore, $\phi^{p,q}$ is an isomorphism for all $p, q \in \mathbb{Z}$.

Since a morphism of double complexes is an isomorphism if and only if it is an isomorphism at every term (p, q) , we conclude that ϕ is an isomorphism. Thus, $F(X^\bullet, -)$ preserves colimits of shape J .

□

Lemma 8.6.14. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 2.3.12), and let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bi-additive functor (Definition 2.4.2). Assume that $\mathcal{C}$ is Ab5 (Definition 2.5.7).

1. Suppose that $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves filtered colimits (Definition 2.2.13) for all objects $X \in \mathcal{A}$. Let $P^\bullet$ be a (cohomologically) bounded above complex of objects of $\mathcal{A}$ that are flat (Definition 4.6.9) with respect to F on the left. The complex $P^\bullet$ is K -flat with respect to F on the left for direct sum total complexes (Definition 8.6.2).
2. Suppose that $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ preserves filtered colimits (Definition 2.2.13) for all objects $Y \in \mathcal{B}$. Let $P^\bullet$ be a bounded above complex of objects in $\mathcal{B}$ that are flat with respect to F on the right. The complex $P^\bullet$ is K -flat with respect to F on the right for direct sum total complexes.

Proof. We show the claimed statement in the case that $P^\bullet$ is a complex of flat objects of $\mathcal{A}$ with respect to F on the left; the other case can be argued symmetrically. Let $M^\bullet$ be an acyclic complex (Definition 3.0.2) of objects in $\mathcal{A}$. Writing $\tau_{\leq k} M^\bullet$ for the brutal truncation of $M^\bullet$, note that $M^\bullet$ is the filtered colimit of these truncations (??):

$$M^\bullet = \varinjlim_k \tau_{\leq k} M^\bullet.$$

Since $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves filtered colimits, so does $F(X^\bullet, -) : \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ (Definition 4.6.6) (Definition 3.0.1) (Definition 4.6.3) by Lemma 8.6.13. Therefore,

$$\mathrm{Tot}^\oplus F(P^\bullet, M^\bullet) \cong \mathrm{Tot}^\oplus F(P^\bullet, \varinjlim_k \tau_{\leq k} M^\bullet) \cong \varinjlim_k \mathrm{Tot}^\oplus F(P^\bullet, \tau_{\leq k} M^\bullet).$$

Since $\tau_{\leq k} M^\bullet$ is bounded above and is of flat objects, the double complex $F(P^\bullet, \tau_{\leq k} M^\bullet)$ (Definition 4.6.6) is bounded above (Definition 4.6.4), and each row is acyclic. Since $\mathcal{C}$ satisfies Ab5, $\mathrm{Tot}^\oplus F(P^\bullet, \tau_{\leq k} M^\bullet)$ is acyclic by Lemma 4.6.8. That $\mathcal{C}$ satisfies AB5 further implies that

$$H^n(\mathrm{Tot}^\oplus F(P^\bullet, M^\bullet)) \cong \varinjlim_k H^n \mathrm{Tot}^\oplus F(P^\bullet, \tau_{\leq k} M^\bullet) = 0,$$

i.e. $\mathrm{Tot}^\oplus F(P^\bullet, M^\bullet)$ is acyclic. Therefore, $P^\bullet$ is K -flat as desired. $\square$

8.7. General conditions for defining derived functors.

Proposition 8.7.1 (see [Sta25, Tag 06XN, 05S8]). Suppose that $F : \mathcal{D} \rightarrow \mathcal{D}'$ is an exact functor (Definition 8.1.3) of triangulated categories (Definition 8.1.2) and that S is a saturated multiplicative system (Definition 8.2.3) in $\mathcal{D}$ compatible with (Definition 8.1.4) the structure of the triangulated category (Definition 8.1.2) on $\mathcal{D}$, i.e. $s : X \rightarrow Y$ is in S if and only if $s[1] : X[1] \rightarrow Y[1]$ is in S and for every commutative diagram in $\mathcal{D}$ where the rows are distinguished triangles:

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \rightarrow & Z & \rightarrow & X[1] \\ \downarrow^{s_1} & & \downarrow^{s_2} & & \exists \downarrow^{s_3} & & \downarrow^{s_1[1]} \\ X' & \xrightarrow{f'} & Y' & \rightarrow & Z' & \rightarrow & X'[1] \end{array}$$

if $s_1, s_2 \in S$, there exists a morphism $s_3 \in S$ making the entire diagram commute.

If there exists a subcollection $\mathcal{I} \subset \mathrm{Ob}(\mathcal{D})$ such that

1. for all $X \in \mathrm{Ob}(\mathcal{D})$ there exists $s : X \rightarrow X'$ in S with $X' \in \mathcal{I}$, and
2. for every arrow $s : X \rightarrow X'$ in S with $X, X' \in \mathcal{I}$ the map $F(s) : F(X) \rightarrow F(X')$ is an isomorphism,

then RF (Definition 8.5.4) is everywhere defined (Definition 8.5.5) and every $X \in \mathcal{I}$ computes RF .

Dually, if there exists a subcollection $\mathcal{P} \subset \mathrm{Ob}(\mathcal{D})$ such that

1. for all $X \in \mathrm{Ob}(\mathcal{D})$ there exists $s : X' \rightarrow X$ in S with $X' \in \mathcal{P}$, and
2. for every arrow $s : X \rightarrow X'$ in S with $X, X' \in \mathcal{P}$ the map $F(s) : F(X) \rightarrow F(X')$ is an isomorphism,

then LF is everywhere defined and every $X \in \mathcal{P}$ computes LF . (♠ TODO: Proposition 8.5.6 should be relevant)

Proposition 8.7.2 (see [Sta25, Tag 070K]). Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Assume that every complex (Definition 3.0.1) of objects in $\mathcal{A}$ has a right resolution (Definition 8.6.1) by a K -injective (Definition 8.6.4) complex. For any exact functor (Definition 8.1.3) $F : K(\mathcal{A}) \rightarrow \mathcal{D}'$ (Definition 8.1.1) of triangulated categories (Definition 8.1.2), the right derived functor (Definition 8.5.8)

$$RF : D(\mathcal{A}) \rightarrow \mathcal{D}'$$

is defined (Definition 8.5.5) and $RF(I^\bullet) = F(I^\bullet)$ for K -injective complexes $I^\bullet$.

Proof. By Proposition 8.7.1, it suffices to show that if $I^\bullet \rightarrow J^\bullet$ is a quasi-isomorphism of K -injective complexes, then $F(I^\bullet) \rightarrow F(J^\bullet)$ is an isomorphism in $\mathcal{D}'$ (♠ TODO: continue). $\square$

8.7.1. Adjunction of derived functors of adjoint functors.

Proposition 8.7.3 ([Sta25, Tag 0FND]). Let $F : \mathcal{D} \rightarrow \mathcal{D}'$ and $G : \mathcal{D}' \rightarrow \mathcal{D}$ be exact functors of triangulated categories (Definition 8.1.3). Assume F is right adjoint (Definition 2.5.1) to G . Let S and S' respectively be multiplicative systems (Definition 8.2.3) for $\mathcal{D}$ and $\mathcal{D}'$ compatible with the triangulated structures (Definition 8.1.4). Let $K \in \text{Ob}(\mathcal{D})$ and $M \in \text{Ob}(\mathcal{D}')$. If RF (Definition 8.5.4) is defined (Definition 8.5.5) at K and LG is defined at M , then there is a canonical isomorphism

$$\text{Hom}_{\mathcal{D}'[(S')^{-1}]}(M, RF(K)) = \text{Hom}_{\mathcal{D}[S^{-1}]}(LG(M), K)$$

which is functorial (Definition 1.2.2) in both variables on the triangulated subcategories (Definition 8.1.2) of $\mathcal{D}[S^{-1}]$ and $\mathcal{D}'[(S')^{-1}]$ where RF and LG are defined.

Proposition 8.7.4 ([Sta25, Tag 0DVC]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ be functors (Definition 1.2.2) of abelian categories (Definition 2.3.12) such that F is a right adjoint (Definition 2.5.1) to G ; in particular, F and G are additive (Definition 2.4.1) (??). Let $K^\bullet$ be a complex (Definition 3.0.1) of $\mathcal{A}$ and let $M^\bullet$ be a complex of $\mathcal{B}$. If RF (Definition 8.5.8) is defined (Definition 8.5.5) at $K^\bullet$ and LG is defined at $M^\bullet$, then there is a canonical isomorphism

$$\text{Hom}_{D(\mathcal{B})}(M^\bullet, RF(K^\bullet)) = \text{Hom}_{D(\mathcal{A})}(LG(M^\bullet), K^\bullet)$$

This isomorphism is functorial (Definition 1.2.2) in both variables on the triangulated subcategories (Definition 8.1.2) of $D(\mathcal{A})$ (Definition 8.3.1) and $D(\mathcal{B})$ where RF and LG are defined.

Proposition 8.7.5 ([Sta25, Tag 09T5]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ be functors (Definition 1.2.2) of abelian categories (Definition 2.3.12) such that F is a right adjoint (Definition 2.5.1) to G . If the derived functors (Definition 8.5.8) $RF : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ and $LG : D(\mathcal{B}) \rightarrow D(\mathcal{A})$ exist, then RF is a right adjoint to LG .

8.8. Having enough K -injectives and flats for sheaves of modules.

Theorem 8.8.1. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition B.0.17) where the underlying site $(\mathcal{C}, J)$ is essentially small. In particular, $\text{Mod}_{\mathcal{O}}$ is an abelian category (Definition 2.3.12) and in fact is a Grothendieck category (Definition 2.5.7) (Theorem 2.5.9).

For every chain complex (Definition 3.0.1) $A^{\bullet}$ in $\text{Mod}_{\mathcal{O}}$ (Definition B.0.18) (resp. ${}_{\mathcal{O}}\text{Mod}$), there is a right resolution (Definition 8.6.1) $A^{\bullet} \rightarrow I^{\bullet}$ such that $I^{\bullet}$ is K -injective (Definition 8.6.4).

Proof. This follows from Theorem 8.6.12. □

Definition 8.8.2. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition B.0.17) whose underlying site (Definition B.0.5) is essentially small.

A left (resp. right) $\mathcal{O}$ -module (Definition B.0.18) $\mathcal{F}$ is called *flat* if the functor $- \otimes_{\mathcal{O}} \mathcal{F} : \text{Mod}_{\mathcal{O}} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$ (resp. $\mathcal{F} \otimes_{\mathcal{O}} - : {}_{\mathcal{O}}\text{Mod} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$) (Definition 7.6.6) (Definition B.0.8) is exact (Definition 2.4.12).

Equivalently, $\mathcal{F}$ is flat if and only if it is flat (Definition 4.6.9) with respect to $\otimes$.

Proposition 8.8.3. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition B.0.17) whose underlying site (Definition B.0.5) is essentially small.

The categories $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ have enough (Definition 4.2.1) flat $\mathcal{O}$ -modules (Definition 8.8.2) on the left. (♠ TODO: prove)

Proposition 8.8.4. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition B.0.17) whose underlying site (Definition B.0.5) is essentially small.

There are enough K -flat resolutions to the left of chain complexes (Definition 3.0.1) in of objects in $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$. More precisely, for any chain complex (Definition 3.0.1) $A_{\bullet} \in \mathbf{Ch}(\text{Mod}_{\mathcal{O}})$ (resp. $\mathbf{Ch}({}_{\mathcal{O}}\text{Mod})$), there is a K -flat (Definition 8.6.2) (with respect to $\otimes : \text{Mod}_{\mathcal{O}} \times {}_{\mathcal{O}}\text{Mod} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$) chain complex $F_{\bullet} \in \text{Mod}_{\mathcal{O}}$ and a left resolution (Definition 8.6.1) $F_{\bullet} \rightarrow A_{\bullet}$.

Proof. Since $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ have enough flats on the left by Proposition 8.8.3, every (homologically) bounded below complex in $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ has a left resolution by a (homologically) bounded below complex of flat objects; a (homologically) bounded below complex of flats is K -flat with respect to $\otimes$ by Lemma 8.6.14. To see that Lemma 8.6.14 applies, note that $\otimes$ is a left adjoint by Theorem 7.6.7 and hence commutes with colimits by ??, and note that $\text{Sh}(\mathcal{C}, J; \text{Ab})$ is AB5. Theorem 8.6.10 then applies to conclude that chain complex of objects in $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ has a left K -flat resolution. □

8.9. Definitions of derived functors for modules of ringed sites.

Notation 8.9.1. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition B.0.17) where the underlying site $(\mathcal{C}, J)$ is essentially small or let $(\mathcal{C}, \mathcal{O})$ be a ringed topos. In particular, $\text{Mod}_{\mathcal{O}}$ is an abelian category (Definition 2.3.12).

Denote by $D(\mathcal{O})$, $D^+(\mathcal{O})$, $D^-(\mathcal{O})$, $D^b(\mathcal{O})$ the derived categories $D(\mathcal{A})$, $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, $D^b(\mathcal{A})$ (Definition 8.3.1) where $\mathcal{A}$ is either the abelian category $\mathcal{O}\text{-Mod}$ or $\text{Mod}_{\mathcal{O}}$.

Definition 8.9.2. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition B.0.17) whose underlying site (Definition B.0.5) is essentially small.

Given (cochain) complexes (Definition 3.0.1) $K^\bullet, L^\bullet$ of (say right) $\mathcal{O}$ -modules (Definition B.0.18), the *internal sheaf/sheafy Hom complex* $\mathcal{H}om^\bullet(K^\bullet, L^\bullet)$, also called the *(product) total sheaf Hom complex*, but not to be confused with the internal hom complex (Definition 8.6.3) $\text{Hom}^\bullet(K^\bullet, L^\bullet)$, is the product total complex (Definition 4.6.5) $\text{Tot}^\Pi(\mathcal{H}om(K^\bullet, L^\bullet))$, where $\mathcal{H}om(K^\bullet, L^\bullet)$ is the double complex associated to (Definition 4.6.6) the sheaf hom functor (Definition 7.6.5)

$$\mathcal{H}om : \text{Mod}_{\mathcal{O}}^{\text{op}} \times \text{Mod}_{\mathcal{O}} \rightarrow \text{Sh}(\mathcal{C}, J; \mathbb{Z})$$

(Definition B.0.18) (Definition B.0.8).

More explicitly, $\mathcal{H}om^\bullet(K^\bullet, L^\bullet)$ is the complex of objects in $\text{Sh}(\mathcal{C}, J; \mathbb{Z})$ given by

- **Objects:** In each degree $n \in \mathbb{Z}$, the sheaf of $\mathbb{Z}$ -modules is the product

$$\mathcal{H}om^n(K^\bullet, L^\bullet) = \prod_{p \in \mathbb{Z}} \mathcal{H}om_{\mathcal{O}}(K^p, L^{p+n}).$$

A local section $f \in \mathcal{H}om^n(K^\bullet, L^\bullet)(U)$ for an object $U \in \mathcal{C}$ is a family of $\mathcal{O}|_U$ -linear morphisms $f_p : K^p|_U \rightarrow L^{p+n}|_U$.

- **Differential:** The differential $d^n : \mathcal{H}om^n(K^\bullet, L^\bullet) \rightarrow \mathcal{H}om^{n+1}(K^\bullet, L^\bullet)$ is defined for a family of local morphisms $f = (f_p)$ by

$$(d^n f)_p = d_L^{p+n} \circ f_p - (-1)^n f_{p+1} \circ d_K^p,$$

where $d_K^p : K^p \rightarrow K^{p+1}$ and $d_L^q : L^q \rightarrow L^{q+1}$ are the cohomological differentials of the complexes.

Definition 8.9.3 (cf. [Sta25, Tag 08J7] for a discussion when $\mathcal{O}$ is assumed commutative). Let $(\mathcal{C}, J, \mathcal{O})$ be an essentially small ringed site (Definition B.0.17) with a pretopology (Definition B.0.5). (♠ TODO: the “pretopology” part should be loosened)

The *derived (sheafy/sheaf) internal hom functor* is the functor $R\mathcal{H}om : D(\mathcal{O})^{\text{op}} \times D(\mathcal{O}) \rightarrow D(\text{Sh}(\mathcal{C}, J; \mathbb{Z}))$ (Notation 8.9.1), also denoted by notations such as $R\mathcal{H}om(\mathcal{M}, \mathcal{N})$, $R\mathcal{H}om(\mathcal{M}, \mathcal{N})$, or $R\mathcal{H}om(\mathcal{M}, \mathcal{N})$, defined as the right derived functor (Definition 8.5.8) of the sheaf hom functor (Definition 7.6.5)

$$\mathcal{H}om : \text{Mod}_{\mathcal{O}}^{\text{op}} \times \text{Mod}_{\mathcal{O}} \rightarrow \text{Sh}(\mathcal{C}, J; \mathbb{Z}).$$

(♠ TODO: 07A5 for justifying why such a thing exists)

(♠ TODO: This is defined because...) Explicitly, given objects $L, M \in D(\mathcal{O})$, $R\mathcal{H}om(L, M)$ is given as follows: let $\mathcal{L}^\bullet$ be any complex of $\mathcal{O}$ -modules representing L and let $\mathcal{I}^\bullet$ be any (right) K -injective (Definition 8.6.4) resolution (Definition 8.6.1) of M , which exists by Theorem 8.8.1. The object $R\mathcal{H}om(L, M)$ is then given as the object of $D(\text{Sh}(\mathcal{C}, J; \mathbb{Z}))$ represented by the internal sheaf hom object (Definition 8.9.2) $\mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)$.

Definition 8.9.4 (cf. [Sta25, Tag 06YU] for a simplified discussion when $\mathcal{O}$ is assumed commutative). Let $(\mathcal{C}, J, \mathcal{O})$ be an essentially small ringed site (Definition B.0.17) with a pretopology (Definition B.0.5). (♠ TODO: the “pretopology” part should be loosened)

The *derived tensor product* is the functor

$$- \otimes_{\mathcal{O}}^L - : D(\text{Mod}_{\mathcal{O}}) \times D({}_{\mathcal{O}}\text{Mod}) \rightarrow D(\text{Sh}(\mathcal{C}, J; \text{Ab}))$$

(Definition B.0.18) (Notation 8.9.1) also denoted by notations such as $\overset{\mathbf{L}}{\otimes}$ defined as the left derived functor (Definition 8.5.8) of the tensor product functor (Definition 7.6.6)

$$- \otimes_{\mathcal{O}} - : \text{Mod}_{\mathcal{O}} \times {}_{\mathcal{O}}\text{Mod} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$$

Explicitly, given objects $L, M \in D(\mathcal{O})$, $L \otimes_{\mathcal{O}} M$ is given as follows: let $\mathcal{L}^{\bullet}$ be any complex of right $\mathcal{O}$ -modules representing L and let $\mathcal{M}^{\bullet}$ be any complex of left $\mathcal{O}$ -modules representing M , such that $\mathcal{L}^{\bullet}$ is K -flat (Definition 8.6.2) with respect to $\otimes_{\mathcal{O}}$ on the left for direct sum total complexes or $\mathcal{M}^{\bullet}$ is K -flat with respect to $\otimes_{\mathcal{O}}$ on the left for direct sum total complexes; note that such complexes exist by Proposition 8.8.4. The object $L \otimes_{\mathcal{O}}^L M$ is then given as the object of $D(\text{Sh}(\mathcal{C}, J; \text{Ab}))$ represented by the total complex $\text{Tot}^{\oplus}(\mathcal{L}^{\bullet} \otimes_{\mathcal{O}} \mathcal{M}^{\bullet})$. (♠ TODO: justify definedness, i.e. why does this actually define the derived functor)

Theorem 8.9.5 (cf. [Sta25, Tag 08J9]). Let $(\mathcal{C}, J, \mathcal{O})$ be an essentially small ringed site (Definition B.0.17) with a pretopology (Definition B.0.5). (♠ TODO: this pretopology condition should be loosened)

Let K, L, M be objects of $D(\mathcal{O})$ (Notation 8.9.1). There is a canonical isomorphism

$$R\mathcal{H}om(K, R\mathcal{H}om(L, M)) \cong R\mathcal{H}om(K \otimes_{\mathcal{O}}^L L, M)$$

(Definition 8.9.3) (Definition 8.9.4) in $D(\mathcal{O})$ functorial in K, L, M . Taking the zeroth sheaf cohomology (or equivalently global sections) of both sides, one obtains the adjunction

$$\text{Hom}_{D(\mathcal{O})}(K, R\mathcal{H}om(L, M)) \cong \text{Hom}_{D(\mathcal{O})}(K \otimes_{\mathcal{O}}^L L, M).$$

Definition 8.9.6. Let $((\mathcal{C}, J), \mathcal{O}_{\mathcal{C}})$ and $((\mathcal{D}, K), \mathcal{O}_{\mathcal{D}})$ be ringed sites (Definition B.0.17).

A *morphism of ringed sites* $f : ((\mathcal{C}, J), \mathcal{O}_{\mathcal{C}}) \rightarrow ((\mathcal{D}, K), \mathcal{O}_{\mathcal{D}})$ is a pair $(f, f^{\#})$ where:

1. $f : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ is a morphism of sites.
2. $f^{\#} : \mathcal{O}_{\mathcal{D}} \rightarrow f_*\mathcal{O}_{\mathcal{C}}$ (??) is a morphism of sheaves of rings (Definition B.0.8) on the site $(\mathcal{D}, K)$.

Equivalently, the morphism $f^{\#}$ a morphism can be described by the adjoint morphism (??) $f^{\#} : f^{-1}\mathcal{O}_{\mathcal{D}} \rightarrow \mathcal{O}_{\mathcal{C}}$ of sheaves of rings on $(\mathcal{C}, J)$, where f^{-1} is the inverse image functor of sheaves.

Definition 8.9.7 (cf. [Sta25, Tag 07A5]). Let $f : (\mathcal{C}, \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathcal{D}, \mathcal{O}_{\mathcal{D}})$ be a morphism of ringed topoi.

The *(total) derived direct image* or *(total) derived pushforward functor* Rf_* is the functor

$$Rf_* : D(\mathcal{O}_{\mathcal{C}}) \rightarrow D(\mathcal{O}_{\mathcal{D}})$$

(Notation 8.9.1) defined as the right derived functor (Definition 8.5.8) of the direct image functor f_* . Explicitly, given an object $K \in D(\mathcal{O}_\mathcal{C})$, $Rf_*(K)$ is given as follows: Let $\mathcal{I}^\bullet$ be any (right) K -injective (Definition 8.6.4) resolution (Definition 8.6.1) of M , which exists by Theorem 8.8.1. The object $Rf_*(K)$ is then given as the object of $D(\mathcal{O}_\mathcal{D})$ represented by the chain complex $f_*(\mathcal{I}^\bullet)$ (Proposition 8.4.2). This is well defined by Proposition 8.7.2.

Definition 8.9.8 (cf. [Sta25, Tag 06YY]). Let $f : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \rightarrow (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi.

The *(total) derived direct image* or *(total) derived pushforward functor* Lf^* is the functor

$$Lf^* : D(\mathcal{O}_\mathcal{D}) \rightarrow D(\mathcal{O}_\mathcal{C})$$

(Notation 8.9.1) defined as the left derived functor (Definition 8.5.8) of the inverse image functor f^* . Explicitly, given an object $K \in D(\mathcal{O}_\mathcal{D})$, $Lf^*(K)$ is given as follows: Let $\mathcal{F}^\bullet$ be any (left) K -flat (Definition 8.6.2) (with respect to $\otimes_{\mathcal{O}_\mathcal{D}}$ either on the left or right, depending on whether we are considering right or left $\mathcal{O}_\mathcal{D}$ -modules) resolution (Definition 8.6.1) of M , which exists by Proposition 8.8.4 (♠ TODO: justify the well definedness)

Theorem 8.9.9. Let $f : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \rightarrow (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi.

The functors Rf_* (Definition 8.9.7) and Lf^* (Definition 8.9.8) are adjoint, i.e. for all $L \in D(\mathcal{O}_\mathcal{C})$ and $M \in D(\mathcal{O}_\mathcal{D})$, we have natural isomorphisms

$$\mathrm{Hom}_{D(\mathcal{O}_\mathcal{C})}(Lf^*M, L) \cong \mathrm{Hom}_{D(\mathcal{O}_\mathcal{D})}(M, Rf_*L)$$

(Definition 8.9.7) (Definition 8.9.8)

♠ TODO: total derived sheaf cohomology

9. ASSIGNMENT 1: DUE FRIDAY, JAN 23

Problem 9.0.1. In the following categories, prove whether there are initial/final objects (Definition 2.3.1) and describe what they are.

1. The category of sets (Definition 1.1.7).
2. The category of groups (Definition 1.1.8).

Read the definition of a group object (Definition C.0.27).

Problem 9.0.2. Let $\mathcal{C}$ be a locally small category (Definition 1.1.2) with a final object (Definition 2.3.1) and let G be a group object of $\mathcal{C}$. Prove that the representable functor (Definition 1.3.12) $h_G : \mathcal{C} \rightarrow \mathbf{Sets}$ in fact factors through $\mathbf{Grp}$ (Definition 1.1.8). In other words, h_G can be regarded as a functor $\mathcal{C} \rightarrow \mathbf{Grp}$, and composing this functor with the forgetful functor $\mathbf{Grp} \rightarrow \mathbf{Sets}$ recovers the original functor $h_G : \mathcal{C} \rightarrow \mathbf{Sets}$.

Problem 9.0.3. Given a commutative ring (Definition C.0.9) R , one can construct a topological space (Definition C.0.5) $\mathrm{Spec} R$, see (Definition C.0.23) (focus on the construction of the topological space, and ignore the discussion on the structure sheaf).

1. Show that there is a functor $\mathrm{Spec} : \mathbf{CommRing}^{\mathrm{op}} \rightarrow \mathbf{Top}$ given by
 - sending a commutative ring R to $\mathrm{Spec} R$, and
 - sending a ring homomorphism $\varphi : R_1 \rightarrow R_2$ to the map $\varphi^* : \mathrm{Spec} R_2 \rightarrow \mathrm{Spec} R_1$ given by $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$.
2. Show that the above functor Spec is not faithful (Definition 1.3.5). (Hint: one should be able to find examples involving finite rings)

Read the definition of a fiber product (Definition C.0.24) of two objects in a category.

Problem 9.0.4. Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be morphisms in the category of sets (Definition 1.1.7). Prove that the fiber product (Definition C.0.24) $X \times_Z Y$ exists by explicitly constructing it (along with canonical morphisms from $X \times_Z Y$ to X, Y, Z), and verifying that your construction possesses the appropriate universal property.

Problem 9.0.5 (The Pullback Lemma). Consider a commutative diagram in a category $\mathcal{C}$:

$$\begin{array}{ccccc} A & \xrightarrow{u} & B & \xrightarrow{v} & C \\ \downarrow f & & \downarrow g & & \downarrow h \\ X & \xrightarrow{p} & Y & \xrightarrow{q} & Z \end{array}$$

Assume that the right-hand square (with corners B, C, Y, Z) is a Cartesian square (Definition C.0.24). Prove that the left-hand square (with corners A, B, X, Y) is a Cartesian square if and only if the outer rectangle (with corners A, C, X, Z) is a Cartesian square.

Read the definition of the category of opens (Definition 7.1.2) of a topological space.

Problem 9.0.6. Let $\mathbb{R}(x)$ denote the field of rational functions of x with coefficients in $\mathbb{R}$. Given an open subset U of $\mathbb{R}$, let $\mathcal{O}(U)$ denote

$$\mathcal{O}(U) = \{f \in \mathbb{R}(x) : f \text{ is defined at all points } x \in U\}.$$

Note that it is a commutative ring under pointwise addition and multiplication: for $f, g \in \mathbb{R}(U)$, we have $(f + g)(x) = f(x) + g(x)$ and $(f \cdot g)(x) = f(x) \cdot g(x)$.

Given an inclusion $U \subseteq V$ of open subsets, note that there is an injective ring homomorphism $\mathcal{O}(V) \hookrightarrow \mathcal{O}(U)$ and note that this describes a functor/diagram (Definition 2.2.6) $\mathcal{O} : \mathbf{Open}(\mathbb{R})^{\text{op}} \rightarrow \mathbf{CommRings}$ (Definition 7.1.2) (Definition 1.2.1) (you do not need to prove this).

Fix a point $x \in \mathbb{R}$. There is a (full) subcategory (Definition 1.3.7) N_x of $\mathbf{Open}(\mathbb{R})$ whose objects are open neighborhoods of x , so there is an induced diagram $N_x^{\text{op}} \rightarrow \mathbf{CommRings}$. Show that the direct limit (Definition 2.2.13)

$$\varinjlim_{U \in N_x^{\text{op}}} \mathcal{O}(U)$$

of this diagram is isomorphic to the following ring:

$$\mathcal{O}_x = \{f \in \mathbb{R}(x) : f \text{ is defined at } x\}.$$

10. ASSIGNMENT 2: DUE FEB 6

Problem 10.0.1. Let R and S be not necessarily commutative rings (Definition C.0.7). Let $\{M_i\}_{i \in I}$ be a small family of R - S -bimodules (Definition 2.1.1). Prove that $\prod_{i \in I} M_i$ and $\bigoplus_{i \in I} M_i$ as constructed in Definition 2.2.4 and Definition 2.2.5 are respectively the categorical product and coproduct (Definition 2.2.1) in the category of R - S -bimodule (Definition 2.1.3). In particular, note that when I is infinite, the product and coproduct do NOT coincide.

Problem 10.0.2. Let R and S be not necessarily commutative rings (Definition C.0.7). Show that the following categories are equivalent (Definition 1.3.4) by producing an explicit equivalence of categories and showing that it is indeed an equivalence of categories:

- The category ${}_R\mathbf{Mod}_S$ (Definition 2.1.3)
- The category $(R \otimes_{\mathbb{Z}} S^{\text{op}}) - \mathbf{Mod}$ (Definition C.0.38) (Definition 2.1.10).

Problem 10.0.3. Let $\mathcal{C}$ be a locally small category (Definition 1.1.2). Let X and Y be objects of $\mathcal{C}$. Show that X and Y are isomorphic if and only if h_X and h_Y (Definition 1.3.12) are naturally isomorphic (Definition 1.3.1) functors (note that, dually, we can also say that X and Y are isomorphic if and only if h^X and h^Y are naturally isomorphic functors).

Problem 10.0.4. Let R, S, T, U be not necessarily commutative rings (Definition C.0.7). Let M be an $R - S$ -bimodule, let N be an $S - T$ -bimodule, and let P be an $T - U$ -bimodule. Show that the tensor products $(M \otimes_S N) \otimes_T P$ and $M \otimes_S (N \otimes_T P)$ are isomorphic using Problem 10.0.3 and the tensor-Hom adjunction (Theorem 2.5.2)¹². In particular, do not write out an explicit isomorphism between the tensor products.

Problem 10.0.5. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be an adjoint pair (Definition 2.5.1) $F \dashv G$ of functors between locally small categories (Definition 1.1.2). Prove that the right adjoint G is faithful (Definition 1.3.5) if and only if the counit morphism (Definition 2.5.1) $\varepsilon_D : FG(D) \rightarrow D$ is an epimorphism (Definition 2.3.8) for every object D of $\mathcal{D}$.

(Dually, the left adjoint F is faithful if and only if the unit morphism $C \rightarrow GF(C)$ is a monomorphism for every object C of $\mathcal{D}$, but I am not asking you to prove this.)

Problem 10.0.6. Prove the following, called the short five lemma. Do not assume the general five lemma (Lemma 2.4.21).

Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Consider the following commutative diagram in $\mathcal{A}$ where the rows are short exact (Definition 2.4.4):

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \longrightarrow & 0 \end{array}$$

¹²In fact, the tensor products are naturally isomorphic, and you should think about what that means precisely, but I am not telling you to prove that the isomorphism is naturally isomorphic

1. If α and γ are monomorphisms (Definition 2.3.8), then β is a monomorphism.
2. If α and γ are epimorphisms, then β is an epimorphism.
3. If α and γ are isomorphisms, then β is an isomorphism.

Problem 10.0.7. Show that the short exact sequence (Definition 2.4.4) (in the category of abelian groups (Definition 1.1.8))

$$0 \rightarrow \mathbb{Z} \xrightarrow{1 \mapsto p} \mathbb{Z} \rightarrow \mathbb{Z}/p \rightarrow 0$$

is not split (Definition 2.4.8).

Problem 10.0.8. Let $R = \mathbb{Z}[t]/(1 - t^2)$. Let $A = R/(1 - t)$ and let $B = R/(1 + t)$. Show that the short exact sequence

$$0 \rightarrow B \xrightarrow{1 \mapsto 1-t} R \rightarrow A \rightarrow 0$$

is

1. split (Definition 2.4.8) as a short exact sequence of abelian groups (Definition C.0.3).
2. not split (Definition 2.4.8) as a short exact sequence of R -modules (Definition 2.1.1).

11. ASSIGNMENT 3: DUE FEB 20

Problem 11.0.1. Let $\mathcal{A}$ be an abelian category (Definition 2.3.12). Let $\{A_{\bullet,i}\}_{i \in I}$ be a small collection of chain complexes (Definition 3.0.1) in $\mathcal{A}$.

1. precisely show that there exists a “canonical” “comparison” morphism

$$\bigoplus_{i \in I} H_n(A_{\bullet,i}) \rightarrow H_n \left(\bigoplus_{i \in I} A_{\bullet,i} \right)$$

coming from the functoriality (Proposition 3.2.7) of H_n , the canonical “inclusion” morphisms for coproducts (Definition 2.2.1)¹³

2. Show that if $\mathcal{A}$ satisfies AB4 (Definition 2.5.7), then

$$\bigoplus_{i \in I} H_n(A_{\bullet,i}) \cong H_n \left(\bigoplus_{i \in I} A_{\bullet,i} \right)$$

(In fact, the canonical comparison morphism would be such an isomorphism, and if you solve this problem “correctly”, then the isomorphism that you produce will be the canonical comparison morphism, but I am not asking you to carefully justify that the isomorphism that you produce is the canonical comparison morphism).

Problem 11.0.2. Suppose that $f, g : B_{\bullet} \rightarrow C_{\bullet}$ are homotopic (Definition 3.5.1) chain maps (Definition 3.0.1) of chain complexes (Definition 3.0.1) in an additive category (Definition 2.3.4). Show that the mapping cones (Definition 3.7.3) $\text{cone}(f)$ and $\text{cone}(g)$ are isomorphic chain complexes.

¹³the canonical morphisms for general coproducts refer to the morphisms $C_i \rightarrow \coprod_{i \in I} C_i$

Problem 11.0.3. Suppose that $f : C_\bullet \rightarrow D_\bullet$ is a chain map of chain complexes (Definition 3.0.1) in an additive category (Definition 2.3.4). Show that f is null-homotopic (Definition 3.5.1) via a chain homotopy s (where $s_n : C_n \rightarrow D_{n+1}$) if and only if there exists a chain map $g : \text{cone}(f)_\bullet \rightarrow D_\bullet$ (Definition 3.7.3) such that $g_n = (\text{id}_{D_n}, s_{n-1})$.

12. ASSIGNMENT 4: DUE MAR 6

Problem 12.0.1.

1. Compute $\text{Tor}_n^{\mathbb{Z}}(\mathbb{Z}/a\mathbb{Z}, \mathbb{Z}/b\mathbb{Z})$ (Definition 4.5.1) for all $n \geq 0$ and integers $a, b \geq 1$.
2. Compute $\text{Ext}_{\mathbb{Z}}^n(\mathbb{Q}, \mathbb{Z})$ (Definition 4.5.2) for all $n \geq 0$.
3. Let $R = \mathbb{Z}[C_2]$ be the group ring of the cyclic group of order 2, $C_2 = \{1, \sigma\}$. Compute $\text{Ext}_R^n(\mathbb{Z}, \mathbb{Z})$ for all $n \geq 0$, where $\mathbb{Z}$ is viewed as a trivial R -module (i.e., $\sigma \cdot 1 = 1$).
4. Compute $\text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}/a\mathbb{Z})$ for $a > 1$.

Problem 12.0.2. Suppose that $A_\bullet$ and $B_\bullet$ are chain complexes (Definition 3.0.1) of R -modules (Definition 2.1.1) where R is a PID (Definition C.0.41). Further assume that $A_\bullet$ is bounded below¹⁴ as a (homological) chain complex, i.e. it is of the form

$$\cdots \rightarrow A_{n_0+2} \rightarrow A_{n_0+1} \rightarrow A_{n_0} \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow \cdots$$

for some integer n_0 .

Say that a_1, b_1, a_2, b_2 are integers such that $H_i(A_\bullet) = 0$ for $i \notin [a_1, b_1]$ and $H_i(B_\bullet) = 0$ for $i \notin [a_2, b_2]$.

1. Show that $H_i(\text{Tot}\oplus(A_\bullet \otimes B_\bullet)) = 0$ (Definition 4.6.5) (Definition 4.6.6) for $i \notin [a_1 + a_2, b_1 + b_2 + 1]$.
2. If R is a field then show that $H_i(\text{Tot}\oplus(A_\bullet \otimes B_\bullet)) = 0$ for $i \notin [a_1 + a_2, b_1 + b_2]$.

Problem 12.0.3. Let R be a commutative ring (Definition C.0.9), and let M be a module over R . Prove that a non-zero divisor (Definition 4.2.25) $x \in R$ is a zero-divisor of M (Definition 4.2.26) if and only if $\text{Tor}_1^R(R/(x), M) \neq 0$ (Definition 4.5.1).

Problem 12.0.4. Let $R = k[x]/x^n$ for a field k and where $n \geq 2$. Let $M = R/(x^a)$ and $N = R/(x^b)$ with $1 \leq a, b < n$. Compute $\text{Ext}_R^i(M, N)$ (Definition 4.5.2) for all $i \geq 0$. (Hint: trying to compute every $\text{Ext}_R^i(M, N)$ for every i using a resolution will probably be very annoying. Consider the following short exact sequence:

$$0 \rightarrow R/(x^{n-a}) \xrightarrow{\cdot x^a} R \rightarrow M \rightarrow 0.$$

)

Problem 12.0.5. Compute $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$.

¹⁴I claim that this hypothesis is not necessary for the truth of the statements in this problem, but also there is a fact in these notes stated in a way that uses this hypothesis.

13. ASSIGNMENT 5 DUE MAR 27

Problem 13.0.1. Let $f : X \rightarrow Y$ be a continuous map of topological spaces.

1. Show that for all $\mathcal{G} \in \text{Sh}(Y, \text{Ab})$ and all $x \in X$,

$$(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$$

2. Show that $f^* : \text{Sh}(Y, \text{Ab}) \rightarrow \text{Sh}(X, \text{Ab})$ is exact.

Problem 13.0.2. Let $i : Z \hookrightarrow X$ be the embedding of a closed subspace (Definition C.0.5) Z of X . Show that $i_* : \text{Sh}(Z, \text{Ab}) \rightarrow \text{Sh}(X, \text{Ab})$ (Definition 7.7.2) (Definition 7.1.6) is an exact functor (Definition 2.4.12).

Problem 13.0.3. Let $(X, \mathcal{O}_X)$ be a ringed space (Definition 7.6.1).

1. Let $\mathcal{F}$ and $\mathcal{G}$ respectively be a right and left sheaves of $\mathcal{O}_X$ -modules (Definition 7.6.3).
 - (a) For any $U \subseteq X$ open (Definition C.0.5), show that $\mathcal{F}|_U \otimes_{\mathcal{O}_X} \mathcal{G}|_U$ (Definition 7.6.6) (Definition 7.7.3) and $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})|_U$ are naturally isomorphic (Definition 1.3.1).
 - (b) For any $x \in X$, show that $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})_x \cong \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$ (Definition 2.1.9) (Definition 7.5.1).
2. Let $\mathcal{F}$ and $\mathcal{G}$ both be (left) sheaves of $\mathcal{O}_X$ -modules (Definition 7.6.3).
 - (a) For any $U \subseteq X$ open, show that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})|_U$ (Definition 7.6.5) (Definition 7.7.3) is naturally isomorphic to $\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U)$.
 - (b) For any $x \in X$, show that there is a canonical morphism $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})_x \rightarrow \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$ (Definition 2.1.8); one should be aware that this canonical morphism is not generally an isomorphism, but is an isomorphism under “nice” circumstances, more specifically when $\mathcal{F}$ is “locally finitely presented”.

Problem 13.0.4. Let $\mathbb{RP}^2$ be the real projective plane. Let $\pi : S^2 \rightarrow \mathbb{RP}^2$, $(x, y, z) \mapsto [x : y : z]$ be the standard quotient map that identifies antipodal points, i.e. $\pi(x) = \pi(-x)$ for all $x \in S^2$. Define a sheaf Λ on $\mathbb{RP}^2$ as follows:

For an open subset $U \subseteq \mathbb{RP}^2$, let $\Gamma(U, \Lambda)$ be the set of continuous functions $f : \pi^{-1}(U) \rightarrow \mathbb{Z}$ (where $\mathbb{Z}$ is equipped with the discrete topology) such that for all $x \in \pi^{-1}(U)$, we have

$$f(-x) = -f(x).$$

Let $\mathcal{U} = \{U_0, U_1, U_2\}$ be the open cover of $\mathbb{RP}^2$ given by

- U_0 : the set of points $[x : y : z]$ such that $|x| > \sqrt{y^2 + z^2}$.
- U_1 : the set of points $[x : y : z]$ such that $|y| > \sqrt{x^2 + z^2}$.
- U_2 : the set of points $[x : y : z]$ such that $|z| > \sqrt{x^2 + y^2}$.

1. Show that Λ is a locally constant sheaf (Definition 7.7.16).
2. Compute the Čech cohomology groups (Definition 7.8.1) $\check{H}^p(\mathcal{U}, \mathbb{RP}^2)$.
3. Prove that $\check{H}^p(\mathcal{U}, \mathbb{RP}^2) \cong \check{H}^p(X, \mathbb{RP}^2)$. You may use without proof the following:

Let X be a contractible, locally path connected, and semi-locally simply connected space (e.g. all of the connected components in all of the finite intersections of them embers of $\mathcal{U}$ satisfy these conditions). All locally constant sheaves on X are constant.

APPENDIX A.

Definition A.0.1. Let C be a category enriched in a monoidal category $\mathcal{V}$. Given an object X of C , the *functor of points* h_X is the functor (Definition 1.2.2)/presheaf (Definition 7.1.1) $C^{\text{op}} \rightarrow \mathcal{V}$ given by $T \mapsto \text{Hom}_C(T, X)$. A functor $C^{\text{op}} \rightarrow \mathcal{V}$ (or equivalently, a presheaf on C valued in $\mathcal{V}$) is said to be *representable* if it is naturally isomorphic (Definition 1.3.1) to some functor h_X of points for an object X of C .

Dually, a functor $C \rightarrow \mathcal{V}$ is called *co-representable* if it is naturally isomorphic to a functor $T \mapsto \text{Hom}_C(X, T)$ for an object X in C .

For instance, we may speak of these notions when $\mathcal{V}$ is the monoidal category **Sets**, i.e. C is a locally small category (Definition 1.1.2).

Definition A.0.2. A *symmetric monoidal category* is a monoidal category $(\mathcal{C}, \otimes, \mathbb{I})$ together with a natural isomorphism (symmetry)

$$\gamma_{X,Y} : X \otimes Y \xrightarrow{\cong} Y \otimes X$$

for all $X, Y \in \mathcal{C}$, such that for all $X, Y, Z \in \mathcal{C}$ the following holds:

- $\gamma_{Y,X} \circ \gamma_{X,Y} = \text{id}_{X \otimes Y}$ (involutivity);
- the **hexagon coherence diagrams** commute:

$$\begin{array}{ccc} (X \otimes Y) \otimes Z & \xrightarrow{\alpha_{X,Y,Z}} & X \otimes (Y \otimes Z) \xrightarrow{\gamma_{X,Y \otimes Z}} (Y \otimes Z) \otimes X \\ \gamma_{X,Y} \otimes \text{id}_Z \downarrow & & \uparrow \text{id}_Y \otimes \gamma_{X,Z} \\ (Y \otimes X) \otimes Z & \xrightarrow{\alpha_{Y,X,Z}} & Y \otimes (X \otimes Z) \end{array}$$

and the analogous hexagon with inverse braiding:

$$\begin{array}{ccc} X \otimes (Y \otimes Z) & \xrightarrow{\alpha_{X,Y,Z}^{-1}} & (X \otimes Y) \otimes Z \xrightarrow{\gamma_{X \otimes Y, Z}} Z \otimes (X \otimes Y) \\ \text{id}_X \otimes \gamma_{Y,Z} \downarrow & & \uparrow \gamma_{X,Z} \otimes \text{id}_Y \\ X \otimes (Z \otimes Y) & \xrightarrow{\alpha_{X,Z,Y}^{-1}} & (X \otimes Z) \otimes Y \end{array}$$

- the **symmetry coherence diagram** commutes:

$$\begin{array}{ccc} X \otimes Y & \xrightarrow{\gamma_{X,Y}} & Y \otimes X \\ & \searrow \text{id}_{X \otimes Y} & \downarrow \gamma_{Y,X} \\ & & X \otimes Y \end{array}$$

A *closed symmetric monoidal category* usually refers to a symmetric monoidal category that is closed as a monoidal category.

Definition A.0.3 (Grothendieck Universe). Let U be a set. We say U is a *Grothendieck universe* (or just a *universe*) if the following conditions hold:

1. If $x \in U$ and $y \in x$, then $y \in U$ (transitivity).

2. If $x, y \in U$, then $\{x, y\} \in U$ (closed under pair formation).
3. If $x \in U$, then the power set $\mathcal{P}(x) \in U$.
4. If $I \in U$ and $(x_\alpha)_{\alpha \in I}$ is a family with each $x_\alpha \in U$, then $\bigcup_{\alpha \in I} x_\alpha \in U$.

A set X is called *U -small* or a *U -set* if $X \in U$.

Definition A.0.4. *Tarski-Grothendieck set theory*, denoted by **TG**, is the theory consisting of the axioms of ZFC together with *Tarski's Axiom of Universes*, which asserts that for every set x there exists a universe (Definition A.0.3) U such that $x \in U$:

$$\forall x \exists U (U \text{ is a Grothendieck universe} \wedge x \in U)$$

Definition A.0.5. A set x is a *hereditarily finite set* if its transitive closure $\text{tc}(x)$ is a finite set.

The collection of all hereditarily finite sets is denoted by V_ω . It is a set.

Definition A.0.6. Given an axiom system containing the Tarski-Grothendieck set theory (Definition A.0.4) axioms, the *hierarchy of Grothendieck universes* consists of the Grothendieck universes (Definition A.0.3) U_n for $n \in \mathbb{N} \cup \{0\}$ constructed as follows:

1. Base case: $U_0 := V_\omega$, the set of all hereditarily finite sets (Definition A.0.5).
2. Successor step: For each $n \geq 0$, U_{n+1} is the unique minimal Grothendieck universe such that $U_n \in U_{n+1}$.

In particular, $U_n \in U_{n+1}$ and $U_n \subset U_{n+1}$ for all n .

Definition A.0.7. Let $\mathcal{U}$ be a fixed Grothendieck universe (Definition A.0.3). The *category of categories (relative to $\mathcal{U}$)* or the *category of U -small categories* is the category defined by:

- The objects are all U -small (Definition 1.1.2) categories (Definition 1.1.1) $\mathcal{C}$.
- The morphisms are functors (Definition 1.2.2) between such categories.

This category is denoted by **$U - \mathbf{Cat}$** or by **$\mathbf{Cat}$** , if U is understood.

In the case that Tarski-Grothendieck (Definition A.0.4) set theory is assumed (so in particular, the axiom of universes is assumed), one often adopts the convention of denoting $U_n - \mathbf{Cat}$ (Definition A.0.6) by **$\mathbf{Cat}$** and $U_{n+1} - \mathbf{Cat}$ (Definition A.0.6) by **$\mathbf{CAT}$** . Thus, for $n = 1$, **$\mathbf{Cat}$** would be the category of small categories, whereas **$\mathbf{CAT}$** would serve as the “category of large categories” which consists of many common categories that are too large to be U_1 -categories such as **$\mathbf{Set}$** , **$\mathbf{Grp}$** , **$\mathbf{Top}$** or even **$\mathbf{Cat}$** .

Definition A.0.8. Let $\mathcal{X}$ be a category. A *concrete category over $\mathcal{X}$* is a pair $(\mathcal{C}, U)$ consisting of a category $\mathcal{C}$ and a faithful functor (Definition 1.3.5) $U: \mathcal{C} \rightarrow \mathcal{X}$. In this context, U is called the *underlying functor* (or *forgetful functor*) of the concrete category.

When $\mathcal{X} = \mathbf{Set}$ (the category of sets), the pair $(\mathcal{C}, U)$ is simply referred to as a *concrete category*. For any object A in $\mathcal{C}$, the set $U(A)$ is called the *underlying set* of A , and for any morphism $f: A \rightarrow B$, the function $U(f): U(A) \rightarrow U(B)$ is called the *underlying function* of f .

APPENDIX B. GROTHENDIECK TOPOLOGIES

Definition B.0.1 ([?, Exposé I Définition 4.1]). Let C be a (large) category (Definition 1.1.1).

1. A *sieve S on the category C* is a full subcategory (Definition 1.3.7) D of C such that for any object U of C there exists an object V of (♠ TODO: correctly parse the definiton)
2. A *sieve S on an object $U \in \text{Ob}(C)$* is a collection of morphisms in C with codomain U that is closed under precomposition by any compatible morphism in C . In other words, S is a sieve if for every $f : V \rightarrow U$ in S and morphism $g : W \rightarrow V$ in C , the composition $f \circ g : W \rightarrow U$ is also in S .

Given a morphism $f : V \rightarrow U$ in a sieve S , we also say that *f factors through U* .

Definition B.0.2. Let $\mathcal{C}$ be a category (Definition 1.1.1) and $U \in \mathcal{C}$ an object. Let $\mathcal{S} = \{f_i : U_i \rightarrow U\}_{i \in I}$ be a family of morphisms with codomain U .

The *sieve generated by $\mathcal{S}$* , denoted $\langle \mathcal{S} \rangle$ or $\langle S \rangle$, is the smallest sieve on U (Definition B.0.1) containing all the morphisms in $\mathcal{S}$.

Explicitly, a morphism $h : V \rightarrow U$ belongs to the generated sieve if and only if h factors through some morphism in $\mathcal{S}$. That is, there exists an index $i \in I$ and a morphism $g : V \rightarrow U_i$ such that

$$h = f_i \circ g.$$

Definition B.0.3. Let C be a category, let $U \in \text{Ob}(C)$, and let S be a sieve on U (Definition B.0.1). For a morphism $f : V \rightarrow U$ in C , the *pullback sieve f^*S* (or *basechange sieve $S \times_U V$*) on V is defined by

$$f^*S = \{g : W \rightarrow V \mid f \circ g \in S\}.$$

In other words, f^*S consists of all morphisms into V whose composite with f belongs to the sieve S on U .

Definition B.0.4. Let $\mathcal{C}$ be a category (Definition 1.1.1). A *basis for a Grothendieck topology* (also called a *Grothendieck pretopology* or simply a *pretopology*) on $\mathcal{C}$ is a collection of families $K(U)$ of morphisms for each object $U \in \mathcal{C}$, called *coverings* or *covering families*, satisfying the following axioms:

1. **(Identity)** For every isomorphism $U' \rightarrow U$, the singleton family $\{U' \rightarrow U\}$ is in $K(U)$.
2. **(Base Change)** If $\{U_i \rightarrow U\}_{i \in I}$ is a covering family in $K(U)$ and $V \rightarrow U$ is any morphism in $\mathcal{C}$, then the fiber products (Definition C.0.24) $U_i \times_U V$ exist, and the family of projections $\{U_i \times_U V \rightarrow V\}_{i \in I}$ is in $K(V)$.
3. **(Composition)** If $\{U_i \rightarrow U\}_{i \in I}$ is in $K(U)$ and for each $i \in I$, $\{V_{ij} \rightarrow U_i\}_{j \in J_i}$ is in $K(U_i)$, then the composite family $\{V_{ij} \rightarrow U_i \rightarrow U\}_{i \in I, j \in J_i}$ is in $K(U)$.

Definition B.0.5 (Grothendieck topology). Let $\mathcal{U}$ be a universe (Definition A.0.3).

1. (See [?, Exposé II, Définition 1.1]) Let $\mathcal{C}$ be a category (Definition 1.1.1). A *Grothendieck topology on $\mathcal{C}$* assigns to each object U of $\mathcal{C}$ a collection $J(U)$ of sieves (Definition B.0.1) $\{U_i \rightarrow U\}_{i \in I}$, each called a *covering sieve of U* , satisfying:

- (a) (Stability under “base change”): If $S \in J(U)$ is a covering sieve of an object U , and $f : V \rightarrow U$ is any morphism in $\mathcal{C}$, then the pullback sieve (Definition B.0.3) f^*S is a covering sieve of U .
- (b) (Local character condition) If S is a sieve on U , and if there exists a covering sieve $R \in J(U)$ such that for all $f : V \rightarrow U$ in R the pullback sieve (Definition B.0.3) f^*S is in $J(V)$, then $S \in J(U)$.
- (c) The maximal sieve is a covering sieve.

Some will refer to a Grothendieck topology as simply a **topology**, not to be confused with the related, but less general, notion of a topology on a set (Definition C.0.5).

- 2. (See [?, Exposé II, 1.1.5]) A **site** is a category $\mathcal{C}$ equipped with a Grothendieck topology.

When we are working with a Grothendieck pretopology (Definition B.0.4) K on a category $\mathcal{C}$, we may regard $\mathcal{C}$ as a site by equipping it with the Grothendieck topology generated by K . In this case, we may call $\mathcal{C}$ equipped with this Grothendieck topology a **site with a pretopology**. Note that the Stacks project notion of “site” is that of a small (Definition 1.1.2) site (i.e. a site whose underlying category is small) with a pretopology, cf. [Sta25, Tag 00V0] for a discussion for this choice, [Sta25, Tag 00VH] for the definition and [Sta25, Tags 0005 and 0015] for the Stacks project’s convention of “categories” (what we call small categories (Definition 1.1.2)) and “big categories” (what we call locally small categories (Definition 1.1.2)). (♠ TODO: talk more about how, at least when the axiom of universes is assumed, topoi coming from general Grothendieck topology are equivalent to those on pretopologies).

- 3. (See [?, Exposé II, Définition 1.2]) Let $(\mathcal{C}, J)$ be a site. A family of morphisms $(U_i \rightarrow U)_{i \in I}$ is called a **covering family of U (with respect to the site/topology)** or a **cover of U (with respect to the site/topology)** if the sieve generated by (Definition B.0.2) the family is a covering sieve of U .
- 4. (See [?, Exposé II, Définition 3.0.1]) Let $(\mathcal{C}, J)$ be a site (Definition B.0.5), where J is a Grothendieck topology on $\mathcal{C}$.

A family G of objects $\mathcal{C}$ is called a **topologically generating family of the site/topology** or a **generating family/collection of the site/topology** if for every object $X \in \mathcal{C}$, there is a covering family $\{X_\alpha \rightarrow X\}_{\alpha \in A}$ of X such that every X_α is a member of G .

Equivalently, the Grothendieck topology J is the smallest Grothendieck topology containing all covers of the U_i . Also equivalently, for any $S \in J(X)$, the sieve S contains a covering family $\{V_i \rightarrow X\}$ such that each morphism $V_i \rightarrow X$ factors through some member of G . (♠ TODO: Verify that these claimed equivalences are indeed equivalences)

Definition B.0.6. 1. An **small site** is a site (Definition B.0.5) whose underlying category is small (Definition 1.1.2).

- 2. A **locally small site** is a site whose underlying category is locally small (Definition 1.1.2).
- 3. Given a Grothendieck universe (Definition A.0.3) $\mathcal{U}$, a **$\mathcal{U}$ -site** is a site (Definition B.0.5) whose underlying category (Definition 1.1.1) is a $\mathcal{U}$ -(locally small) category (Definition 1.1.2) and that has a $\mathcal{U}$ -small topologically generating family (Definition B.0.5); a Grothendieck topology making the site a $\mathcal{U}$ -site is called a **$\mathcal{U}$ -topology**.

A $\mathcal{U}$ -site is called **$\mathcal{U}$ -small** if its underlying category is $\mathcal{U}$ -small (Definition 1.1.2).

Definition B.0.7. Let (X, τ_X) be a topological space. The **small site associated to X** or **the site of open covers of X** or **the canonical site on $\text{Open } X$** is the category $\text{Open}(X)$ of

open subsets (Definition 7.1.2) of X with inclusion morphisms, equipped with the canonical Grothendieck topology (Definition B.0.5) generated by the Grothendieck pretopology (Definition B.0.4) whose covering families $\{U_i \rightarrow U\}_{i \in I}$, for $U \in \text{Open}(X)$ are families of morphisms in $\text{Open}(X)$ such that $\bigcup_{i \in I} U_i = U$. In other words, $\{U_i \rightarrow U\}_{i \in I}$ is a covering for the pretopology if it is an open coverings.

Definition B.0.8 (Sheaf on a site). Let $(\mathcal{C}, J)$ be a site (Definition B.0.5). Let $\mathcal{A}$ be a (large) category (Definition 1.1.1).

1. A presheaf (Definition 7.1.1) $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ (Definition 1.2.1) is called a *sheaf on the site $(\mathcal{C}, J)$ valued in $\mathcal{A}$* if, for every object U of $\mathcal{C}$ and every covering sieve (Definition B.0.5) $S \in J(U)$, the limit (Definition 2.2.8)

$$\varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(V),$$

exists and the canonical natural morphism

$$\mathcal{F}(U) \rightarrow \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(V)$$

is an isomorphism. Here, $\mathcal{D}_S \hookrightarrow \mathcal{C}/U$ is the full downward-closed subcategory such that $\text{Ob}(\mathcal{D}_S) = \{(f : V \rightarrow U) : f \in S(V)\}$,

In particular, when we are working with a Grothendieck pretopology (Definition B.0.4) K on a category $\mathcal{C}$, we may speak of sheaves on the site whose Grothendieck topology is the one generated by K .

2. Given sheaves $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ on the site $(\mathcal{C}, J)$, a *morphism between the sheaves* is a morphism (Definition 7.1.1) between $\mathcal{F}$ and $\mathcal{G}$ as presheaves.
3. Let U be a universe (Definition A.0.3). A *U -sheaf* typically refers to a U -presheaf that is a sheaf for a U -site. In other words, a U -sheaf is a sheaf on a site whose underlying category is U -locally small (Definition 1.1.2) and which has a U -small topologically generating family such that the sheaf is valued in U -sets.
4. The *sheaf category/category of $\mathcal{A}$ -valued sheaves on $\mathcal{C}$* is the (large) category defined as the full subcategory of $\text{PreShv}(\mathcal{C}, \mathcal{A})$ whose objects are the sheaves on $\mathcal{C}$ with values in $\mathcal{A}$. Common notations for the sheaf category include $\text{Shv}(\mathcal{C}, \mathcal{A})$, $\text{Shv}(\mathcal{C}, J, \mathcal{A})$, $\text{Sh}(\mathcal{C}, \mathcal{A})$, $\text{Sh}(\mathcal{C}, J, \mathcal{A})$. If the value category $\mathcal{A}$ is clear from context, then notations such as $\text{Shv}(\mathcal{C})$, $\text{Shv}(\mathcal{C}, J)$, $\text{Sh}(\mathcal{C})$, $\text{Sh}(\mathcal{C}, J)$ are also common.

Definition B.0.9 (Topos). There are multiple notions of a topos depending on the context (geometric vs. logical).

1. A *Grothendieck topos* (or *sheaf topos*) is a category (Definition 1.1.1) equivalent (Definition 1.3.4) to the category of sheaves (Definition B.0.8) of sets on a small site. That is, there exists a small site $(\mathcal{C}, J)$ such that the category is equivalent to $\text{Sh}(\mathcal{C}, J, \mathbf{Sets})$ (Definition 1.1.7).

If a Grothendieck universe (Definition A.0.3) $\mathcal{U}$ is available, then a *$\mathcal{U}$ -(Grothendieck/sheaf) topos* is a category equivalent to (Definition 1.3.4) the category of sheaves of $\mathcal{U}$ -sets (Definition A.0.3) on a $\mathcal{U}$ -small site (Definition B.0.6) $(\mathcal{C}, J)$ [?, Exposé IV Définition 1.1].

2. An **elementary topos** is a category which has all finite limits (Definition 2.2.8), is cartesian closed, and has a subobject classifier.

Unless otherwise specified, a **topos** refers to a Grothendieck/sheaf topos. The plural for topos is **topoi**. *Remark:* Every Grothendieck topos is an elementary topos, but the converse is not true (e.g., the category of finite sets is an elementary topos but not a Grothendieck topos).

Definition B.0.10. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be small sites (Definition B.0.6). let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a morphism of sites.

The **morphism of topoi induced by u** is the topos morphism $f : \text{Sh}(\mathcal{D}, K; \mathbf{Sets}) \rightarrow \text{Sh}(\mathcal{C}, J; \mathbf{Sets})$ given by $f_* = u_*$ (??), which is defined by precomposition, and $f^* = a \circ \text{Lan}_u$ where Lan_u is left Kan extension and a is sheafification (Definition B.0.14).

It turns out that there is indeed an adjunction $f^* \dashv f_*$ (♠ TODO: why; apparently, there is a presheaf level adjunction between Lan and u^* .) and that f^* is indeed left exact (♠ TODO: why).

Therefore, whenever we talk about a property or a notion concerning a topos morphism, that property or notion may be sensibly extended to concern sites.

The notion of a topos morphism induced by a morphism of small sites should not be mistaken for the notion of a topos morphism induced by (Definition B.0.12) a cocontinuous flat functor of small sites.

See also Definition B.0.13.

Remark B.0.11. In fact, for every morphism $f : \mathcal{E} \rightarrow \mathcal{F}$ of topoi, there are some sites $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ (which may even be taken to be small) whose categories of sheaves of sets are $\mathcal{E}$ and $\mathcal{F}$ and there is some morphism $u : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$ whose induced topos morphism is f . However, this assignment is not injective.

Definition B.0.12. Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a cocontinuous functor of small sites that is also flat. The **topos morphism induced by u** is the morphism of topoi

$$f : \text{Sh}(\mathcal{C}, J) \rightarrow \text{Sh}(\mathcal{D}, K)$$

defined by the following pair of adjoint functors (Definition 2.5.1):

- The **inverse image** $f^* : \text{Sh}(\mathcal{D}, K) \rightarrow \text{Sh}(\mathcal{C}, J)$ is the inverse image/precomposition functor u^* , given by $f^*(\mathcal{H}) = \mathcal{H} \circ u$. Because u is cocontinuous, $\mathcal{H} \circ u$ is automatically a sheaf (Definition B.0.8); because u is flat, f^* is left exact (Definition 2.4.11).
- The **direct image** $f_* : \text{Sh}(\mathcal{C}, J) \rightarrow \text{Sh}(\mathcal{D}, K)$ is the right adjoint (Definition 2.5.1) to the precomposition functor. (♠ TODO: why does this exist? Apparently, this right adjoint is the right kan extension) (♠ TODO: Also apparently, there is an adjoint functor theorem that states that a colimit-preserving functor between Grothendieck topoi always has a right adjoint)

The notion of a topos morphism induced by a cocontinuous flat functor of small sites should not be mistaken for the notion of a topos morphism induced by a morphism of small sites (Definition B.0.10) (which is in particular a continuous flat functor).

See also Definition B.0.13.

Definition B.0.13. Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.2.2) of small sites (Definition B.0.6) that is continuous (Definition C.0.6), cocontinuous, and flat. The *(essential) morphism of topoi induced by u* usually refers to the morphism of topoi induced by u as a (Definition B.0.12) cocontinuous and flat functor, and not the morphism of topoi induced by u as a (Definition B.0.10) morphism of sites. (♠ TODO: is morphism of sites = continuous + flat functor?)

More explicitly, this morphism of topoi is the morphism

$$f : \mathrm{Sh}(\mathcal{C}, J) \rightarrow \mathrm{Sh}(\mathcal{D}, K)$$

whose

- Inverse image f^* is precomposition/inverse image u^* of u as a cocontinuous functor, not as a continuous functor
- Direct image f_* is right adjoint to f^* .

Furthermore, since u is continuous, f^* has a left adjoint denoted by $f_!$ and called by names such as the *lower shriek of f* , the *exceptional direct image of f* , or the *extension by zero of f* , and which is given by (Definition B.0.10) $f_! = a \circ \mathrm{Lan}_u$.

To summarize, there are adjunctions $f_! \dashv f^* \dashv f_*$.

Nevertheless, one can talk about the morphism of topoi in the other direct — regarding u as a continuous and flat functor, one has a topos morphism

$$g : \mathrm{Sh}(\mathcal{D}, K) \rightarrow \mathrm{Sh}(\mathcal{C}, J)$$

in the other direction whose

1. Inverse image g^* is given by $a \circ \mathrm{Lan}_u$
2. Direct image g_* is given by precomposition/direct image u_* of u as a continuous functor, not as a cocontinuous functor.

Furthermore, since u is cocontinuous, g_* has a right adjoint, denoted by $g^!$ and called by names such as the *upper shriek of g* or the *exceptional inverse image of g* .

To summarize, there are adjunctions $g^* \dashv g_* \dashv g^!$.

Moreover, in the above notations, $f_! = g^*$, $f^* = g_*$, and $f_* = g^!$.

Definition B.0.14. Let $\mathcal{C}$ be a site (Definition B.0.5) and let $\mathcal{A}$ be a (large) category (Definition 1.1.1).

Assuming that the presheaf (Definition 7.1.1) category $\mathrm{PreShv}(\mathcal{C}, \mathcal{A})$ (and hence the sheaf (Definition B.0.8) category $\mathrm{Shv}(\mathcal{C}, \mathcal{A})$) is locally small (Definition 1.1.2) (or U -locally small if a Grothendieck universe (Definition A.0.3) U is available), a *sheafification functor* refers to a functor

$$a : \mathrm{PreShv}(\mathcal{C}, \mathcal{A}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathcal{A})$$

that is left adjoint (Definition 2.5.1) to the inclusion functor

$$i : \mathrm{Shv}(\mathcal{C}, \mathcal{A}) \hookrightarrow \mathrm{PreShv}(\mathcal{C}, \mathcal{A}).$$

If such a sheafification functor exists, then it is unique up to unique natural isomorphism. Given a presheaf P , the sheafification $a(P)$ is also sometimes called the *sheaf associated to P* . See Theorem B.0.15 for common conditions under which sheafification exists.

Theorem B.0.15. cf. [?, Exposé II, Théorème 3.4]

1. Let U be a universe. Let $\mathcal{C}$ be a U -site (Definition B.0.5). A sheafification functor (Definition 7.3.2)

$$a : \mathrm{PreShv}(\mathcal{C}, U\text{-}\mathbf{Sets}) \rightarrow \mathrm{Shv}(\mathcal{C}, U\text{-}\mathbf{Sets}).$$

exists.

2. Let $\mathcal{C}$ be a site whose underlying category is locally small (Definition 1.1.2) and which has a topologically generating family (Definition B.0.5) that is a set (rather than a proper class). A sheafification functor

$$a : \mathrm{PreShv}(\mathcal{C}, \mathbf{Sets}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathbf{Sets})$$

exists.

(♠ TODO: It may be the case that I need to assume $\mathcal{A}$ to be locally small; if so, recheck all the statements that use this.)

3. (see e.g. [nLa25f, 3], [KS06, Theorem 17.4.7]) Let $(\mathcal{C}, J)$ be a site (Definition B.0.5) on an essentially small category $\mathcal{C}$. Suppose that the category $\mathcal{A}$ is complete, cocomplete (Definition 2.2.11), that small filtered colimits (Definition 2.2.13) in $\mathcal{A}$ are exact (Definition 2.4.11), and that $\mathcal{A}$ satisfies the IPC-property. A sheafification functor (Definition B.0.14)

$$a : \mathrm{PreShv}(\mathcal{C}, \mathcal{A}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathcal{A})$$

exists.

(♠ TODO: state as a fact that the category of groups and k -algebras are complete, cocomplete, with small filtered colimits that are exact) This is true for instance of $\mathcal{A} = \mathbf{Set}, \mathbf{Grp}, k\text{-}\mathbf{Alg}$ for a field k , or $\mathbf{Mod}_R$ for a (not necessarily commutative unital) ring R (Definition C.0.7).

Remark B.0.16. If the presheaf is valued in nice “algebraic category”, e.g. groups, abelian groups, rings, modules over a ring, etc., then the sheafification is again valued in that category. (♠ TODO: Make this more precise.)

Definition B.0.17. (♠ TODO: there are places where sites and sheaves of rings on them are used, but it would be better to just have them be ringed sites.)

A *ringed site* is a site (Definition B.0.5) $(\mathcal{C}, J)$ with a small topological generating family (Definition B.0.5) equipped with a sheaf (Definition B.0.8) of (not necessarily commutative) rings $\mathcal{O}$. If the Grothendieck topology J is clear in context, one may even write that $(\mathcal{C}, \mathcal{O})$ is a ringed site.

A *morphism of ringed sites*

$$((\mathcal{C}, J), \mathcal{O}) \rightarrow ((\mathcal{C}', J'), \mathcal{O}')$$

consists of a morphism of sites $f : (\mathcal{C}, J) \rightarrow (\mathcal{C}', J')$ and a morphism of sheaves (Definition B.0.8) of rings $f^\# : \mathcal{O}' \rightarrow f_* \mathcal{O}$.

Definition B.0.18. 1. Let $\mathcal{C}$ be a site (Definition B.0.5), and let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition B.0.8) of (not necessarily commutative) rings (Definition C.0.7) on $\mathcal{C}$.

- (a) An $(\mathcal{A}, \mathcal{B})$ -bimodule (or a bimodule over $(\mathcal{A}, \mathcal{B})$) is a sheaf (Definition B.0.8) $\mathcal{M}$ of abelian groups on $\mathcal{C}$ equipped with a left $\mathcal{A}$ -module structure given by a morphism of sheaves (Definition B.0.8) of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$ -module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every object U in $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$\lambda_U(a, \rho_U(m, b)) = \rho_U(\lambda_U(a, m), b)$$

holds in $\mathcal{M}(U)$. In standard multiplicative notation where $\lambda(a, m)$ is denoted $a \cdot m$ and $\rho(m, b)$ is denoted $m \cdot b$, this condition is the associativity axiom

$$(a \cdot m) \cdot b = a \cdot (m \cdot b).$$

In particular, for every object $U \in \mathcal{C}$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$ -bimodule (Definition 2.1.1).

- (b) Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$ -bimodules. A $\text{homomorphism of } (\mathcal{A}, \mathcal{B})\text{-bimodules}$ (or an $(\mathcal{A}, \mathcal{B})$ -linear morphism) is a morphism of sheaves of abelian groups $f : \mathcal{M} \rightarrow \mathcal{N}$ such that for every object U of $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$ -bimodules, with morphisms being morphisms of sheaves of abelian groups compatible with both the left $\mathcal{A}$ -action and the right $\mathcal{B}$ -action, by $\mathcal{A}\text{-}\mathcal{B}\text{-Mod}$ or sometimes by ${}_{\mathcal{A}}\text{Mod}_{\mathcal{B}}$ (♠ TODO: talk about how bimodules can be identifies with left/right modules)

2. Let $(\mathcal{C}, J)$ be a site (Definition B.0.5). Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on $(\mathcal{C}, J)$ (Definition B.0.8), i.e. $((\mathcal{C}, J), \mathcal{O})$ is a ringed site (Definition B.0.17).

- (a) An $(\text{left/right/two-sided}) \mathcal{O}\text{-module}$ consists of the following data:

- A sheaf $\mathcal{F}$ of abelian groups on $(\mathcal{C}, J)$,
- for every object $U \in \mathcal{C}$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$ -module on $\mathcal{F}(U)$,

such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

is $\mathcal{O}(U)$ -linear when the $\mathcal{O}(U)$ -action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \rightarrow \mathcal{O}(V)$$

induced by f .

(b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$ -modules (Definition B.0.18).

A **morphism of $\mathcal{O}$ -modules** $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves (Definition B.0.8) of abelian groups such that, for every object $U \in \mathcal{C}$, the component map

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is $\mathcal{O}(U)$ -linear, i.e. it satisfies

$$\varphi_U(r \cdot s) = r \cdot \varphi_U(s) \quad \text{for all } r \in \mathcal{O}(U), s \in \mathcal{F}(U).$$

The collection of all $\mathcal{O}$ -modules together with their morphisms of $\mathcal{O}$ -modules forms the **category of $\mathcal{O}$ -modules**, denoted **Mod($\mathcal{O}$)**. The Hom functors $\text{Hom}_{\text{Mod}(\mathcal{O})}$ of this category may be denoted by **Hom $_{\mathcal{O}}$** .

In case that a sheafification functor (Definition B.0.14)

$$\text{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \text{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists, a left, right, two-sided $\mathcal{O}$ -module (and morphisms thereof) is equivalent to a $(\mathcal{O}, \mathbb{Z})$ -bimodule, $(\mathbb{Z}, \mathcal{O})$ -bimodule, and $(\mathcal{O}, \mathcal{O})$ -bimodule (and morphisms thereof) respectively, where $\mathbb{Z}$ is the constant sheaf of the integer ring $\mathbb{Z}$. In particular, the category of $(\mathbb{Z}, \mathbb{Z})$ -bimodules is equivalent to the category of sheaves of abelian groups on $\mathcal{C}$.

Definition B.0.19. Let $\mathcal{E}$ be a topos (Definition B.0.9), and let $p : \mathbf{Set} \rightarrow \mathcal{E}$ be a point of the topos. By definition, p is a geometric morphism, so it consists of an adjoint pair of functors (p^*, p_*) where the inverse image functor $p^* : \mathcal{E} \rightarrow \mathbf{Set}$ is left exact.

1. The inverse image functor $p^* : \mathcal{E} \rightarrow \mathbf{Set}$ is called the **stalk functor associated with the point p** .
2. For any object (sheaf) $F \in \mathcal{E}$, the set $p^*(F)$ is called the **stalk of F at the point p** .

Definition B.0.20 (Locally constant sheaf on a site). Let $(\mathcal{C}, J)$ be a site (Definition B.0.5).

1. Let $\mathcal{A}$ be a (large) category, and let A be an object of $\mathcal{A}$. (♠ TODO: If such a sheafification functor exist, does a sheafification functor exist when restricted to an object U ?) Assume that a sheafification functor (Definition B.0.14)

$$a : \text{PreShv}(\mathcal{C}, J, \mathcal{A}) \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$$

(Definition B.0.8) (Definition 7.1.1) exists (e.g. see Theorem B.0.15) . Let U be an object of $\mathcal{C}$.

A sheaf $\mathcal{F}$ on $\mathcal{C}$ with values in $\mathcal{A}$ is said to be a **locally constant sheaf on U with value A** if there exists a covering sieve (Definition B.0.5) $\{U_i \rightarrow U\}$ of every object U in $\mathcal{C}$ such that for each i , the restriction $\mathcal{F}|_{U_i}$ is isomorphic to the constant sheaf $\underline{A}$ on the slice site $\mathcal{C}_{/U_i}$.

If $\mathcal{C}$ has a final object, then we may say that $\mathcal{F}$ is a **locally constant sheaf with value A** if it is a locally constants sheaf on the final object with value A .

2. Let $\mathcal{O}$ be a sheaf of (not-necessarily commutative) rings on $\mathcal{C}$. A **locally constant $\mathcal{O}$ -module** is a $\mathcal{O}$ -module (Definition B.0.18) $\mathcal{F}$ such that there exists a covering sieve

(Definition B.0.5) $\{U_i \rightarrow U\}$ for every object U in $\mathcal{C}$ such that for each i , the restriction $\mathcal{F}|_{U_i}$ is isomorphic, as an $\mathcal{O}|_{U_i}$ -module, to a constant $\mathcal{O}|_{U_i}$ -module¹⁵.

We additionally say that $\mathcal{F}$ is

- (a) *locally free of rank r over $\mathcal{O}$* if there exists a covering $\{U_i \rightarrow U\}$ such that $\mathcal{F}|_{U_i}$ is isomorphic as a $\mathcal{O}|_{U_i}$ -module to $(\mathcal{O}|_{U_i})^{\oplus r}$ for each i .
- (b) *free of rank r over $\mathcal{O}$* if $\mathcal{F} \cong \mathcal{O}^{\oplus r}$ as $\mathcal{O}$ -modules.
- (c) *of finite type* if there exists a covering $\{U_i \rightarrow U\}$ such that $\mathcal{F}|_{U_i}$ generated by finitely many sections over U_i as an $\mathcal{O}|_{U_i}$ -module. In other words, there is an epimorphism

$$(\mathcal{O}|_{U_i})^{\oplus n_i} \rightarrow \mathcal{F}|_{U_i}$$

of $\mathcal{O}|_{U_i}$ -modules for each i .

In case that $\mathcal{O}$ is the constant sheaf on $\mathcal{C}$ associated to A for some (not-necessarily commutative) ring A , a locally constant $\mathcal{O}$ -module is simply called a *locally constant A -module*.

APPENDIX C. MISCELLANEOUS DEFINITIONS

Definition C.0.1. Let X and Y be sets. A *map* (or *function*) from X to Y is a rule f assigning to each element $x \in X$ exactly one element $f(x) \in Y$. We write $f : X \rightarrow Y$.

We say that X is the *domain* and that Y is the *codomain of f* .

Definition C.0.2 (Monoid). A *monoid* is a semigroup $(M, \cdot)$ such that there exists an element $e \in M$, called the *identity element*, with the property:

$$e \cdot a = a \cdot e = a \quad \text{for all } a \in M.$$

Equivalently, a monoid is a monoid object (Definition C.0.26) in the category of sets (Definition 1.1.7).

Definition C.0.3 (Groups). A *group* is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a binary operation, subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

2. (Identity element) There exists an element $e \in G$ such that for all $g \in G$,

$$e \cdot g = g \cdot e = g.$$

3. (Inverse element) For all $g \in G$ there exists an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

The element e is called the *identity element of G* , and g^{-1} is called the *inverse of g* .

¹⁵The global sections rings $\Gamma(\mathcal{O}|_{U_i})$ (Definition 7.1.5) of the sheaves $\mathcal{O}|_{U_i}$ on the slice sites $\mathcal{C}/_{U_i}$ exist because each $\mathcal{C}/_{U_i}$ has a final object (Definition 2.3.1)(??), so we may speak of constant $\mathcal{O}|_{U_i}$ -modules.

Equivalently, a group is a monoid (Definition C.0.2) with inverse elements.

Equivalently, a group is a group object (Definition C.0.27) in the category of sets (Definition 1.1.7).

A group $(G, \cdot)$ is often simply written as G , when the notation for the binary operation $\cdot$ is clear.

An **abelian group** or synonymously, a **commutative group**, is a group $(G, \cdot)$ whose binary operation $\cdot$ is **abelian** or **commutative**, i.e. satisfies

$$g \cdot h = h \cdot g$$

for all $g, h \in G$.

An abelian group is equivalent to a $\mathbb{Z}$ -module.

Definition C.0.4 (Group homomorphism). Let $(G, \cdot)$ and $(H, *)$ be groups (Definition C.0.3). A map $f : G \rightarrow H$ is called a **group homomorphism** if for all $g_1, g_2 \in G$ one has

$$f(g_1 \cdot g_2) = f(g_1) * f(g_2).$$

The collection of all groups with the group homomorphisms forms a locally small (Definition 1.1.2) category (Definition 1.1.1), called the **category of groups**.

If f is bijective, then f is called a **group isomorphism**. Equivalently, a group isomorphism is an isomorphism (Definition 1.1.13) in the category of groups.

Definition C.0.5 (Topology). Let X be a set. A **topology on X** is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i \in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \dots \cap U_n \in \mathcal{T}$.

If $\mathcal{T}$ is a topology on X , the pair $(X, \mathcal{T})$ is called a **topological space**. Members of $\mathcal{T}$ are called **open sets**.

A subset $C \subseteq X$ is **closed** (also called a **closed subset**) if its complement $X \setminus C$ is an open set in $\mathcal{T}$.

One very often refers to X as a topological space, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as **Top**(X), **Top**(X), or **Top**(X).

Definition C.0.6. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition C.0.5). A map $f : X \rightarrow Y$ is called **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is an open set in X , that is,

$$\forall V \in \mathcal{T}_Y, f^{-1}(V) \in \mathcal{T}_X.$$

Equivalently, f is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .

A *map of topological spaces* usually refers to a continuous map between the topological spaces.

The set of continuous maps from X to Y is sometimes denoted by $C(X, Y)$. Other standard notation include $\text{Hom}_{\text{Top}}(X, Y)$ or $\text{Top}(X, Y)$ coming from more general notation for morphisms between objects in a category (Definition 1.1.1).

Definition C.0.7. A *ring* is a triple $(R, +, \cdot)$ where

1. $(R, +)$ is a commutative group (Definition C.0.3), and
2. $(R, \cdot)$ is a monoid (Definition C.0.2).
3. $\cdot$ is distributive over $+$, i.e. for all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

Equivalently, a ring is a triple $(R, +, \cdot)$ where $+, \cdot : R \times R \rightarrow R$ are binary operations satisfying

1. $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ for all $a, b, c \in R$
2. There exists an element $0 \in R$ such that $a + 0 = a = 0 + a$ for all $a \in R$.
3. For every $a \in R$, there exists an element $-a \in R$ such that $a + (-a) = 0 = (-a) + a$ for all $a \in R$.
4. There exists an element $1 \in R$ such that $a \cdot 1 = a = 1 \cdot a$ for all $a \in R$.
5. For all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

The operation $+$ is often called *addition* and the operation $\cdot$ is often called *multiplication*. Accordingly, the identity element 0 of $+$ is often called the *additive identity* and the identity element 1 of $\cdot$ is often called the *multiplicative identity*.

Remark C.0.8. Some writers might not require a ring to have a multiplicative identity element, i.e. would define a ring so that $(R, +)$ is a commutative group, $(R, \cdot)$ is a semigroup, and $\cdot$ is distributive over $+$. Such writers would call the notion of ring in Definition C.0.7 a *unitary ring* to emphasize the existence of the multiplicative identity 1 .

Definition C.0.9. A *commutative (unital) ring* is a ring (Definition C.0.7) $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. “commutative” algebraists or number theorists), a *ring* refers to a commutative ring as above.

Definition C.0.10. Let $(R, +, \cdot)$ be a not-necessarily commutative ring (Definition C.0.7). A *unit* or *invertible element of R* is an element $u \in R$ such that there exist an element $v \in R$ such that

$$uv = 1 = vu.$$

Such an element v is called the *multiplicative inverse of u* and is often denoted by u^{-1} . If it exists, then it is unique.

The set of units of R forms a group (Definition C.0.3), often denoted by $R^\times$ or R^* , under the multiplication operation $\cdot$. It is called the *group of units* or *unit group* of R .

Definition C.0.11. Let $(R, +, \cdot)$ be a not-necessarily commutative ring (Definition C.0.7). It is called a *division ring*, a *skew field*, or an *sfield*, if it is a nontrivial ring in which every nonzero element $a \in R$ is a unit (Definition C.0.10).

Definition C.0.12 (Field). A *field* is commutative division (Definition C.0.11) ring (Definition C.0.9). In other words, a field is a commutative ring for which all nonzero elements have a multiplicative inverse (Definition C.0.10).

Definition C.0.13. Let $(R, +, \cdot)$ and $(S, +, \cdot)$ be rings (Definition C.0.7), not assumed to be commutative. A function $f : R \rightarrow S$ is called a *ring homomorphism* if for all $r_1, r_2 \in R$ the following properties hold:

1. $f(r_1 + r_2) = f(r_1) + f(r_2)$,
2. $f(r_1 r_2) = f(r_1) f(r_2)$,
3. $f(1_R) = 1_S$ where 1_R and 1_S denote the multiplicative identities in R and S , respectively.

A ring homomorphism is said to be a *ring isomorphism* if it is invertible as a map of sets.

An *R-ring* refers to a ring S equipped with a ring homomorphism $f : R \rightarrow S$.

We note that a ring homomorphism $f : R \rightarrow S$ yields a natural left R -module (Definition 2.1.1) structure on S and a natural right R -module structure on S respectively as follows for $r \in R$ and $s \in S$:

$$\begin{aligned} r \cdot s &= f(r) \cdot s \\ s \cdot r &= s \cdot f(r). \end{aligned}$$

However, these left and right module structures need not yield a two-sided R -module structure.

Definition C.0.14 (Vector space over a field). Let $(k, +, \cdot)$ be a field (Definition C.0.12). A *vector space over k* or a *k -vector space* is a triple $(V, +, \cdot)^{16}$ where

1. $(V, +)$ is an abelian group, and
2. $\cdot$ is a map $k \times V \rightarrow V$, called *scalar multiplication*

such that the following axioms hold for all $a, b \in k$ and all $u, v \in V$:

1. (Compatibility with field multiplication)

$$(ab) \cdot v = a \cdot (b \cdot v).$$

2. (Identity scalar)

$$1 \cdot v = v.$$

3. (Distributivity over vector addition)

$$a \cdot (u + v) = a \cdot u + a \cdot v.$$

¹⁶Note that $+$ and $\cdot$ are abuse of notation here as these are already used for the addition and multiplication of $\cdot$.

4. (Distributivity over scalar addition)

$$(a + b) \cdot v = a \cdot v + b \cdot v.$$

Definition C.0.15. Let F be a field (Definition C.0.12), and let V and W be F -vector spaces (Definition C.0.14). A function $T : V \rightarrow W$ is called a *(homo)morphism of vector spaces over F* , or an *F -linear map*, if for all $u, v \in V$ and all $a, b \in F$, we have

$$T(au + bv) = aT(u) + bT(v).$$

The set of all such morphisms from V to W is denoted by

$$\text{Hom}_F(V, W).$$

Definition C.0.16. Let F be a field (Definition C.0.12), and let V be an F -vector space (Definition C.0.14). A subset $B \subseteq V$ is called a *basis of V* if: (i) B is linearly independent over F , and (ii) B spans V .

If B is a basis, we define the *dimension of V over F* (or *rank of V over F*), denoted by

$$\dim_F(V),$$

(♠ TODO: cardinality) to be the cardinality of B . This value is uniquely determined by V and F .

Definition C.0.17 (Partially ordered set). 1. A *partially ordered set* (or *poset*), or *ordered set* is a pair $(P, \leq)$ where P is a set and

$$\leq : P \times P \rightarrow \{\text{true}, \text{false}\}$$

is a binary relation on P satisfying the following axioms for all $a, b, c \in P$:

- *Reflexivity*: $a \leq a$,
- *Antisymmetry*: if $a \leq b$ and $b \leq a$, then $a = b$,
- *Transitivity*: if $a \leq b$ and $b \leq c$, then $a \leq c$.

The relation $\leq$ is called an *order* or a *partial order*

2. A partially ordered set $(P, \leq)$ is called a *directed partially ordered set* if for every pair $a, b \in P$, there exists $c \in P$ such that

$$a \leq c \quad \text{and} \quad b \leq c.$$

3. A partially ordered set $(P, \leq)$ is called a *codirected partially ordered set* (or *downward directed poset*) if for every pair $a, b \in P$, there exists $d \in P$ such that

$$d \leq a \quad \text{and} \quad d \leq b.$$

Lemma C.0.18. Let $(P, \leq)$ be a nonempty poset (Definition C.0.17).

1. Regarding P as a category whose objects are the elements of P and such that there is a unique arrow $a \rightarrow b$ if and only if $a \leq b$, the category is filtered.
2. Every nonempty small (Definition 1.1.2) thin filtered category (Definition 2.2.12) corresponds to a poset in this way.
3. Moreover, the poset P is directed (Definition C.0.17) if and only if the category is filtered. The poset P is codirected (Definition C.0.17) if and only if the category is cofiltered.

Definition C.0.19. Let S be a set. The *free group generated by S* is a pair $(F(S), \iota)$ consisting of a group (Definition C.0.3) $F(S)$ and a function $\iota : S \rightarrow F(S)$, satisfying the following universal property: for any group G and any function $f : S \rightarrow G$, there exists a unique group homomorphism $\varphi : F(S) \rightarrow G$ such that the diagram commutes (i.e., $\varphi \circ \iota = f$). The standard notation for the free group on S is $F(S)$ or sometimes $\langle S \rangle$. Elements of $F(S)$ are uniquely represented as reduced words in the alphabet $S \cup \{s^{-1} \mid s \in S\}$.

Definition C.0.20. Let S be a set. The *free abelian group generated by S* is the abelian group consisting of all formal linear combinations of elements of S with integer coefficients, such that only finitely many coefficients are nonzero. This group is denoted by $\mathbb{Z}[S]$ or alternatively as the direct sum

$$\mathbb{Z}^{(S)} := \bigoplus_{s \in S} \mathbb{Z}s$$

It satisfies the universal property that for any abelian group A and any function $f : S \rightarrow A$, there exists a unique group homomorphism $\psi : \mathbb{Z}[S] \rightarrow A$ extending f .

Definition C.0.21 (Homotopy groups of a pointed topological space). For any pointed topological space (Definition 1.1.10) (X, x_0) and integer $n \geq 0$, the *n -th homotopy set of X at x_0* , denoted $\pi_n(X, x_0)$, is defined as the set of all homotopy classes (rel. ∂I^n) (Definition C.0.31) of paired maps (Definition C.0.45)

$$f : (I^n, \partial I^n) \rightarrow (X, \{x_0\}),$$

where $I^n = [0, 1]^n$ and ∂I^0 is taken to be the singleton I^0 instead of the literal boundary thereof. Equivalently, one can also define it as the set of all homotopy classes of pointed maps (Definition 1.1.10)

$$f : (S^n, s_0) \rightarrow (X, x_0),$$

where S^n is the n -sphere (♠ TODO:) and $s_0 \in S^n$ is a choice of base point. In other words, $\pi_n(X, x_0)$ is by definition $[(S^n, s_0), (X, x_0)]_*$ (Definition C.0.31).

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of pointed maps, and for $n \geq 2$, it is abelian. As such, for $n \geq 1$, it is appropriate to refer to $\pi_n(X, x_0)$ as the *n th homotopy group of (X, x_0)* .

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

For every $n \geq 0$, the assignment $(X, x_0) \mapsto \pi_n(X, x_0)$ is functorial — in fact

1. For $n = 0$, it is a functor $\mathbf{Top}_* \rightarrow \mathbf{Sets}$ (Definition 1.1.7),
2. For $n = 1$, it is a functor $\mathbf{Top}_* \rightarrow \mathbf{Grp}$ (Definition 1.1.8),
3. For $n \geq 2$, it is a functor $\mathbf{Top}_* \rightarrow \mathbf{Ab}$ (Definition 1.1.8)).

Given a map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces, the induced map $\pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$ is often denoted by f_* .

We further note that the set $\pi_n(X, x_0) = [(S^n, s_0), (X, x_0)]_*$ is identifiable with $\pi_0(\mathrm{Map}_*((S^n, s_0), (X, x_0)))$.

Definition C.0.22. Let R be a (not necessarily commutative) ring (Definition C.0.7). A proper two-sided ideal $P \subseteq R$ is called a **prime ideal** if the following equivalent conditions holds:

1. If I, J are left ideals and $IJ \subseteq P$, then $I \subseteq P$ or $J \subseteq P$.
2. If I, J are right ideals and $IJ \subseteq P$, then $I \subseteq P$ or $J \subseteq P$.
3. If I, J are two-sided ideals and $IJ \subseteq P$, then $I \subseteq P$ or $J \subseteq P$.
4. If $x, y \in R$ with $xRy \subseteq \mathfrak{p}$, then $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$.

A proper left/right/two-sided ideal $M \subsetneq R$ is called **maximal** if there exists no other left/right/two-sided ideal $J \subseteq R$ such that $M \subsetneq J \subsetneq R$. Equivalently,

- a left/right ideal M of R is maximal if and only if the quotient module R/M (Definition 2.1.5) is a simple left/right R -module.
- a two-sided ideal M of R is maximal if and only if the quotient ring R/M is a simple ring.

Definition C.0.23 (Affine scheme). Let A be a commutative ring with unity (Definition C.0.9). Define the set $\text{Spec}(A)$ to be the set of all prime ideals (Definition C.0.22) of A . Equip it with the **Zariski topology**, which is the topology (Definition C.0.5) whose closed sets are given by **vanishing loci**

$$V(I) = \{\mathfrak{p} \in \text{Spec}(A) : I \subseteq \mathfrak{p}\}$$

for ideals $I \subseteq A$. Define the sheaf $\mathcal{O}_{\text{Spec}(A)}$, called the **structure sheaf of $\text{Spec } A$** , by

$$\mathcal{O}_{\text{Spec}(A)}(U) = \{\text{locally defined fractions of elements of } A \text{ on } U\},$$

for each open set $U \subseteq \text{Spec}(A)$. More precisely, $\mathcal{O}_{\text{Spec}(A)}(U)$ can be described as the set of functions $s : U \rightarrow \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ such that

1. For each $\mathfrak{p} \in U$, the value $s(\mathfrak{p})$ lies in the localization $A_{\mathfrak{p}}$
2. For each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ containing $\mathfrak{p}$ and elements $a, f \in A$ such that for all $\mathfrak{q} \in V$,
 - $f \notin \mathfrak{q}$, so that the fraction $\frac{a}{f}$ is well-defined in $A_{\mathfrak{q}}$.
 - $s(\mathfrak{q}) = \frac{a}{f}$ in $A_{\mathfrak{q}}$.

It is the case that the stalk (Definition 7.5.1) at $\mathfrak{p} \in \text{Spec}(A)$ is canonically the localization $A_{\mathfrak{p}}$. Then $(\text{Spec}(A), \mathcal{O}_{\text{Spec}(A)})$ is a locally ringed space (Definition 7.7.19), called the **affine scheme associated to A** .

Moreover, given $f \in A$, we define the locus $D(f)$ by

$$D(f) = \text{Spec } A \setminus V((f)) = \{\mathfrak{p} \in \text{Spec } A : f \notin \mathfrak{p}\}$$

Definition C.0.24. Let $\mathcal{C}$ be a category (Definition 1.1.1), let Z be an object, and let X, Y be objects of $\mathcal{C}$ over Z , i.e. morphisms $X \rightarrow Z$ and $Y \rightarrow Z$ are fixed. A **cartesian product of X and Y over Z in $\mathcal{C}$** (or **fiber product** or **pullback diagram**) is an object, often denoted by $X \times_Z Y$, with **projection morphisms** $X \times_Z Y \rightarrow X$ and $X \times_Z Y \rightarrow Y$ that are universal. More precisely, for any object T of $\mathcal{C}$ and morphisms $f_X : T \rightarrow X$, $f_Y : T \rightarrow Y$ such that

the compositions $T \xrightarrow{f_X} X \rightarrow Z$ and $T \xrightarrow{f_Y} Y \rightarrow Z$ agree, there exists a unique morphism $u : T \rightarrow X \times_Z Y$ such that the following diagram commutes:

$$\begin{array}{ccccc}
 T & & & & \\
 \text{\scriptsize f_X} \swarrow & & & & \searrow \\
 & X \times_Z Y & \longrightarrow & X & \\
 \text{\scriptsize f_Y} \searrow & \downarrow & & \downarrow & \\
 & Y & \longrightarrow & Z &
 \end{array}$$

(Note: In the original image, there is a curved arrow from T to X labeled f_X , a curved arrow from T to Y labeled f_Y , and a dotted arrow from T to $X \times_Z Y$ labeled u .)

Equivalently, $X \times_Z Y$ is the limit (Definition 2.2.8) of the diagram (Definition 2.2.6)

$$\begin{array}{ccc}
 & X & \\
 & \downarrow & \\
 Y & \longrightarrow & Z
 \end{array}$$

in $\mathcal{C}$.

The commutative diagram

$$\begin{array}{ccc}
 X \times_Z Y & \longrightarrow & X \\
 \downarrow & & \downarrow \\
 Y & \longrightarrow & Z
 \end{array}$$

may be referred to as a **cartesian square**.

Definition C.0.25. Let $\mathcal{C}$ be a (large) category (Definition 1.1.1). A **semigroup object in $\mathcal{C}$** is an object $A \in \mathcal{C}$ such that the product (Definition 2.2.1) $A \times A$ exists in $\mathcal{C}$ together with a morphism

$$\mu : A \times A \rightarrow A,$$

called the **multiplication morphism** such that the associativity diagram

$$\begin{array}{ccc}
 A \times A \times A & \xrightarrow{\mu \times \text{id}_A} & A \times A \\
 \text{id}_A \times \mu \downarrow & & \downarrow \mu \\
 A \times A & \xrightarrow{\mu} & A
 \end{array}$$

commutes.

The semigroup object structure (A, μ, η, ι) is said to be **abelian** or **commutative** if the morphisms $\mu : A \times A \rightarrow A$ and $\mu \circ \tau_{A,A} : A \times A \rightarrow A$ coincide, where $\tau_{A,A} : A \times A \rightarrow A \times A$ is the symmetry morphism swapping the two factors.

Definition C.0.26. Let $\mathcal{C}$ be a (large) category (Definition 1.1.1) with a final object (Definition 2.3.1). A **monoid object in $\mathcal{C}$** is a semigroup object (Definition C.0.25) (A, μ) together with a **unit morphism**

$$\eta : 1 \rightarrow A$$

such that the products (Definition 2.2.1) $1 \times A$ and $A \times 1$ exist and the unit diagrams

$$\begin{array}{ccc} 1 \times A & \xrightarrow{\eta \times \text{id}_A} & A \times A \\ & \searrow \text{pr}_2 & \swarrow \mu \\ & A & \end{array}$$

$$\begin{array}{ccc} A \times 1 & \xrightarrow{\text{id}_A \times \eta} & A \times A \\ & \searrow \text{pr}_1 & \swarrow \mu \\ & A & \end{array}$$

commute.

Definition C.0.27. Let $\mathcal{C}$ be a (large) category (Definition 1.1.1) with a final object (Definition 2.3.1).

A **group object in $\mathcal{C}$** is a monoid object (Definition C.0.26) (A, μ, η) together with a **inverse morphism**

$$\iota : A \rightarrow A$$

such that the diagrams

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \times A \\ & \searrow \eta \circ !_A & \downarrow \mu \circ (\text{id}_A \times \iota) \\ & & A \end{array}$$

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \times A \\ & \searrow \eta \circ !_A & \downarrow \mu \circ (\iota \times \text{id}_A) \\ & & A \end{array}$$

commute, where $\Delta : A \rightarrow A \times A$ is the diagonal and $!_A : A \rightarrow 1$ is the unique morphism.

An **abelian group object in $\mathcal{C}$** is a group object as above that also satisfies... (♠ TODO:)

A **morphism of group objects in $\mathcal{C}$** is ...

The category of group objects may be denoted by notations such as **Grp**($\mathcal{C}$). The category of abelian group objects may be denoted by notations such as **Ab**($\mathcal{C}$).

Definition C.0.28 (Power Set). Let A be a set. The **power set of A** , denoted by **$\mathcal{P}(A)$** , is the set of all subsets of A :

$$\mathcal{P}(A) := \{ B \mid B \subseteq A \}.$$

Equivalently, every element of $\mathcal{P}(A)$ is itself a set B satisfying $B \subseteq A$. Under the axiom of power set, note that the $\mathcal{P}(A)$ exists.

Definition C.0.29. Let R be a (not-necessarily commutative) ring with unity (Definition C.0.7). An *R -algebra* is a ring A together with a ring homomorphism (Definition C.0.13)

$$\varphi : R \rightarrow A$$

into the center $Z(A)$ of A (so that $\varphi(r)$ commutes with every element of A for all $r \in R$), such that $\varphi(1_R) = 1_A$. The ring homomorphism φ is called the *structure map* of the algebra.

Equivalently, an R -algebra consists of a ring A endowed with a two-sided R -module (Definition 2.1.1) structure for which the scalar multiplication satisfies

$$r \cdot (ab) = (r \cdot a)b = a(r \cdot b) \quad \text{for all } r \in R, a, b \in A.$$

In particular, any ring homomorphism between commutative rings (Definition C.0.9) specifies an algebra structure.

Definition C.0.30 (Homotopy of maps of topological spaces). Let X and Y be topological spaces and let $K \subseteq X$ be a subset. Let $C(X, Y)$ denote the set of all continuous maps $f : X \rightarrow Y$.

1. A *homotopy between two maps $f, g \in C(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $x \in K$ and $t \in [0, 1]$,

$$H(x, t) = f(x) = g(x).$$

If such an H exists, we say f and g are *homotopic relative to K* , and we write $f \simeq g \text{ rel } K$ or $f \simeq_K g$; this is an equivalence relation.

In the case that $K = \emptyset$, a homotopy relative to K is simply called a *homotopy*. We write $f \simeq g$ if a homotopy between f and g exists.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.1.10) and let $K \subseteq X$ be a subset with $x_0 \in K$. Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ satisfying $f(x_0) = y_0$.

A *homotopy of based maps $f, g \in C_*(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

in particular fixing the basepoint throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

If such an H exists, we say f and g are *based homotopic relative to K* , and we write $f \simeq g \text{ rel } K$ or $f \simeq_K g$. This is an equivalence relation.

When $K = \{x_0\}$, one calls a homotopy of based maps relative to K simply as a *homotopy of based maps* or *based homotopy*. We write $f \simeq g$ if such a homotopy exists.

Definition C.0.31 (Homotopy class of maps relative to a subset). Let X and Y be topological spaces (Definition C.0.5) and let $K \subseteq X$. Let $C(X, Y)$ denote the set of all continuous maps (Definition C.0.6) $f : X \rightarrow Y$.

1. Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class relative to K* or that f and g are *homotopic relative to K* if there exists a homotopy relative to K (Definition C.0.30)

$$H : X \times [0, 1] \rightarrow Y$$

such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and

$$H(k, t) = f(k) = g(k) \quad \text{for all } k \in K, t \in [0, 1].$$

The *homotopy class of maps relative to K* containing a map $f : X \rightarrow Y$ is denoted by $[f]_K$.

Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\emptyset$.

The *homotopy class of maps* containing a map $f : X \rightarrow Y$ is denoted by $[f]$.

The set of homotopy classes of maps may often be denoted by $[X, Y]$.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.1.10) and let $K \subseteq X$ be a subset containing x_0 . Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ with $f(x_0) = y_0$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class relative to K* or that f and g are *homotopic relative to K* if there exists a homotopy of based maps relative to K

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

particularly ensuring the basepoint is fixed throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

In the case that $K = \{x_0\}$, one simply says that f and g are *homotopic (as based maps)*.

The *homotopy class relative to K* containing $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]_K$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\{x_0\}$.

The *homotopy class* containing a map $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]$.

The set of homotopy classes of pointed maps $(X, x_0) \rightarrow (Y, y_0)$ may often be denoted by $[(X, x_0), (Y, y_0)]$ or by notations such as $[X, Y]_*$, $[X, Y]_\bullet$, or $[X, Y]$ if the base points are clear.

Definition C.0.32. The *homotopy category of topological spaces*, denoted $\mathbf{hTop}$, is the category whose objects are topological spaces (Definition C.0.5) and whose morphisms are homotopy classes (Definition C.0.31) of continuous maps (Definition C.0.6). In other words, for objects X and Y , the set of morphisms is defined as $\text{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y] = C(X, Y) / \simeq$.

Proposition C.0.33. Composition in the homotopy category of topological spaces (Definition C.0.32) is well-defined. If $f_1, f_2 : X \rightarrow Y$ are homotopic and $g_1, g_2 : Y \rightarrow Z$ are homotopic, then the compositions $g_1 \circ f_1$ and $g_2 \circ f_2$ are homotopic as maps from X to Z . That is, $[g] \circ [f] = [g \circ f]$ is independent of the choice of representatives.

Theorem C.0.34. There exists a canonical functor $Q : \mathbf{Top} \rightarrow \mathbf{hTop}$ which is the identity on objects and maps each continuous map f to its homotopy class $[f]$. This functor is full and essentially surjective.

Definition C.0.35. Let $k \in \mathbb{N}_0 \cup \{\infty\}$ be fixed. Let $(M, \mathcal{A}_M)$ and $(N, \mathcal{A}_N)$ be C^k -manifolds with boundary (Definition C.0.37) of dimensions n and m , respectively, where M, N are topological manifolds with boundary and $\mathcal{A}_M$ and $\mathcal{A}_N$ are C^k -atlases whose charts map to open subsets of the closed half-spaces $\mathbb{H}^n$ and $\mathbb{H}^m$.

A *C^k -morphism* (or *C^k -map*) between M and N is a continuous map (Definition C.0.6)

$$f : M \rightarrow N$$

such that for every $p \in M$ there exist charts $(U, \varphi) \in \mathcal{A}_M$ with $p \in U$ and $(V, \psi) \in \mathcal{A}_N$ with $f(p) \in V$ satisfying

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \rightarrow \psi(V)$$

is a C^k -map between open subsets of the closed half-spaces $\mathbb{H}^n$ and $\mathbb{H}^m$, i.e.,

$$\psi \circ f \circ \varphi^{-1} \in C^k(\varphi(U \cap f^{-1}(V)), \psi(V)).$$

If f is a homeomorphism and its inverse $f^{-1} : N \rightarrow M$ is also a C^k -morphism, then f is called a *C^k -diffeomorphism*. We let $C^k(M, N)$ denote the space of C^k -maps $M \rightarrow N$. We let $C^k(M)$ denote the space of *C^k -functions*, i.e., the C^k -maps $M \rightarrow \mathbb{R}$.

In particular, we may speak of these notions when M and N are C^k -manifolds without boundary (Definition C.0.37).

Remark C.0.36. The notations $C^k(M, N)$ (and $C^k(M)$) agrees with the usual notations $C^k(M, N)$ and $C^k(M)$ in the case that M is an open subset of $\mathbb{R}^n$.

Definition C.0.37. Let $k \in \mathbb{N}_0 \cup \{\infty\}$ be fixed. An *n -dimensional C^k/k -differentiable (real) manifold with boundary (resp. without boundary)* is a pair $(M, \mathcal{A})$, where M is a topological manifold with boundary (resp. without boundary) of dimension n and $\mathcal{A}$ is a C^k -atlas on M .

The atlas $\mathcal{A}$ is usually taken to be maximal with respect to C^k -compatibility, meaning it contains every C^k -chart compatible with all charts in $\mathcal{A}$.

Note that a C^0 -manifold is simply a topological manifold and that a C^∞ -manifold is synonymously referred to as a *smooth/differentiable (real) manifold*.

Definition C.0.38 (Opposite ring). Let $R = (R, +, \cdot, 0, 1)$ be a ring (Definition C.0.7) with addition $+$, multiplication $\cdot$, additive identity 0 , and multiplicative identity 1 (not necessarily commutative).

The *opposite ring of R* , denoted R^{op} , is the ring with the same underlying set R and the same addition $+$ and additive identity 0 , but with multiplication defined by

$$r \star s := s \cdot r$$

for all $r, s \in R$.

That is, multiplication in R^{op} is the multiplication of R reversed in order.

If R is commutative (Definition C.0.9), then R and R^{op} are naturally isomorphic to each other.

Definition C.0.39 (Subspace topology and subspace). Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$ a subset. The topology induced on Y by $\mathcal{T}$, called the *subspace topology*, is defined as

$$\mathcal{T}_Y := \{U \cap Y : U \in \mathcal{T}\}.$$

The pair $(Y, \mathcal{T}_Y)$ is called a *subspace of $(X, \mathcal{T})$* .

Definition C.0.40 (Separation axioms). Let $(X, \mathcal{T})$ be a topological space (Definition C.0.5).

- $(X, \mathcal{T})$ is T_0 (*Kolmogorov*) if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is T_1 (*Fréchet*) if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is T_2 or *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is *regular* if it is T_1 and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is T_3 (regular Hausdorff) if it is T_1 and regular.
- $(X, \mathcal{T})$ is *completely regular* if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is $T_{3\frac{1}{2}}$ (completely regular Hausdorff) if it is T_1 and completely regular.
- $(X, \mathcal{T})$ is *normal* if it is T_1 and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.
- $(X, \mathcal{T})$ is T_4 (normal Hausdorff) if it is T_1 and normal.
- $(X, \mathcal{T})$ is T_5 (completely normal Hausdorff) if it is T_1 and completely normal.
- $(X, \mathcal{T})$ is *perfectly normal* if every closed set is a G_δ (countable intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is T_6 (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

Definition C.0.41 (Principal ideal ring/domain (PID)). Let R be a commutative unital ring (Definition C.0.9). Then R is a *principal ideal ring (PIR)* if every ideal of R is principal. If R is additionally an , then R is said to be a *principal ideal domain (PID)*

Definition C.0.42 (Finitely generated modules and bimodules). Let R and S be (not necessarily commutative) rings (Definition C.0.7).

1. An R - S -bimodule M is *finitely generated* if it has a finite spanning set.
2. A left/right/two-sided R -module is *finitely generated* if has a finite spanning set, or equivalently if its natural bimodule structure (Definition 2.1.1) is finitely generated.

Definition C.0.43. An *equivalence relation on a set X* is a binary relation on X that is reflexive, symmetric, and transitive.

Definition C.0.44. A *pair of topological spaces* (or simply a *pair*) (X, A) consists of a topological space (Definition C.0.5) X and a subspace (Definition C.0.39) $A \subseteq X$. If $A = \emptyset$, the pair $(X, \emptyset)$ is often identified with the space X itself.

These form a category, see Definition C.0.46.

Definition C.0.45. Let (X, A) and (Y, B) be pairs of topological spaces (Definition C.0.44). A *map of pairs*

$$f : (X, A) \rightarrow (Y, B)$$

is a continuous map (Definition C.0.6) $f : X \rightarrow Y$ such that the image of A is contained in B , i.e., $f(A) \subseteq B$.

A map of pairs may also be called a *paired map*.

Definition C.0.46. The *category of pairs of topological spaces*, often denoted by $\mathbf{Top}_{\text{pair}}$ or $\mathbf{Top}_{\text{rel}}$ (rel for relative), is the category (Definition 1.1.1) defined by:

- **Objects:** Pairs (Definition C.0.44) of topological spaces (X, A) .
- **Morphisms:** Maps of pairs (Definition C.0.45) $f : (X, A) \rightarrow (Y, B)$.
- **Composition:** The composition of morphisms is the usual composition of continuous maps, which naturally preserves the subset relation: if $f(A) \subseteq B$ and $g(B) \subseteq C$, then $(g \circ f)(A) \subseteq C$.
- **Identity:** For each object (X, A) , the identity morphism is the identity map $\text{id}_X : (X, A) \rightarrow (X, A)$.

It is a locally small category (Definition 1.1.2).

Definition C.0.47. Let X be a topological space (Definition C.0.5) and let $\{U_i\}_{i \in I}$ be an open cover of X . An *open refinement of $\{U_i\}_{i \in I}$* is an open cover $\{V_j\}_{j \in J}$ of X such that for every $j \in J$, there exists an $i \in I$ with $V_j \subseteq U_i$.

Definition C.0.48. Let X be a topological space (Definition C.0.5) and let $\{U_i\}_{i \in I}$ be an open cover of X . The open cover $\{U_i\}_{i \in I}$ is said to be *locally finite* if for every point $x \in X$, there exists an open neighborhood W of x such that $W \cap V_j \neq \emptyset$ for only finitely many indices $j \in J$.

Definition C.0.49. A topological space is *paracompact* if it is Hausdorff (Definition C.0.51) and every open cover of X admits an open refinement (Definition C.0.47) that is locally finite (Definition C.0.48).

Theorem C.0.50. The following are paracompact (Definition C.0.49) topological spaces:

1. Metric spaces (Stone's theorem)
2. Topological manifolds
3. Compact Hausdorff (Definition C.0.51) spaces
4. CW-complexes (Definition 3.3.27)

Definition C.0.51 (Separation axioms). Let $(X, \mathcal{T})$ be a topological space (Definition C.0.5).

- $(X, \mathcal{T})$ is T_0 (*Kolmogorov*) if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is T_1 (*Fréchet*) if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is T_2 or *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is *regular* if it is T_1 and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is T_3 (regular Hausdorff) if it is T_1 and regular.
- $(X, \mathcal{T})$ is *completely regular* if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is $T_{3\frac{1}{2}}$ (completely regular Hausdorff) if it is T_1 and completely regular.
- $(X, \mathcal{T})$ is *normal* if it is T_1 and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.
- $(X, \mathcal{T})$ is T_4 (normal Hausdorff) if it is T_1 and normal.
- $(X, \mathcal{T})$ is T_5 (completely normal Hausdorff) if it is T_1 and completely normal.
- $(X, \mathcal{T})$ is *perfectly normal* if every closed set is a G_δ (countable intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is T_6 (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

Definition C.0.52 (Quasi-Compact Scheme). A *quasi-compact scheme* X is a scheme (Definition 7.7.22) such that the underlying topological space (Definition C.0.5) of X is quasi-compact. That is, every open cover of X admits a finite subcover.

Equivalently, X is quasi-compact if it can be covered by finitely many affine open subschemes $\text{Spec } A_i$ (Definition C.0.23).

Definition C.0.53 (Quasi-separated morphism of schemes). Let $f : X \rightarrow Y$ be a morphism of schemes.

(♠ **TODO: closed immersion**) The morphism f is *quasi-separated* (resp. *separated*) if the diagonal morphism

$$\Delta_f : X \rightarrow X \times_Y X$$

is quasi-compact (resp. a closed immersion). In this case, we can also say that X is a *quasi-separated Y -scheme* or a *quasi-separated scheme over Y* . A *quasi-separated scheme* refers to a quasi-separated scheme over $\text{Spec } \mathbb{Z}$ (Definition C.0.23).

Intuitively, f is quasi-separated if the intersection of two affine open subsets in X which map into an affine open of Y can be covered by finitely many affine opens.

Definition C.0.54. A topological space (Definition C.0.5) is *contractible* if it is homotopy equivalent (Definition 3.4.2) to a point.

Definition C.0.55. (♠ **TODO: contractible**) Let X be a topological space (Definition 1.1.9). X is *locally contractible* if every point $x \in X$ has a basis of contractible open neighborhoods. Since every contractible space is simply connected, every locally contractible space is locally simply connected. (♠ **TODO: simply connected space**)

Definition C.0.56. A topological space (Definition 1.1.9) X is *semi-locally simply connected* if for every point $x \in X$, there exists an open neighborhood U of x such that the inclusion-induced homomorphism (Definition C.0.4) on fundamental groups (Definition C.0.21)

$$i_* : \pi_1(U, x) \rightarrow \pi_1(X, x)$$

is the trivial homomorphism.

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